

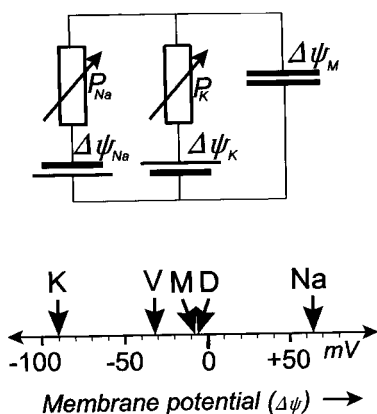


Fig. 3.25. An electrical circuit as a model illustrating the $\text{Na}^+ - \text{K}^+$ diffusion potential of a cell as the result of a sodium ($\Delta\psi_{\text{Na}}$), and a potassium ($\Delta\psi_{\text{K}}$) battery. In the lower part of the figure, possible potential alterations are illustrated for the case of human erythrocytes in a solution containing 145 mM NaCl and 5 mM KCl. K $\Delta\psi_{\text{K}}$, V valinomycin induced diffusion potential, M potential of untreated erythrocytes corresponding to $\Delta\psi_{\text{M}}$, D position of the Donnan potential, Na $\Delta\psi_{\text{Na}}$

considered as storage batteries, or electrical accumulators having opposite polarities. The permeability characteristics of the ions are expressed in this model as conductivities of the variable resistors, or potentiometers through which these accumulators are discharged. If the resistance is low, then a large discharge current would flow, and if the accumulator is not recharged continuously, it would soon be empty. In fact, the permeabilities P_{Na} and P_{K} are usually so low that the electrochemical gradient of the living cell will persist for hours or even days. The effective membrane potential in this model is represented by the voltage difference across the capacitor $\Delta\psi_{\text{M}}$. This capacitor represents the capacity of the membrane (see Sect. 2.5.5). If P_{Na} and P_{K} have about the same value, then $\Delta\psi_{\text{M}}$ will be very small. If they differ, a membrane potential ($\Delta\psi_{\text{M}}$) will be established according to Eqs. (3.4.11) and (3.4.12).

Figure 3.25 demonstrates membrane potentials which can be induced in human erythrocytes. In this case the Nernst potentials for potassium and sodium give the limits of these possible shifts. They range approximately between -95 and $+65$ mV. The actual membrane potential of these cells *in vivo* is found to be -9 mV (M), and is only a little greater than the Donnan potential (D) which would result if the cell achieved a thermodynamic equilibrium (see Fig. 3.17). If the cells are treated with valinomycin, the membrane potential falls to about -35 mV (V). Valinomycin is an ionophore that is rapidly incorporated into the membrane causing a highly selective increase of potassium permeability. It will not reach the limiting value of the Nernst potential of potassium, because the values of P_{Cl} and P_{Na} are not negligible, as was assumed for Eq. (3.4.11). However, it is shifted in this direction.

Even if these sorts of potential alterations are possible without a significant change of concentration profiles, they must in fact be accompanied by a certain transmembrane shift of charges. It is easy to show that this charge flux is extremely small. For this we calculate the charge transfer across the membrane capacitor, which is required to adjust these potential differences ($\Delta\psi_{\text{M}}$ in Fig. 3.25). Let us ask the question: how many charges must be displaced in the cell

membrane with a specific capacity of 10^{-2} F m^{-2} (see Sect. 2.5.5) in order to generate a transmembrane potential $\Delta\psi_M = 0.1 \text{ V}$?

Equation (2.5.19) gives the corresponding relation for a capacitor. This enables us to calculate this surface charge density (σ) as a function of the transmembrane potential ($\Delta\psi$) and specific capacity (C_{sp}):

$$\sigma = C_{\text{sp}} \Delta\psi = 10^{-3} \text{ C m}^{-2}. \quad (3.4.13)$$

This value can be converted into charge equivalents of ions, using the Faraday constant (F):

$$\frac{\sigma}{F} = \frac{10^{-3}}{9.65 \cdot 10^4} \approx 10^{-8} \text{ charge equivalents} \cdot \text{m}^{-2}.$$

The resulting charge density, so far, is very small. Considering a certain geometry of the cell, for example a sphere, or in the case of a neuron, a cylinder, one can easily transform this number into a concentration shift. This type of calculation shows that the amount of ions which really must be exchanged to induce an action potential is negligibly small in relation to the total amount of ions in the cell.

This example demonstrates a most important element in the functional arrangement of the living cell: an ion pump driven by metabolic energy accumulates electrochemical energy by generating a concentration gradient of sodium and potassium. This electrochemical energy can be converted into electrical energy, alerting the membrane permeabilities (for example: P_K and P_{Na}). In this way a wide-ranging control of the electric field in the cell membrane is possible. It must be emphasized that this control is possible without any sizeable input of energy and can be realized in milliseconds.

Such permeability changes can be induced by the cell itself as well as by external influences.

As mentioned before, there are many ion-selective transporters in the cell which are controlled by internal calcium concentrations, by internal pH, by mechanical tension of the membrane, or by modifications of other parameters. Diffusion potentials may also result from an interaction between the cell and specific drugs, or may be triggered locally through mechanical contacts with surfaces or particles, such as viruses. These alterations of membrane potentials caused by local permeability changes can induce electric fields in the plane of the membrane itself.

In the next section we will consider the action potential of nerve cells as a classical example of the feedback loop between an electric field and ionic permeability in more detail. Recently, the interest in the transmembrane potential of the cell as a regulator of cellular events has greatly increased. This concerns the size of the membrane potential in various cells, as well as its time dependence. Although action potentials have a special significance in signal transfer in neurons, they occur also in many other cells.

In Fig. 3.26 quite interesting correlations of membrane potential and the state of various animal cells are illustrated. In contrast to cells with active proliferation rates like cancer cells or cells of embryos, indicating a transmembrane potential between -10 and -30 mV, non-dividing cells, like neurons or skeletal muscle cells show membrane potentials between -70 to -90 mV. The transmembrane potential of cells which pass through a state of proliferation falls before mitosis takes place. It is not yet clear whether this reflects a regulatory mechanism of the cell, or whether it is only a phenomenon that accompanies such a mechanism.

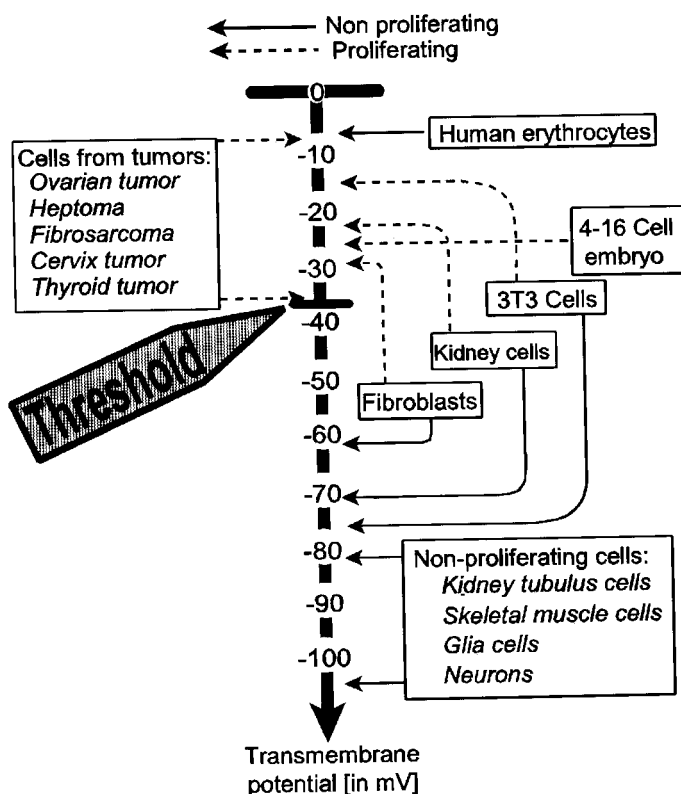


Fig. 3.26. The transmembrane potential of normal animal cells (right) and transformed tumor cells (left). It can be seen that proliferating cells indicate a membrane potential which is above the threshold value of -37 mV. Cells transiently arriving at a proliferating state lower their absolute amount of the potential. The human erythrocyte, as a non-nucleated cell with special physiological functions appears to be an exception. (Drawn according to values from Bingelli and Weinstein 1986)

In fact, in many cases alterations in the electrical field of a membrane seem to be of functional importance. The following mechanisms may cause this:

- The transverse component of an electrical field in the membrane may affect the functional state of intrinsic molecules. Dipole orientations, for example, may modify the function of transport or other functional proteins, phase transitions in the lipid components of the membrane can be influenced by the field, or a transverse shift of small charged molecules can occur.
- The lateral component of the field can cause a displacement in its mosaic structure. This could lead to a local change in the mechanical properties of the membrane causing vesiculation, spike formation etc.
- The electric field can influence local ionic concentrations, as well as local pH values in close proximity to the membrane which, in turn, could affect transport processes, biochemical reactions at the membrane surface and receptor properties.

Further reading: Role of membrane potential: Glaser 1996; voltage dependent transports: Keynes 1994; Sigworth 1994

3.4.3

The Action Potential

In the previous section we described the possibility of cells using the electrochemical gradient of potassium and sodium ions which is built up by active transport, to trigger various amounts of membrane potential simply by changing their permeabilities. This mechanism is expressed most clearly in nerve and muscle cells. This is why excitation phenomena were detected first in these cells.

Particular progress was achieved following the rediscovery of the giant axon of the squid in 1937 by J. Z. Young, and its subsequent introduction for biophysical measurements by K. S. Cole. The use of these giant axons have made it possible to apply the method of the voltage clamp. In this technique, the membrane resistance is determined at various fixed transmembrane potentials, generated by microelectrodes. Recently, using patch-clamp techniques it has been possible to investigate the kinetics of these permeability alterations in extremely small membrane areas.

Because of the large number of existing textbooks on nerve physiology we will concentrate here just on some biophysical principles of the process of nerve excitation.

The action potentials of various nerve and muscle cells as illustrated in Fig. 3.27, can be qualitatively explained using the electrical scheme of Fig. 3.25 which was discussed in the previous section. The non-excited nerve shows a very low sodium permeability (P_{Na}), its resting potential, therefore, was determined chiefly by the diffusion potential of potassium which is negative. After excitation the membrane permeability for ions increases abruptly, whereas the sodium permeability rises more quickly than that of potassium. For a short time,

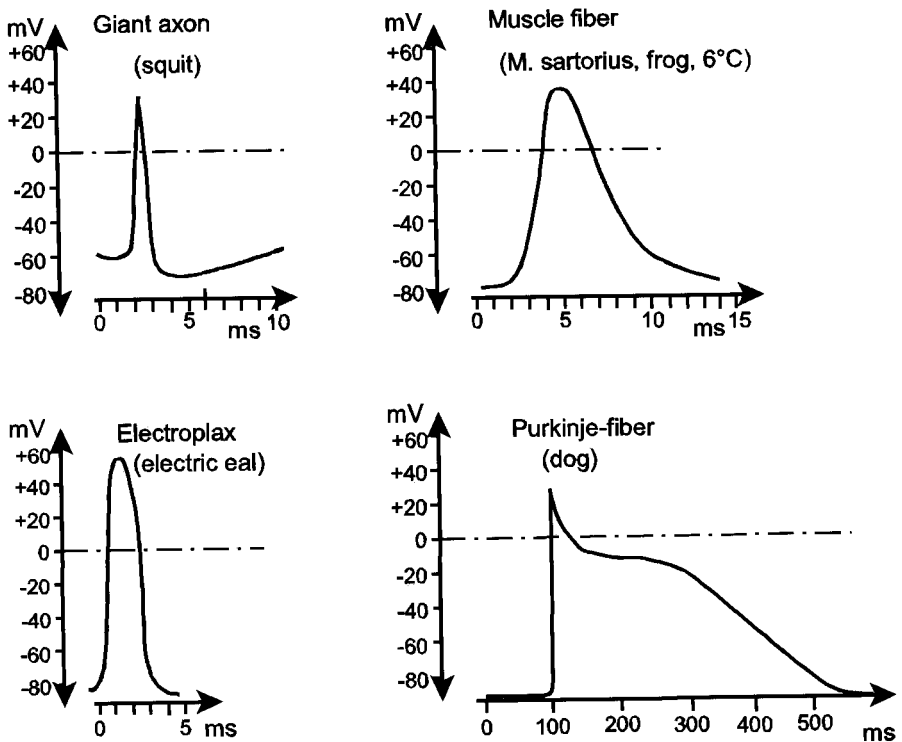


Fig. 3.27. Examples of various action potentials. (After Penzlin 1991)

therefore, the diffusion potential of sodium becomes dominant. This has the opposite polarity to the potassium potential which explains the spike of the action potentials.

As we demonstrated in the previous section, the amount of charge which is needed for this kind of depolarization is extremely low. This was checked by flux measurements in excited nerves. During the generation of an action potential, therefore, no significant alterations of the internal ion concentration occur. A nerve can generate action potentials for a long time after the ion pumps have been blocked. Only after hours does the electrochemical battery of the cell become empty.

In 1952 Hodgkin and Huxley proposed a theoretical model of this process, based on intensive experimental investigations. Its form is of a purely kinetic nature and does not contain information about concrete molecular mechanisms taking place in the membrane.

The basic equation describes the kinetics of the current in an electrical circuit, similar to the scheme in Fig. 3.25. The current density (j) in such a system can be described by the following equation:

$$j = C' \frac{d(\Delta\psi_M)}{dt} + (\Delta\psi_M - \Delta\psi_K)G'_K + (\Delta\psi_M - \Delta\psi_{Na})G'_{Na}. \quad (3.4.14)$$

$\Delta\psi_M$ is the electrical membrane potential, whereas the symbols $\Delta\psi_K$ and $\Delta\psi_{Na}$ indicate the Nernst potentials of potassium and sodium according to Eqs. (3.4.11) and (3.4.12). C' is the capacity of the membrane, and G'_K and G'_{Na} the potassium and sodium conductivities, always corresponding to a unit of area in the membrane. The conductivity of the membrane for individual ions cannot be measured electrically but can be obtained from experiments in which the kinetics of radioactive tracer ions are measured.

The first term of Eq. (3.4.14) gives the current density which leads to the charge of the membrane capacitor (Fig. 3.25). The following terms represent the current densities associated with potassium and sodium fluxes.

The conductivities G'_K and G'_{Na} are not constant, but are functions of the electric field in the membrane, that is of the membrane potential. The potentiometers in Fig. 3.25, therefore, are controlled directly by $\Delta\psi_M$. From the molecular point of view this means that these conductivities are the result of voltage dependent channels. It is therefore necessary to make statements about field dependents of these conductivities, i.e. the functions $G'_K(\Delta\psi_M)$ and $G'_{Na}(\Delta\psi_M)$.

To describe the behavior of these channels, a statistical approach is used. It is assumed that the channels can obtain two discrete states: "open", or "closed". The phenomenological conductivities (G'_K , G'_{Na}) then represent the average of the functional states of a large number of such channels. If all of the channels are open then the maximal conductivities $G'_{K \max}$ and $G'_{Na \max}$ are established.

Furthermore, it is assumed that the potassium channel will be open when exactly four events take place simultaneously, all having the same probability of occurrence (n). The real nature of these events is not explained. It could be, for example, the presence of four potassium ions near the entrance of the channel.

This assumption leads to the following equation:

$$G'_K = G_{K \max} n^4. \quad (3.4.15)$$

The probability n is a function of time and can be characterized by rate constants α_n and β_n as follows:

$$\frac{dn}{dt} = \alpha_n(1 - n) - \beta_n n. \quad (3.4.16)$$

Concerning the sodium permeability, it is assumed that the channel will be open when three events, each having the probability m occur simultaneously, and if another inhibitory event having the probability h has not taken place. This leads to the expression:

$$G'_{Na} = G_{Na \max} m^3 h. \quad (3.4.17)$$

For the parameters m and h the following kinetic equations can also be written:

$$\frac{dm}{dt} = \alpha_m(1 - m) - \beta_m m, \quad (3.4.18)$$

$$\frac{dh}{dt} = \alpha_h(1 - h) - \beta_h h. \quad (3.4.19)$$

The voltage dependence of the channels is proposed to be the result of influences on the rate constants α and β :

$$\begin{aligned} \alpha_n &= \frac{0.01(\Delta\psi + 10)}{e^{\frac{\Delta\psi + 10}{10}} - 1}, & \beta_n &= 0.125e^{\frac{\Delta\psi}{80}}; \\ \alpha_m &= \frac{0.1(\Delta\psi + 25)}{e^{\frac{\Delta\psi + 25}{10}} - 1}, & \beta_m &= 4e^{\frac{\Delta\psi}{18}}; \\ \alpha_h &= 0.7e^{\frac{\Delta\psi}{30}}, & \beta_h &= \frac{1}{e^{\frac{\Delta\psi + 30}{10}} + 1}. \end{aligned} \quad (3.4.20)$$

(In these equations, the potentials are in mV!)

These equations were obtained from a purely empirical approach.

It is easy to see that if the relations given in Eq. (3.4.20) are substituted into Eqs. (3.4.16), (3.4.18), and (3.4.19), a system of non-linear differential equations will be obtained. The solution of these equations can be substituted into Eqs. (3.6.15) and (3.6.17), and eventually, into the basic Eq. (3.4.14). An analytical solution of this system of differential equations is not possible. Computer simulations of these equations, however, indicate a good accordance with experimental results.

Figure 3.28 shows the calculated time courses for the changes in sodium and potassium conductivities at different membrane potentials. This also corresponds well with experimental findings. These curves illustrate the mechanism described above for the generation of an action potential. These conductivities from Fig. 3.28 illustrate the time dependent changes of the potentiometers shown in Fig. 3.25, whereas the conductivities are directly proportional to the permeabilities. Within the first millisecond following the stimulus, the sodium potential is dominant because of the rapid increase in G'_{Na} (and thus P_{Na}). This will then be counteracted by the increasing potassium potential.

Beside the direct mechanism of membrane excitation in nerve cells, the translation of this point of excitation along the axon of a nerve cell is of particular interest. In unmyelinated axons of nerves the process of pulse transmission is based on a lateral spreading of excitability by the electric field of excitation itself. A local action potential triggers the neighbor proteins in the membrane. In fact the impulse can proceed only in one direction, because of the refractory period of several milliseconds which the proteins need after an excitation to become excitable again.

Figure 3.29 illustrates the advantage of this kind of impulse propagation in relation to the transmission of a voltage pulse in an electric cable. In contrast to

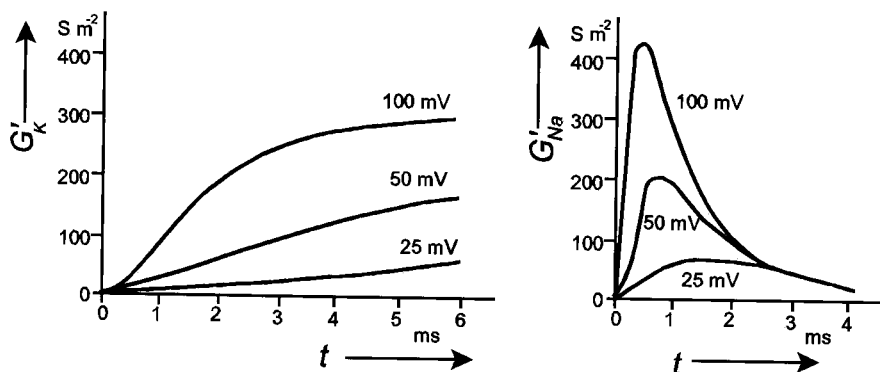
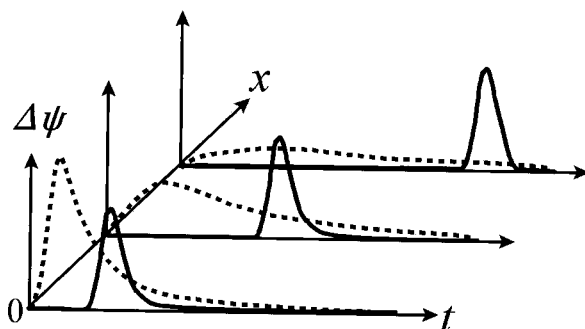


Fig. 3.28. The time dependence of the conductivities G'_K and G'_{Na} for various membrane potentials, corresponding to the theory of Hodgkin and Huxley. (After Waterman and Morowitz 1965)

Fig. 3.29. The time course of a voltage pulse which is put at time $t = 0$ at the point $x = 0$, by transmission in an isolated cable (dotted line) and in an unmyelinated nerve (solid line)



the cable, the time characteristics of the nerve pulse remain constant, even after a certain distance of transmission. On the other hand of course, the absolute velocity of pulse transmission in a cable is much faster than in an axon of a nerve. This advantage of simple electrical conductivity is used in many vertebrate, and in a few invertebrate axons which are surrounded by a fatty sheath, composed of myelin. Such nerves are called *myelinated*. This sheath is interrupted at intervals of some millimeters by so-called *nodes of Ranvier*. In the myelinated regions simple electric conductivity of the pulse occurs, as in a cable. The nodes of Ranvier represent membrane areas which are excitable in a normal way. If a certain node of Ranvier is excited, then the pulse propagates by simple electric conduction along the myelinated length and excites the subsequent node. This so-called *saltatory conduction* is a way of pulse amplification leading to a faster transport of information.

Further reading: Ferreira and Marshall 1985; Hille 1992; Markin et al. 1987; Sigworth 1994

3.5

Electric Fields in Cells and Organism

3.5.1

The Electric Structure of the Living Organism

In Section 2.3.3 we pointed out that the term “structure” must be used in a very broad sense, not limiting it to visible details of a biological system. There exists not only a morphological structure of the biological system, but also structures which are based on other physical properties. Using the definition of the electrical potential as given in Section 2.4.1, and considering its distribution in a protein, a cell, an organ etc., what we will call an “electric structure” of a biological system becomes clear. In the following we will consider chiefly the electric structure on the supracellular and organismic level of organization. Firstly, however, we will take a look at this problem in general.

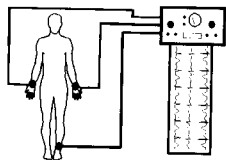
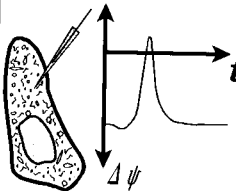
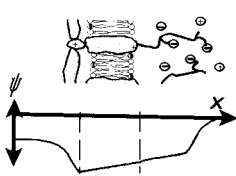
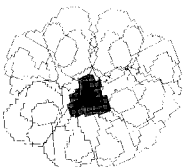
As shown in Fig. 3.30, from the physical point of view a hierarchic order of structures is possible which fully corresponds to the usual classification in biology. In Section 2.1, we learned that atomic interactions can be explained using the approaches of wave mechanics. The Schrödinger equation allows us to calculate the electric parameters in the dimensions of atoms and therefore determines the energies of chemical bonds and molecular interactions.

Considering supramolecular structures like membranes, we arrived at the next level of hierarchy. In this case we considered charges and dipoles as elements of the system, and calculated structures like electric double layers, using the Poisson-Boltzmann equation of statistical thermodynamics [Eq. (2.4.13)]. We discussed electrical structures at this level of organization in Section 2.5.5.

In a next step we considered the cell as a thermodynamic system, consisting of various phases with particular properties. We found that differences exist in the electrical potential between these phases, and directed our attention to the transmembrane potential in particular. To calculate this, we used the approaches of phenomenological thermodynamics. The Nernst-Planck equation [Eqs. (3.3.32), (3.3.33)], for example, allows us to calculate flows of ions and the resulting diffusion potentials. The role of the transmembrane potential was discussed mainly in Sections 3.4.2 and 3.4.3.

In the following text we enter a further region in the hierarchic structure. We will consider electric fields in the extracellular space, in tissues and organs. This is the region of classical electrodynamics. In this case the Maxwell equations allow us to calculate electric fields in inhomogeneous dielectrics. The question arises: how does the field spread inside the body, which consists of organs with different conductivities, like bones, soft tissue, air filled cavities etc? We will come back to this question again in Section 4.4., where we will discuss the influence of electromagnetic fields on biological systems.

Fig. 3.30. The hierarchic system of the electric structure of the living organism and the corresponding physical approaches.

Organism		Elektrodynamics <i>Maxwell equations</i>
Cell		Phenomenological thermodynamics <i>Nernst-Planck equation</i>
Supramolecular structures		Statistical thermodynamics <i>Poisson-Boltzmann equation</i>
Atomar and molecular structures		Wave and quantum mechanics <i>Schrödinger equation</i>

3.5.2

Electric Fields in the Extracellular Space

It has long been known that electric fields exist in the living body. Such fields are measured as electrocardiograms (ECG), electromyograms (EMG), and electroencephalograms (EEG) in medical diagnosis using simple metal electrodes. In the case of ECG potential differences between the limbs are in the order of several millivolts. In the case of EEG, because of the isolating role of the skull, they are much lower.

Furthermore, it is possible to find DC potentials between various parts of the animal body and even in plants. One has, however, to be very careful to avoid measuring electrode artefacts since in these cases absolute values in the order of a few millivolts are measured, in contrast to relative alterations of the potential as in case of EEG, ECG and EMG. Many measurements of these potentials have suffered due to use of inaccurate electrodes.

In the last 20 years it has been possible to measure electrostatic fields in the vicinity of cells using the vibrating probe techniques (see Fig. 3.32). In this case a

glass microelectrode is used, the top of which consists of a metallic sphere with a diameter of only a few micrometers. This electrode vibrates at a frequency of about 500 Hz causing this sphere to be displaced periodically at an amplitude of 10–30 μm . Using a very sensitive low noise amplifier it is possible to measure voltage differences near $1 \text{ nV} = 10^{-9} \text{ V}$ between these two points. In this way, DC electric fields around a cell or an organ can be detected. A large number of various biological objects have already been investigated with this method, such as single cells (Fig. 3.32), various growing and wounded parts of plants, follicles of insect eggs, growing embryos, muscle fibers and various other tissues and organs.

The origin of extracellular electric fields and currents in biological organisms can be different. One must differentiate between fields which are generated directly by living cells, and others which are induced by mechanical deformations of tissue and motions of fluids inside the body and, in swimming fishes, also on their outer surface.

The transmembrane potential, i.e. the electrical membrane potential difference between the internal and the external phase of a cell usually does not induce an external field. This does happen however, if the transmembrane potential of the same cell differs at different microscopic locations of the membrane. Another possibility of influencing the extracellular space is by the rheogenic pumps which induce a certain transmembrane current. In both cases, the extracellular induced electric currents and fields are the result of electrochemical nonequilibria.

Of course, lateral electric fields also may be the result of a lateral mosaic of surface charges (see. Sect. 2.5.5). These fields, however, like the strong electric fields in a double layer, are part of an electrochemical equilibrium system. Consequently, they do not generate electrical currents and do not influence the system outside the electrical double layer.

The axon of a nerve cell is the best example for the case where local differences in the membrane potential will induce external fields. If an action potential is propagated along the axon, small areas of this membrane will subsequently become depolarized. In contrast to the parts of the axon with resting potential, where the external membrane is positively charged in relation to the inner one, the polarity of membrane areas with an action potential is opposite. The result is an electrical current from one point of the membrane, laterally, to another one (Fig. 3.31). Such local depolarization occurs not only in nerves, but also in muscle and other cells.

The second possibility of inducing an extracellular electric field is depicted as an example in Fig. 3.32. In this case, a rheogenic calcium pump is localized at a particular point in the membrane of an egg of a fucoid seaweed *Pelvetia*. This pump induces a flow with a current density up to 0.3 A m^{-2} . This current flows in the direction in which the first rhizoid of the cell will grow. Experiments in which such eggs were exposed to artificial fields indicate that, in fact, the field determines the direction of growth.

Examples of the above mentioned extracellular electric fields not generated by living cells are those which are induced by piezoelectricity (Sect. 2.2.2), and by

Fig. 3.31. Schematic illustration of a snapshot of an excited nerve. The dark areas represent the actual region of depolarized membrane (see also Fig. 3.27)

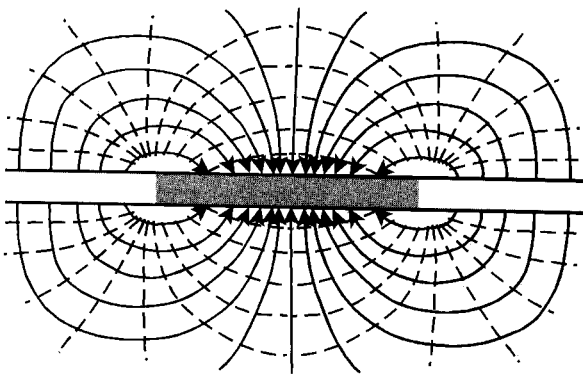
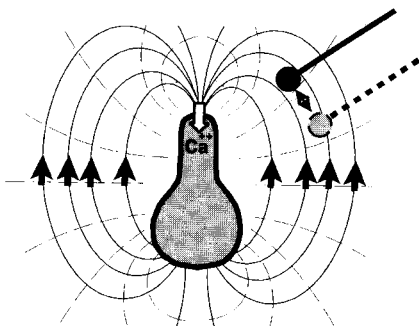


Fig. 3.32. Current (—), and equipotential lines (---) near a growing egg of the brown algae *Pelvetia*, caused by a local rheogenic calcium transport, which marks the location where the rhizoid is formed. The external field was measured using the vibrating probe method. The corresponding electrode in two extreme positions during vibration is shown in relation to the size of the cell in approximately correct dimensions. (According to data from Jaffe 1979)



electrokinetic effects (Sect. 2.5.4). Both of them occur in bones and cartilage during in vivo deformations. Piezoelectric potentials result from deformations of charged materials, in this case chiefly by deformation of collagens. Furthermore, such deformations cause a flow in the narrow channels of the bone, carrying negative surface charges. This additionally causes streaming potentials. In contrast to the electric fields in the whole body induced by nerves and muscles, these fields are rather low at larger distances. On the other hand, they seem to be important locally in the process of bone growth and bone remodeling in vivo. This circumstance is used in therapies to stimulate bone repair by applying electric field pulses or ultrasound.

A special source of electric currents and fields are the so-called *wound potentials*. The reason for them is a permanent electrostatic potential difference between different cavities in the body, caused for example by pumps, differences in electrolyte composition and others. If the isolating properties of an epithelium are damaged by a wound, then a current between the phases of two cavities occurs and a lateral field. In most cases the wound represents the positive polarity in this battery. In Section 4.4.2 we will indicate how cells may use this field by way of galvanotaxis to tend the wound.

In a special adaptation, electric fishes use extracellular fields induced by excited cells. In the electric organs specialized muscle cells are organized in so-

called electroplaques, where the voltage of several cells adds up to values of 800 V. Such high voltage is only possible, though, in fresh water fishes like the electric eel. Marine fishes do not reach this value because of the high conductivity of sea water. In this case the current density, and therefore the required power would be too high. In such systems it is not the arrangement of these elements in series which leads to a higher voltage, but rather a parallel arrangement which increases the current. So-called strong electric fishes use the induced electric fields to catch their prey, or to defend themselves.

In contrast to this, weak electric fishes use their electric fields for orientation. For this the electric organ discharges potential and electric field patterns in the water around the fish and on the skin surface itself. The fish therefore becomes an oscillating dipole, inducing an external electric field. This field is disturbed by the surrounding objects. Bodies with a conductivity higher than the environment, such as animals or submersed plants, focus the field, others with lower, or even no conductivity, like for example stones, induce regions with divergent field lines. Over each cycle of the electric organ discharge, the fish measured the electric field amplitude at any point near its surface. As a result, the relatively simple electric organ discharge leads to a fairly uniform pattern of electrosensory stimuli along the body surface, which may facilitate central processing of electrosensory images. (For electroreception see Sect. 4.4.2.)

These considerations indicate that the role of extracellular electric fields in the body is rather heterogeneous. This results from the different electric field strengths and current densities which have been measured in various kinds of growing tissues, embryos etc., which amounts to values of up to 1 A m^{-2} . Wound potential can induce current densities which are even larger. The electric fields near cells, indicated by use of the vibrating probe techniques are in the order of 0.1 V m^{-1} . This method, however, has no high resolution. It only allows the measuring of fields at a distance of more than $30 \text{ }\mu\text{m}$ from the source. One can assume that the field close to the membrane of an excited nerve, for example, may be much higher. The field strengths and the current densities of usual electrocardiograms (ECG), or electroencephalograms (EEG) at a certain distance from the heart, or in the brain, are obviously much lower.

Recently, the role of extracellular electric fields has been discussed in the context of processes of differentiation and growth. So, for example, wound potentials may lead to galvanotaxis of embryonal cells and therefore influence the process of healing (for galvanotaxis, see Sect. 4.4.2). Electrophoresis of growth factors in DC fields in the body probably also occur, controlling processes of morphogenesis (for theoretical models of morphogenesis see Sect. 5.3.2).

Very little is known about the electric fields in the direct vicinity of cells and in intercellular clefts. The mosaic of surface charges creates a static field, and additionally rheogenic transporters induce various nonequilibrium situations which may lead to electric currents. Furthermore, streaming potentials and streaming currents can be induced which could become important locally. In this respect, the control of lateral translocations of membrane proteins is discussed (see also Sect. 2.5.5).

The use of electric fields in the body in medical diagnosis raises the question: how do electric fields, generated for example by the beating heart, spread in the body and, vice versa, what can an electrocardiogram tell a doctor about the function or dysfunction of the heart? Modern techniques of ECG measurement allow a reasonable reconstruction of the electric field in the body using parallel recordings of the ECG simultaneously in different parts of the body. These fields were not induced by single cells but rather by cellular arrangements excited simultaneously. In the case of an ECG, definite parts of the cardiac muscle are excited in the rhythm of the beating heart. Consequently, the heart becomes an oscillating dipole, the orientation of which changes according to the orientation of the excited parts of the muscle.

The first person to propose a method to evaluate ECGs was Willem Einthoven who was awarded the Nobel Prize for this in 1924. He proposed that it should be possible to localize the excited parts of the heart leading the potentials from three points of the body, which are more or less equally distant from the heart. This so-called *Einthoven triangle* uses leads from both arms and the left leg. This kind of recording is used even today for routine measurements of the ECG, though an exact correlation between the resulting dipole vectors of the field with the anatomical position of various heart muscles is impossible because of the electrical heterogeneity of the body.

In Fig. 3.33 this method is illustrated schematically. The three cardiograms, led by the corresponding points, indicate in a periodically repeated sequence P, Q, R, S, and T waves. They represent the sequence of excitation of different parts of the heart muscle. It starts with the excitation of the atria (P wave). If all the atrial fibers are in the plateau phase, the PQ segment is reached. Subsequently, the excitation spreads over the ventricles, beginning on the left side of the ventricular septum, spreading toward the apex (QRS waves), finally reaching the ventricular recovery phase (T wave). As a result of the projection of these curves corresponding to the geometry of an equilateral triangle, a rotating vector appears, the origin of which lies in the crossing point of the three axis. The arrowhead moves periodically along the dashed line.

Further reading: The role of extracellular fields: Hotary and Robinson 1990; Nuccitelli 1995; the electric field of ECG: Reilly 1998; the electric field in bones: MacGinitie 1995

3.5.3

Passive Electrical Properties of Tissue and Cell-Suspensions

To describe the electric structure of cells and organisms in view of electrostatics, or of ionic currents induced by rheogenic pumps, parameters such as membrane resistance, conductivity of cytoplasm etc. were introduced. Furthermore, the membrane capacity was used for explanation of membrane potentials in Section 3.4.2. In contrast to membrane potentials that will require active biological processes, the latter parameters can be summarized as *passive electric properties*.

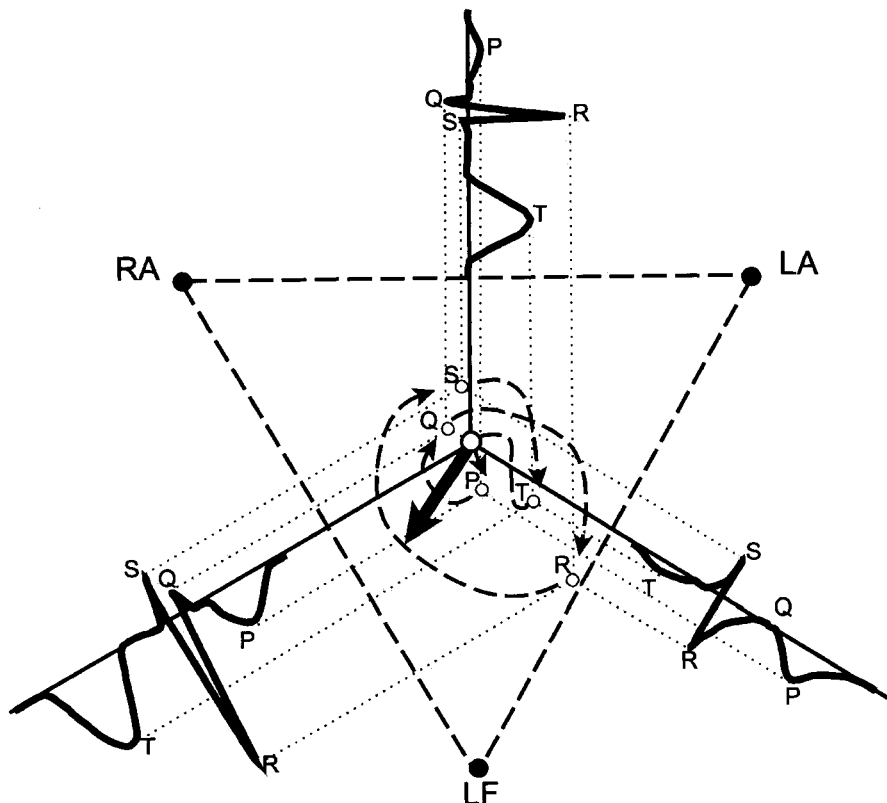


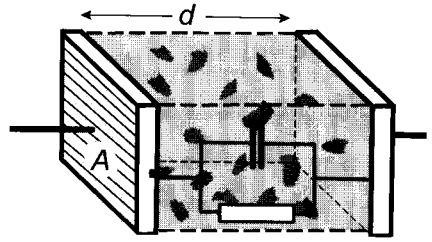
Fig. 3.33. Construction of a vector cardiogram according to Einthoven's triangle. Using the time course of potentials curves, orientated according to the sides of an equilateral triangle, a rotating dipole can be constructed in the center of it. P, Q, R, S, T – means the corresponding waves of the ECG. RA – the lead from the right arm, LA – from the left arm, and LF – from the left foot

In this Section we will consider consequences of passive electric properties for the frequency dependence of effects in AC fields.

To derive the basic parameters and equations, let us first consider the electrical properties of a plate capacitor filled with material of certain conductivity and dielectric constant (Fig. 3.34). This system can be described by an equivalent circuit consisting of a conventional capacitor and a resistor in parallel. This system, therefore, has an effective AC resistance which is called *impedance*, whereas its reciprocal is the AC conductance or *admittance* (Y^*).

Applying an AC voltage, the current will flow through the resistor and periodically recharge the capacitor. The admittance of this system, therefore, is determined on the one hand by the static conductance (G) of the resistor, and on the other hand by the displacement current passing through the capacitor, which depends on the frequency (ω). The admittance (Y^*) of this RC circuit is therefore:

Fig. 3.34. A capacitor consisting of two parallel plates of area A and a mutual distance d filled with an inhomogeneous medium described by conductivity and a dielectric constant as an example for the impedance of a biological tissue



$$Y^* = G + j\omega C. \quad (3.5.1)$$

This equation can be modified using the geometrical parameters of the electrode: area (A) and mutual distance (d), together with the material constants specific conductivity (g) and the permittivity ($\epsilon\epsilon_0$):

$$G = \frac{A}{d}g \quad \text{and} \quad C = \frac{A}{d}\epsilon\epsilon_0. \quad (3.5.2)$$

Introducing these relations into Eq. (3.5.1), one gets:

$$g^* = g + j\epsilon\epsilon_0\omega, \quad \text{whereas:} \quad Y^* = \frac{A}{d}g^*. \quad (3.5.3)$$

In this equation g^* is the complex specific admittance of the system.

Why is the imaginary number $j = \sqrt{-1}$ used in this equation, transforming some of these parameters into complex numbers marked by the superscript *? The reason for this is a particular time dependence of these parameters which must be taken into account.

In describing the behavior of an RC circuit in an AC field, the amplitude of the sine current as well as the phase shift are important. To understand this, let us consider an AC voltage having the following time function:

$$U = U_{\max} \sin \omega t. \quad (3.5.4)$$

An applied voltage (U) therefore oscillates with an angular frequency ($\omega = 2\pi\nu$) at an amplitude of $U_{\max}$. According to Ohm's law [Eq. (3.1.51)], in a circuit containing only a resistance, the current (I) possesses the same time behavior:

$$I = I_{\max} \sin \omega t. \quad (3.5.5)$$

As already mentioned, in an RC circuit, the displacement current of the capacitor must also be taken into account. The charge (q) of a capacitor is determined by the following equation:

$$q = UC. \quad (3.5.6)$$

Since the current is defined as the time derivative of charge, the displacement current (I_C) is:

$$I_C = \frac{dq}{dt} = \frac{d(UC)}{dt} = C \frac{dU}{dt}. \quad (3.5.7)$$

Introducing Eq. (3.5.4) into Eq. (3.5.7) and leaving $C = \text{constant}$, one gets:

$$I_C = C \frac{d(U_{\max} \sin \omega t)}{dt} = CU_{\max} \omega \cos \omega t = CU_{\max} \omega \sin\left(\frac{\pi}{2} + \omega t\right). \quad (3.5.8)$$

Further, considering that:

$$I_{C \max} = CU_{\max} \omega, \quad (3.5.9)$$

it follows:

$$I_C = I_{C \max} \sin\left(\frac{\pi}{2} + \omega t\right). \quad (3.5.10)$$

Comparison of Eqs. (3.5.5) and (3.5.10) clearly indicates a phase shift of $\pi/2$ in between current and voltage.

Consequently, in Eq. (3.5.1) in the case of an AC current, the time course of the capacitive component of the sum must be considered. This circumstance is taken into account using a Gauss plane containing imaginary numbers.

The term ωC in Eq. (3.5.1) is multiplied by the imaginary number $j = \sqrt{-1}$ and therefore plotted on the ordinate in a Gauss plane (Fig. 3.35).

In the same way as the complex specific admittance (g^*) in Eq. (3.5.3), a complex dielectric constant (or relative permittivity; ϵ^*) of the system can be introduced. Starting with the equation for the effective AC resistance of a capacitor ($R = 1/\omega C$), one can introduce a complex capacitance (C^*) as follows:

$$C^* = \frac{Y^*}{j\omega}. \quad (3.5.11)$$

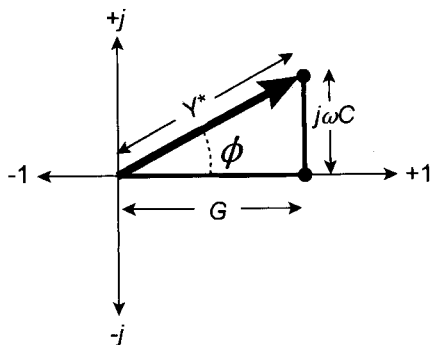


Fig. 3.35. Representation of the conductance (G) of the resistor and the admittance of the capacitor ($j\omega C$) of the analog circuit from Fig. 3.34 in the Gauss plane of complex numbers. The admittance of the system (Y^*) corresponds to the length of the resulting vector. ϕ is the resulting phase shift

Inserting the parameters of [Eqs. (3.5.1) and (3.5.2)] into this equation, and considering $C^* = \varepsilon^* \varepsilon_0 A/d$, and $1/j = -j$, one gets:

$$\varepsilon^* = \varepsilon - j \frac{g}{\omega \varepsilon_0}. \quad (3.5.12)$$

This derivation of the basic equations [Eqs. (3.5.3) and (3.5.12)] allows us to understand the properties of complex dielectrics in AC fields.

These parameters are used to calculate, for example, the distribution of AC fields in the body as illustrated in Fig. 3.30, which is important in medical and environmental biophysics. We shall come back to this problem in Section 4.4.3.

The electrical properties of biological tissues containing a multitude of various cells with various sizes and intercellular clefts are, of course, too complicated to be described by a simple RC circuit as we did before. An enormous network of elementary RC elements with different parameters would be required to build an appropriate model. This, of course, is impossible for practical use. For this reason one uses the same simple RC circuit which is depicted in Fig. 3.34 except that the properties of the resistor as well as of those of the capacitor are now frequency dependent. This simply results from the dielectric heterogeneity of the system, even though the individual dielectric parameters of any particular constituent of the systems, like water, lipid membranes etc. can be considered as frequency independent.

Figure 3.36 indicates mean values of the complex dielectric constants (ε^*) and the specific admittance (g^*) of various tissues in a broad frequency range. Such data can be measured by electrodes as illustrated schematically in Fig. 3.34. It can be seen that the specific admittance increases with increasing frequencies. The main reason for this is the increase of the membrane admittance. On the other hand, the dielectric constants of the tissue at low frequencies have enormous values up to the MHz range. These

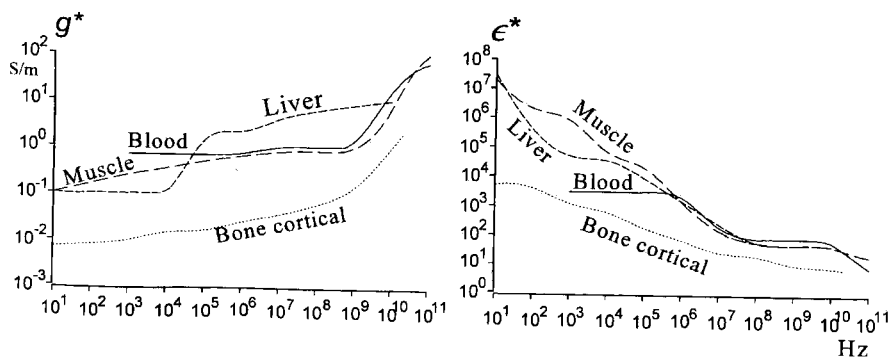


Fig. 3.36. Complex specific admittances and complex dielectric constants of various tissues as functions of frequency. (According to averaged data from Gabriel et al. 1996)

constants drop below the standard value of the dielectric constant of water ($\epsilon \approx 80$) only at microwave frequencies.

These curves do not continuously increase or decrease with frequencies, but change in a characteristic step-wise fashion. The reasons for this, formally, are the properties of various kinds of RC-circuits, which are included in the system. These frequency regions, in the first instance have formally been designated as α -, β -, and γ -dispersions with increasing frequency (see Fig. 3.37).

Various phenomena are summarized as α -dispersion, which for cell-sized objects occur in the frequency range below 10 kHz. Mostly, reactions in the electric double layer of cell membranes, such as various electrokinetic phenomena (see Sect. 2.5.4) e.g. the deformation of the ionic clouds, are responsible for this. Also, artefacts related to polarization phenomena of the measuring electrodes must be expected in the α -range. It is difficult to determine the individual contributions of these different phenomena.

The β -dispersion is mostly based on processes connected with the membrane as a dielectric barrier. In general, these are related to *Maxwell-Wagner dispersions* which are caused by structural inhomogeneities of the material, such as tissue inhomogeneities, cells and organelles. Some authors subdivide the β -region into the β_1 -, and the β_2 -range which refer to the dispersion of the cell membrane (β_1) and cytoplasm (β_2) polarization respectively.

The γ -dispersion at higher frequencies is caused by so-called *Debye relaxations* of various molecular dipoles. At frequencies of the γ -dispersion region, even the resistivity of the internal and external milieu of the cell cannot simply be described by ohmic resistances (see small RC-circuits in Fig. 3.38).

Assuming that the dispersion of water dipoles occurs at 18.7 GHz, one might define for this a special δ -region. This, in fact, is true only for free water. Recent measurements indicate that the dispersion of bound water may occur just in the γ -dispersion region. This is important because of possible effects of high frequency electromagnetic fields on biological tissue (Sect. 4.4.4).

Further reading: Gabriel et al. 1996; Gimsa and Wachner 1998; Pethig 1979; Pethig and Kell 1987; Polk and Postow 1996; Riu 1999; Schwan 1957

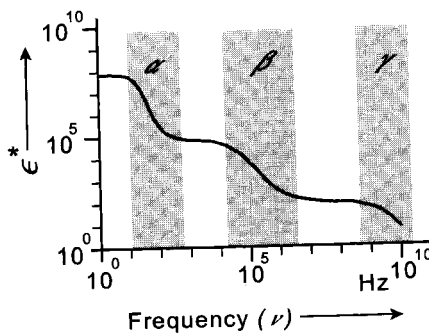


Fig. 3.37. Schematic illustration of the frequency regions of α -, β -, and γ -dispersion as demonstrated for the frequency dependence of the complex dielectric constant of a biological tissue

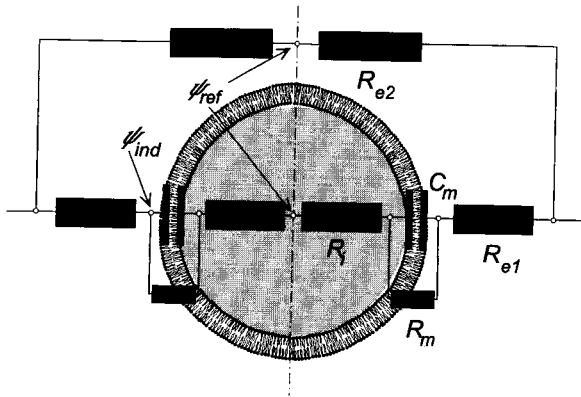


Fig. 3.38. A simplified analog circuit demonstrating current ways through, and around a spherical cell. R_{e1} and R_{e2} – resistances of the external medium, R_i – resistance of the cytoplasm, R_m – membrane resistance, C_m – membrane capacity, ψ_{ref} – reference potential at the symmetry plan of the system, ψ_{ind} – induced potential at the membrane surface. At higher frequencies RC properties must be attributed to the resistances R_{e1} , R_{e2} , and R_i too (internal and external resistors were split to demonstrate the line of the reference potential ψ_{ref})

3.5.4

Single Cells in External Electric Fields

Figure 3.38 indicates a simplified electronic circuit, describing the passive electric properties of a spherical cell in an electrolyte solution. As in Fig. 3.25, the membrane is represented by a capacitor (C_m) in parallel with a resistor (R_m) simulating the membrane conductivity. In DC fields and low frequency AC fields the conductivities of external and internal media can be described by simple resistors (R_{e1} , R_{e2} , R_i).

The high specific conductivity of the cytoplasm and the external medium on the one hand, and the high resistance of the membrane on the other results in R_m being more than seven orders of magnitude higher than R_e or R_i . Applying Kirchhoff's law, a low frequency current therefore does not flow through the cell, but around it.

This is demonstrated schematically in Fig. 3.39 where a spherical cell is shown with the electrical properties as described above. Since in the membrane the field cannot be resolved at this scale, it will be discussed in the next figure. Because of the high membrane resistance, at DC fields and extremely low frequency AC fields, the induced field inside the cell is negligible, and the external field becomes deformed. As already discussed in Section 3.5.3 with increasing frequencies of the external field, the membrane capacitor will become short circuited. This leads to a change of the external field distribution (Fig. 3.39B and C). If there is no large difference in the permittivities between the medium and cell plasma, the amount of field penetrating the cell depends only on the relation of the inside-outside conductivities. In any case, the field penetration increases

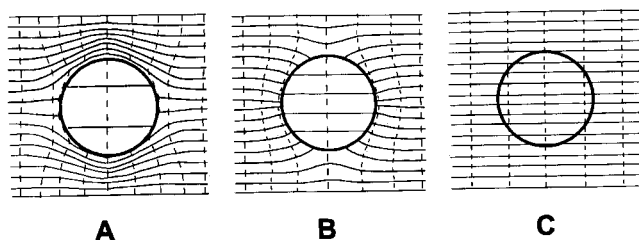


Fig 3.39. Current lines (—), and equipotential lines (- - -) for a spherical cell in a homogeneous electric AC field. In contrast to Fig. 3.40 the polarization charges at the membrane are not depicted. A the cell in a low frequency, B and C – in a high frequency AC field. For B the conductivity of the external medium is lower than that of the internal one, in C the cell is surrounded by a physiological medium where the permittivities of the internal and external milieus are the same

at high frequencies. Therefore, membrane polarization decreases and polarization of the cytoplasm increases.

To discuss the influence of external fields on the membrane potential, let us now consider a spherical cell in a DC field (Fig. 3.40). Neglecting the conductivity of the membrane, which in fact is very low, it can be understood that the external field leads to an accumulation of charges, especially in membrane areas that are orientated perpendicular to the current lines. This yields a deformation of the external electric field (Fig. 3.39) and also produces charging of the membrane capacitor. The polarization of the membrane induces an opposite intracellular field which significantly mitigates the field influence from outside (short arrows inside the cell). Because of the low field strength inside the cell, polarization of the membranes of cell organelles can be neglected at least in DC, or low frequency AC fields.

The transmembrane potential difference $\Delta\psi_{\text{ind}}$ which is induced by an external field, as depicted in Fig. 3.40, corresponds to the difference $\psi_{\text{ref}} - \psi_{\text{ind}}$ of Fig. 3.38. Therefore, there is no potential difference across the resistor R_p and no current flows inside the cell.

Because of the deformation of the external field by the cell itself (Fig. 3.39A), the potential ψ_{ind} is somewhat different from the potential at an identical x -coordinate away from the cell. This was taken into account by introducing the resistor R_{e1} (Fig. 3.38) and by bending the lines of the extracellular potential function near the membranes in the lower part of Fig. 3.40. The extent of this deviation depends on the shape of the cell. In case of a spherical cell, $1.5r$ has to be considered.

The polarity of the induced potential difference ($\Delta\psi_{\text{ind}}$) on both sides of the cell is the same as that of the external field, whereas the undisturbed in-vivo transmembrane potential $\Delta\psi_M$ is always polarized inside-out. Therefore, on one side the cell is polarized in the same direction as the induced potential difference ($\Delta\psi_{\text{ind}}$), and on the other side in an opposite direction. This means that the resulting potential differences ($\Delta\psi_1$, $\Delta\psi_2$) differ from one another on both sides of the cell. The transmembrane potential at locations in the cell perpendicular to

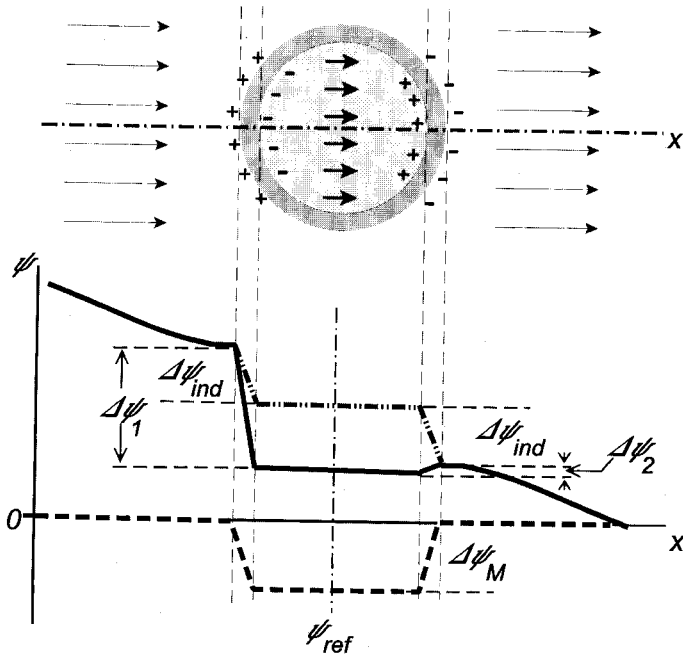


Fig. 3.40. The distribution of charges and potentials at a spherical cell in an external DC field. In the *upper* part the *arrows* indicate direction and strengths of the field vectors. The charges near the membrane are the result of polarization by the external field. In the *lower* part of the figure the potential profile over the *x*-axis through the center of the cell is depicted: - - - membrane potential without an external field influence including the undisturbed *in vivo* transmembrane potential ($\Delta\psi_M$); - · - · - potential induced by the external field, not considering the biological membrane potential, with the induced potential difference ($\Delta\psi_{ind}$); — actual potential function, as the sum of both functions, and the resulting membrane potential differences on both sides of the cell ($\Delta\psi_1$, $\Delta\psi_2$). ψ_{ref} is the reference potential according to Fig. 3.38. For simplicity, in contrast to Fig. 2.48, the electric double layer is not taken into account

the current lines will not be influenced by the external field at all. For this reason, in Fig. 3.38 the reference level for the induced potential (ψ_{ref}) is identical inside and outside the cell.

To calculate this induced membrane potential (ψ_{ind}), a small column orientated in the field direction which processes the cross-sectional area A , and cut out along a line through the cell's center can be considered. The characteristic length up to which cell polarization enhances the external medium is $r_e = 1.5r$. Furthermore, we shall consider that the Ohmic current flow through the membrane resistor R_m (Fig. 3.38) can be neglected. Considering an external field of the strength E , the potential difference outside the cell, from point ψ_{ref} to $\psi_{1.5r}$ amounts to $1.5rE$. The same potential must drop over the column. Using the proportionality between the impedance, as effective AC

resistance, and the potential drop, and considering the impedance of the membrane capacitor: $Z^* = 1/Y^* = -j/\omega C$, one can write:

$$\frac{\Delta\psi'_{\text{ind}}}{1.5Er} = \frac{-j/\omega C}{-j/\omega C + R_i + R_{e1}}, \quad (3.5.13)$$

where ψ'_{ind} , in contrast to ψ_{ind} means the induced potential just in the membrane section of this column. Introducing:

$$R_i = \frac{r}{g_i A}; \quad R_{e1} = \frac{0.5r}{g_e A} = \frac{r}{2g_e A}; \quad C_m = \frac{C}{A}, \quad (3.5.14)$$

and noticing that:

$$|j + a| = \sqrt{1 + a^2}, \quad (3.5.15)$$

one gets:

$$\Delta\psi'_{\text{ind}} = \frac{1.5Er}{\sqrt{1 + \left[r\omega C_m \left(\frac{1}{g_i} + \frac{1}{2g_e} \right) \right]^2}}. \quad (3.5.16)$$

This is the maximum of the membrane potential induced by the external field at the point of the spherical cell, where the membrane is perpendicular to the field direction. At an angle perpendicular to the field (Fig. 3.40) the induced transmembrane potential vanishes. To consider the induced potential at all points of the sphere, the radial coordinate α can be introduced, and the [Eq. (3.5.16)] must be multiplied by $\cos \alpha$:

Let us consider a further aspect: what is the time constant (τ) of the charge of the membrane capacitor? Again, applying the scheme of Fig. 3.41 we can write:

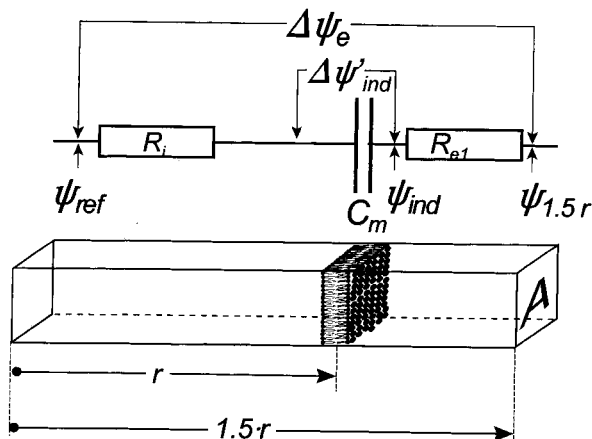
$$\tau = (R_i + R_{e1})C = r C_m \left(\frac{1}{g_i} + \frac{1}{2g_e} \right). \quad (3.5.17)$$

Introducing this into [Eq. (3.5.15)] and considering the angular coordinates, one gets the equation describing the induced potential (ψ_{ind}) for every location:

$$\Delta\psi_{\text{ind}} = \frac{1.5Er \cos \alpha}{\sqrt{1 + (\omega\tau)^2}} = \frac{1.5Er \cos \alpha}{\sqrt{1 + (2\pi\nu\tau)^2}}. \quad (3.5.18)$$

Using common cell parameters, like for example: $r = 10^{-5}$ m, $C_{\text{sp}} = 10^{-2}$ F m $^{-2}$, $g_i = 0.5$ S m $^{-1}$, and $g_e = 1$ S m $^{-1}$ in Eq. (3.5.17) a relaxation time of $\tau = 2.5 \cdot 10^{-7}$ s can be obtained. Introducing this parameter into Eq. (3.1.18) it is easy to demonstrate that for low frequency AC fields ($\nu < 10^5$ Hz), the

Fig. 3.41. A column with the area A , cut out from the cell (Fig. 3.38), and the electric scheme for a corresponding circuit. The resistor R_m is so high that it is disregarded in this scheme. (Corresponding to the approach of Gimsa and Wachner 1999)



denominator of this equation will approach 1. For DC fields and low frequency AC fields, therefore, this equation reduces to:

$$\Delta\psi_{ind} = 1.5Er \cos \alpha. \quad (3.5.19)$$

As already mentioned, this, as well as [Eq. (3.5.18)], is correct only for very high membrane resistance which is justified in most cases. Using this equation one can for example calculate that in a spherical cell with a diameter $r = 10 \mu\text{m}$ a transmembrane potential of $\Delta\psi_{rel} = 1.5 \text{ V}$ will be induced at position $\alpha = 0^\circ$ by an external low frequency, or DC field of $E = 100 \text{ kV m}^{-1}$.

Further reading: Gimsa and Wachner 1999; Grosse and Schwan 1992

3.5.5

Manipulation of Cells by Electric Fields

Using various types of electric fields, it is possible to move, separate, fuse, perforate, or deform cells. Nowadays, these methods are broadly applied in biotechnology. In contrast to the effects of weak electric and electromagnetic fields, which we will discuss later (Sect. 4.4), in these techniques rather strong fields are used, the energy input of which is significantly larger than the energy of thermic noise.

Let us first consider the influence of strong electric field pulses on cells, leading to considerable hyper-, or hypopolarization of the membrane. We discussed the electrostatic properties of the cell membrane in Sections 2.4.1, 2.5.4, 3.4.2, and 3.5.1. There we focused on the high electric field strength in the cell membrane *in vivo*, and its role in regulation of structure and distribution of membrane components. An externally applied DC field increases the membrane potential ($\Delta\psi_1$ in Fig. 3.40) on one side of the cell, and consequently, raises the

internal electric field strength in the membrane at this point. If the transmembrane potential reaches an amount of approximately 1 V, the internal field strength will destabilize the membrane, resulting in a so-called *electric break down*.

The application of such strong fields in solutions with considerable conductivities, as in a physiological milieu will, of course, induce significant Ohm's heating. The electrical power absorbed in a medium per volume equals $E^2 g$ [see Eq. (4.4.7)]. This parameter, by the way, is the so-called specific absorption rate (SAR) which we will use in Section 4.4.3. Considering a specific conductivity of $g = 1 \text{ S m}^{-1}$, and the above-mentioned field strength of $E = 10^5 \text{ V m}^{-1}$, a power density of 10^{10} W m^{-3} or 10^7 W kg^{-1} will be absorbed. In fact, electric fields with such a high field strength can be applied only for a short period of time (1 to 500 μs).

In case of a so-called electric break down the membrane loses its property as a diffusional barrier; its electrical resistance breaks down, which may even lead to a full destruction of the cell by subsequent lysis. However, by selecting correct treatment parameters, i.e. using a moderate shape, amplitude and duration of the pulse at optimal temperature and electrolyte conditions, a reversible electric break down can be induced. For this, or also for processes of electroinjection it is very important to avoid a Donnan-osmotic cytolysis of the cells (see Sect. 3.2.3).

This kind of treatment of cells is used for so-called *electroinjection* or *electropermeabilization*. When cells suspended in a solution or in a suspension of macromolecules or small particles are pulsed in this way, the short moment of the induced membrane destabilization is enough to allow for membrane penetration of particles with sufficient probability. Additionally, such disturbances of the membrane structure can also increase the ability of the cell to absorb particles by phagocytosis. Electroinjection is used, for example, to introduce drugs, genetic materials, or even small organelles into cells.

If two cells are attached to each other during electric pulse application, or during a short time after the pulse, the points of induced membrane instability may interact, leading to a fusion of the lipid phases of both cells. Subsequently, osmotically governed process of fusion of the cytoplasm may follow.

This process of so-called *electrofusion* of cells has recently been explained by several mechanisms which cannot be discussed here in detail. It is just necessary to point out that membrane fusion and subsequent cell fusion may be triggered by electric pulses. However, these processes need a longer time than the pulse duration itself.

Electrofusion of cells is broadly applied in biotechnology. In contrast to the induction of cell fusion by some chemical agents, electrofusion has the advantage of being much more selective. Using sufficiently small electrodes, it is possible to fuse even two particular cells. Meanwhile, methods have been developed to bring specific cells into contact with each other. Usually, cells are arranged for fusion by dielectrophoretic aggregation.

This leads us to cell translocations, caused by polarization of cells in AC fields. In the following we will consider in particular the effects of dielectrophoresis and electrorotation.

Dielectrophoresis is the translation of objects in inhomogeneous AC fields (Fig. 3.42). *Electrorotation* is the rotation of particles in a rotating electric field (Figs. 3.43, 3.44).

Fig. 3.42. Dielectrophoresis of cells in an inhomogeneous AC field between a flat and a pointed electrode

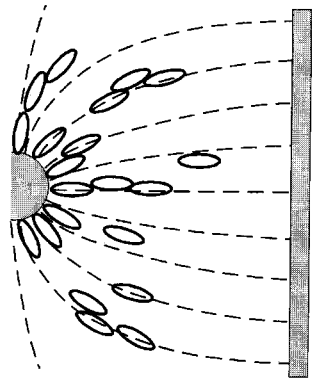
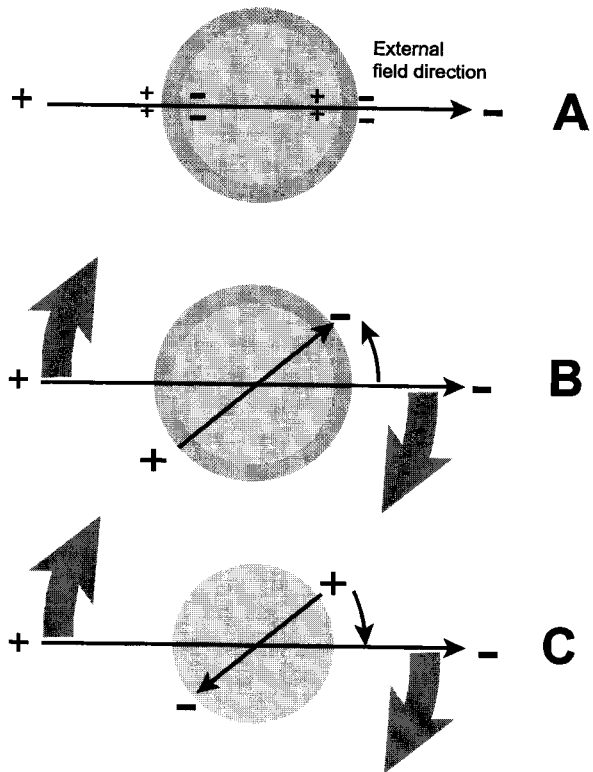


Fig. 3.43. In the same way as depicted in Fig. 3.40, an external field leads to charge separations near the membrane (A). In accordance with this polarization, an effective dipole can be constructed for the low frequency region, where the membrane has a low conductivity (B). In contrast to this, in high frequency fields, where the membrane capacitor is fully short circuited, the dipole is formed only by the polarization of the cytoplasm (C). In both cases the induced dipole follows the rotating external field with some delay (*large arrows*). This delay in one case (B) leads to a repelling, in the other case (C) to an attractive force resulting in anti-field rotation (B), or in cofield rotation (C)



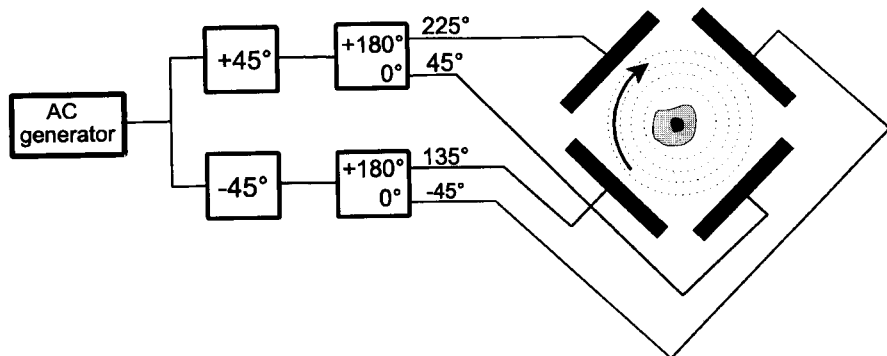


Fig. 3.44. Principle scheme of a four-electrode system for electrorotation. A generator produces an AC signal with adjustable frequency. This signal is split into four signals of particular phase relations. Using a four-electrode chamber, a field rotating at the frequency of the generator is obtained. The rotation of cells or other objects in the chamber is much smaller than that of the field. Therefore, the rotation can be observed under a microscope

Let us first consider the process of dielectrophoresis. A homogeneous electric field may induce a dipole and orientate it, but it will not lead to a translocation. In an inhomogeneous field, however, both charges of the dipole are located at different positions with different field strengths. Therefore, the forces which are acting on both charges in opposite directions are no longer balanced. The direction of the dielectrophoretic translation depends on the kind of polarization of the cell. As depicted in Fig. 3.45, there exist frequency regions of negative as well as positive dielectrophoresis. The driving force of dielectrophoresis is proportional to the square of the applied field strength. This results from the double role of the field: on the one hand it induces the dipole, on the other hand, it drives the object.

In Fig. 3.42 the dielectrophoresis of yeast cells is shown schematically in an inhomogeneous field between the peak of a needle, shown as a small hemispheric electrode, and a flat electrode opposite. The strength of the applied AC field increases in the direction of the needle, as indicated by the increasing density of the current lines. The figure shows the accumulation of the cells near this electrode as a result of positive dielectrophoresis. The polarization of the cells additionally leads to their mutual attraction and subsequently to a sort of *pearl chain formation*. This effect also occurs in homogeneous fields.

The interaction of the external field with the induced cellular dipole may also lead to cell deformations. Depending on the frequency of the applied fields, compression or stretching of the cell with respect to the field direction might occur. These forces also increase proportional to the square of the applied field strength and can be applied to investigate mechanical properties of cells and membranes.

In contrast to dielectrophoresis, which can be described by the real part of the induced dipole, electrorotation is related to its imaginary component (see

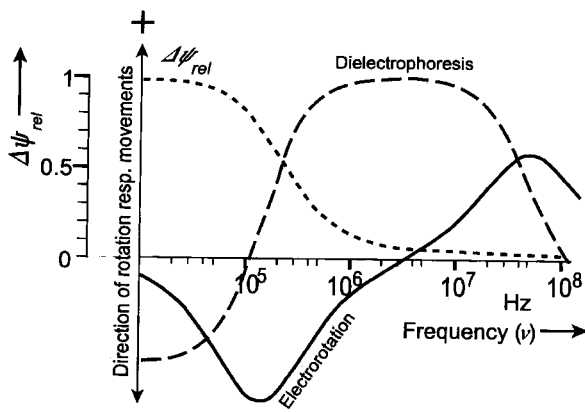


Fig. 3.45. An example for electrorotation (—) and dielectrophoresis (- - -) of a cell in a solution of low conductivity dependent on the applied frequency. Positive values of the electrorotation spectrum mean spinning in the direction of field rotation (cofield rotation), or in the case of dielectrophoresis a movement into the direction of higher field intensity, and vice versa. Additionally, the parameter $\Delta\psi_{rel} = [1 + (2\pi\nu)^2]^{-1/2}$ is depicted as function of frequency (····) according to Eq. (3.5.18) standing for the influence of the induced membrane potential. (With the help of Wachner and Gimsa)

Fig. 3.35). Figure 3.43 shows that a permanent torque acts due to the permanent phase shift between the dipole field and the rotating field which is directed opposite to the direction of the external field (*antifield rotation*), or in the same direction (*cofield rotation*). In Fig. 3.45 the frequency dependence of the electrorotation of a cell in a low conductivity medium is depicted. It shows that the antifield rotation occurs at frequencies of about 100 kHz, followed by a point of zero rotation and finally, in the MHz region a maximum of cofield rotation occurs at about 50 MHz.

To calculate these two types of movement, model considerations are applied. For the simplest approach a single shell model is used which considers a dielectrically homogeneous sphere, covered by a membrane. It represents a spherical cell without electrically significant organelles (Fig. 3.38). Non-nucleated swollen erythrocytes can be modelled by this approach. In this case the radius of the cell and the membrane thickness as geometrical parameters are included in the corresponding equations, as well as the conductivities and permittivities of the membrane, the cytoplasm, and the external medium. These model considerations indicate that the peak of the antifield rotation in its frequency position, and its amplitude, reflects the membrane properties of the cell. This first characteristic frequency is determined by the time constant given by Eq. (3.5.17). If the property of the membrane as a diffusion barrier is changed, this peak vanishes. The maximum of the cofield rotation, i.e. the second characteristic frequency, indicates the conductivity and permittivity of the cytoplasm in relation to that of the external milieu.

Electrorotation is an approved method for measuring dielectric properties of individual living cells. There are a number of investigations indicating that electrorotation can measure alterations of cell properties which are induced by drugs, toxic agents, virus attacks, cell activations and other events. Under special conditions even properties of organelles like nuclei or vacuoles can be measured. Automatic video systems and methods of dynamic light scattering are applied to register the induced movement.

In contrast to electrorotation, which is mostly used for cell analyses, dielectrophoresis can also be applied for preparative cell separation and other biotechnologically interesting techniques. As already mentioned, the formation of cell chains, or cell-cell attachments by dielectrophoresis, for example, is used to perform selective electrofusion of cells by applying short pulses of high field intensity. Recently, special micro devices have been constructed to investigate single cells by electrorotation and to manipulate them by dielectrophoretic translations. According to the large relative surface of such chambers in relation to their volume, heating of the samples was minimized. This allows one to apply dielectrophoresis and electrorotation in physiological solutions of relatively high conductivities. As demonstrated in Fig. 3.46, it is possible to produce microscopic field traps, in which cells, lifted by dielectrophoretic force can be held in a stable position without any surface contact. Using electrode arrangements inducing traveling waves (Fig. 3.47), it is also possible to translocate cells on microchips. This new technique opens enormous possibilities for biotechnological applications.

Further reading: For electrorotation and dielectrophoresis: Georgiewa et al. 1998; Gimsa and Wachner 1998; Fuhr et al. 1996; Fuhr and Hagedorn 1996; electromanipulation and electrofusion: Cevc 1990; Lynch and Davey 1996;

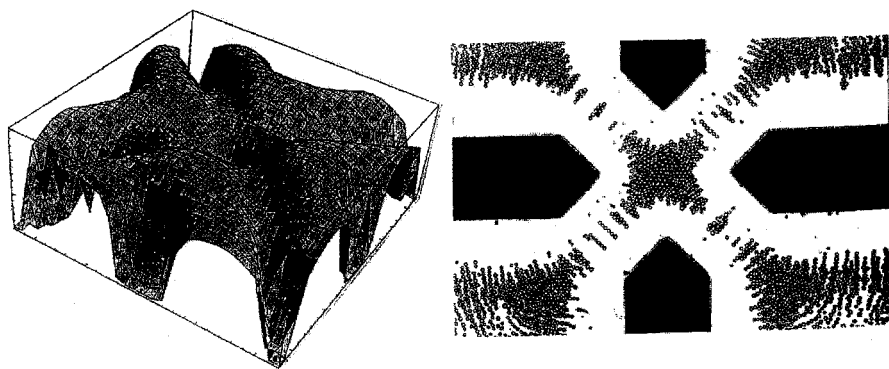
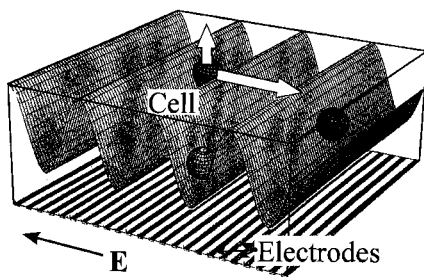


Fig. 3.46. On the *left*: surface of constant force potential in a high frequency field trap. Four electrodes (see also *left*) produce a time average electric field funnel. On the *right*: Trapping, levitation and pearl chaining of latex particles at 1 MHz in the chamber according to the calculated force distribution. (After Fuhr et al. 1996)

Fig. 3.47. Cells moving in a system of interdigitated travelling-wave electrodes. Note that the direction of field propagation is opposite to the direction of cell motion according to negative dielectrophoresis. (After Fuhr et al. 1996)



Zimmermann 1996; dielectric cell deformation: Sukhorukov et al. 1998; Ziemann et al. 1994

3.6 Mechanical Properties of Biological Materials

Biomechanics is a branch of biophysics which explains the mechanical properties of anatomical stability, the movements of limbs as well as the mechanics of walking, flying, swimming, the mechanics of blood flow, of mechanoreceptors and many others. In fact it is one of the oldest branches of biophysics. It has developed in strong connection with the development of the physical mechanics itself. In the period of renaissance the pioneers of mechanics, like Gallileo Galliley, René Descartes, Isaac Newton and many others were always also interested in the mechanics of animals. The first classical book on biomechanics, Alfonso Borelli's *De Motu Animalium* was printed in Rome in 1680. It already contained basic physical considerations on swimming, flying, and movement of animals as well as various calculations of moments of human limbs and the spine under conditions of loads. As we already mentioned in Section 1, this book marked the beginning of medical physics, which was called at that time iatro-physics. Furthermore, D'Arcy Thompson's book *On Growth and Form*, published first in 1917 must be mentioned. This book analyzed for the first time basic processes of cell mechanics, shape formations, and many other biomechanical fields.

In recent decades biomechanics has become increasingly important in diagnostics and therapeutics, especially the biophysics of blood circulation (hemorheology). The biomechanics of the skeleton, as well as of limbs and joints are also of central interest. These investigations of complex mechanical processes require a profound knowledge about the viscoelastic properties of blood and tissues. The investigation of these parameters in vitro as well as in vivo has become an important task in biomechanics.

A further interest in special questions of biomechanics comes from sport. Here various kinds of complex body motions are of interest to optimize its

outcome. There is also interest in ecological aspects of biomechanics, for example life in moving fluids, wind resistance of plants etc.

Further reading: On biomechanics in general: Alexander 1983; Biewener 1992; Fung 1993; Mazumdar 1992; Niklas 1992; Skalak and Chien 1987; Vogel 1983

3.6.1

Some Basic Properties of Fluids

Many parameters characterizing the physical property of a fluid are defined for the case of laminar flow. A flow is called *laminar* if particles of the fluid move in parallel layers to each other (more on properties of laminar flow in Sect. 3.7.1). Figure 3.48 illustrates examples of three kinds of laminar flows. In contrast to a turbulent flow, processes in laminar flows can be calculated by linear thermodynamics approaches (see Sect. 3.1.3, Fig. 3.3).

To understand the approaches of fluid mechanics, and to characterize laminar fluid profiles, we firstly need to introduce the *velocity gradient* (γ), also known as *shear rate*. It is the derivation of the streaming velocity (v) with respect to a coordinate (z), perpendicular to the direction of the flow.

$$\gamma = \frac{dv}{dz}. \quad (3.6.1)$$

The measure of this velocity gradient, therefore, is s^{-1} .

The shear rate of a moving fluid is at a maximum near a solid surface, or in the case of a tube, near the wall. Far away from the surface this gradient becomes zero.

If two parallel plates slowly move, one in relation to the other, they generate a laminar flow in between, the velocity gradient of which is always constant, and is equal to the relative velocity of the plates, divided by their mutual distance ($\gamma = \Delta v / \Delta z$).

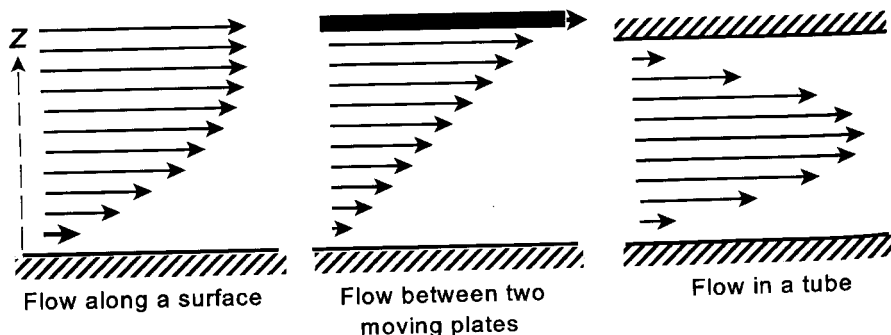


Fig. 3.48. Various examples of laminar velocity profiles

The force (F) driving a plate with a surface A in the case of laminar flow is proportional to the velocity gradient (γ) between the two plates:

$$F = \eta \gamma A. \quad (3.6.2)$$

In this equation a friction coefficient (η) is introduced which is called *viscosity*. The viscosity therefore determines the force which is required to move a plate with an area of 1 m^2 , at a 1 m distance from a parallel surface at a velocity of 1 m s^{-1} , if a laminar flow of a liquid between these surfaces is induced. Thus, the measuring unit for the viscosity is: $\text{N} \cdot \text{s} \cdot \text{m}^{-2}$, or: $\text{Pa} \cdot \text{s}$. Sometimes an older unit P (Poise) is used, where: $1 P = 0.1 \text{ N s m}^{-2}$.

Parallel to the viscosity (η), another parameter, the *fluidity* ($\phi = 1/\eta$) is used, as well as the *kinematic viscosity* ($\nu = \eta/\rho$), a parameter containing the density (ρ) of the fluid. The force deforming a body in a streaming fluid with a velocity gradient (γ) is given by the *shear stress* (τ):

$$\tau = \eta \gamma. \quad (3.6.3)$$

The viscosity depends to a high degree on the temperature. Especially for aqueous solutions, this is caused by the cluster structure of water (see Sect. 2.4.2). In contrast to the viscosity of pure water at $T = 0^\circ \text{C}$, which is 1.79 mPa s , at 25°C it amounts to only 0.89 mPa s , and at 100°C , finally 0.28 mPa s (see Fig. 2.22 in Sect. 2.42).

In order to characterize the influence of dissolved or suspended substances on the viscosity of a fluid, the following derived parameters are used:

$$\text{Relative viscosity: } \eta_{\text{rel}} = \frac{\eta}{\eta_w},$$

$$\text{Specific viscosity: } \eta_{\text{sp}} = \eta_{\text{rel}} - 1,$$

$$\text{Reduced viscosity: } \eta_{\text{red}} = \frac{\eta_{\text{sp}}}{c} \quad (\text{in: } \text{l mol}^{-1}),$$

$$\text{Intrinsic viscosity: } [\eta] = \lim_{c \rightarrow 0} \eta_{\text{red}} \quad (\text{in: } \text{l mol}^{-1}),$$

where η is the viscosity of the solution of suspension, η_w is the viscosity of the pure solvent, and c is the molar concentration of the solute.

The viscosity increases with increasing concentration of the dissolved or suspended substances. As already pointed out in the definition of the intrinsic viscosity $[\eta]$, even the reduced viscosity of a solution is a function of the concentration. The intrinsic viscosity contains information on the structure and the molecular mass of a substance.

For diluted suspensions of small rigid spherical particles the Einstein relation can be applied:

$$\eta_{\text{sp}} = 2.5 V_{\text{rel}} \quad (\text{for: } V_{\text{rel}} < 0.1). \quad (3.6.4)$$

The relative volume (V_{rel}) is the volume of all particles in the suspension together in relation to the volume of the suspension. For suspension of cells

(sperms, erythrocytes etc.) the term *cytocrit* (or specifically, *spermatocrit* or *hematocrit*) is used. It should be emphasised that neither the absolute size of an individual particle, nor the homogeneity of the diameters of all particles in the suspension are of importance for this relation. The Einstein equation, however, is correct only for very diluted suspensions.

Newtonian fluids are those in which viscosity is independent of the velocity gradient ($\dot{\gamma}$). In contrast, *non-Newtonian fluids* alter their viscosity depending on this parameter.

In Fig. 3.49 the behavior of various kinds of non-Newtonian fluids are demonstrated. *Dilatant fluids* are mostly suspensions of solids, like quartz particles. These particles produce entropy by mutual friction as much as the shear rate of fluid increases. The *Bingham-plastic* behavior occurs, for example, in suspensions of non-spherical particles. In this case velocity gradients lead to their orientation, which decreases the viscosity of the suspension. At certain points, if the particles are orientated at a maximum these suspensions behave like Newtonian fluids. The same behavior is to be expected if the particles tend to aggregate in the resting fluid, but disaggregate at low shear stress.

The most common property of biological fluids is the *pseudoplastic* behavior. It occurs for example in blood (Fig. 3.51) and many other biological fluids with heterogeneous composition. Different components of these fluids, for example blood cells, proteins and other macromolecules, aggregate, orientate, and deform at various shear gradients. The resulting function, therefore, does not come to a saturation level at reasonable values of $\dot{\gamma}$.

These shear-induced processes of course need some time to become established. In the case of spontaneous induction of a shear gradient, a time function of the viscosity can be observed. The diminution of the viscosity of a fluid after a sudden decrease of the velocity gradient is called *thixotropy*. The inverse indicates *rheoplectic* behavior.

These properties of fluids must be taken into account when choosing instruments to measure the viscosity. Some of the most common methods are depicted in Fig. 3.50. In a capillary viscosimeter (Fig. 3.50A) the time is

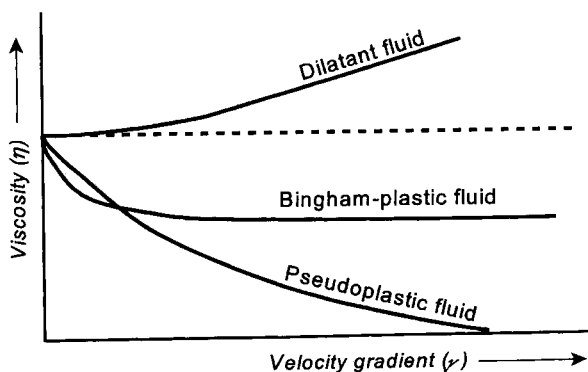
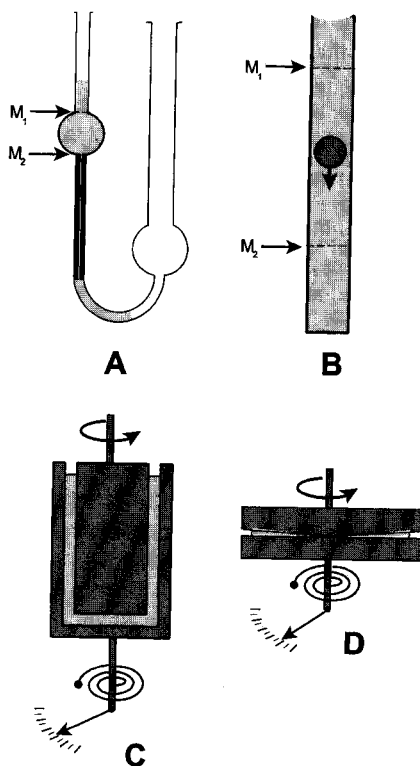


Fig. 3.49. The dependence of the viscosity (η) on the velocity gradient ($\dot{\gamma}$) for a Newtonian fluid (---), and for various types of non-Newtonian fluids (—). (After Glaser 1986)

Fig. 3.50. Various setups to measure the viscosity of fluids. **A** capillary viscosimeter after Ostwald, **B** viscosimeter with falling sphere, **C** coaxial type rotational viscosimeter, **D** cone type rotational viscosimeter



measured which a given fluid needs to pass through a capillary under a known pressure. For this, a certain volume of the fluid is placed in the left part of the U-tube and the time is measured, the fluid must pass the two marks M_1 and M_2 . The time measured is proportional to the viscosity. In this way, the viscosity can be measured after calibration of the setup by a fluid with known viscosity. This so-called Ostwald viscosimeter was later modified by Ubbelohde in such a way that a further vertical tube arranged at the end of the capillary produces an interruption of the fluid. Another way to measure the viscosity of a fluid is the use of falling bodies (Fig. 3.50B). Both methods have the advantage of being simple and, in relation to other equipment, cheap to purchase. On the other hand, however, they are only suited for measuring the viscosity of Newtonian fluids because the velocity gradient of the streaming fluid in the capillary, as well as between the falling sphere and the wall of the tube, are not at all constant (see Fig. 3.48).

To investigate the viscosity of non-Newtonian fluids, measuring instruments are required which allow one to establish well-defined shear gradients. As demonstrated in Fig. 3.48, this is possible between plates moving parallel to each other. This principle is used in so-called rotational viscosimeters. In this case the fluid is placed in the gap between two coaxial cylinders (Fig. 3.50C), or between

a flat plate and a truncated cone (Fig. 3.50D). By moving one part of this equipment, the viscosity of the fluid transmits a torque which can be measured. Usually, one part is rotated with adjustable speed, and the torque of the opposite part is measured by a sensitive instrument. By changing the speed and the thickness of the gap, one can produce various velocity gradients. In the case of the rotating cone the tangential velocity is compensated for by an increasing width of the gap. Therefore, in this case a constant velocity gradient is also established. Curves like that of Figs. 3.49 and 3.51 are produced using such instruments.

3.6.2

The Viscosity of Biological Fluids

Through the development of sensible rotational viscosimeters it has been possible to investigate viscous properties of a large number of biological materials, such as blood, lymph, various secretions, synovial fluids, and many others. This has led on the one hand to a better understanding of the processes of blood flow, of the mechanics of joints etc., and on the other hand it has become a useful indicator in diagnostics.

In Fig. 3.51 the viscosity of a suspension of red blood cells is depicted as a function of the velocity gradient. In contrast to the blood plasma which appears to be a Newtonian fluid, these suspensions, as well as the whole blood, indicate pseudoplastic thixotropic behavior. The reason for this is complex: native erythrocytes, suspended in blood plasma, aggregate at low shear rates. These aggregates are not very stable and disaggregate at somewhat higher shear rates. This disaggregation lowers the viscosity of the suspension. A further increase of the shear rate leads to a deformation of the erythrocytes, furthermore decreasing

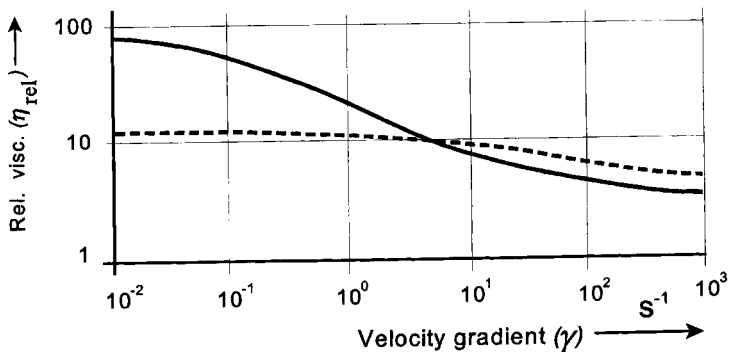


Fig. 3.51. The relative viscosity (η_{rel}) depending on the shear rate ($\dot{\gamma}$) of human blood (—), and heat-hardened erythrocytes, which were resuspended in plasma (---). The differences between the curves in the region of low shear rates are achieved by the aggregation of the native red blood cells, the differences at higher shear rates, by differences in cell deformability. (After Lerche and Bäumler 1984)

the viscosity. Looking at the behavior of erythrocytes which were hardened by fixation, neither the aggregation, nor deformation by shear stress occurs. Living erythrocytes in the shear gradient become elongated ellipsoids which are orientated along the streaming vectors. With increasing shear stress they become more and more elongated. Whereas slow elongations are reversible, up to a certain degree of elongation, irreversible alterations in the membrane occur. Eventually hemolysis occurs as a result of maximal shear stress.

The energy of the streaming fluid which leads to deformation of the cells corresponds to the shear stress (τ) according to Eq. (3.6.3). This means that it depends not only on the shear rate ($\dot{\gamma}$), but additionally on the viscosity (η) of the fluid. To investigate shear induced shape deformations, therefore, high viscosity solutions are usually used, for example, solutions of high molecular weight dextrans of various concentrations.

For biomechanical problems in orthopaedics the properties of the synovial fluid which is located in the joint cavity and surrounded by the joint capsule are of particular interest. In the language of tribology, as the study of the effects of friction on moving machine parts, the mechanism of joints are a kind of depot lubrication with a porous surface. The expression "depot lubrication" points to the synovial fluid which is accumulated in the bursa of the joints. "Porous surface" refers to the articular cartilage which covers the cortical bone in the joint capsule in a 0.3–0.5 mm thick layer. The joints are not only burdened by movement, i.e. by a shear stress of the synovial fluid, but additionally in some cases by a considerable static pressure. It must be guaranteed that these loads do not press the synovial fluid out of the gap. In fact synovial fluid has thixotropic pseudoplastic properties. A large viscosity of this fluid in the joints (η between 1 and 40 Pa s) prevents its exclusion from the gap by hydrostatic pressure. If the joint is moving, however, shear rates up to 10^5 s^{-1} appear. In this case, the viscosity of the synovial fluid falls to 10^{-2} Pa s , leading to a highly effective lubrication of the joint. This particular property of the synovial fluid is caused by a special structure of proteoglycans, which are high molecular weight glycoproteins with an intriguing structure.

This leads us to considerations of the viscosity of microscopic, or even supramolecular structures. It must be pointed out that the definition of viscosity, as given in the previous Section 3.6.1, comes from continuum physics. It does not take into account the behavior of molecules and supramolecular composition of the liquids. It just considers a homogeneous, continuous fluid without any structures. Already, the non-Newtonian behavior discussed above indicates some limitations of this assumption. Moreover, problems arise if we consider the viscoelastic properties of cells and their constituents. At the beginning of the twentieth century, at a time when the physical properties of cells were first discussed, many investigators measured the viscosity of the cytoplasm in relation to various biological functions. This question from the modern point of view is outdated because of our knowledge of the highly organized structure of this region. Even in cells with pronounced cytoplasmic streaming, as in amoebas or plant cells, the question of the origin of these movements and of the molecular mechanisms driving these flows, is of more

central interest than phenomenological models using plasma viscosity in general.

This problem of viscosity in molecular and supramolecular dimensions is important to a high degree in relation to the “viscosity” of the cell membrane (see also Sect. 2.5.3). Using molecular probes, various physical parameters of the membrane can be measured which allow calculation of viscosity parameters. So, for example, a lateral diffusion constant of fluorescent labels in the membrane can be measured and, using the Einstein equation [Eq. (2.3.39)], the viscosity of the surrounding medium can be calculated. In the same way, with specific spin-probes which are bound to particular membrane molecules, the electron-spin resonance techniques (ESR) allow determination of their rotation diffusion constant, which is also a function of the viscosity of the surroundings.

In all cases the viscosity which is determined by these methods is a typical effective parameter. In the same way as, for example, the hydration radius of an ion (see Sect. 2.4.2), this parameter depends on the techniques and the physical phenomenon used for its determination. One should not consider these “quasi-viscosities” in the same sense as the phenomenological viscosity of a fluid which we discussed before. Moreover, the mechanical anisotropy of the membrane must be taken into account, which means that the mobility of particles is not equal in all directions of the space.

Further reading: Leyton 1974; Skalak 1987

3.6.3

Viscoelastic Properties of Biomaterials

The simplest kind of deformation of a body is its stretching. In Fig. 3.52 this is illustrated by a stress-strain diagram. The parameters are defined as follows:

$$\text{Stress: } \sigma = \frac{F}{A} \quad (\text{in: } \text{N m}^{-2}), \quad (3.6.5)$$

$$\text{Strain: } \varepsilon = \frac{\Delta l}{l}, \quad (3.6.6)$$

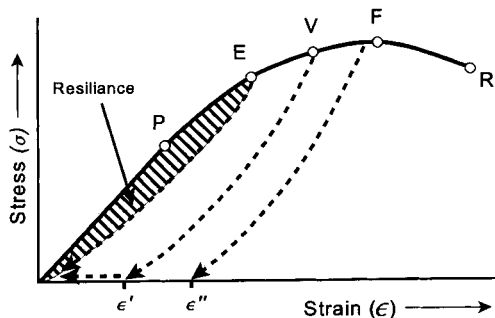


Fig. 3.52. Stress-strain diagram. *P* – limit of proportionality, *E* – limit of elasticity, *V* – limit of reversible viscoelastic deformation, *F* – point of floating deformation, *R* – rupture point, ε' , ε'' – residual strains, *dashed area* – resilience

where F is the force, A is the cross-sectional area of the body, l is its length and Δl is the difference between the resting and the extended material. The strain (ε) is just a relation and therefore does not have a measuring unit.

In the region of minimal strain up to the limit of proportionality (P) Hooke's law is valid stating that the strain (ε) is directly proportional to the stress (σ). The ratio of these two properties is called the *modulus of elasticity*, or *Young's modulus* (Y).

$$Y = \frac{\sigma}{\varepsilon}, \quad (3.6.7)$$

where Y has the units N m^{-2} or Pa. Mostly the unit $1 \text{ MPa} = 10^6 \text{ Pa}$ is used.

The linear relationship between stress and strain does not hold for large stress behind the point P. The deformation however, is reversible up to the elastic limit (E). This means that the body will spontaneously and quickly return to its original length when the deforming force is removed. In this case, however, the relaxation curve (dotted line) did not follow the extension curve (full line). The area between these two lines is called *resilience*. In general the area in this plot, i.e. the product of stress and strain has the unit of energy. The resilience therefore represents the thermal energy which is dissipated during the process of extension and relaxation. In terms of irreversible thermodynamics it is the dissipation function (Φ) of this process [see: Eq. (3.1.64)]. It results from the viscous friction within the body and is an expression of the non-ideal behavior of the system.

If the strain is taken beyond the elastic limit (E), the body begins to deform irreversibly. Up to the limit of viscoelastic deformation (point V), the body quickly relaxes to some residual strain (ε') which eventually may slowly vanish. Overcoming this point, an irreversible deformation (ε'') persists after the stress has been removed. A further extension leads to the point where the body begins to show spontaneous flowing elongation, in spite of further increase in stress until it rips up at point R.

This stress-strain diagram indicates that the deformation of a body outside the limit of proportionality depends not only on elastic, but also on viscose properties of the material. For this the term *viscoelasticity* is used. Biological tissue shows viscoelasticity to a great extent. In this case the stationary stress-strain function, as demonstrated in Fig. 3.52, as well as the kinetics of deformation become important.

It is possible to simulate the strain behavior of elastic and viscoelastic bodies by mechanical systems made up of elasticity elements (springs, Fig. 3.53A) and viscosity or damping elements (pistons in cylinders, Fig. 3.53B). In contrast to the spring, which, if it is massless, elongates immediately if a force is applied, the viscous damping element elongates with a constant velocity and remains in position if the force vanishes. A damping element combined in series with a spring is called a *Maxwell element* (Fig. 3.53C). In the case of an applied rectangular function of force, the spring elongates immediately, as in the case of Fig. 3.53A, then a further continuous elongation of the system proceeds. If the

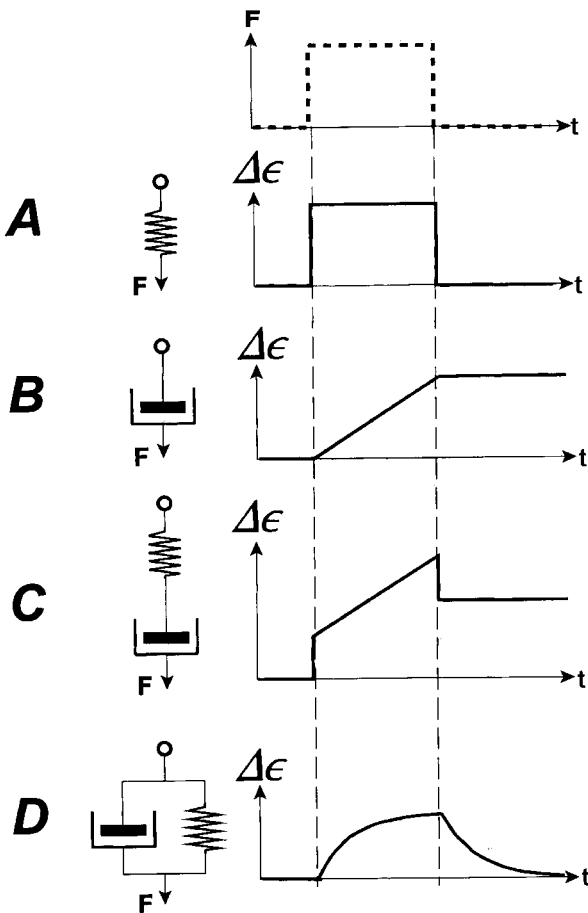


Fig. 3.53. Mechanical models to demonstrate kinetic properties of viscoelastic elements: A – elastic element, B – damping element, C – Maxwell element, D – Voigt element. The graphs at the right side indicate the strain $[\Delta\epsilon(t)]$ of these systems, if the force $[F(t)]$ is applied at their ends corresponding to the rectangular function (.....) in the graph depicted at the top

force vanishes, the spring contracts but the viscous part of the Maxwell element remains elongated. If a spring and a damping component are connected in a parallel arrangement we obtain a *Voigt element* (Fig. 3.53D). In this way a sudden increase of the force (F), leads to an exponential elongation of the system which will contract in the same way if the force vanishes.

Real systems must be regarded as being made up of both Maxwell and Voigt elements in various combinations. Correspondingly, complicated strain-relaxation graphs and complex kinetic behavior is to be expected. Even complicated viscoelastic systems respond elastically to short-term mechanical stress, provided that the time constant of the viscous elements are large in comparison to the time factor of the mechanical impulse. The schematic graphs of Fig. 3.53 with their different combined mechanical elements are somehow similar to the electrical RC circuits in Figs. 3.25 and 3.38. In fact there are some similarities in

their kinetic treatment. In analogy to the electrical impedance which we have discussed in Section 3.5.3, a kind of mechanical impedance can be formulated as the response of a viscoelastic system on time-varying mechanical forces. Measurements of this kind give information on basic mechanical properties of biological tissue.

Comparison of the elastic properties of several materials (Table 3.2) shows that they differ in many degrees of magnitude. In general, two groups of materials with typical elasticity properties can be found – rubber and elastin with very low parameters of Y on the one hand, and steel, bone and flax on the other hand, with an elasticity modulus which is ten-thousand or even a hundred-thousand times larger. Corresponding differences occur in the limits of elasticity.

The reason for this difference are two completely different molecular processes leading to these deformations. In the case of the strain of a steel wire, the atomic spacings in the crystal lattice are found to change. The atoms are moved from their equilibrium position of minimal internal energy and are raised to a higher energy level. The free energy that is stored up in the strained state is released during relaxation. This is the reason for the high elasticity modules of this type of behavior which is called *steel elasticity*.

In contrast to this, materials with a low elasticity modulus ($Y = 0.1\text{--}10\text{ MPa}$) show so-called *rubber elasticity*. A rubber thread, of course, can be extended much more easily and to a much larger extent than a steel wire. We discussed this type of elasticity in Section 2.3.6 (Fig. 2.17). This is the characteristic property of macromolecular materials. The strain leads to a partial deconvolution of the molecules, and as a result, to a reduction of their entropy content. The subsequent relaxation is the result of the increase in entropy back to its maximum according to the second law of thermodynamics. This mechanism is the reason why this type of stretching mechanism is also called *entropy elasticity*. In addition, it explains the strong temperature dependence of the elasticity modulus of these kinds of materials.

Most biological materials indicate rubber elastic properties. Steel elasticity occurs in bones, and to some extent in materials with low water content such as hair, wood or dry plant materials, as for example flax (Table 3.2). In these cases a devolution of the macromolecules is not possible.

Table 3.2. Examples of elasticity properties of some materials

Material	Modulus of elasticity Y , in MPa	Limit of elasticity ϵ_E	Tear resistance σ_R in MPa
Steel	$2 \cdot 10^5$	$1.2 \cdot 10^{-3}$	$5 \cdot 10^2$
Rubber	1	3	
New Zealand Flax (<i>Phormium tenax</i>)	$1.5 \cdot 10^4$	$1.4 \cdot 10^{-2}$	$2 \cdot 10^2$
Bone	10^4	10^{-2}	10^5
Elastin	0.6		

A number of structural proteins like, for example collagen, elastin or resilin are responsible for the rubber elastic properties of tendons and ligaments. These substances also play an important role in the storage of mechanical energy in some periodic or suddenly occurring movements. Although these processes also occur in mammals, they are investigated best in insect jumping and flying. In jumping the muscles apply tension to a system of tendons which take up the energy and subsequently release it, by means of a trigger mechanism, to achieve an increase in power. In this way power, i.e. transformed energy per unit time can be generated which would be impossible to obtain directly from the muscle. In fact, this is the same mechanism that we use for archery. To jump, a grasshopper needs a power of about 5 kW per kg muscle. This would exceed the maximal output of a muscle by ten times. Similarly, during flight a grasshopper stores about 20–30% of the energy of the oscillating movements of the wings using passive elastic elements.

The following special characteristics of the viscoelastic behavior of cells and tissues should be noted:

Occurrence of different regions of elasticity in the stress-strain diagram. Frequently cells and elastic fibers in a tissue are interconnected with each other, forming a network. In this case the viscoelastic properties of the system result not only from the intrinsic properties of these fibers, but first of all from the construction of this network, and in the viscosity of the fluid within it. Such a network can be stretched easily as long as it is deformable as a whole. This is the first region in the stress-strain diagram with low Young's modulus. If the components of the network are eventually fully orientated by the applied stress, a further strain is possible only by stretching the fibers themselves. This means that the Young's modulus increases suddenly. The resulting stress-strain diagram, therefore, does not indicate a flattening of the curve as in Fig. 3.52, but on the contrary it becomes steeper at a certain degree of strain. This sort of behavior occurs, for example, in the walls of blood vessels, where the situation becomes even more complicated due to the activity of the smooth muscles.

Regulation of elasticity behavior. The viscoelastic properties of the network as described above can be controlled by biological processes. This, for example, is possible by loosening and fastening the points of connections between the components. The result would be a sort of viscous prolongation of the body. Another way to control the viscoelastic behavior is the alteration of the water content of the tissue. Changing the cell volume or the intercellular space in the network would change its elasticity modulus. This sort of control is best investigated in the case of the viscoelastic properties of the uterine cervix at various periods of pregnancy.

Mechanical anisotropy. In many biological systems the elasticity modulus depends on orientation. This property is best investigated in bones. Depending on the angle of measurement, Young's modulus in bones can vary by a factor of two. The reason for this is the particular structure of the bone. The struts of the cancellous bone, the so-called trabeculae, are orientated according to its loading in vivo. They are orientated according to the trajectories of pressure and tension

of the bone in the skeleton. This is the result of an adaptation, which was first investigated in 1892 by J. Wolff, who formulated a law of bone remodeling.

The special mechanical properties of the cell membrane has been discussed in detail in Section 2.5.3 (Fig. 2.41). A lipid membrane can be considered as a two-dimensional crystal. The head groups of the phospholipids show similar behavior to the atoms in the 3-dimensional crystal of steel. Therefore, they show a kind of steel elastic behavior in the plane. The elasticity modulus of these membranes is high and their limit of rupture low. The membrane proteins with their rubber elastic properties have no significant influence on this behavior. In contrast to technical materials like rubber sheets, the biological membrane can be easily deformed in an isoplanar way, but cannot resist expansion. This property is important for the dynamics of cell shape and for cell swelling.

Further reading: On viscoelastic properties of biomaterials: Bereiter-Hahn et al. 1987; Fung 1993; Schmid-Schönbein et al. 1986; Skalak and Chien 1987

3.6.4

The Biomechanics of the Human Body

Investigations of biomechanical conditions of the human body, its movement, its carriage under conditions of loading etc. are important tasks in orthopaedics, surgery, sports and occupational safety. What kinds of loads of the muscles and joints result from a particular loading and a particular movement? How can pathologic deviations be cured by surgical operations? How can diseased bones and joints be replaced by artificial materials? In this case immunological tolerance must be realized as well as the adaption of the applied material to the viscoelastic properties of the living bone the artificial joints will be connected to.

Recently, greater effort has been directed toward the construction of mathematical models of the locomotor system including human walking movements. With computer simulation programs, hopefully surgical corrections can be performed with optimal success. Of course, in this context sports efforts should also be mentioned which optimize various techniques of jumping, running etc.

In contrast to the structure of plants which can be modeled statically by systems of flexionally and torsionally loaded bonded structures, for animal systems dynamic body models are required. They are composed of combinations of elements that are stable in relation to pressure and bending with tendons and muscles as elements of tensile stability and contraction. This is illustrated in Fig. 3.54. The head of the horse, for example, is held by a system of elastin tendons, the ligamentum nuchae, as the tensile element and the cervical part of the spinal column as the component in compression. In the abdominal region of quadrupeds the compression element is located dorsally and the tension element is located ventrally. The carriage of the body is stabilized by tendons and it is maintained in an upright position without an additional supply of energy.

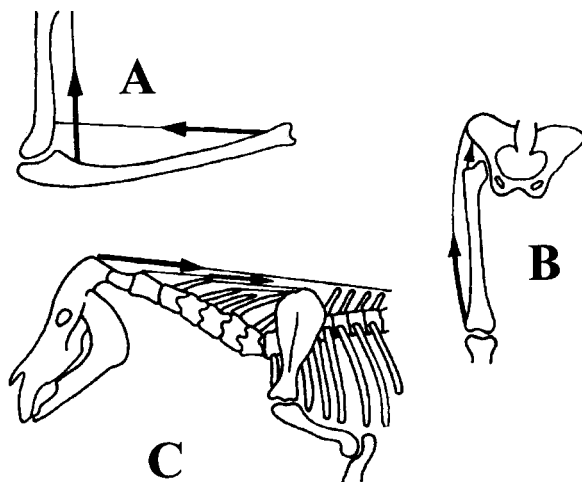


Fig. 3.54. Stabilization of human forearm (A) and thigh (B), as well as the head of a horse (C) by combination of bending and compression stable bones, and tension stable muscles and tendons. (After Glaser 1989)

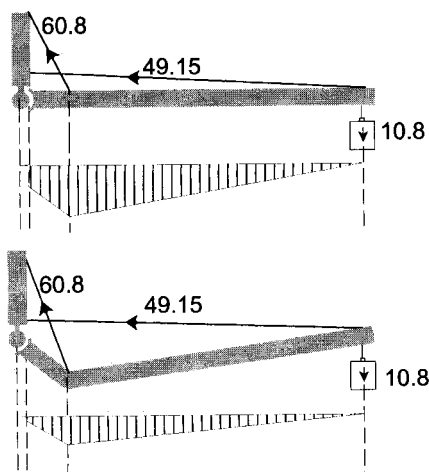


Fig. 3.55. Two steps on the way to optimization of the bending load of the forearm. The two muscles which hold the forearm bone must generate forces of 60.8 N and 49.15 N to compensate the loading force of 10.8 N at its end. These muscle forces are identical in both cases. The bending forces of the bone, expressed in the graphs below, however, are quite different. Compare this scheme with the real situation in Fig. 3.54. (Redrawn after Pauwels 1980)

During evolution, the systems of combined elements which are stable against compression, together with those for tension, have been developed toward optimal use of muscle force, as well as toward maximal stability of the supporting bones. The attachment point of the muscles and tendons determines the vectors of tension and compression in the bone. We already mentioned in the previous section that this induces the orientated growth of the struts in the cancellous bone.

Figure 3.55 shows an example of optimization of the shape of the human forearm. The bending of the bone and the shift of the muscle attachment have led to a significant reduction of the bending force. Similar principles of optimization can also be found in other parts of the muscle-skeleton apparatus.

The bending or the torque of a body can be attributed to stretching and compression of the material. As a measure of the bending, the *radius of bending* (R), or its reciprocal, the *curvature of bending* ($K = 1/R$) is used. If a homogeneous beam or bar bends, a compression of the concave side and a stretching of the convex side occurs. Between these two regions there must be a neutral plane that is neither compressed nor stretched.

Let us consider a section of a bar which is bent between two points that subtend an angle α at the center of curvature (Fig. 3.56). Let R be the radius of curvature measured to the neutral plane. If the angle is expressed in radians then the length (l) of the section at the distance from the neutral plane is:

$$l(x) = \alpha R_i = \alpha(R + x). \quad (3.6.8)$$

If the length of the section along the neutral plane is l_n (at $x_n = 0$; $R_n = R$) then the strain (ε) of any plane parallel to the neutral plane can be calculated by Eq. (3.6.6):

$$\varepsilon(x) = \frac{\Delta l}{l} = \frac{l(x) - l_n}{l_n} = \frac{x}{R}. \quad (3.6.9)$$

As the distance x is measured from the neutral plane, ε can have both positive and negative values. Negative strain in this context means compression of the material and occurs when $x < 0$.

The combination of Hooke's law [Eq. (3.6.7)] and the above relation enables the stress to be calculated:

$$\sigma(x) = Y \frac{x}{R}. \quad (3.6.10)$$

The differential of moment of force (dM) is used to find the force which is necessary to bend the bar. It is calculated from the product of the force (F) and the leverage distance from the neutral plane (x), and it is also related to the differential of an area (dA):

$$dM = x dF = x \sigma dA = \frac{x^2 Y}{R} dA. \quad (3.6.11)$$

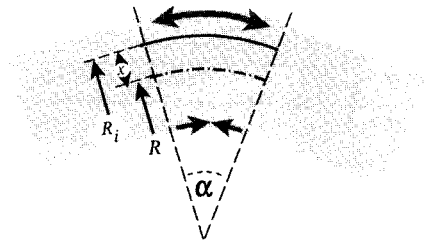


Fig. 3.56. The bending of a bar

The bending moment of the bar is obtained by integration of this equation. If the bar is homogeneous, the modulus of elasticity (Y) is not a function of x .

$$M = \frac{Y}{R} \int x^2 dA = \frac{Y}{R} I_A. \quad (3.6.12)$$

In this equation the integral expression has been replaced by the second moment of area (I_A). This is a measure of the bending resistance of a bar made out of a material having a modulus of elasticity (Y). It can be seen that, for a given bending moment, the smaller the value of I_A , the more the bar bends. This means that if the bending moment does not change and the modulus of elasticity remains constant, the second moment of area is directly proportional to the radius of curvature.

It is easily understood from everyday experience that the second moment of area of a bar depends on the plane in which it is bent. A bar having a flat section bends more easily across the flat than it does on edge. This is determined by the position of the neutral plane. It passes through the center of gravity of a cross-section of the bar, and is always perpendicular to the radius of curvature.

The second moment of area for a given cross section, for example a cross section of a tubular bone, can be determined graphically. First it is necessary to find the position of the center of gravity of this kind of profile. This can be done by suspending a cardboard model of the cross-section by a thread. The center of gravity is the crossover point of all lines which can be drawn in a vertical direction from the points of suspension. After the position of the neutral plane has been fixed, the cross section can be subdivided into rectangular areas (Fig. 3.57). The following approximation can be obtained from the definition of the second moment of area:

$$I_A \approx \sum_{i=1}^n x_i^2 \Delta A_i. \quad (3.6.13)$$

The individual rectangular areas (ΔA_i) are calculated, the distance of their centers from the neutral plane (x_i) are measured, and the products of these two measurements are summed as required by [Eq. (3.6.13)]. The accuracy of this method will be increased if the areas are made as small as possible. The units of the second moment of area are m^4 .

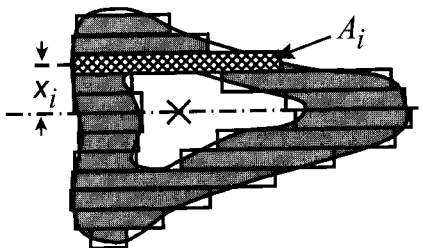


Fig. 3.57. Determination of the second moment of area of a tubular bone by graphical method. The profile of the bone is approximated by many rectangles with individual areas A_i and distances from the neutral plane (-----) x_i . * - center of gravity

For bars with particular geometrical profiles, analytical expressions for I_A are derived. For a compact cylindrical rod with radius r , for example, we have:

$$I_{A(\text{cylinder})} = \frac{\pi}{4} r^4. \quad (3.6.14)$$

The second moment of area of a tube with an inner radius r_1 , and an outer radius r_2 is:

$$I_{A(\text{Tube})} = \frac{\pi}{4} (r_1^4 - r_2^4). \quad (3.6.15)$$

This equation can be used to show the advantage of a structure made up of hollow tubes, an optimizing principle that has occurred in the construction of bones and some plant stems.

In a similar way to bending, torsional deformation can also be estimated. In contrast to bending, in the case of torsion, all regions are stretched as much as they are away from a central axis of gravity. A *polar moment of area* (I_P) can be formulated if the area elements (dA) are multiplied by the square of the radial distance (r) from the center of gravity, and subsequently, the products are summed. In an integral formulation this means:

$$I_P = \int r^2 dA. \quad (3.6.16)$$

The polar moment of area of a tube is:

$$I_{P(\text{Tube})} = \frac{\pi}{2} (r_1^4 - r_2^4). \quad (3.6.17)$$

These statements form the basis for calculating the stability of plants, animals and humans. However, it can be seen from the above equations that some simplifying assumptions have to be made. In particular it must be noted that the modulus of elasticity (Y) may vary with the position and also, as illustrated by the properties of bone, with the direction of the applied force. If the viscoelastic properties are also taken into account, the calculations become even more complicated.

Further reading: Biostatics of humans and animals: Fung 1993; Pauwels 1980; Schmid-Schönbein et al. 1986; Skalak and Chien 1987; biomechanics of plants: Niklas 1992; special problems of paleo-biomechanics: Rayner and Wootton 1991

3.7

Biomechanics of Fluid Behavior

Streaming of viscous fluids occurs at all levels of biological organization. It starts with water flow through pores of membranes, streaming of cytoplasm in plants,

streaming of blood in vessels, and includes the flow of water and air around animals, i.e. the problems of flying and swimming. In this section we will concentrate on some medically important problems of hemodynamics, or hemorheology. We will further include some aspects of flying and swimming because this is one of the classical problems of biomechanics.

3.7.1

Laminar and Turbulent Flows

When a liquid flows along the surface of a thin plate, a flow profile is formed over this boundary which changes its character as the distance (l) from the leading edge of the plate (at: $l = 0$) increases. This effect is illustrated in Fig. 3.58.

Directly at the surface of the plate there is a trapped layer, i.e. a non-moving film of air or liquid. At increasing distance from the plate (z), the velocity of the flow increases. Layers ("lamina") of the fluid slide over one another. This boundary layer is characterized by the velocity gradient ($dv/dz \neq 0$). There is no sharp transition between this layer and the region of undisturbed flow ($dv/dz = 0$, $v = v_\infty$). Usually the limit of this laminar boundary layer is denoted by the distance $z = \delta_L$, when $v = 0.99v_\infty$ (see Fig.

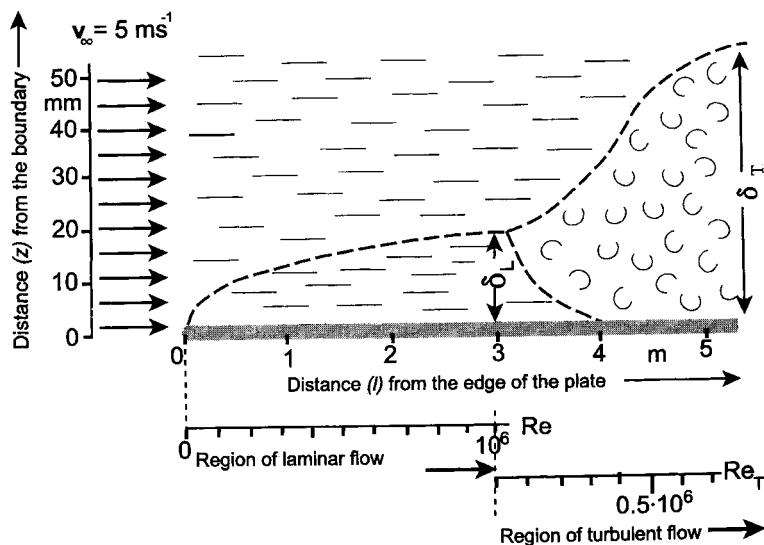


Fig. 3.58. Formation of laminar and turbulent flow profiles near a planar plate. As an example, the numbers correspond to a flow of air with a velocity $v_\infty = 5 \text{ m s}^{-1}$. The thickness of the laminar (δ_L); and the turbulent boundary layer (δ_T) are calculated according to the equations in Table 3.3

3.59). As can be seen in Fig. 3.58, δ_L gets larger as the distance from the leading edge of the plate increases.

As the boundary layer gets thicker it becomes increasingly unstable. At a critical point, this leads to the spontaneous appearance of turbulence. Turbulent flow is the consequence of non-linear processes (see Sect. 3.1.3, Fig. 3.3). The resulting whirls can be considered as a kind of a dissipative structure. Phenomenologically, the transition from laminar to turbulent flow is accompanied by an increase in friction and a substantial increase in the thickness of the boundary layer ($\delta_L < \delta_T$).

In Table 3.3 some basic equations are listed for calculating parameters of laminar and turbulent flow. These equations are based partly on empirical observations, particularly those for turbulent flow.

It is not possible to determine exactly the critical point at which the transition from laminar to turbulent flow takes place. In fact, this is a stochastic process of state transition of the system which occurs as a result of increasing destabilization. The position of the critical point where the laminar flow abruptly transforms into a turbulent one can only be calculated as a probability. It depends on the flow velocity (v), the viscosity (η), the density of the medium (ρ) and a characteristic streaming distance (l). These parameters are connected in the so-called *Reynolds number* (Re) which plays a crucial role in rheology:

$$Re = \frac{lv\rho}{\eta} = \frac{lv}{\nu}. \quad (3.7.1)$$

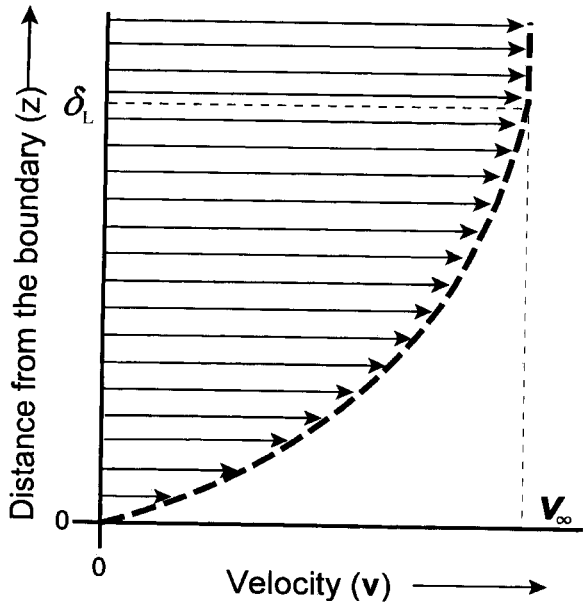


Fig. 3.59. Laminar velocity profile near a surface. The velocity vectors are depicted at various distances (z) from the surface; at $z = 0$, $v = 0$; δ_L is the thickness of the laminar boundary layer

Table 3.3. Equations for parameters of laminar and turbulent flow near boundaries. Symbols (see also Fig. 3.59): v – velocity, z – distance from the boundary, ρ – density of the medium, η – viscosity, Re – Reynolds number; subscripts: 0 – at the boundary, ∞ – in the bulk phase

	Laminar flow (subscript L)	Turbulent flow (subscript T)
Shear stress $\tau(z)$	$\tau_0 (1 - z/\delta_L)$	
τ_0	$0,332\rho v_\infty^2 (Re_L)^{-1/2}$	$0,023\rho v_\infty^2 (Re_T)^{-1/5}$
Velocity $v(z)$	$2v_\infty/\delta_L (z - z^2/2\delta_L)$	$v_\infty (z/\delta_T)^{1/7}$
Thickness δ	$5l_L (Re_L)^{-1/2}$	$0,376l_T (Re_T)^{-1/5}$
Force of surface friction $F_0(l)$	$0,664\rho v_\infty^2 l_L (Re_L)^{-1/2}$	$0,0366\rho v_\infty^2 l_T (Re_T)^{-1/5}$

In this equation the kinematic viscosity ($\nu = \eta/\rho$) is used which has already been introduced in Section 3.6.1.

The Reynolds number is a typical parameter of the theory of similarity. Bodies of identical shape show identical flow behavior, for flow conditions with the same Reynolds number. This is independent of, whether the body is large or small, or whether it is moving in water or in air.

The critical Reynolds number characterizing the transition from laminar to turbulent streaming of a flow parallel to a flat surface, as illustrated in Fig. 3.58, is about $Re = 10^6$. For a sphere with a flow around it, this transition occurs at $Re \approx 10^3$ (see Fig. 3.3). The critical Reynolds numbers of streamlined bodies with so-called laminar profiles are somewhere between these limits, depending on their exact shapes.

Flow inside a cylindrical tube can be characterized in a similar way. If the radius of the tube is taken for the characteristic length l in Eq. (3.7.1) then the critical value of the Reynolds number is about 10^3 . Turbulent flow inside a tube means that the entire flow in it has become destabilized.

The following values for the kinematic viscosity can be used to calculate the Reynolds number for $T = 291$ K:

$$\nu_{\text{Water}} = 1.06 \cdot 10^{-6} \text{ m}^2 \text{ s}^{-1},$$

$$\nu_{\text{Air}} = 14.9 \cdot 10^{-6} \text{ m}^2 \text{ s}^{-1}.$$

If these values are substituted into [Eq. (3.7.1)] it is seen that, in the example shown in Fig. 3.58, turbulence would occur not at $l = 3$ m, but at $l = 0.2$ m from the leading edge, if water is flowing across the plate instead of air, as depicted in this figure. A comparison of flying and swimming objects can only be made under conditions of equal Reynolds numbers. It is quite senseless simply to relate velocities to the length of an object (which unhappily is often done in popular scientific publications much to the amazement of the readers!).

Table 3.4 shows some typical Reynolds numbers for the movement of quite different animals. In Section 3.7.3 we will show that, although they may possibly have optimal laminar flow shapes, the large, fast swimming fishes and aquatic animals exceed the critical Reynolds number for laminar flow.

Table 3.4. Characteristic Reynolds numbers of various moving animals

	Characteristic length (<i>l</i>) m	Characteristic velocity (<i>v</i>) m · s ⁻¹	Reynolds number (<i>Re</i>)
<i>Paramecium caudatum</i>	$2.1 \cdot 10^{-4}$	$1.1 \cdot 10^{-3}$	$1.8 \cdot 10^{-1}$
Mosquito (<i>Ceratopogonidea</i>)	$0.9 \cdot 10^{-2}$	$2.5 \cdot 10^{-1}$	$1.5 \cdot 10^2$
Chaffinch	$3.6 \cdot 10^{-2}$	$2.1 \cdot 10^1$	$5.4 \cdot 10^4$
Crane	$2.6 \cdot 10^{-1}$	$2.8 \cdot 10^1$	$5.0 \cdot 10^5$
Water bug (<i>Dytiscus</i>)	$3.0 \cdot 10^{-2}$	$3.0 \cdot 10^{-1}$	$8.4 \cdot 10^3$
Stickleback (marine)	$1.0 \cdot 10^{-1}$	$7.2 \cdot 10^{-1}$	$5.5 \cdot 10^4$
Shark	$1.5 \cdot 10^0$	$5.2 \cdot 10^0$	$6.1 \cdot 10^6$
Dolphin (<i>Stenella spec.</i>)	$2.1 \cdot 10^0$	$9.3 \cdot 10^0$	$1.5 \cdot 10^7$
Blue whale	$3.3 \cdot 10^1$	$1.0 \cdot 10^1$	$2.6 \cdot 10^8$

3.7.2

Biomechanics of Blood Circulation

Biomechanical calculations of blood flow are based on the one hand on physical equations describing the flow through a tube, and on the other hand, processes must be taken into account which lead to the expansion of the vessels by internal pressure.

Laminar flow through a tube may be thought of as the mutual movement of concentric hollow cylinders. During this movement, each of these cylinders with a radius r and a thickness dr experience a frictional force (F_F) which is proportional to the velocity gradient (dv/dr), their surface area ($2\pi rl$), and the viscosity of the fluid (η):

$$F_F = 2\pi r l \eta \frac{dv}{dr}. \quad (3.7.2)$$

The driving force (F_D) behind such a flow can be obtained from the pressure difference (Δp) and the cross-sectional area of the cylinder (πr^2):

$$F_D = \pi r^2 \Delta p. \quad (3.7.3)$$

In the case of stationary movement, both forces are balanced: $F_F = F_D$. Let the radius of the tube be r' , and assume that there is a trapped boundary layer of liquid [$v(r') = 0$]. Connecting Eqs. (3.7.2) and (3.7.3), and integrating them according to dv one gets an equation for the velocity profile of this flow:

$$v(r) = \frac{\Delta p}{4l\eta} (r'^2 - r^2). \quad (3.7.4)$$

Thus, $v(r)$ is a parabolic function, whereas $v_{\max}$ is the velocity at the center of the tube (see Figs. 3.48, 3.60). In order to calculate the volume flux (J_V) through the tube, the function $v(r)$ must be integrated over the entire profile. This finally leads to the *Hagen-Poiseuille equation*:

$$J_V = \frac{\pi \Delta p r^4}{8 l \eta}. \quad (3.7.5)$$

The units of J_V are $\text{m}^3 \text{s}^{-1}$. Thus, the flow through a tube is proportional to the fourth power of its radius. This means, for example, that slight widening or narrowing of the blood vessels can cause large changes in the blood flow.

These fundamental laws of physical rheology can only be used as a first approximation for what really happens in blood flow through the vessels. The following particularities must be taken into account, leading to an extension of these approaches:

Blood is a non-Newtonian fluid. I.e. its viscosity (η) depends on the shear rate ($\dot{\gamma}$) of the flow (see Sect. 3.6.2, Fig. 3.51). However, integrating Eq. (3.7.2) in order to derive Eq. (3.7.4) we considered the viscosity as constant. The parabolic velocity profile according to Eq. (3.7.4) therefore cannot be applied for blood. Using a function $\eta(\dot{\gamma})$ another velocity profile would result, not identical with a simple parabolic distribution.

Blood is not a homogeneous liquid, but a suspension of cells. In capillaries, the diameters of which are of the same order of magnitude as the diameter of erythrocytes, the velocity profile of the plasma is determined by the moving cells which become strongly deformed in these narrow and branched vessels. This is a problem of *microrheology* of circulation. In large vessels, on the other hand, the so-called *Fahraeus-Lindqvist effect* occurs. This leads the erythrocytes to concentrate in regions of minimal shear stress, namely in the center of the vessel. This means that the viscosity of the blood which we found to depend on the hematocrit (Sect. 3.6.1) increases in this region, but decreases near the wall of the vessel. This leads to a lowering of the streaming resistance of the total blood flow. On the other hand, of course, the streaming profile is changed dramatically. The parabola becomes flattened at the center of the vessel and

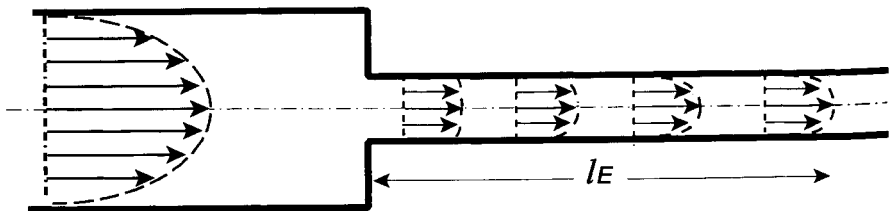


Fig. 3.60. Laminar flow in a tube: a parabolic velocity profile is changing during a sudden narrowing of its radius (entrance effect). Only at a distance l_E is a new parabolic profile established again

steeper near the walls. Furthermore, this effect also leads to a distribution of different sorts of blood cells. In fact, the intensity of the force, shifting the cells by the Fahraeus-Lindqvist effect into regions of lower shear stress, depends on their size. As a result, smaller cells like blood platelets are not as influenced by this effect as erythrocytes with a larger diameter. This leads to a concentration of the platelets near the walls of the vessel. The Fahraeus-Lindqvist effect can be understood as the result of the Prigogine principle of minimal entropy production, as described in Section 3.1.4. It is valid for linear approaches, thus also for laminar flow (see: Sect. 3.6.1). Minimal entropy production for the case of blood flow means that the cells should concentrate at locations of minimal frictional energy dissipation, namely at locations of minimal shear rate.

The diameter of the blood vessels differs along the system of circulation. If a tube suddenly becomes narrow, a so-called *entrance effect* occurs (Fig. 3.60). This means that first the velocity profile of the narrow part of the tube corresponds to that of the central part of the broad tube. Only after a certain distance from the place of narrowing (l_E in Fig. 3.60), will a new profile be established. Usually this occurs at: $l_E = 0.06r\text{Re}$, where r is the radius of the narrow tube. This effect becomes important at the entrance of blood into the aorta. Furthermore, it occurs in the case of air flow in the lungs.

Blood flow is not stationary, but pulsed. Therefore the condition $F_F = F_D$ is no longer valid. This fact is only important for arterial flow. The pulse waves of the heart partly become damped by the elasticity of the walls of the vessels, but they nevertheless proceed as oscillations of pressure and velocity of the blood flow up to the arterioles. One must differentiate between the true velocity of the blood streaming ($\mathbf{v}$) on the one hand, and the velocity of pulse propagation ($\mathbf{v}_p$) on the other hand. This can be calculated using the Moens-Korteweg equation, which is derived for cylindric tubes with thin walls:

$$\mathbf{v}_p = \sqrt{\frac{Yd}{2r\rho}}. \quad (3.7.6)$$

In contrast to the flow rate ($\mathbf{v}$) [Eq. (3.7.4)], the pulse propagation rate ($\mathbf{v}_p$) depends on the density (ρ) of the medium, but not on the viscosity (η). On the other hand, the thickness (d), and the elasticity modulus (Y) of the wall of the vessels are included. This equation does not say anything about the damping of the pulse waves.

To relate various pulsing flows, similar to the Reynolds number, another parameter of similarity (α) is introduced:

$$\alpha = r\sqrt{\frac{\omega\rho}{\eta}}. \quad (3.7.7)$$

In this equation, besides the density (ρ) and the viscosity (η), the pulse frequency (ω) is used. For values $\alpha \geq 3$, the pulse propagation velocity becomes

equal to the streaming velocity ($v_p = v$). In the aorta this parameter is higher than that limit ($\alpha = 15$). In the arteria femoralis this limit is reached ($\alpha = 3$). In these vessels, therefore, the pulses are propagated with the same velocity as the total flow rate of the blood. Additionally, at points of branching of the vessels, reflections of the waves can occur.

These considerations already raise the question of elasticity of the blood vessels. We will start to answer these questions again with simple physical approaches. How is the radius of a tube changed by the strain of the tube wall? For the simplest case, let us use Hooke's law [Eq. (3.6.7)], considering an elastic tube with a radius r , and a circumference $2\pi r$. In this case the stress (σ) of the wall of the tube can be written as follows:

$$\sigma = Y \frac{\Delta r}{r}. \quad (3.7.8)$$

We defined the stress (σ) as the stretching force (F) per area (A) of the stretched material [Eq. (3.6.5)]. Let the thickness of the wall be d , and its length l , so that the cross-sectional area of the stretched wall becomes $A = d \cdot l$. Therefore:

$$\sigma = \frac{F}{ld}. \quad (3.7.9)$$

Defining σ' as stretching force per length, it gives:

$$\sigma' = \frac{F}{l} = \sigma d. \quad (3.7.10)$$

Now, how large would the stress (σ) be in the wall of a tube with an internal pressure p ? This can be calculated using the Laplace equation for a tube with radius r :

$$\sigma' = pr. \quad (3.7.11)$$

Equations (3.7.8) to (3.7.11) enable us to derive an equation to calculate the increase of the radius of a tube (Δr) as a function of the internal pressure (p):

$$\Delta r = \frac{pr^2}{Yd}. \quad (3.7.12)$$

Let us remember that we started with the assumption that the vessel wall shows a linear elastic behavior of the material according to Hooke's law. The relation between stress and strain therefore would correspond to Young's modulus [Eqs. (3.6.7) and (3.7.8)]. In fact the wall of the vessels exhibit a particular viscoelastic behavior. The strain is additionally controlled by smooth muscles. This interaction of passive and active properties of the vessel wall is of great importance for the regulation of blood flow.

All these calculations consider the blood flow in the system of circulation as laminar. Is this correct or is there turbulence in human blood flow? The numbers in Table 3.5 indicate that in fact, in large arteries critical Reynolds numbers can occur. We mentioned in Section 3.7.1 that the laminar flow in tubes becomes unstable at Reynolds numbers near 1000. This limit, however, is only correct for smooth, rigid and absolutely cylindrical tubes. In blood vessels, various factors help to stabilize the laminarity even beyond this limit, others induce turbulent flow at lower Reynolds numbers. Factors inducing turbulence are for example the branching of the vessels and inhomogeneities of their walls. These include arteriosclerotic alterations, or alterations following surgical operations. In general the system of blood circulation can be considered as biomechanically optimized.

Ultrasound techniques are mainly used to investigate flow properties of blood vessels *in vivo*. Furthermore, investigations are undertaken in tubes made of transparent materials which copy particular regions of human vessels (Fig. 3.61).

The model system of Fig. 3.61 shows that near the branching of vessels special properties of flow behavior occur. The critical Reynolds numbers in these regions are lower than in an unbranched vessel. This depends on the angle of bifurcation (α in Fig. 3.61). For $\alpha = 180^\circ$ the laminar flow becomes critical at $Re = 350$. If the angle is only 165° , the critical point is at $Re = 1500$. Additionally, the critical Reynolds number depends on the relation of the radius of the branches.

These few aspects of biophysics of blood rheology show us how complicated this situation is *in vivo*. This branch of biomechanics is developing quickly. This tendency is promoted by the fast progress in surgery of blood vessels on the one hand, and on the construction of artificial cardiac valves, of artificial hearts as well as of various systems of extra-corporal circulation.

Further reading: Fung 1984, 1993; Gabelnick and Litt 1973; Mazumdar 1992; Schneck 1980; Skalak and Chien 1987

Table 3.5. Some rheological parameters of human blood circulation. (Data from Talbot and Berger 1974)

Vessel	Average velocity (m s^{-1})	Diameter (m)	Average wall shear rate (s^{-1})	Reynolds number (Re)
Aorta	$4.8 \cdot 10^{-1}$	$2.5 \cdot 10^{-2}$	155	$3.4 \cdot 10^3$
Artery	$4.5 \cdot 10^{-1}$	$4 \cdot 10^{-3}$	900	$5 \cdot 10^2$
Arteriole	$5 \cdot 10^{-2}$	$5 \cdot 10^{-5}$	8000	$7 \cdot 10^{-1}$
Capillary	$1 \cdot 10^{-3}$	$8 \cdot 10^{-6}$	1000	$2 \cdot 10^{-3}$
Venule	$2 \cdot 10^{-3}$	$2 \cdot 10^{-5}$	800	$1 \cdot 10^{-2}$
Vein	$1 \cdot 10^{-1}$	$5 \cdot 10^{-3}$	160	$1.4 \cdot 10^3$
Vena cava	$3.8 \cdot 10^{-1}$	$3 \cdot 10^{-2}$	100	$3.3 \cdot 10^3$

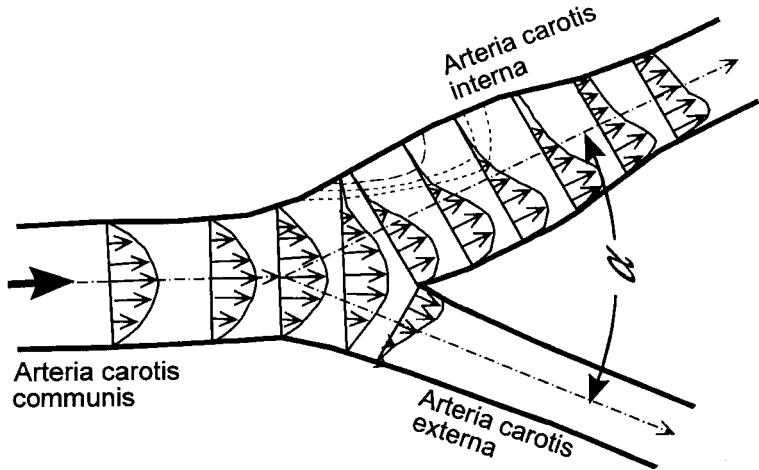


Fig. 3.61. Velocity profile in a model of the human carotis with 70% flow through the arteria carotis interna. α – angle of bifurcation. (After Schneck 1980 modified)

3.7.3

Swimming and Flying

We learned that the Reynolds number enables us to relate flow properties of water with those of air. This gives us the possibility of relating mechanisms of swimming to that of flying.

When a body moves relative to its surrounding medium then a *surface*, or *skin friction* occurs, and furthermore a drag which is caused by the shape of the body, the so-called *profile*, or *form drag*. The surface friction arises from phenomena which occur in the boundary layer and which have been discussed in Section 3.7.1. The form drag, on the other hand, is related to the volume of the surrounding medium that is displaced by the moving body. Regions around the body can become influenced which are much larger than those occupied by the boundary layer in a fluid flowing parallel to a flat plate (Fig. 3.58). The critical Reynolds number, indicating that the laminar boundary layer becomes destabilized, is determined, to a great extent, by the shape of the body. This explains why these two components of drag generation, namely the surface drag on one hand, and the form drag on the other hand, cannot be treated separately from one another.

The boundary layer around a moving body is to a great extent influenced by local pressures caused by differences in the local velocities (v) at different locations. Because of the law of conservation of energy, the kinetic energy of the moving medium ($1/2\rho v^2 V$), and the static energy of compression (pV) must be equal at all points of the space:

$$p + \frac{1}{2}\rho v^2 = \text{const.} \quad (3.7.13)$$

This constant is determined by the following conditions: at $v = 0$ the hydrostatic pressure must be: $p = p_0$, where p_0 is the static pressure of the environment. This leads to the *Bernoulli equation*:

$$p = p_0 - \frac{1}{2}\rho v^2. \quad (3.7.14)$$

The local pressure (p) acting on the surface of a body which moves in relation to a fluid, therefore results from the hydrostatic pressure (p_0) of the environment, lowered by the parameter $1/2\rho v^2$. If the local velocity v is great enough, then p can even become negative.

Figure 3.62 indicates the flow profile and the pressure distribution around a streamlined body, i.e. a body with minimal form drag. Additional to the pressure which is given by the Bernoulli equation [Eq. (3.7.14)] an impact pressure at the point occurs where the velocity vector is perpendicular to the body surface. At points where the velocity of flow is greatest, which in the diagram is indicated by maximal density of the lines of flow near the middle of the body, there will be a local reduction in pressure. These local pressures lead to forces which are always directed perpendicular to the surface of the body. At points where this dynamic pressure becomes negative, i.e. the forces are directed away from the surface, a disruption of the flow from the surface of the body can occur. This results in a wake of large eddies that is a region of turbulent flow which occupies a space much larger than that corresponding to the thickness of the turbulent boundary layers given in Fig. 3.58 and Table 3.3 (page 222).

The following flow patterns can occur by increasing the velocity:

- laminar flow around the body,
- turbulent flow in a boundary layer,
- turbulent, disrupted flow forming a wake.

From step to step the drag resistance is increasing.

In Table 3.4 (page 223) Reynolds numbers are indicated for some swimming animals. The small animals, in spite of having unfavorable shapes, are in the region where the flow is always laminar, but for animals where the Reynolds number is in the order of 100 and higher, a hydrodynamic optimization of the shape of the body is required. In contrast to the sphere which allows laminar

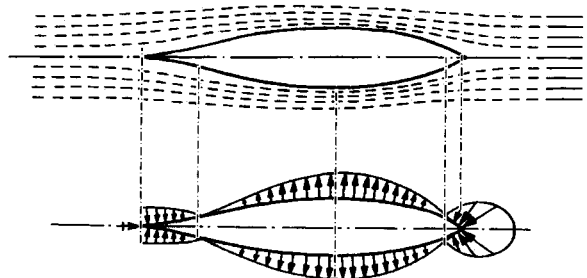


Fig. 3.62. Flow profile and pressure distribution around a moving streamlined body. (After Hertel 1963)

flow up to $Re \approx 1000$, the critical Reynolds number for a streamlined body is, of course, higher. The body shape of fast swimming animals really does appear to be optimal in this respect. Nevertheless, the Reynolds numbers of fast fishes and dolphins, and also of many birds, lie in the supercritical range. In such cases adaptations can be found that, at least, impede the disruption of the turbulent flow. This, for example, is achieved by the early induction of microturbulences at the surface of the body. Surface structures such as feathers or scales bring this about.

Much has been written in papers on bionics about the specific adaptations of the dolphin which enable it to swim extremely fast. Apparently there is a viscoelastic damping layer of the skin and, furthermore, the ability to induce folds at the body surface by muscular activity, both of which prevent the occurrence of latent instabilities in the flow.

The real drag of a swimming fish is difficult to determine. The simplest way to measure this would be to pull dead or anesthetized fish at a given velocity through water. As a result, the values for the drag determined in these experiments are so high that it would be impossible for the musculature of an actively swimming fish to overcome them. The fact that a fish nevertheless does swim with such a velocity is known as *Grays paradox*. The solution to this apparent contradiction is as follows: in contrast to technical constructions, as for example vessels which consist of a driving component (screw), and a frictional component (body), in case of the fish both elements are integrated. A fish diminishes its friction during active swimming. An actively swimming fish has a lower drag than one that is towed passively through the water.

The part of the body which is involved in propulsion of fishes can be very different. Depending on the relative length and flexibility of the tail, three main types can be differentiated. The *anguilliform type* of movement, for example the eel, involves the whole body for propulsion. Most fishes show the *carangiform type* of propulsion (Fig. 3.63). In this case tapering tails of medium length allow fast and dexterous swimming. They are able to accelerate quickly, reaching a high speed after a short time. In the case of *ostraciiform type* of movement, named after the trunk of coffer fish, only the fins, like propellers, propagate the fish. These fishes are only able to swim slowly.

In contrast to the mechanism of swimming, where we considered mechanisms of drag reduction, in the biomechanics of flying, lift is of central interest.

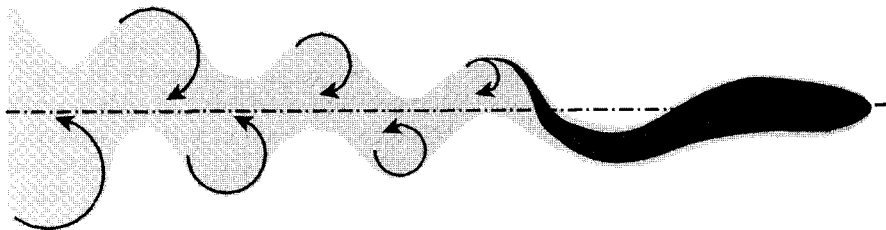


Fig. 3.63. Carangiform mechanism of propulsion of a fish with the vortex street behind

First, we must distinguish between passive gliding and active flying. Various insects (especially butterflies), birds and mammals can glide, as can a number of plant seeds and fruits. Many types of aerodynamic mechanisms are employed by these vegetable objects, ranging from simple adaptations to reduce the rate of descent by means of rotating wings, for example the propeller seed of the maple tree, up to the stable gliding flying-wing of the seeds of the climbing pumpkin species *Zanonia macrocarpa*.

Active flying by animals is achieved through a most varied assortment of mechanisms. Those animals which can rise into the air without some initial help and then move forwards or backwards from this position can be considered as ideal fliers. These are the properties of a helicopter and are only possible in animals which have a large wing area relative to their body weight. It can be deduced from Fig. 3.64 that the wing area does not increase in direct proportion to body mass, but rather to the power of $2/3$. The wing loading of heavier animals is usually greater than $1 \text{ kg} \cdot \text{m}^{-2}$ but lower than in many insects.

The wing loading, i.e. the relation between body weight and wing area, is crucial for the power which is required to hover. Table 3.6 shows calculated values for the power outputs which would be required for stationary hovering flight for some animals. In fact, due to the real efficiency of that mechanism which may be lower, as proposed in the calculation, the required muscular power might be even greater than these figures indicate. The upper limit of the specific muscular performance has already been reached in the case of humming

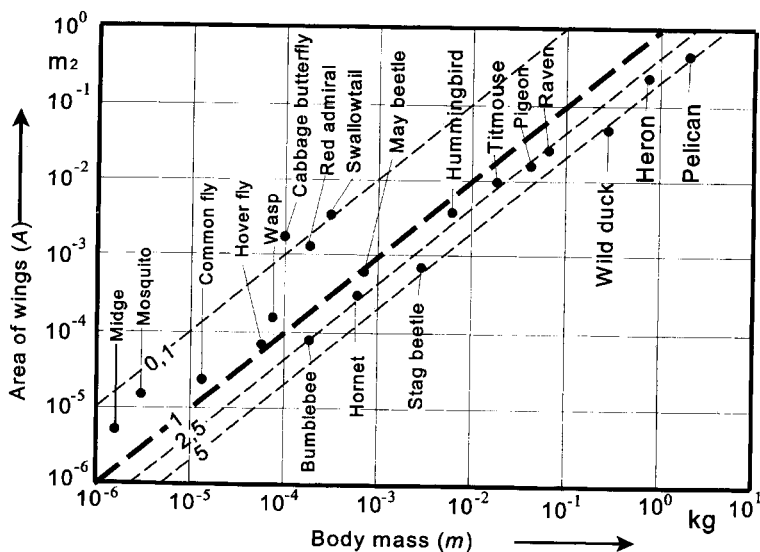


Fig. 3.64. Area of wings (A , in m^2), body mass (m , in kg), and wing loading (dashed lines with parameters in $\text{kg} \cdot \text{m}^{-2}$) of various insects and birds. The points enable us to draw an allometric function $A = 1.5 m^{2/3}$. (Data from Hertel 1963)

Table 3.6. Wing loading and the required specific power for stationary hovering flight in various animals

Animal	Wing loading (kg m^{-2})	Specific power (in relation to the body mass) (W kg^{-1})
Cabbage butterfly	0.04	2.6
Dragon-fly	0.08	3.7
Bee	0.71	1.0
Humming bird	1.01	3.8

birds. Larger birds are not able to hover on their own although some of them, like the kestrel can do so with the help of the wind.

Figure 3.65 shows schematically the wing movement of a hovering hummingbird. The wings move horizontally and each covers about a third of the horizontal area around the bird. The angle of attack is continuously adjusted so that lift is generated as the wing moves forwards as well as backwards. In contrast to humming birds, the wings of flies and bees move in a vertical

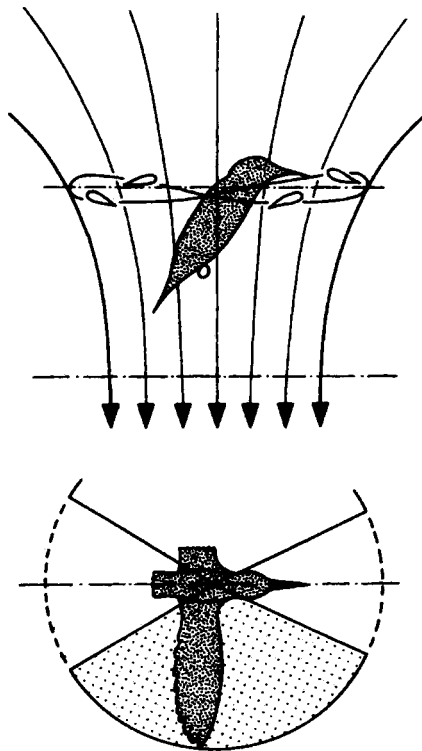
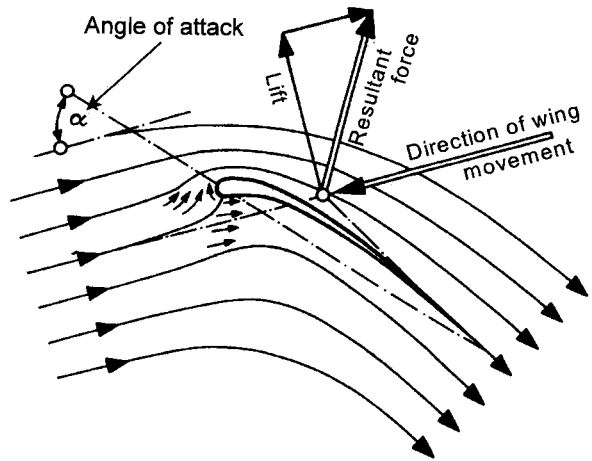


Fig. 3.65. Wing position and air flow of a hovering hummingbird. (After Hertel 1963)

Fig. 3.66. Airstream and forces of a bird's flight during a stroke forwards and downwards. (After Hertel 1963)



direction. The wings are also twisted but only the downstroke is used to generate lift. In case of birds flying forward, the wings are moved up and down. The angle of attack is also regulated, but additionally the wings are bent or partly folded on the upstroke.

In contrast to the humming bird, all other birds require some assistance to get off the ground. Just as an aircraft must reach a take-off speed to ensure that the wings are generating sufficient lift, so must birds. This is achieved by running, jumping or dropping. Nevertheless, the take-off speed of birds is remarkable low. This means that maximum lift at minimal Reynolds number is required. Figure 3.66 shows the force diagram for a wing that is being thrust forwards and downwards. If the incident flow remains constant (v_0) then the lift generated by the wing increases with the degree of curvature of the wing section as well as with the angel of attack (α). The amount of increase of both these parameters is, however, limited. If they exceed a critical value, the flow is disrupted and turbulence develops. This will not only cause a loss of thrust but will at the same time destabilize the entire system. Such an occurrence would cause an airplane to crash. There are a number of biological adaptations preventing such a disaster and at the same time ensuring maximum lift at low speeds.

Further reading: Alexander 1983; Azuma 1992; Hoppe et al. 1984; Leyton 1975; Videler 1993; Vogel 1994; Webb and Weihs 1983

Physical Factors of the Environment

The subject of this section, namely the influence of physical factors on biological systems belongs to environmental biophysics on the one hand and, on the other hand, is important for applications in medicine and biotechnology.

The input of energy from the environment into biological systems can lead to different consequences:

- It can directly interact with the energetics of the organism. This can occur in a specific way, as in the case of photosynthesis, or nonspecifically, for example, just by heating the organism.
- Very small amounts of energy, occasionally being absorbed at particular targets, can significantly disturb biological mechanisms of control. The place of interaction in this case is not a specific receptor, but rather a crucial point in the network of reactions. An example of this is the induction of mutations by ionizing radiation.
- Environmental energy can act as a carrier of information. In this case specific receptors are activated by negligible amounts of energy input.

According to the so-called *Grotthus-Draper principle* it is not the energy penetrating the organism which is effective, but that part of the energy which is absorbed in the system. Investigations into the mechanism of interaction, therefore, start with the question: how, and where is this kind of energy absorbed? Eventually it must be clarified whether alterations are induced by the absorbed energy, and how these influence the biological system. In some cases, biological cascades of amplification exist. The theory of biological amplification was first developed in radiation biology. It is, however, also crucial for a number of other physical influences.

When discussing possible influences of weak physical energies on biological systems, its relation to thermic noise (kT , Sect. 2.3.4) is frequently stressed. This is correct from the physical point of view, assuming the following circumstances are taken into account:

- To relate the energy input to the energy of thermic noise, the primary mechanism must be known which leads to the absorption of this kind of energy in the biological system.
- It must be considered that cooperative effects may be included in these processes. Such interactions are quite probable in more highly organized

molecular systems. This sort of cooperation can significantly promote low energetic influences.

- It must be considered that the organism has developed a multitude of filter systems against thermic noise along the path of evolution. The best example of this is the human ear.

In this context it should be pointed out that the organism has and is adapting through evolution to all physical parameters of its environment. So, for example, the double strand of the DNA is an adaptation to overcome single strand breaks by ionizing radiation. Additionally, a large number of mechanisms have been developed to repair these kinds of damage. On the other hand, during evolution an astonishing number of mechanisms have been developed to sense minimal amounts of energy, in the form of receptors for vibration, for temperature, for light and for other kinds of radiation.

4.1

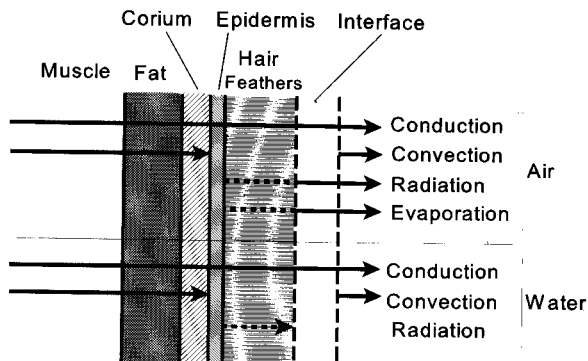
Temperature

In Section 2.3.5 the Arrhenius equation was introduced to describe the temperature dependence of chemical reactions. We learned that every elementary reaction step depends on an activation energy and will be accelerated by an increase in temperature. In this way, even concentrations of metabolites in a steady-state system can be changed. The degree of these changes depends on the extent to which two opposing fluxes are affected by temperature alterations. If, for example, in an open system the temperature dependence of the rate of decomposition or of the efflux of a certain component is predominant, then with rising temperature the corresponding steady-state concentration decreases, and vice versa. It follows that in complex systems, an increase in temperature can lead to an increase, as well as to a decrease in the overall activity. In biological systems the denaturation of proteins is mainly responsible for the reduction of their activity at elevated temperatures. The complete organism as well as isolated enzyme systems show optimal temperatures of operation. In homoiothermic animals the optimal temperature is regulated carefully. Recently, differences in the temperature response between normal versus cancerous tissues has been used in hyperthermy therapy of cancer. The temperature regulation of an organism is closely connected to its entropy balance [see Eq. (3.1.63)].

In Fig. 4.1 different paths of heat transport between an organism and its environment are illustrated. In general, a distinction has to be made between conduction, convection, radiation and heat loss due to evaporation of water. It is possible to define a thermal flux (J_Q) which is the thermal energy that is exchanged per unit of time across a particular area of the surface.

Thermal conductivity is governed by the same kind of equations which we have used to calculate the diffusion of a substance. In this way the thermal flux (J_{Qc}) resulting from conductivity can be included into the flux matrix [Eq. (3.1.58)], and consequently it can be coupled with transport of matter (so-called

Fig. 4.1. Various paths of heat transport between the body and its environment (After Precht 1955)



thermodiffusion, see Sect. 3.4.1). For a temperature gradient in the x -direction, the following linear flux equation can be written

$$J_{Qc} = -\lambda \frac{dT}{dx}. \quad (4.1.1)$$

The negative sign indicates the direction of the thermal flux from higher to lower temperature. The factor λ in this equation is the thermal conductivity, also called the thermal coefficient. It has the units $\text{J m}^{-1} \text{s}^{-1} \text{K}^{-1}$. In Table 4.1 some examples of thermal conductivity are listed, showing the good thermal insulating properties of a layer of fat.

Convection allows a very intensive heat transport at physiological temperatures. As shown in Fig. 4.1, convection in homoiothermic animals is mediated internally through blood circulation, and externally through the flow of air or water. At constant flow, the amount of heat flow transported by convection is proportional to the temperature differences between the two phases. The flow rate on the one hand, and the geometric conditions of the boundary layer on the other hand are crucial for the effectiveness of this kind of heat transfer. The level of convection depends on the thickness of unstirred layers of air or water near the body, and on the flow behavior at the boundary layer (see Sect. 3.7.1). This insulating layer can be improved artificially by hairs or feathers.

Thermal radiation is a kind of electromagnetic radiation as illustrated in Fig. 4.13. To investigate thermal radiation, a so-called black body is used. This is a physical body that will completely absorb all of the infrared radiant energy that falls on it. The black body is also suitable for measuring the radiation which is emitted by an object at a given temperature. In contrast to the other kinds of heat transfer, thermal radiation does not depend on the temperature difference between the body and the environment but rather on the absolute temperature of the emitting body itself, whereas the temperature (T) occurs in the fourth power. The amount of heat emitted by a black body (J_{QR}) obeys the Stefan-Boltzmann law:

$$J_{QR} = \sigma T^4. \quad (4.1.2)$$

Table 4.1. Thermal conductivity for various tissues and other biological materials. (After Precht et al. 1955)

	Thermal conductivity $\text{J m}^{-1} \text{s}^{-1} \text{K}^{-1}$
Tissue in vitro	
Human skin: epidermis and corium	0.34
Epidermis (pig)	0.21
Fat (pig)	0.17
Muscle (pig)	0.46
Living tissues in situ	
Human skin, poor blood flow	0.31–0.33
Human skin, strong blood flow	1.5
Human muscle, no blood flow	0.46
Human muscle, normal blood flow	0.53
Human muscle, strong blood flow	0.63
Hair and feathers	
Wool (loose)	0.024
Feathers	0.024
Rabbit fur	0.025
Other substances	
Air	0.023
Water (20 °C)	0.59
Silver	420.0

The proportionality constant is $\sigma = 5.67 \cdot 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$. Additionally, a correction factor is used if the surface of real bodies does not behave like a black body. For the surface of biological objects it lies between 0.9 and 1. The value depends on the pigmentation of the organism.

Evaporation of water is an important mechanism of temperature regulation of animals living in air. The evaporation of 1 g of water will cause the body to lose about 2.4 kJ. The rate of evaporation depends on the surface structure, on the wind velocity, and on the difference in the vapor pressure between the surface of the body and the surroundings. Because the vapor pressure is temperature dependent, evaporation can still take place when the air is saturated with moisture provided that the ambient temperature is below the body temperature.

The properties of biological systems at extremely low temperatures is of special interest for the techniques of cryopreservation. In this respect, two processes are critical for the survival of cells: the structural damage of the cell by microscopic ice crystals, and the alteration of the osmotic equilibrium of the cell during processes of freezing and the retrieval of them from frozen storage. Similar to other processes of crystallization, the formation of ice crystals starts with certain crystal nuclei in the cell. The rate of cooling determines the number and the size of the developing crystals which can be crucial for the survival of the cell. To allow water to flow quickly out of the cell, cryoprotective agents are used, such as for example glycerol or dimethyl sulfoxide (DMSO). They depress the freezing point in parallel, so that ice crystals begin to form only at about

-5 °C. If, at -5 °C to -15 °C the cell is sufficiently dehydrated, ice crystals will form in the surrounding medium but not in the cell interior. Dehydration of the cells means shrinkage and concentration of the internal solutes. This must be restored in the period of thawing. The size of biological objects which are suitable for cryopreservation is determined by the ability to control the rate of heat exchange during freezing as well as during warming.

Further reading: For temperature regulation: Stanier et al. 1984; for cryobiology: Franks 1985

4.2

Pressure

When aquatic animals move in a vertical direction they are exposed to greatly differing pressure. For an increase in depth of 10 m, the pressure increases by about 0.1 MPa. Deep-sea organisms live under conditions where the hydrostatic pressure can be as high as 100 MPa. Seventy percent of the surface of the earth is covered by oceans with an average depth of 3800 m. Fifty percent of the surface of our planet is covered by water at least 4000 m deep. In fact this is the depth with the greatest number of different species.

Pressure an important physiological parameter not only for aquatic organisms. Humans can also be exposed to changes in environmental pressure. This particularly concerns the field of occupational medicine. It can involve conditions where the pressure is either increased or reduced.

Basically, the following mechanisms should be taken into account when considering the effects of hydrostatic pressure on biological systems:

- volume changes in gas-filled cavities,
- changes in the solubility of gases in blood and tissue water,
- changes in the structure of water and corresponding consequences on hydrophobic bonds.

These mechanisms become important at very different levels of hydrostatic pressure. While the first two mechanisms concern low pressures, changes in water structure, and consequently hydrophobic bonds, only occur at extremely high levels of hydrostatic pressure.

The reduced oxygen solubility in the blood which occurs at great heights, is particularly important for alpinists and pilots. This oxygen deficit can be compensated for by an increase in the partial pressure of oxygen in the inspired air. If humans remain under an increased pressure, for example divers, the reverse effect occurs, namely an increase in the amount of dissolved gas in the blood. This may lead to a dangerous situation during the return to normal atmospheric pressure, especially following a long stay at a great depth. If the decompression is too fast then gas bubbles are formed in the blood (mainly N₂). This can cause aeroembolism also called the bends or caisson disease.

The exposure of biological systems to high pressure can also shift the equilibrium positions of physiological reactions according to the Le Chatelier principle if the reaction leads to alterations of the partial molar volume of the reactants. This will occur mostly in reactions producing a gas. Moreover, it is important in processes associated with modifications of the water structure. We already discussed these circumstances in Section 2.4.2. Although this effect only occurs at pressures greater than 10 MPa, and is therefore important only for deep-sea research, it is particularly useful as a method for studying cellular processes.

Further reading: Péqueux and Gilles 1985

4.3

Mechanical Oscillations

Mechanical oscillations on the one hand can influence biological systems directly by input of considerable energy, and on the other hand they can be recorded by animals with the help of highly specified receptors and serve as physical carriers of information. Oscillations in the air at frequencies between 16 Hz and 20 kHz are perceived by humans as sound. Oscillations with lower frequency, in fact those lower than 20 Hz are called *infrasound*, and those with frequencies higher than 20 kHz are called *ultrasound*. Furthermore, vibrations can influence biological systems. Oscillations of materials of various frequencies which may induce sound, but which influence biological systems mostly by direct contact, are described as vibrations.

4.3.1

Vibration

The effects of vibration are of particular interest in occupational medicine. It concerns induction of oscillations in the human body by various vibrating tools and devices, vibrating seats of tractors, trucks and cars etc. In addition to affecting vegetative functions, leading to kinetosis, or for example to seasickness, a high and long-lasting vibration exposure may contribute to low back pain, or other acute or even latent disorders or injuries.

Generally, the human body becomes vibrated through contact with a vibrating solid object, standing or sitting on it, holding it etc. The intensity of the vibration as well as the efficiency of its transfer to the human body and its own resonance frequency are of interest. From the physical point of view this is the problem of oscillations of resonators consisting of springs and dampers, enforced by external vibrations. In the simplest case, the body will be excited by a simple harmonic oscillation in the form:

$$x = x_{\max} \sin \omega t, \quad (4.3.1)$$

where x means the displacement, oscillating with the angular frequency ω and a maximum amplitude $x_{\max}$. Deriving this equation in respect to time (t), the vibration velocity (v) is obtained by:

$$v = \frac{dx}{dt} = \omega x_{\max} \cos \omega t = v_{\max} \cos \omega t. \quad (4.3.2)$$

In practice, man is not influenced by vibrations consisting of simple sinus waves but rather by random vibrations, consisting of a sum of many frequencies differing in amplitude as well as in phase behavior. Using Fourier analysis they can be regarded as a sum of many sinusoidal oscillations.

Investigating the induction of vibrations by contact with vibrating materials, biological systems cannot be considered simply as consisting of rigid materials. We have already discussed viscoelastic properties of biological materials in Section 3.6.3 and demonstrated some viscoelastic models in Fig. 3.53. Such models can be used to construct vibratory biomechanical models of the human body, taking in to account the inertia of masses.

The kinetic properties of similar systems will be discussed later in Section 5.1.2, where we consider the behavior of various systems after a single extension (Fig. 5.3). Here, enforced oscillations of such systems are of interest. Every oscillating body has its own characteristic resonance frequency which is indicated in Figs. 5.3C and 5.4. If the frequency of external vibration inducing these oscillations is much smaller than this resonance frequency, then the system follows exactly the amplitude and phase of the applied oscillations. When the applied frequency is near that of the resonance frequency, or equals it, then the amplitude of the enforced oscillation is greater than that of the applied oscillations. An undamped system in this case would always increase its amplitude until it breaks down (Fig. 5.4, curve 4). Depending on the degree of damping, in the resonance region the maximal amplitude of vibration will be lowered, but nevertheless it can be larger than the amplitude of the applied vibration. A further increase of the external frequency leads to a decrease in the coupling efficiency up to zero, at $\omega \Rightarrow \infty$. In all cases the frequency of the enforced vibration corresponds to the external frequency, but phase shifts may occur.

The oscillation of a mechanical system can be treated as analogous to the electrical oscillations of an RCL circuit (see also Sect. 3.5.3). Complex parameters are used to describe damping and elastic parameters of these mechanical systems. In analogy to electrical systems a *mechanical impedance* of the system can be defined. It is the relation between vibrational force and vibrational velocity.

This coupling of external vibrations to various parts of the body can be measured directly. Figure 4.2 shows the movement of head, shoulders and hips of a person sitting on a vertically vibrating seat. The ordinate shows the relation between the amplitude of the vibration of the designated points of the body, and the amplitude of vibration of the support. It is shown that at frequencies below 2 Hz, the body simply follows the movement of the seat, i.e. the degree of amplification is 1. The fundamental resonance frequency of a human body is

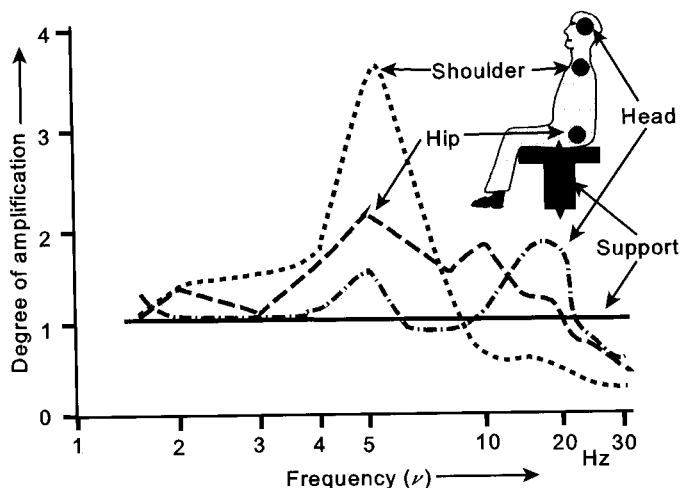


Fig. 4.2. The amplification factor for vertical oscillations at various points of the body of a man sitting on a vibrating support as function of the frequency. (Redrawn according to Dieckmann 1984)

about 5 Hz. This is the frequency of maximal displacement of all three reference points of the body in Fig. 4.2. The head furthermore shows a second maximum at about 20 Hz.

The biological effects of long-lasting vibrations mostly consist of a kind of wear of the joints. The actual forces set up in the joints during vibration can only be measured directly in special cases. They cannot be calculated simply by physical models, using passive mechanical properties of anatomical constituents of the body. It must be taken into account that forces are generated by the muscles trying to compensate the external stress. This, however, in some cases can lead to an opposite effect, namely to an additional strain on the joints. In general, one must consider that the mechanical impedances of parts of the body are influenced by neuro-physiological processes.

In the frequency range below 2 Hz the effects of vibration of the body cannot be simply attributed to mechanical processes. Particularly, the frequency of 0.3 Hz is responsible for sea-sickness, a special kind of so-called kinetosis. All these processes are caused by neuro-physiological influences, and are induced by the vestibular system detecting accelerations of the head. This can lead to alteration in blood circulation with all subsequent physiological consequences.

The mechanisms by which humans and animals can perceive vibrations are a special field of research. Humans, for example, can detect a vibration of 200 Hz with an amplitude of only 0.1 μm ! The sensibility of insects and spiders can be several orders greater.

Further reading: Fritz 1998; Qassem et al. 1994; Seidel and Heide 1986; Wilson 1989

4.3.2

Sound

It has been known for a long time that alterations of external signals are perceived more sensitively, the lower the absolute intensity of the signal is. Let us take a sound intensity (I , in W m^{-2}). The intensity difference is dI and a relative difference: dI/I results. This relative alteration of the physical signal, should thus be proportional to the perceived difference of the sound intensity level (dL). This rule which is valid for all kinds of sense organs is called the *Weber-Fechner law*. The integration of the corresponding differential equation leads to the following logarithmic relation:

$$L = 10 \lg \frac{I}{I_0} = 10 \lg \frac{p^2}{p_0^2} = 20 \lg \frac{p}{p_0}. \quad (4.3.3)$$

The dimensionless unit for the sound intensity level (L) is the decibel (dB).

The integration which led to Eq. (4.3.3) needs an initial condition which is included in the parameter I_0 . This is the reference sound intensity which is for sound transmitted in air: $I_0 = 10^{-12} \text{ W m}^{-2}$. It corresponds to the human hearing threshold level for a reference frequency of 1000 Hz, i.e. for the frequency of maximal sensitivity of the human ear. Furthermore, the sound pressure (p) is introduced, the square of which is proportional to the intensity (I). The reference pressure, correspondingly to I_0 is defined as: $p_0 = 2 \cdot 10^{-5} \text{ Pa}$.

Figure 4.3 indicates a frequency diagram of human auditory sense. The sound intensity level (L in dB), and the sound pressure (p in Pa) are plotted against the frequency. Additionally, the unit phon is used. This is a unit for measuring the apparent loudness of a sound, equal in number for a given sound to the sound intensity in dB of a sound having a frequency of 1000 Hz when, in the judgment

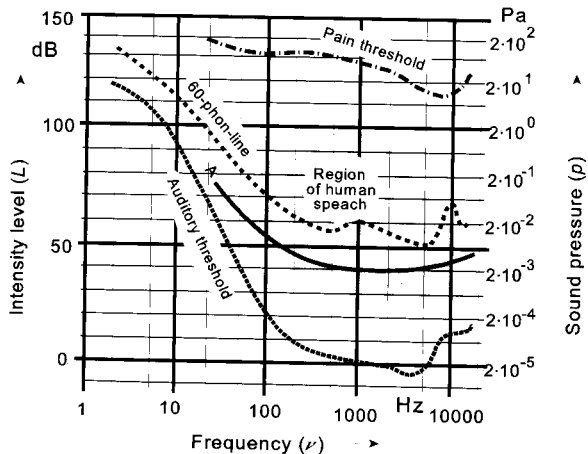


Fig. 4.3. Response of human ear to sound intensity. The curves are extended up to the frequencies of infrasound ($\nu < 20 \text{ Hz}$) according to data from Møller 1984. A-weighting scale according to Fig. 4.5

of a group of listeners, the two sounds are of equal loudness. The 60-phon curve therefore crosses the 60 dB level at the frequency of 1000 Hz. This indicates that a 50 Hz sound, for example, with an objective sound intensity level of 120 dB subjectively seems to have the same loudness as a 1 kHz sound of only 60 dB. The phon unit therefore is a subjective measure which is related to a normal human ear. It is also useful to relate not only various pure tones (sine waves), but also sounds (true musical sounds with overtones) and noise (random mixture of frequencies). This evaluation, however, depends not only on the loudness but also on duration. Sounds with a duration lower than 1 s will be subjectively underestimated.

The use of the unit phon is not sufficient to determine an objective noise exposure of humans in their environment. Figure 4.4 shows some examples of noise production of various sources. It has been known for a long time that in case of permanent exposure, vegetative diseases are possible at 60 dB. The level for dangerous noise at low frequency is at 80 dB, and for high frequencies at 90 dB. Sound intensities which can induce acute injury start above that region.

From the legal point of view, noise is a sound which is able to disturb, to endanger, to cause inconvenience, or to put some one at a significant disadvantage. It is complicated but absolutely necessary to measure this legally definite noise in a physical way.

A simple measurement of an overall sound intensity in dB is too crude to be used for practical environmental protection because of the broad variations of the character of noise. Usually, weighing scales are used to evaluate a particular noise. These scales are determined empirically and fixed by standardization (Fig. 4.5). They are always normalized for 1 kHz and take into account the frequency dependence of the sensibility of the human ear. With the help of these scales it is possible to take into account the lower sensibility of the human ear to frequencies which are lower or higher than 1 kHz. Of course, such scales are

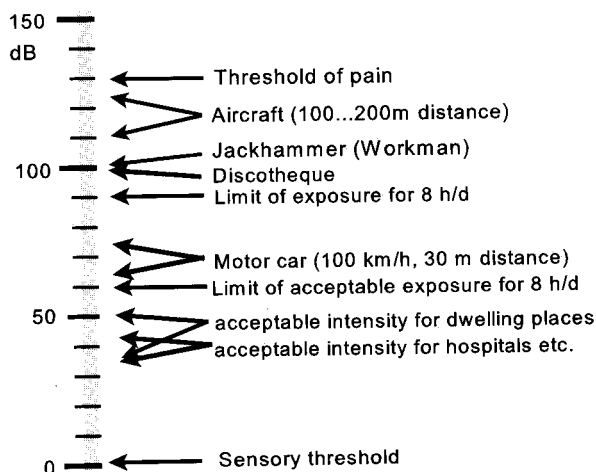
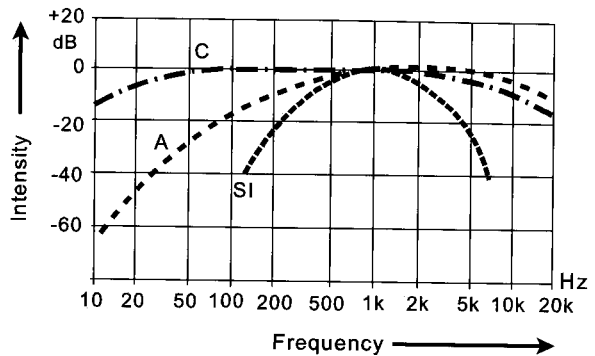


Fig. 4.4. Example of noise production of various sources

Fig. 4.5. Weighting scales for practical noise measurement (description in the text)



dependent on the real character of the measured noise and on the question being asked. There are differences in the frequency mixture, depending on whether the noise comes from engines, from a discotheque, from traffic or from other sources. It also makes a difference as to whether the noise limit of a working place in a factory, in an apartment or in a hospital is to be determined. Mostly the weighing scale A (Fig. 4.5) is used. It takes into account in particular the low sensibility of the human ear to lower frequencies (Fig. 4.4). A very flat scale B (not drawn) is not used anymore. Instead the C-scale is used for noise in a very broad frequency range. The SI scale particularly evaluates the disturbances of human communication.

Further reading: Harris 1991; Wilson 1989

4.3.3

The Biophysics of Hearing

Hearing on the one hand means the reception and analysis of sound, on the other hand the localization of its source. The process of sound localization is based on the analysis of the time delay for the sound reaching the two ears, and on evaluation of the differences in the corresponding intensities. Humans are able to localize a source of sound with an angular resolution of 3° . This means that the minimal time resolution of the differences of arriving auditory signals must be about $35 \mu\text{s}$. This is mainly a question of neurophysiology. In the following text we will for the most part concentrate on the problem of sound perception and frequency analysis.

To differentiate between neurobiological and biomechanical mechanisms of sound analysis, the following experiments are instructive: two pure tones are transmitted by a headphone. In one case, both of them are simultaneously transmitted into both earpieces, in a second case, one tone is transmitted into the right, the other into the left earpiece. If both tones are mistuned, having nearly the same frequency, first order beats occur, i.e. the amplitude of the resulting tone becomes low frequency modulated. This sort of mistuned sound

is recognized, if both tones are fed together into both earphones. The resulting first order beat is indicated in the inner ear. In contrast to this, tones with nearly harmonic frequencies, i.e. mistuned fourth, or third etc., produce beats of second order, namely a sort of frequency modulation. In this case, recognition is also possible, if each tone is fed into one of the two earphones. Then no physical interference of the tones can occur. They come together only in the brain.

Figure 4.6 schematically shows the organization of the human ear. Anatomically as well as functionally it can be divided into three parts: outer, middle and inner ear. Each of them participates in the process of sound analysis. Additionally, of course, the sound will be analyzed in the brain as well.

Beside the true nerve impulse from the excited hair cells, the auditory nerve also transmits so-called microphonic potentials. These are obviously generated passively as a result of vibrations of the inner ear as inhomogeneous dielectrics, and are not caused by nerve excitations. It remains unclear as to whether they have any physiological relevance or not.

The outer ear acts as a simple acoustic amplifier with certain frequency characteristics and a particular directional selectivity. The resonance frequency of the human outer ear is between 2 and 3 kHz, i.e. at the maximum of human sound sensitivity (Fig. 4.3).

In the middle ear the auditory ossicles (malleus, incus, stapes) which are suspended by flexible ligaments transfer the vibrations of the eardrum having an area of 65–85 mm² to the oval window which is only 3.2 mm². This system works passively, without energy supply as a kind of mechanical impedance transformer. It transforms vibration of air into vibration of the lymph fluids, which resemble water as far as density and viscosity are concerned ($\eta = 1.97 \text{ mPa s}$). During this transduction a lowering of amplitude and an increase in pressure occurs. In the case of the human ear the pressure increases by a factor of about 17, for cats by 60. Recently it was found that regulation of

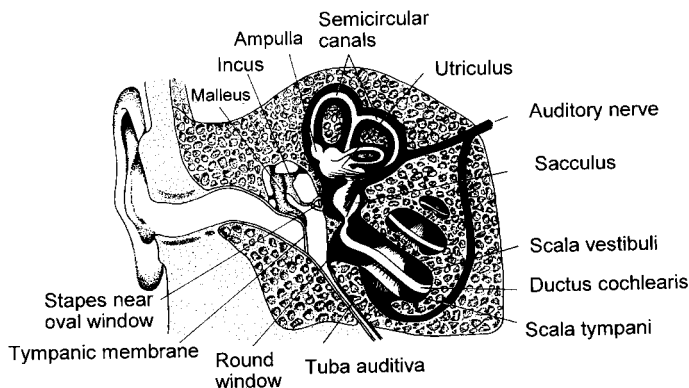


Fig. 4.6. The human ear consisting of the outer ear, the middle ear between the eardrum and the oval window, and the inner ear. (After Penzlin 1991, redrawn)

the properties of this impedance transformation can be controlled by ligaments, protecting the ear against overloading.

The actual sound detection takes place in the inner ear. This is a helical coiled canal that, along its length, is divided by Reissners membrane into two cavities, the scala vestibuli and the scala tympani. The perilymph fluid of the scala vestibuli and scala tympani is separated from the endolymph fluid of the scala media. The sound pressure is transmitted from the oval window to the perilymph fluid and proceeds to the scala tympani (Figs. 4.6, 4.7). At the end of this, the round window buffers the sound waves. There is no reflection of the sound and therefore no standing waves can occur.

Reissners membrane and the basilar membrane transmit the sound pressure of the perilymph fluid everywhere to the endolymph fluid of the scala media. The sensory cells themselves are arranged on the basilar membrane, or better: in the cleft toward the canalis spiralis under the tectorial membrane (Fig. 4.7).

Herrmann von Helmholtz assumed that each point on the basilar membrane along the cochlea is sensitive to a particular pitch. This single-point theory of auditory perception has been confirmed by George von Békésy (Nobel prize winner 1961), but not in the sense of the auditory theory of Helmholtz who postulated a resonance theory, claiming that various places on the basilar membrane indicate resonance properties for various frequencies. There is, however, no experimental evidence for this.

On the other hand, the basilar membrane exhibits very specific viscoelastic properties. In humans the basilar membrane is 35 mm long and gets 10 fold wider over this diameter away from the oval window. At the same time, its elastic properties change. As shown in Fig. 4.8, the stiffness of the basilar membrane is reduced by about 100-fold with increasing distance from the oval window.

Investigations into the vibration characteristics of this system are very complicated. A first approach to this behavior can be made using the model shown in Fig. 4.9. A pendulum transmits its movement to a rotatable rod to

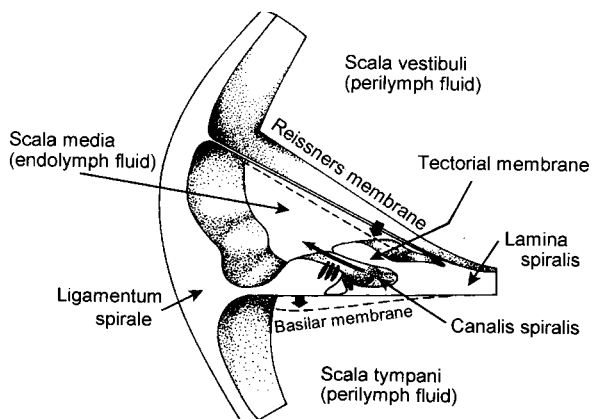


Fig. 4.7. A section through one turn of the cochlea. The dotted lines indicate the extension of the membranes induced by local sound pressure. (After Penzlin 1991, redrawn)

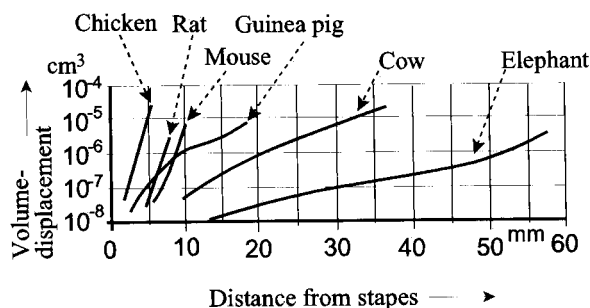


Fig. 4.8. The elasticity of the cochlear partition of various animals as a function of the distance from the stapes. As a measure, the volume displacement is used which is produced in a 1 mm segment by a pressure of 1 cm of water. (After Békésy, in Waterman and Morowitz 1965)

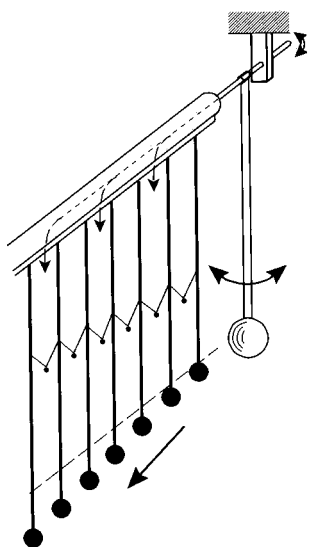


Fig. 4.9. Pendulum resonator model to demonstrate travelling waves in the inner ear. (After Békésy, in Waterman and Morowitz 1965)

which other pendulums, made of balls suspended by threads of various lengths, are attached at equal intervals. These, through the backwards and forwards movements of the axle are caused to move at a given frequency, which in most cases is not the resonant frequency of the individual pendulum. Additionally, the pendulums are coupled to one another in series by threads weighted with small ball bearings. The degree of coupling depends on the point of attachment of these threads.

This model demonstrates the following properties of the inner ear: the sound waves spread so rapidly in the perilymph fluid that, similar to the effect of the axle in the model, there is a simultaneous stimulation along the whole of the membrane. The coupling of the pendulums corresponds with the position-dependent viscoelastic properties of the basilar membrane.

If the model is activated by simple harmonic oscillation, travelling waves are generated which always move from the stiffer to the more compliant region

along the row of pendulums. The amplitudes of these waves are not constant. They reach a maximum at certain points of their movement in the direction of the arrow (Fig. 4.10), depending on the frequency of activation. Connecting these maxima, an envelope curve results, the maximum of which depends on the frequency of activation and on the properties of the pendulums, particularly the viscoelastic properties of the basilar membrane and the scala vestibuli. The higher the frequency, the closer the maximum of this envelope curve is to the oval window. Figure 4.11 indicates the positions of maximum stimulation along the cochlear partition of the elephant. In the case of complicated tones, envelope curves occur with several maxima. At the locations of maximal amplitude, stimulations of the sensory cells occur. So in this simple mechanical way a frequency analysis of the sound is already possible.

These sensory cells, the so-called hair cells, are located at the basilar membrane. They contain at their surface bundles of fine hairs, so-called stereocilia in a hexagonal arrangement. These stereocilia differ in length and thickness. Some of them attend the tectorial membrane on the opposite side of the cleft. They are excited by the oscillating flow in the cleft between the scala

Fig. 4.10. Cochlear partition vibrations at two instants within a cycle, and the corresponding envelope curve of a vibration of 200 Hz. (After Békésy, in Waterman and Morowitz 1965)

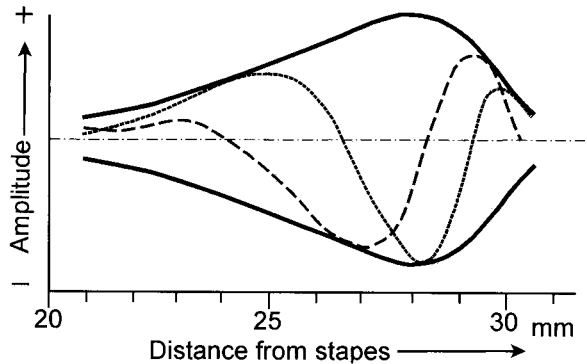
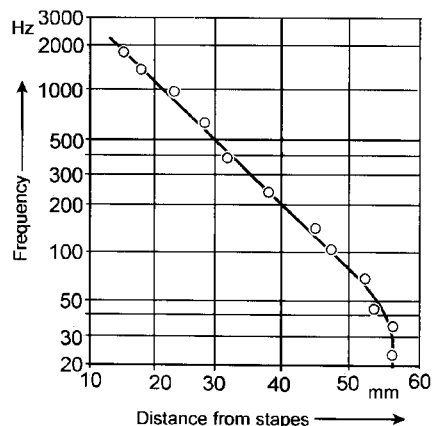


Fig. 4.11. Position of maximum stimulation along the cochlear partition of the elephant. (After Békésy, in Waterman and Morowitz 1965)



media and the canalis spiralis, as well as by oscillations of the distance between the opposite surfaces. The excitation of the hair cells occurs by mechanical coupling between the stereocilia and ion channels of the cell membrane. In this way the mechanical vibration will be transformed into an electrochemical, i.e. an electrical signal.

In this context the enormous spread of the measuring range of our auditory system must be remembered. We learn from Fig. 4.3 that, for example, at 1 kHz the sound pressure for the auditory threshold of pain is about 10^7 times larger than the sound pressure for the auditory threshold. Let us suppose that this pressure is proportional to the maximal amplitude of the displacement of the basilar membrane, or to the amplitude of the oscillating flow in the cleft between the scala media and the canalis spiralis. This means that a variation of this amplitude could occur between about $1 \text{ nm} = 10^{-10} \text{ m}$ near the auditory threshold, and $1 \text{ mm} = 10^{-3} \text{ m}$ near the pain threshold! Near the maximal intensity, damaging the system must be avoided, near the minimal displacement, mechanisms must exist which protect the system against thermic noise.

One of the mechanisms against thermic noise is the mutual coupling of the stereocilia. These hairs of a cell are connected together at their ends by fine filaments. This kind of connection follows a particular pattern. The bending properties of the system of stereocilia in this way become anisotropic. The stereocilia are much more sensitive in the direction of the proposed oscillating flow than to the random bending induced by thermic noise.

Summarizing this chapter we can conclude that the process of auditory analysis of sound frequency is realized in several steps: it starts with the filtration quality of the outer ear and proceeds step by step with the frequency characteristics of the impedance transducer of the middle ear, the frequency characteristics of the travelling waves, the resonances of stereocilia and sensory cells, to the neuronal mechanisms of sound analysis.

Further reading: Hudspeth 1985 and 1989; Markin and Hudspeth 1995; Skalak and Chien 1987; Waterman and Morowitz 1965; especially for biophysics of music: Roederer 1979

4.3.4

Infrasound

Infrasound consists of oscillations of frequencies below 20 Hz, whereby 0.1 Hz is taken as the lower limit. Because of their large wavelength they have some special properties in relation to audible sound. For example they are only poorly damped by stone walls, buildings etc. It is therefore difficult to shield from this kind of noise. Sometimes, it even happens that in spaces with corresponding resonance frequencies amplification occurs. This happens, for example, in cars, or housing spaces with a typical resonant frequency of 2–8 Hz.

There are natural sources of infrasound, like wind, waterfalls, sea surf etc. Technically, infrasound is produced by vibrations of various engines. There is considerable exposure to infrasound in cars if the airstream induces internal

resonances. Sometimes the 100 dB limit is exceeded. In recent years special infrasound loudspeakers have been used in discotheques in order to induce psychosomatic resonance of the participants. Because of the above mentioned problems of shielding they may considerably disturb the neighborhood.

Despite the fact that infrasound in general is a non audible sound, it nevertheless leads to unspecific auditory sensations even at frequencies below 16 Hz, if it has sufficient intensity (see Fig. 4.3). Sometimes, even clinking and overtones can be heard. In general, however, infrasound interacts with air-filled cavities in the human body, like lungs, nasal cavities, frontal sinus, middle ear, gut etc. At intensities above 160 dB dramatic damage of the middle ear is to be expected. Infrasound of intensities between 140–155 dB leads to acute psychophysical injuries like headache, difficulties in breathing and stress sensations. Below 100 dB no physiological influences could be found, but nevertheless psychosomatic reactions occur which are regarded as a disturbance. These occur near the threshold of perception.

Further reading: Møller 1984

4.3.5

Biophysics of Sonar Systems

It has been known since the experiments of the Italian monk and scientist Lazzaro Spallanzani in 1794 that bats have the ability to orientate in complete darkness and to avoid obstacles. It was demonstrated at that time that a bat could still fly safely when blinded, but was quite helpless when its ears were plugged. Only in 1938 was D. R. Griffin able to show by direct measurements that bats are able to emit ultrasound, and detect the echoes reflected from objects in their surroundings to determine their location. In the following years many investigations have been carried out, and it has been found that not only bats, but also rodents, aquatic mammals, birds, fish and insects use echolocation by high frequency sound or ultrasound.

Because bats have been investigated in this respect more than other species, we will start with this example to explain the biophysical basis of echolocation. Figure 4.12 shows an oscillogram of a typical orientation signal of a bat. It is characterized by the following physical parameters: frequency (ν), amplitude, duration (Δt), and of course, the distances between this, and the next signal.

The frequency of the emitted sound depends on the size of the object which is to be localized. An effective echo can be expected only from targets which are equal, or even larger in size than the wavelength of the sound. The relation between wavelength (λ) and the frequency (ν) is given by:

$$\lambda = \frac{v}{\nu}, \quad (4.3.4)$$

where v is the velocity of sound waves. In air under atmospheric conditions $v = 331 \text{ m s}^{-1}$. A sound of 25 kHz therefore has a wavelength $\lambda = 331/25\,000 = 0.01324 \text{ m}$, i.e. 1.324 cm.

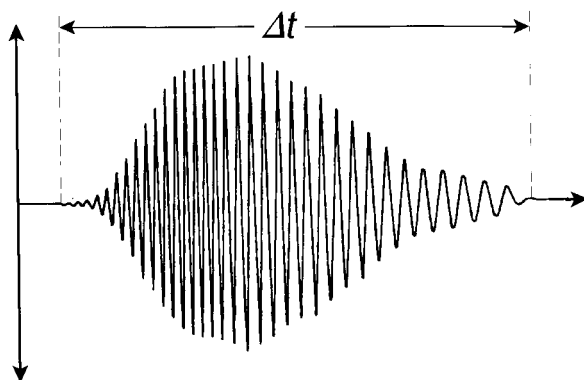


Fig. 4.12. Example of a typical FM modulated signal of a bat

The frequencies emitted by bats differ from species to species. Usually, this amounts to between 25 kHz and 150 kHz. Bats which use their sonar system in order to catch insects must emit signals of higher frequencies than those just orientating in dark caves.

The amplitude of the signal, i.e. its intensity, is determined by the required range of the sonar system. Naturally, the auditory system of the bats must be sufficiently sensitive to receive the corresponding echo. The power of the emitted sound of the big brown bat (*Eptesicus fuscus*), for example, is about 10^{-4} W, whereas as minimum, a power of 10^{-16} – 10^{-14} W corresponds to the auditory threshold. At 5–10 cm distance from the head of this bat, the sound pressure amounts to 2–6 Pa. According to Eq. (4.3.3) this means an intensity of 100–110 dB. We learned in Section 4.3.2, Fig. 4.4 that, transformed into audible sound, this would be a considerable noise.

This large difference between the emitted versus the received sound causes problems for the animal when listening during calling. The maximal duration of a signal (Δt), therefore, is more or less limited by the time the sound needs to arrive at the target and come back, i.e. by the distance of the target.

These then are the basic parameters of an echolocation signal. Looking at the sonogram in Fig 4.12 carefully, more details can be detected. The frequency, for example, as well as the amplitude of the signal are time dependent. It is a frequency modulated sound, a so-called *FM signal*. In the case of the little brown bat (*Myotis lucifugus*) the signal starts with a frequency of 100 kHz and goes down by an octave at the end of the 2 ms long signal. In contrast, the horseshoe bats use signals of 50–65 ms with constant frequencies (*CF signals*). Mostly, however, these CF signals have a short frequency modulated final part. The character of the emitted signals depends on the species, on situations, and even on the individual animal.

The frequency of modulation seems to be an important property of the signal, especially for measurement of distances. In some instances it appears that by superimposition of the FM part of the end of the signal with the initial CF part of the returning echo, specific beat frequencies occur which are evaluated by the animal. Furthermore, it has been suggested that a technique

known as pulse compression in radar technology, may be utilized by bats. For this, short frequency components of the echo are delayed by a special acoustic filter, a so-called optimal filter which depends on their frequency. The higher frequency parts at the beginning of the signal are delayed to a larger extent than the lower frequency parts at the end of the signal. This process of impulse compression provides an instantaneous cross correlation between the emitted and the received signals. Pulse compression is a way to explain the ability of the animal to measure the distance even in cases when the echo time is less than the duration of the emitted sound. This principle would also minimize disturbances by the Doppler effect. It is unknown which parts of the auditory system could carry out this function or what the molecular basis for this could be.

On the one hand, the Doppler effect disturbs the bat when measuring distances, on the other hand, it is used to measure relative velocities between the animal and its surroundings. For this a Doppler-shift compensation occurs. If a flying bat approaches an object at a certain speed, the echo from that object will be Doppler-shifted upward to a higher frequency than the emitted signal. Subsequently, the bat lowers the frequency of the next emitted CF component by an amount that is nearly equivalent to the Doppler shift in the preceding echo. The amount of the required frequency is used by the bat to calculate its relative velocity.

Many questions on information processing in bat echolocation are still open. These include mechanisms to determine the direction of the object as well as information about the size, shape and material of the target.

Echolocation is also known in some birds. Alexander von Humboldt has already described a South American cave-dwelling bird, the oil-bird or guacharo (*Steatornis caripensis*) which flew around the head of the explorer uttering "screams of fear". Today it is known that these cries serve as signals in an echolocation mechanism. The sound is pitched at frequencies of 6–10 kHz, which means that it can be heard by humans. These birds are herbivores, but they require their sonar system in order to orientate in large dark caves. The frequency employed corresponds to wavelengths between 3–5 cm which means that the sound will be reflected even from small rock projections.

Sonar systems are found not only in airborne species, but also in several aquatic animals. Dolphins, for example, emit short click signals with a duration mostly of only 40–50 ms having a frequency up to 130 kHz with repetition after 20–40 ms. It is still unknown where and how this sound in the animal is produced. Due to the larger velocity of sound in water ($v = 1500 \text{ m s}^{-1}$), the wavelength of a sound in water is about 4.5 times larger than that of the same frequency in air. This would require a higher frequency for echolocation in water for targets of the same size. Additionally the specific sound distribution in water must be taken into consideration. The transmission loss for a point source of sound in free space in general is:

$$\text{Transmission loss} = 10 \lg \frac{I_0}{I_r} = 10 \lg r^2 + \alpha r, \quad (4.3.5)$$

where I_0 is the intensity at $r = 0$, I_r is the intensity at distance r , and α is the coefficient of absorption.

The first part of the sum describes the simple geometrical distribution of the sound, whereas the second term concerns the absorption of sound in the medium. Especially in sea water, because of its content of magnesium sulfate, the coefficient α is frequency dependent. It amounts 0.002 dB m^{-1} for 20 kHz and rises up to 0.3 dB m^{-1} for 100 kHz. This means that in contrast to airborne animals, the frequency of the emitted signal for sea water animals is additionally a function of the distance.

Further reading: Au 1993; Griffen 1958; Pollak and Casseda 1989; Popper and Fay 1995

4.3.6

The Effects of Ultrasound

Ultrasound consists of longitudinal waves, i.e. periodic alterations of local pressure at a frequency range between 16 kHz and $10^9 \text{ Hz} = 1 \text{ GHz}$. Sometimes for mechanical waves higher than 1 GHz the term hypersound is used. Technically, ultrasound can be generated by electro-acoustic converters. Because of its short wavelength it can be focused by appropriate reflectors and diffraction lenses. In recent decades ultrasound applications in medical diagnostics and therapy have become increasingly relevant.

The propagation rate of ultrasound in water is 1500 m s^{-1} , in solid bodies about 4000 m s^{-1} , and in water-rich biological tissues it is around 1479 m s^{-1} . Corresponding to Eq. (4.3.4), ultrasound of $\nu = 10 \text{ MHz}$ in tissues has a wavelength of 0.15 mm. This ensures a satisfactory resolution for sonographic images. It can be increased further by using ultrasound of higher frequencies, however this will decrease their penetration into the body. In fatty tissue, for ultrasound of 0.8 MHz a half-value thickness of penetration of 3.3 cm has been measured; in muscles it amounts to only 2.1 cm. This parameter decreases with increasing frequencies. This is the reason why for prenatal observations and mammographic diagnostics, frequencies of only 1–5 MHz are used. In ophthalmologic diagnosis 10–50 MHz are used because in this case a higher resolution is required. For therapeutic applications, ultrasound of a high power density in the frequency region of 0.2–2 MHz is used.

The image generating techniques of medical diagnostics evaluate differences of the mechanical impedance of various membranes, cells or tissues, leading to reflections, diffractions and refractions of the ultrasound waves. Investigations into blood flow with ultrasound are based additionally on the Doppler shifts of the reflected ultrasound.

The influence of ultrasound on cells and tissues on the one hand is caused by the appearance of local pressures, on the other hand on local accelerations. As a concomitant phenomenon a certain heating occurs. According to the frequency of the ultrasound, alternating positive and negative pressures appear locally,

leading to stretch or compression of the material. Homogeneous liquids have a considerable resistance to the disruption effect. In order to pull pure water apart in such a way that two parallel surfaces are formed which are separate from one another, a negative pressure is required of 1500 MPa. Only 150 MPa is necessary to form a spherical cavity. This effect is called *cavitation*. In fact, the local negative pressures which are required for cavitation in real solutions and suspensions are lower. This depends on the concentration of nuclei of cavitation. For short pulses of ultrasound, cavitations in tissue are observed above 10 MPa. This determines the limit of the ultrasound intensity in therapeutic applications.

These caverns are short-lived spaces, filled by the gas which is dissolved in the liquid, and by the vapor of the liquid itself. With increasing pressure, these caverns collapse and vanish completely, or they remain as microscopic crystallization nuclei for the next cavitation. The diameter of these caverns depends on the intensity and frequency of the applied ultrasound. Their mutual distance is also frequency and wavelength dependent. For a frequency of 20 kHz in water, it amounts to about 0.15 mm and becomes lower if the frequency increases. Cavitation can lead to audible noise, finally to “boiling” of the liquid. The sound intensity which is required to produce cavitations increases with the viscosity of the liquid, and decreases with the amount of dissolved gas.

In biological systems this cavitation leads to the destruction of supramolecular structures and cell organelles. Furthermore, various physicochemical processes, for example diffusion of matter, can be influenced. In cases of high intensity, even water radicals can appear, leading to indirect effects similar to that of ionizing radiation.

For the therapeutic application of ultrasound, its property is used to heat the tissue. This effect is called *diathermy*. It depends on the frequency mechanical impedance of the tissue, also being a function of frequency. Caused by diffractions or reflections in the tissue, the thermic effects can be heterogeneous. It is possible that hot spots occur in the tissue, i.e. microscopic locations of increased temperature.

Although ultrasound by definition is non-audible sound, it was found that the human ear may sense tones even up to 40 kHz, although with very low sensibility. Nevertheless, there have been discussions as to whether the weighing scales, discussed in Section 3.4.2 (Fig. 4.5) must be extended up to this frequency.

To prevent risks from ultrasound radiation, the increase of local temperature in tissue must not exceed the limit of 1.5 °C, and local pressures must not exceed 8 MPa. WHO recommend a limit of a mean intensity of 3.10^4 W m^{-2} . The real values applied in diagnostic methods are far below this limit. Therefore one may exclude any risk for health, especially since X-ray methods are being replaced by ultrasound.

Further reading: Nygborg 1985

4.4

Static and Electromagnetic Fields

In this section we will first discuss static fields, i.e. fields with zero frequencies, leading to AC fields of low frequency, and finally to the region of microwaves. Thermic radiation has already been mentioned briefly in Section 4.1, and visible light in context with excitation processes and photosynthesis in Section 2.2. Effects of ionizing radiation will be explained later.

This chapter is based on the previous explanations about cells in static and AC electric fields (Sect. 3.5), but instead of strong electric fields, used for cell manipulations (Sect. 3.5.6) we will concentrate here on fields in the natural and technical environment and on their use in therapeutic treatments.

Figure 4.13 shows the whole spectrum of electromagnetic radiation. Wavelength (λ), frequency (ν) and quantum energy (E) are connected together by the usual relations: $c_0 = \nu \cdot \lambda$ and $E = h \cdot \nu$ (c_0 – speed of light in vacuum, h – Planck's constant). The energy of ionization of water: $E = 12.56$ eV usually is taken as the beginning of the region of ionizing radiation.

4.4.1

The Static Magnetic Field

A magnetic field with the *magnetic field strength* H (in $A \cdot m^{-1}$) causes a *magnetic induction*, or *magnetic flux density* B in a body (in Tesla: $T = V \cdot s \cdot m^{-2}$) of:

$$B = \mu\mu_0 H,$$

(4.4.1)

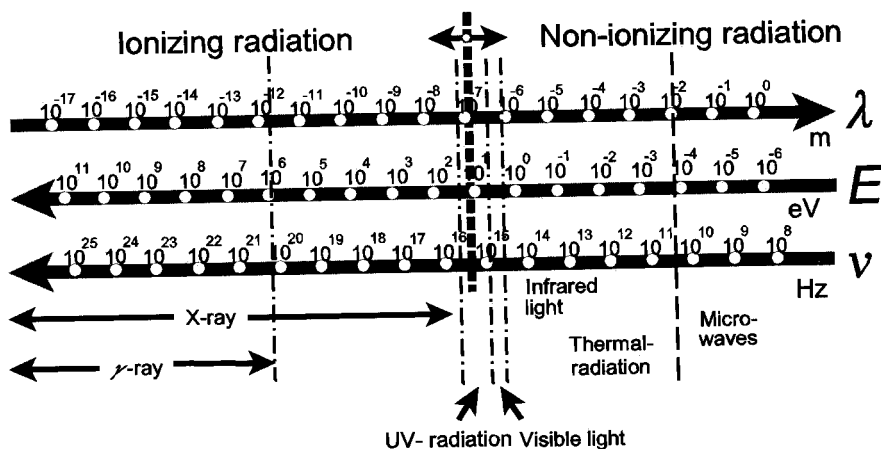


Fig. 4.13. Spectrum of electromagnetic radiation with wavelength (λ), quantum energy (E), and frequency (ν)

where $\mu_0 = 1.256 \cdot 10^{-6} \text{ V s A}^{-1} \text{ m}^{-1}$ is the *magnetic permeability of vacuum*, and μ is the *magnetic permeability number* – a dimensionless number in relation to vacuum, where $\mu = 1$. The deviations of the magnetic permeability for non-ferromagnetic materials from 1 are very small. Therefore the *magnetic susceptibility* $\chi = \mu - 1$ is introduced.

Magnetic properties of various materials are classified according to their properties as being:

$$\begin{aligned} \text{diamagnetic:} & \quad \mu < 1; \chi < 0, \\ \text{paramagnetic:} & \quad \mu > 1; \chi > 0, \\ \text{ferromagnetic:} & \quad \mu \gg 1; \chi \gg 0. \end{aligned} \quad (4.4.2)$$

As shown in Table 4.2, cells and tissue in general have diamagnetic properties. An exception are desoxygenized erythrocytes, which are paramagnetic due to the property of the central Fe-atom in the hemoglobin molecule. If it binds oxygen, it changes from a paramagnetic to a diamagnetic state.

Highly organized structures, like the cell membrane, show magnetic anisotropy. Its magnetic susceptibility depends on the direction in which it is measured. This anisotropy, $\Delta\chi$, is defined as follows:

$$\Delta\chi = \chi_{\parallel} - \chi_{\perp}, \quad (4.4.3)$$

where $\chi_{\parallel}$ and $\chi_{\perp}$ are the corresponding susceptibilities of directions parallel and perpendicular to characteristic directions of the structure. Anisotropic magnetic structures, like for example cell membranes or chromosomes, can be orientated in strong magnetic fields.

In special magnetostatic bacteria, and in low amounts also in animal and human tissue, ferromagnetic particles, so-called *magnetites* have been found.

Table 4.2. The magnetic susceptibility of air, water and various biological materials

	After Maniewski (1991) * 10^{-6}	After Khenia et al., in: Maret et al. (1986) * 10^{-6}
Air	+0.34	+0.264
Water	-9.05	-9.04
Arterial blood	-9.1	-9.3
Oxygenized erythrocytes		-9.03
Venous blood	-8.4	-7.8
Desoxygenized erythrocytes		+3.88
Lungs (breathed in)	-3.9	
Lungs (breathed out)	-4.1	
Lungs (15% air content)		-4.2
Muscle	-9.0	-9.03
Liver	-8.8	-8.26
Bone	-10	-10

Magnetites of magnetostatic bacteria can be obtained in sufficient concentrations to be analyzed chemically and by crystallography. These are cuboidal crystals of Fe_3O_4 (a mixture of FeO and Fe_2O_3) with a size of 50–200 nm. The largest of them, at least, are surrounded by a membrane forming *magnetosomes*. The density of these particles is 5.1 g cm^{-3} . Because of their extremely small size, they consist of only one single magnetic domain which means that they consist of a single elementary magnet. These magnetite crystals are synthesized by the cell from iron bound in a chelate complex. From the crystallographic point of view, they are different from artificially produced ferrite particles or from ferrite particles absorbed from a polluted external medium. The magnetosomes of tissue cells are apparently similar to those of bacteria.

In these magnetostatic bacteria magnetic particles are arranged in a $3 \text{ }\mu\text{m}$ -long chain which is fixed along the axis of motility of the bacterium. The magnetic momentum of this chain is sufficient to orient the bacterium in the geomagnetic field. Magnetostatic bacteria are usually anaerobic inhabitants of the sediment. Cells with appropriate polarization, and in regions away from the equator, swim along the geomagnetic vectors into the anaerobic depths. This orientated movement is called *magnetotaxis*. Bacteria with wrong magnetic polarizations swim in the wrong direction and die off in oxygen-containing surface water. During the division of the cells by mitosis, the magnetic chains will also be divided into two parts, and the daughter cells get a part of the correctly polarized chain. Adding further magnetosomes after division, to complete the chain, these newly synthesized particles automatically become polarized according to the inherited part of the chain. It is possible to reverse the polarity of these chains in artificial magnetic fields of at least 10 mT.

The magnetic field of the earth is demonstrated in Fig. 4.14. The vector of this field can be subdivided into a vertical component, which has its maximum at the poles, coming up to a field strength of $48\text{--}56 \text{ A m}^{-1}$, corresponding to a magnetic flux density of $60\text{--}70 \text{ }\mu\text{T}$. The horizontal component has its maximum near the equator, with $24\text{--}32 \text{ A m}^{-1}$, i.e. $34\text{--}40 \text{ }\mu\text{T}$. The intensity of the geomagnetic field shows various alterations, which are caused chiefly by changes of sun-spot activity. Furthermore, small variations during the day, during the

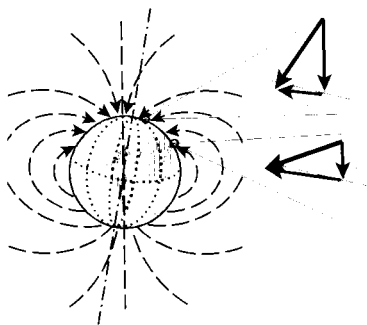


Fig. 4.14. Geomagnetic field lines, and their inclinations at two particular locations

month, and during the year occur. Through history the magnetic pole of the earth has shifted a considerable distance. Additionally, at some locations significant geologically caused inhomogeneities of the geomagnetic field can be measured.

The question as to whether animals can orientate in geomagnetic fields has long been discussed. Strictly speaking, the passive orientation of magnetostatic bacteria cannot be considered as the result of geomagnetic receptors. It seems sufficiently established, however, that various animals really do have the ability of magnetic orientation. In this respect, one must distinguish between several kinds of use of the magnetic sense: compass-orientation (migrant birds, rodents living in the earth), magnetic mapping (bees, pigeons, whales), and probably also magnetic timing. At least in birds the magnetic orientation seems to be connected with optical orientation. Whether human behavior can be correlated with the occurrence of magnetic storms, as is sometimes claimed, is quite questionable.

Although behavioral experiments clearly indicate that many animals are able to orientate with the help of the geomagnetic field, it remains absolutely unclear which mechanisms may be responsible for the reception of these extremely weak magnetic fields. The findings that in some cases light is necessary to allow birds to orientate in the geomagnetic field have led to speculations as to whether transitions of the electron spin could occur during photoactivation processes of rhodopsin in the retina. A further hypothesis claims that this reception is somehow connected with magnetosomes in the tissue. Long magnetite chains which could overcome thermic noise, however, were found only in bacteria. In bees large magnetite crystals were found near nerve endings of particular abdominal hairs. It has been speculated that this ferromagnetic material could focus the magnetic field and amplify it in this way locally. In the case of marine animals, geomagnetic orientation could be realized using the effect of electrical induction. This requires a relative high speed of the animals and a considerable size. In this case a conducting loop occurs between the body of the animal and the surrounding seawater. Moving in the geomagnetic field, a current could be induced in this loop, and electric potential differences could be measured by electroreceptors (see Sect. 4.4.2). It does not seem so evident that similar mechanisms would be possible in terrestrial animals. Sometimes, the vestibular system in mammals and even humans has been discussed in this respect. This, of course, remains quite speculative because of the small size of the induction loop.

Recently, detection of biogenic magnetic fields on the one hand, and applications of strong magnetic fields on the other have become more and more important for medical diagnoses. Using sensitive measuring instruments with superconducting materials (SQUIDS, superconductive quantum interference devices) it is possible to measure magnetic fields which are induced by extracellular electric currents around excited tissue, as well as minimal alterations in the magnetic susceptibility and magnetic anisotropy in the body.

To record magneto-cardiograms, -encephalograms, and -myograms, magnetic fields weaker than 1 pT must be measured. This is seven orders of magnitude lower than the geomagnetic field! These methods are more costly

than the corresponding records of ECG or EEG, but they are much more instructive. As we already pointed out, the magnetic permeability (μ) of various materials does not deviate very much from $\mu = 1$. This means that the magnetic field in the body will not become disordered significantly, in contrast to the electric field. Therefore it is much easier to localize the oscillating electric dipole in nerves and muscles, inducing the magnetic field (see also the explanation of the ECG in Sect. 3.5.2, Fig. 3.33).

Furthermore, the application of highly sensitive measuring instruments for magnetic fields can be used in occupational medicine to analyze accumulation of iron dust in the lungs. The measurement of the magnetic susceptibility of the lungs can also be used in lung diagnostics. After inhalation of a small amount of a ferromagnetic label, the rate of magnetic reorganization of these particles can be recorded, which allows conclusions concerning the activity of the ciliated epithelium of the lung.

The construction of superconducting magnets with bores large enough to accommodate human bodies have made it possible to apply *nuclear magnetic resonance* (NMR) in medical diagnostics. This enables one to measure ionic composition and metabolic activity of various tissues and organs. The method of whole-body tomography, or nuclear magnetic imaging (NMI) makes possible detailed pictures with great diagnostic value, for example in cancer diagnostics. During these measurements, the patient is placed in a magnetic field with a flux density of 0.1–3 T, and field gradients in the order of 0.01 T m^{-1} . Signals from nuclei, such as protons differing in their relaxation times are used to create these images.

Furthermore, during the last decade the method of *transcranial magnetic stimulation* has become available. In this case, one activates the primary motor cortex and the motor roots in conscious man by magnetic millisecond pulses of an intensity of 1.5–2 T. This technique has proved sensitive enough to illustrate early abnormalities of central motor conduction in various neurological diseases such as multiple sclerosis, amyotrophic lateral sclerosis, cervical spondylotic myelopathy, degenerative ataxias or hereditary spastic paraplegias.

These kinds of enforced application of strong magnetic fields in medical diagnostics and therapy, as well as increased occupational exposure to magnetic fields raises the question of mechanisms of interaction between these fields and the biological system. In general, the following effects are possible:

- Orientation of particles in static magnetic fields,
- Translation of particles in inhomogeneous static magnetic fields,
- Action of Lorentz forces on moving charged particles in magnetic fields (e.g. ions, molecules, cells in moving blood),
- Induction of electric currents in the electrolyte milieu of the body in case of movements in a magnetic field or in gradients of a magnetic field.

In inhomogeneous magnetic fields, diamagnetic bodies move in the direction of decreasing field strength, whereas para-, and especially ferromagnetic particles are driven in the opposite direction. In context with the fixation of safety

standards, these mechanisms are important only in relation to ferromagnetic implants. The orientation of molecules against the Brownian movement requires extremely high magnetic fields. So, for example, the orientation force which is induced by a magnetic field of 13 T on a magnetically anisotropic DNA molecule at room temperature amounts only to about 1% of the energy of thermic noise. Therefore it cannot be expected that field strengths as used in medical diagnostics and therapy would lead to translations of cells based on differences of their magnetic susceptibility (see Table 4.2, page 257).

On the other hand, it is possible to induce cellular movements in special arrangements. The passive shift of cells in strong inhomogeneous magnetic fields is called *magnetophoresis*. This, for example, can be applied to separate paramagnetic desoxygenated erythrocytes from diamagnetic lymphocytes (Table 4.2). For this, a net made of thin iron wire is used, which is placed in a polyethylene tube. Applying an external magnetic field of about 2 T, strong inhomogeneities are induced near the wires, attracting the erythrocytes. Lymphocytes, in contrast to this, pass the net.

The method of *immunomagnetic cell separation* is much more effective. For this, latex particles are used, containing supraparamagnetic iron which indicates ferromagnetic properties only during the presence of an external static magnetic field. These particles are loaded with antibodies against cells which are needed to be separated. After the attachment of these particles to the cells, they can be separated easily even at moderate magnetic gradients.

Charges moving in a magnetic field are mechanically influenced by a so-called Lorentz force. This force also acts on ions and other charged particles under the influence of strong magnetic fields in the flowing blood. In fact, this influence is slight at magnetic field strengths as applied in medicine. The inverse of this effect is the induction of an electric current if a conductor crosses lines of a magnetic field. This can occur in limbs moving in the magnetic field of an NMI engine. The result is painful nerve excitations.

Magnetic fields of 10 mT, oscillating with a frequency of 10 Hz, can lead to excitations of the optical system. As a result, so-called *magnetophosphenes* appear, i.e. the production of visual sensations. The mechanism is the same as in the above-mentioned method of magnetostimulation.

Magnetic fields with flux densities much higher than one Tesla can influence biochemical reactions in isolated enzyme systems and consequently various cellular processes, like proliferation or differentiation. Cellular reactions, which are connected with phase transitions of membrane lipids are apparently influenced most. This is caused by the above-mentioned anisotropy of lipids.

Based on established results on the influence of static magnetic fields, safety standards have been recommended. According to this, the average magnetic flux density of 200 mT must not be exceeded by research personnel or others working in high magnetic fields during an 8-h working day. The limit of possible permanent exposure of the population is 40 mT. A short-lasting exposure of humans without ferromagnetic implants and without cardiac pacemakers is possible in a field of maximum 2 T. Exposure of extremities is allowed up to 5 T. This is the limit for NMI equipment. Subjects with implants or cardiac

pacemakers, therefore, are excluded from NMI diagnosis or other magnetic treatments.

Further reading: Influence of magnetic fields in general: Polk and Postow 1996; Maret et al. 1986; magnetic orientation of animals: Able 1994; Wiltshko and Wiltshko 1995; magnetosomes: Dunin-Borkowski et al. 1998; Kirschvink et al. 1985; Vainshtein et al. 1998; magnetophoreses: Higashi et al. 1993; immuno-magnetic separation: Moore et al. 1998; Olsvik et al. 1994; magnetic stimulation: George et al. 1998; deNoordhou 1998; Wassermann 1998

4.4.2

The Electrostatic Field

Aspects of bioelectricity have already been discussed in context with various other circumstances in previous sections of this book: After some basic parameters and laws of electrostatics (Sect. 2.4.1), principles of membrane electrostatics (Sect. 2.5.5) have been explained. Subsequently, we came to transmembrane potentials (Sects. 3.4.2, 3.4.3) and to discussions of electric fields in cells and organisms (Sects. 3.5.1, 3.5.2). Finally, passive electric properties of cells and tissues have been considered (Sects. 3.5.3, 3.5.4), as well as various techniques of cell manipulation using electric fields (Sect. 3.5.5). Now the question arises: how do electric fields in the environment influence cells and organisms? In fact, this can be considered as an interference in the sophisticated electrical structure of the organism, as discussed in Section 3.5.1.

In this respect it must be pointed out that a strong division between electrostatic fields and oscillating AC fields is artificial. In fact, there is a continuous transition between static fields changing for example in a circadian rhythm or in periods of hours or minutes, and very low frequency AC fields. The same holds for static fields switched on and off after some time, and pulsed electric fields (Fig. 4.20). Using Fourier analysis, every kind of pulse can be considered as a sum of sine waves of various frequencies. Many aspects discussed in this chapter, therefore, are important also for low frequency AC fields.

Terrestrial animals live in the electrostatic field of the earth. Its field strength near the earth's surface at sea-level amounts to $100\text{--}200\text{ V m}^{-1}$. Below a thundercloud it can rise up to $20\,000\text{ V m}^{-1}$. The electric field of the earth changes during the day and night and also through the year.

DC fields in the vicinity of technical devices are not very important from the point of view of health protection. They can, however, lead to considerable charging of objects in their vicinities. When touched, such electrically charged surfaces can lead to painful discharges.

The specific conductivity of dry air near the earth's surface about $2.5 \cdot 10^{-14}\text{ S m}^{-1}$. Therefore, at a field strength of 200 V m^{-1} a current density of $5 \cdot 10^{-12}\text{ A m}^{-2}$ results. The conductivity of air depends on the presence of charged particles. These particles have the misleading name *air ions*. In fact,

these are particles of quite heterogeneous nature and size, beginning with true ions of nitrogen or sulfate compounds, up to various microscopic dust particles containing surface charges. The content of such "ions" in air in high altitudes amounts to 10^9 ions per m^3 . It varies across a broad range depending on many conditions. In cities it may increase up to 10^{11} ions per m^3 . There are many speculations about a possible biological role of these charges, but in fact no correlation with any kind of disorder has been established yet.

The biological influence of environment fields depends on their transfer into the body. The electrical conductivity of the human body is about 10^{14} times larger than that of the air. The dielectric constant exceeds that of the air by a factor of 50. This leads to considerable distortion of the field lines around the body (Fig. 4.15). The electric field strength near the surface of the body can significantly increase, especially at the head or at the end of a raised hand.

The external electric field leads to a displacement of internal charges in the body. An external electric field E_e in air ($\varepsilon_e = 1$) induces in a spherical, non conducting body ($g_i = 0$) the following internal field E'_i :

$$E'_i = \frac{3E_e}{2 + \varepsilon_i} \quad \text{for } g_i = 0. \quad (4.4.4)$$

If the body is conducting ($g_i \neq 0$), this internal field will immediately be neutralized by an internal current. Taking this process into account, one gets the real field strength E_i as a function of time (t): 0

$$E_i = \frac{3E_e}{2 + \varepsilon_i} e^{-\frac{g_i t}{\varepsilon_i \varepsilon_0}} = \frac{3E_e}{2 + \varepsilon_i} e^{-kt}, \quad \text{where } k \equiv \frac{g_i}{\varepsilon_i \varepsilon_0}. \quad (4.4.5)$$

Using appropriate parameters ($g_i = 0.6 \text{ S m}^{-1}$, $\varepsilon_i = 50$, $\varepsilon_0 = 8.84 \cdot 10^{-12} \text{ C V}^{-1} \text{ m}^{-1}$), a corresponding rate constant $k = 1.35 \cdot 10^9 \text{ s}^{-1}$ can be obtained. This means that an internal electric field which is induced by a single rectangular pulse will vanish by a half life time of $\ln 2 / 1.35 \cdot 10^9 = 5.13 \cdot 10^{-10} \text{ s}$. A significant field in the body can therefore be induced only in the case of higher frequency AC, or pulsed fields. We will come back to this point in Section 4.4.3.

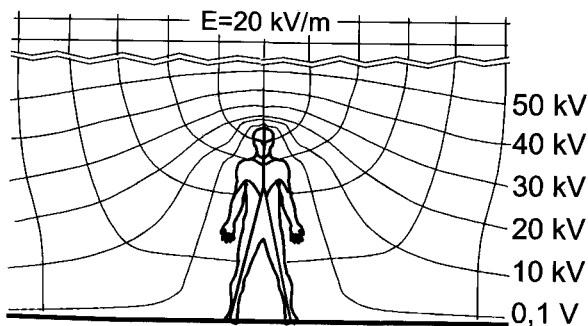


Fig. 4.15. Distortion of an electrostatic field near a standing man. (After Leitgeb 1990, redrawn)

This consideration indicates that an electrostatic field in the air environment may influence only the surface of the body. This chiefly concerns electrostatic erection of hairs. In the case of low frequency AC fields, a vibration of hairs can be sensed. The threshold where humans sense electrostatic fields in the environment is near 10 kV m^{-1} .

In contrast to the conditions in air, electric fields in water are transferred to biological organism to a much higher degree. Vice versa, electric fields of the body are transferred directly into the aquatic environment. This is the basis of the effectiveness of electrical organs of fishes in defense and orientation (see Sect. 3.5.2). Water-living animals have developed electroreceptors which apparently make it possible to localize living prey, or are used for mutual communication and for electro-orientation, or finally, as explained in Section 4.4.1 in some cases they could also be responsible for orientation in the geomagnetic field.

The electroreceptors of some aquatic animals have enormous sensibility. Behavioral evaluation of the lowest field perceived by rays was found to be $0.1\text{--}0.2 \text{ mV m}^{-1}$. In most other fishes, however, it is higher by several orders of magnitude. In the same way as for electrical organs, and in the case of electroreceptors the different conductivity of sea water and fresh water must be taken into account. In the case of fresh water fishes (Fig. 4.16 above) because of the low external conductivity, the field strength in the environment can be measured directly as a potential drop across the skin. Marine fishes use so-called *ampullae of Lorenzini* as amplifier systems (Fig. 4.16 below). These are long, well-insulated canals in the body, filled with a highly conductive jelly. At one end these ampullae are in electrical contact with sea water. In this way they transduce the electrical potential from particular points of the surface of the fish almost without loss to the point of their mutual contact. The external field E_e , therefore, will be amplified to an internal field E_i at the distance between the endings of the ampullae, where the electroreceptors are localized.

The molecular mechanism of them is not clear. It seems that there are similarities to thermoreceptors.

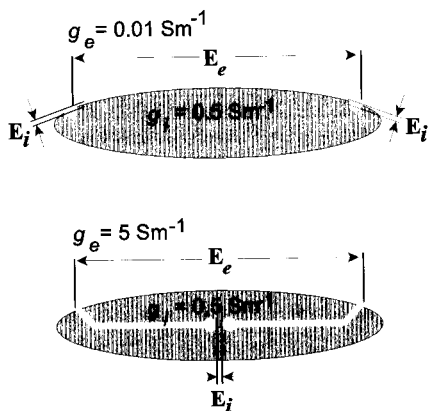


Fig. 4.16. Electroreceptors in freshwater fish (above), and seawater fish (below). The canals in the lower picture indicate the ampullae of Lorenzini (g -specific conductivity of water and tissue). The external electric field E_e becomes amplified to the inner electric field E_i near the receptors

Various calculations indicate that extremely high sensibility of the electro-receptors is possible, even considering thermic and electrical noise. As mentioned in Section 4.4.1 these receptors are probably also used to measure fields which are induced in the electrical loop between fish and environment, if the geomagnetic field lines are crossed.

We already pointed out that there are various biogenic electrical fields in the human body which are probably not simple passive, concomitant phenomena of some electrophysiological processes, but which may have functional relevance. In this context in Section 3.5.2 we mentioned wound potentials, which can create local electric fields of hundreds of V m^{-1} , or electric fields in loaded bones. This is the basis for various attempts of electrotherapy to control biological processes like cell motion, cell differentiation or cell growth etc. by applied electric fields.

One of these cellular properties is *galvanotaxis*. This means an orientation of the motility of living cells by static electric fields with field strengths of about $100\text{--}500 \text{ V m}^{-1}$. These kinds of orientation have been observed not only in various protozoans, sperms, and zoospores, but also in many kinds of cells moving slowly on the substratum, like granulocytes, fibroblasts, osteoblasts, nerve cells etc. This property must not be confused with electrophoresis, which is a passive movement of charged particles in an electric field (see Sect. 2.5.4). In contrast to electrophoresis, galvanotaxis predicts an active movement of the cells which usually will not be accelerated by the applied field, but simply orientated. In contrast to electrophoresis, galvanotaxis, being an active biological process, needs a much longer lag phase. Cells may move by galvanotaxis for many seconds or even minutes in one direction even if the polarity of the applied field has already been changed. Most cells, even when charged negatively, move by galvanotaxis toward the cathode.

The mechanism leading to galvanotaxis of tissue cells is rather unclear. Whereas in ciliata the orientation of their motion is apparently controlled by the transmembrane potential, for other cells other mechanisms are proposed. Using fluorescent labels it is possible to indicate that membrane-bound receptors shift under the influence of an external field laterally in the cell membrane, and concentrate on one side of the cell. This does not seem to be the result of individual electrophoresis of these molecules but rather a reaction to local electro-osmotic flow, induced by the interaction of the external field with all charges of the membrane surface.

A further property of cells in electrostatic fields is *galvanotropism*. This means the influence of an external DC field on the direction of cell growth. So, for example, nerve cells in electric fields of $0.1\text{--}1 \text{ kV m}^{-1}$ form dendrites preferentially at the side of the cell which is directed toward the cathode. In this case, local modifications of the membrane potential are probably responsible for these reactions, but the above-mentioned lateral translation of membrane proteins can also be responsible for this. This property has been applied in therapy to try to stimulate nerve regeneration.

Further reading: Electric fields in the environment: Reiter 1992; Polk and Postow 1996; air ions: Chary and Kavet 1987; electroreception: Kalmijn 1997;

Petracchi and Cercignani 1998; Atema et al. 1987; galvanotaxis: Gruler 1995; Machemer-Röhnisch et al. 1996; Zhao et al. 1999; galvanotropism: Rajnicek et al. 1988

4.4.3

Electromagnetic Fields in the Human Environment

As the result of technical and industrial development, an increasing number of man-made electromagnetic fields have appeared in the human environment. The classification for the frequency ranges, as listed in Fig. 4.17 (ELF, ULF, VLF, LF, MF, HF, VHF, UHF, SHF, EHF) corresponds to the terminology of electronic engineers. Sometimes fields in the GHz range are summarized as “microwaves” (MW), or the generalized term “radiofrequency” (RF) is used for a broad frequency range. Some authors call frequencies below 3 Hz “ultra low frequencies”.

According to Maxwell's equations, electromagnetic fields have a magnetic and an electric vector. With increasing frequencies, it is more and more difficult to separate experimentally the magnetic from the electric components of the fields. At ELF and ULF fields, it depends on the techniques of field application, whether mostly an electric, or preferentially a magnetic interaction will occur. To illustrate this, three methods are shown in Fig. 4.18 to apply low frequency fields as examples. If, for example, the object is positioned between the poles of an iron core, it is influenced mainly by the magnetic component of the field, induced in the iron by the current in the coil (Fig. 4.18A). If the object is between two plates of a capacitor in air (Fig. 4.18c), not being in touch with

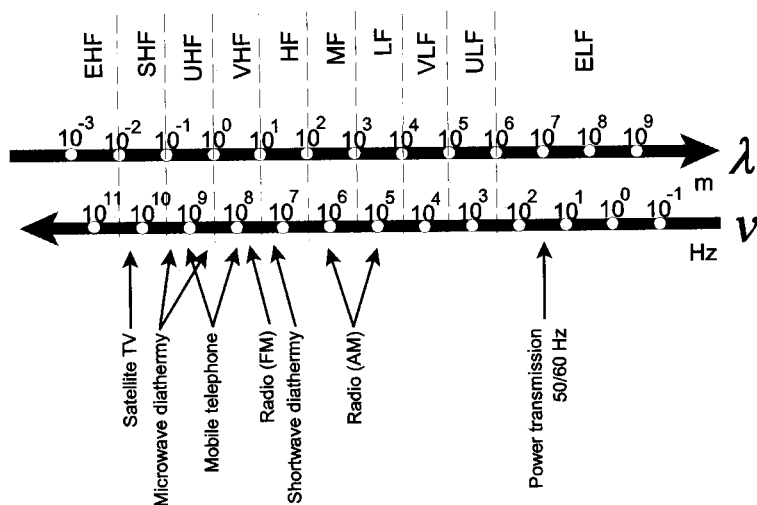
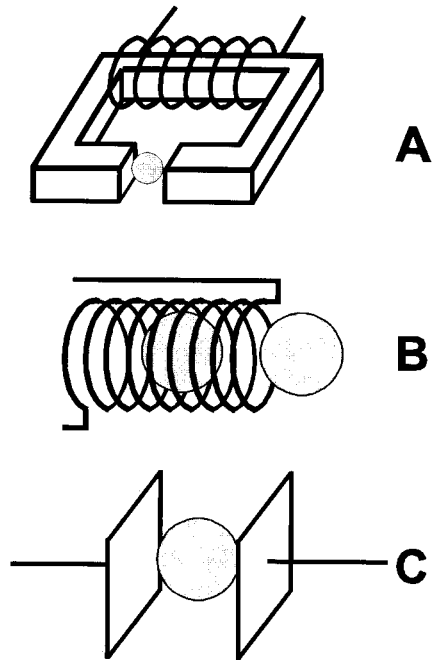


Fig. 4.17. The spectrum of technically used frequencies (HF = high frequency, LF = low frequency, E = extremely, S = super, U = ultra, V = very, M = mean)

Fig. 4.18. Examples of three techniques of experimental application of electromagnetic fields (explanation in text)



them, almost only electric influences occur. In the same way, AC fields can also be applied, using electrodes connected via physiological solutions with biological objects, or which are directly inserted in them. Using coils (Fig. 4.18B), the magnetic, as well as the electric component of the field interacts with the object. There is no difference whether the object is positioned inside the coil or in its vicinity. This situation resembles the general kind of exposure of humans near technical devices.

These techniques of field application are used in research as well as in therapy. In the case of short wave diathermy (Table 4.3), the VLF fields are applied by electrodes with direct electric contact with the tissue. In contrast, microwave diathermy in the UHF and VHF range uses coils to transmit the

Table 4.3. Frequencies of electromagnetic fields used in medical treatment and their penetration (δ) in biological tissue with specific conductivity $g = 0.6 \text{ S m}^{-1}$

	Frequency ν (MHz)	Wave length λ (m)	Skin depth δ (m)
Shortwave diathermy	13.56	22.05	0.1765
	27.12	11.03	0.1248
	40.68	7.35	0.1019
Microwave diathermy	433.92	0.69	0.0312
	915	0.33	0.0215
	2450	0.12	0.0131

energy into the body. Sometimes additional iron cores are applied to increase the magnetic output. The eddy currents induced in this way in the body lead to the desired diathermy effects.

If the population is exposed to technical facilities, additional circumstances have to be considered which may lead to a dominance of the magnetic or the electric vector of the field. The electric field of a particular position near a power line, for example, depends on the construction of the line, i.e. on the voltage of the transported energy, which is constant. The magnetic field emitted by this line, however, varies depending on the current flowing through it, which is always changing during the day. Furthermore, usually several wires are combined in one line, carrying different currents with phase differences. The magnetic field of a power line therefore can vary considerably. Similar aspects must also be taken into account in other electric devices.

Furthermore, it must be considered that electric fields are shielded by buildings, trees etc. Magnetic fields, in contrast to this diminish only with increasing distance from the source. This leads to the dominance of magnetic fields in buildings and at places shielded in other ways.

These properties of the electromagnetic fields are important for low frequency AC fields, the wavelengths of which (see Fig. 4.17) are large in relation to other geometrical conditions such as size of the objects, or distances. Such a splitting between magnetic and electric field regions is not possible in higher frequency fields. Although up to UHF frequencies it is technically possible to apply fields with pronounced electrical or magnetic components, in practical use of technical devices, both components are connected together.

At higher frequencies, the so-called *skin effect* becomes important. This means that the current density in a conductor induced by a high frequency electromagnetic field falls off rapidly below the surface. This effect determines the depth of penetration of high frequency electromagnetic fields in the body. This is characterized by the *skin depth* (δ) which can be calculated for practical use by:

$$\delta = \sqrt{\frac{1}{\pi \mu \mu_0 g \nu}}. \quad (4.4.6)$$

This skin depth (δ) is the distance over which the field decreases to $1/e = 0.368$. Inserting into Eq. (4.4.6) the magnetic permeability of vacuum: $\mu_0 = 1.256 \cdot 10^{-6} \text{ V s A}^{-1} \text{ m}^{-1}$, the magnetic permeability number $\mu = 1$, and a mean conductivity of the tissue: $g = 0.6 \text{ S m}^{-1}$ (see Fig. 3.38), one understands that this effect will become important only at high frequencies (ν). As listed in Table 4.3, the skin depth plays an important role in therapeutic use of electromagnetic fields. The values shown in this table should only be taken as examples because of the dielectric heterogeneity of the human body.

These differences in the physical properties of low versus high frequency fields are also reflected in the use of measuring units for dosimetry. In the case of low frequency fields the magnetic component is expressed in A m^{-1} , or

mostly as magnetic flux density in T, and the electric field strength is measured in V m^{-1} . Multiplying the electric field strength (E) by the specific conductivity (g) of the medium, one gets the current density (j) in A m^{-2} . This parameter is important for evaluating possible excitation phenomena in the body.

At higher frequencies the parameters: *plane power density* (W m^{-2}), and the *specific absorption rate* (SAR) in W kg^{-1} , or W m^{-3} are used. In contrast to the plane power density, which just depends on technical parameters of the field-emitting device, the SAR is the energy which is absorbed in a unit of mass or volume of the body per unit of time. For low frequency fields, using Ohm's law, this means:

$$P = jE = \frac{j^2}{g} = E^2 g. \quad (4.4.7)$$

Knowing the SAR and the heat capacity of the tissue, thermic effects of absorbed electromagnetic fields can be calculated. In fact, the calculation of SAR distribution in a heterogeneous dielectric body, like for example the human head, is very complicated.

As shown in Fig. 4.17 for radiation protection, two frequency regions are of particular interest: the low frequency electromagnetic fields of power transmission and the high frequency fields of various kinds of electronic communication. The worldwide discussions about possible hazards of these fields have led to various technical modifications to minimize exposure. In the low frequency range, one can show that at common distances from electrical devices, a whole body exposure of $30 \mu\text{T}$ will never be exceeded for the general public. In most cases exposure is much lower. Directly beneath a power line, the magnetic flux densities even in worst cases usually do not exceed the $10 \mu\text{T}$ limit. At a distance of 50 m it is already lower than $0.3 \mu\text{T}$. In some industrial appliances, as for example inductive heaters with very high current density, magnetic flux densities of 10 mT may occur.

From the physical point of view, low frequency fields can interact in different ways with the human body. As already pointed out, the magnetic component of a low frequency field fully penetrates the body. In general, it is possible that the magnetic field can interact with the biological system directly. On the other hand, it can act via the induced eddy currents. The current density produced in this way depends on the frequency of the field, on its intensity, and on the geometrical parameters of the induction loop. This is determined by the size or diameter of the body in a projection plane perpendicular to the magnetic field vector, but also by anatomy and the impedance of the corresponding tissue. Differences in the conductivity of the organs in the body can lead to preferential conductivity loops. Such loops may occur via special arrangements of blood vessels or even between electrically communicating cells. The cell itself as a space of high conductivity, surrounded by an isolating membrane, can act as a conducting loop, whose diameter is, however, very small. This in fact complicates the calculation of the real field conditions in the body even if the environmental field parameters are known.

In Section 4.4.2 we considered the possibility of an external electric field interacting directly with a body, inducing charge displacements [Eqs. (4.4.4) and (4.4.5)]. The field strength induced in the body in this way depends on the frequency of the AC field, the size of the body, its conductivity and its dielectric constant.

In the practice of environmental and occupational protection of humans against electromagnetic fields, various physical approaches are used to calculate the resulting current densities in the human body. The difficulties of these calculations and their uncertainties are easy to understand, considering the discussion above. For this reason, two types of parameters are used in safety standards: *basic restrictions*, these are current density in the body, or SAR at high frequency fields and *reference levels* describing the corresponding parameters (electric or magnetic field strength or the plane power density introduced above) in the environment. If a measured value in the human environment exceeds the corresponding reference level, it does not necessarily follow that the basic restriction will also be exceeded. However, an investigation should be carried out whenever a reference level is exceeded.

Besides technically induced fields, humans are also exposed to natural electromagnetic fields in the environment. These fields have wide-spread frequencies, starting with variations of the electrostatic field of the earth, as discussed in Section 4.4.2, in a region far below 1 Hz. Pulses induced by lightning are above 10 Hz. These worldwide discharges induce permanent low frequency electromagnetic oscillations in the atmosphere which can be measured even in clear weather conditions. In this case the so-called *Schumann resonance* is important, a global electrical resonance frequency of 7.8 Hz with corresponding harmonics at 13.8, 19.7, 25.7 and 31.7 Hz. At higher frequencies, so-called *atmospherics* (or *spherics*) can be measured. These are pulses induced by various discharges in the atmosphere, containing electromagnetic frequencies up to tenths of kHz. Their magnetic flux density does not exceed 20 nT. The intensity of these *atmospherics* depends on various atmospheric conditions. It is speculated that the weather sensitivity of humans could be based on this influence. Recently, experiments have been carried out to test volunteers with artificial *spherics* simulation systems.

Further reading: Dosimetry and safety standards: Polk and Postow 1996; therapeutic use of fields: Low and Reed 1992; natural fields and *spherics*: König et al. 1981; Reiter 1992; Schienle et al. 1995

4.4.4

Biological Effects of Electromagnetic Fields

The question as to how electromagnetic fields may affect biological systems is of particular interest for problems of health protection and at the same time it opens new possibilities of therapeutic use. Health protection studies are focused mainly on two frequency regions with quite different kinds of biological interaction: the low frequency range of the power frequency (50 or 60 Hz) and

high frequency ranges which are used for devices of radiocommunication and diathermic heating (Fig. 4.17).

Applications of electromagnetic fields for medical purposes already have a long tradition. From a scientific point of view, however, it must be considered as an area still in development. Various excitation methods are now scientifically established which work with low frequency pulses, for example transcutaneous electric nerve stimulation (TENS), as well as high frequency field applications for diathermy, as listed in Table 4.3 (page 267). Many other attempts have been made to use electromagnetic fields in bone and wound healing, nerve regeneration etc., but up to now mostly without sufficient scientific understanding.

As explained already in Section 3.5.4, low frequency AC fields and field pulses modify the transmembrane potential proportional to the external field strength and the length of the cell in the direction of the field vectors [Eq. (3.5.19)]. In the case of muscle or nerve cells this can lead to excitation processes. The excitation of long nerve or muscle cells by a given external electrical field strength is easier, if the effective length of the cell in the direction of the electric field vector is larger. The word "effective" means that it is not simply the length of the cell which is responsible for this potential alteration, but rather the length of its conductive area surrounded by the isolating membrane. If two cells, for example, are connected by gap junctions with high electric conductivity, then the "effective" length means the extension of this whole system. Because of the time constant of the excitation process, i.e. the chronaxy, the threshold of cells is lower for low frequency, rather than for high frequency fields (Fig. 4.19).

Using the parameters of Fig. 4.19 and inserting them in Eq. (4.4.7) one can easily see that for low frequency fields the SAR, even at the highest threshold in this figure, namely absolute danger to life, amounts only to 0.001 W kg^{-1} . This value is very small in relation to the basic metabolic rate, i.e. the heat production of the resting human body of 1 W kg^{-1} which increases during active movement by a factor of 10. It can be related directly to the dissipation function (Φ) which

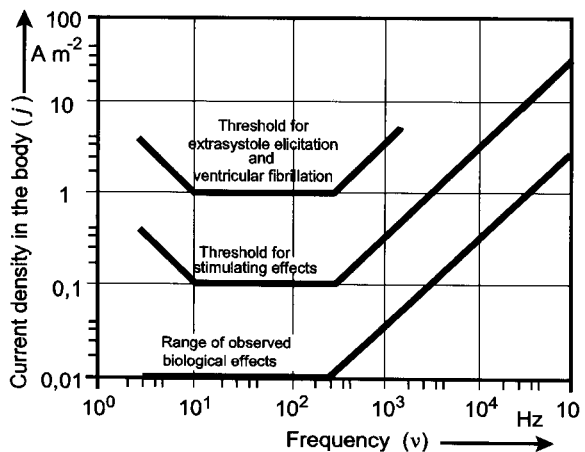


Fig. 4.19. Threshold values of the electric current density in the human body in the low frequency region. (After Bernhardt 1988 simplified)

we discussed in Section 3.1.4 [Eq. (3.1.64)] which for small mammals is even larger (see Fig. 3.7).

This means that in the frequency range up to 100 kHz excitations of nerves and muscles are the dominating mechanisms of interaction, but not thermic effects. The excitation properties shown in Fig. 4.19 lead to recommendations that in the frequency range between 4 Hz and 1 kHz occupational exposure should be limited to fields that induce current densities of less than 10 mA m^{-2} in the body. For the general public an additional factor of 1/5 is applied.

At higher frequencies this situation reverses. As seen in Fig. 4.19 the threshold of nerve excitation increases, corresponding with the decrease in the induced membrane potential ($\Delta\psi_{\text{ind}}$) with increasing frequencies, as depicted in Fig. 3.45. In the frequency region $\nu > 10^5 \text{ Hz}$, diathermic heating becomes dominant. The basic restrictions are therefore determined by the SAR (for definition see Sect. 4.4.3). Established biological and health effects could occur at a rise in body temperature of more than 1°C . This temperature increase results from exposure of individuals under moderate environmental conditions to a whole body SAR of about 4 W kg^{-1} . This means that we now reach the magnitude of the basic metabolic rate. Of course, it is not the mean metabolic rate of the whole body which must be considered when evaluating health risks, but rather that of the most active organs like muscles or brain which may be higher by one order of magnitude. Using an additional safety factor for whole body irradiation with frequencies between 100 kHz and 10 GHz an average whole body SAR limit of 0.08 W kg^{-1} is recommended.

These safety factors are important chiefly for two reasons: firstly, the questions arise: are there low frequency effects below the level of nerve and muscle excitation? And: are their high frequency reactions below the level of diathermic heating? The second reason concerns the problems of microdosimetry.

Let us start with this second question. In the previous section we discussed the problems of local conducting loops in the body in case of eddy currents induced by low frequency fields. But microdosimetry is also complicated for high frequency fields. As we learned in Section 3.5.3, the biological system must be considered as a heterogeneous dielectric. This means that even for a constant irradiation with high frequency fields, because of the skin effect, and differences in impedance of various cells and various tissues, and partly because of reflections of the waves by bone surfaces etc. a heterogeneous spatial energy absorption occurs. So-called "hot spots" appear, i.e. small or even microscopic regions of heating. This means not only local temperature differences, but also permanent temperature gradients occur. Furthermore, one must consider that at points of increased physiological heat dissipation, for example in muscles or in the brain, an enforced blood circulation occurs to control the temperature. The hot spots generated by absorption of high frequency electromagnetic fields, however, do not correspond necessarily to the locations of high physiological heat production. Therefore it might be possible that in organs with lower circulation a particular SAR may lead to a stronger temperature increase than in other parts of the body with stronger blood flow. A typical example for this is the lens of the eye, a small part without any blood supply.

Biological effects of low frequency electromagnetic fields below the threshold of excitation, as well as high frequency fields below thermic alterations have been the subject of many investigations. Nevertheless, knowledge in this field is quite incomplete. Most investigations have been undertaken in a pragmatic way, limited to the particular frequencies of power supplies, or other technical situations. Because of the political relevance of safety standards on the one hand, and of commercial aspects of therapeutic applications on the other, many investigations unfortunately lack scientific seriousness. Many results of this kind of research indicating some field effects on protein expression, on cellular metabolism, on neuronal reactions, on proliferation and differentiation, up to processes of carcinogenesis must be reproduced and verified. Epidemiological investigations do not show any significant effect and mostly fail because of the limited number of people exposed in a definite way. In fact it seems clear that this kind of non-ionizing electromagnetic radiation, in contrast to X-rays or other types of ionizing radiation, does not induce any kind of mutation. So far, there has been no serious reason to decrease the safety level as discussed above.

As already mentioned, further research is necessary not only for health protection, but also in the direction of possible medical applications of electromagnetic fields. For use in therapy much stronger fields can be applied than limited by the guidelines for radiation protection. Apart from the already established methods of electric and magnetic nerve and muscle stimulation, as well as methods of shortwave and microwave diathermy (see Table 4.3, page 267), a large number of more or less scientifically based methods are used to promote bone healing, nerve regeneration or other therapies. Research in this region is quite divergent, using various techniques of field applications (Fig. 4.18) and various kinds of pulses and sine fields (Fig. 4.20). These range from application of continuous sine fields (Fig. 4.20A), pulsed sine fields (PEMF; Fig. 4.20B), modulated sine fields up to various types of unipolar and bipolar electric pulses (Fig. 4.20C, D). Sometimes coils are used (Fig. 4.18B) in order to induce eddy currents in the tissue. The analysis of the real electrical conditions near the cells is mostly unknown, especially if rectangular magnetic pulses are used.

Mean theoretical approaches of biophysical mechanisms of biological field effects can be summarized briefly as follows:

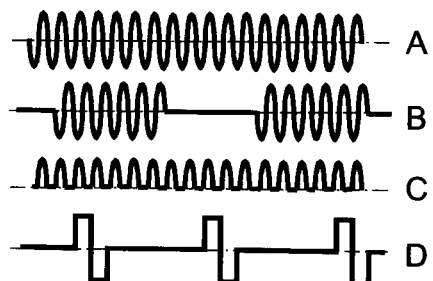


Fig. 4.20. Some examples of AC-fields used in research and therapy

- It is unlikely that one single primary reaction of electromagnetic fields will be found in analogy to the ionization process of X-rays. Probably non-ionizing electromagnetic fields interact with biological systems in many different ways depending on their intensity, frequency and modulation, as well as on physiological properties of the biological system.
- The basic question, especially in the low frequency region, is whether the magnetic, or the electric component of the field is responsible for the primary reaction. There are experiments supporting both kinds of interactions. Clarification of this question is of fundamental importance for radiation protection and dosimetry, as explained in Section 4.4.3.
- Possible targets of magnetic interactions are radicals occurring during various biochemical processes as intermediate products. The recombination of these radicals will generally take place only if the valence electrons in the two radicals are in a singlet state. The effect of the magnetic field consists in the influence of the spin precessions that affect these processes of recombination. The question is whether weak magnetic fields are able to produce this reaction with sufficient probability. Another kind of magnetic hypothesis to explain possible effects is connected with the occurrence of magnetites in the tissue (see Sect. 4.4.1).
- There are various hypotheses of possible electrical interaction mechanisms. In low frequency regions oscillations of the membrane potential occur which could come into resonance with transport processes or oscillating binding reactions of ions with receptors or transport proteins. In the same way, an influence of the electrical field on electrochemical processes at the membrane surface is possible.
- High frequency fields can interact with proteins by a kind of quantum-mechanical resonance. This furthermore requires a coherent behavior of proteins, organized in the strong electrostatic field of the membrane. The theory of this mechanism predicts sharp frequency windows in the field effects. Recently the role of bound water near macromolecular surfaces has been discussed. In contrast to free water, the dipoles of which are oscillating at the frequency of 18.7 GHz (Sect. 3.5.3), the resonance frequency of bound water can be shifted down to frequencies in the order of 10^7 Hz. Alteration of the membrane structure can affect various macromolecular structures and reactions (Sect. 2.4.2).

Research in this field is in fact interdisciplinary to an extremely high degree: it starts with the techniques of generating a well-defined electromagnetic field in the biological system. Medical physics must then help to calculate the field distribution in the body. Biophysics is responsible for explanation of primary effects on molecules and membranes. Subsequently, molecular and cell biology should explain possible reactions of the cells to these effects, and finally, medicine can investigate whether these effects could lead to diseases or at least health risks, or whether they could be used in therapy.

These effects obviously require a systemic rather than a molecular approach. One the one hand, field distribution can be calculated knowing only the

dielectric properties of anatomical as well as of cellular organization. On the other hand, the effects of weak fields, in particular, can be understood only by taking into account cooperative processes and mechanisms of biological amplification.

Further reading: General aspects of field effects: Bernhardt et al. 1997; Matthes et al. 1997; Polk and Postow 1996; Reilly 1998; Repacholi 1998; Repacholi et al. 1999; Wilson et al. 1990; aspects of health protection: International Commission on Non-Ionizing Radiation Protection 1988; history of medical applications: Rowbottom and Susskind 1984

4.5 Ionizing Radiation

Studies on effects of ionizing radiation on living organisms have become necessary following the introduction of X-rays in medical therapy and diagnostics at the beginning of the twentieth century, and have acquired a special relevance in the so-called nuclear age. In this book only a few important aspects of radiation biophysics can be focused on.

In the spectrum of electromagnetic waves (Fig. 4.13), ionizing radiation begins at a quantum energy in the order of 10 eV. Usually the energy of ionization of the water molecule, which amounts to 12.46 eV, is taken as a borderline. This corresponds to a range of wavelengths in the upper limit of ultraviolet light. UV-radiation per definition starts with a wavelength of 380 nm which corresponds to a quantum energy of 3.29 eV. This region is divided into three subregions. Ultraviolet-A (UVA) with a wavelength up to 315 nm, corresponding to 3.94 eV, followed by ultraviolet-B (UVB) up to 280 nm, corresponding to 4.42 eV, and finally the region of ultraviolet C (UVC) lasting up to the quantum energy of water ionization. Because the quantum energy of ultraviolet radiation is lower than the energy of ionization of water, it is usually considered as a kind of non-ionizing radiation. Nevertheless, mechanisms of interaction occur which are similar to those of ionizing radiation.

Corpuscular radiations, such as α -, β -radiation, as well as accelerated elementary particles and ions of various mass numbers are also considered as ionizing radiation.

4.5.1 Nature, Properties and Dosimetry of Radiation

Ionizing radiation in our environment is emitted by extraterrestrial sources, by various technical devices, and by spontaneous decay of certain atomic nuclei. Even if the quality and quantity of human exposure to ionizing radiation has been altered by technical development, a natural exposure of all biological organisms has always existed and has been a part of environmental conditions for evolution.

X-rays and γ -radiation are explained as a part of electromagnetic waves in Fig. 4.13. Let us now characterize briefly some properties of corpuscular radiations.

α -rays are of fast-moving helium nuclei with a mass number = 4 and an atomic number = 2 (${}^4_2\text{He}$). The helium nucleus consists of two neutrons and two protons and thus carries two positive charges. Because of this strong charge, there is a correspondingly large interaction between α -rays and the elementary particles of matter. The energy of α -rays emitted by radionuclides is generally quite high, namely of the order of several millions of electron volts (MeV). As the α -rays pass through matter, this energy is dissipated as a result of ionizing processes and the particles are slowed down. For example, the energy which is required to generate one pair of ions in air is 34.7 eV. It can be easily calculated that a radiation energy of several 10^6 eV is sufficient to induce about 10^5 ionizations. Consequently, α -particles will leave behind a straight-line track of limited length in irradiated matter which consists of ionization products. This can be visualized by special methods.

β -rays consist of fast moving electrons. These particles carry a single negative charge and an extremely small mass. Although it is possible to generate β -radiation of very high energy by means of particle accelerators, the energy of β -radiation from radioactive nuclides on average is lower than that of the α -radiation. In contrast to α -radiation, the energies of electrons of β -emitting isotopes are not at all equal to each other but are spread across a specific range. Therefore the terms *mean energy* and *maximum energy* are used when referring to the β -emission of radionuclides. The distance to which a β -particle penetrates matter will, naturally, correspond with its energy. The ionization tracks of β -particles in contrast to those of α -particles, do not follow a straight line but their path becomes increasingly curved towards the end. In addition, the density of ionization increases as the energy of the particle decreases so that ionization is more densely packed with low energy radiation than with high energy radiation and, for the same reason, this effect can also be seen towards the ends of the tracks.

Neutron radiation occupies a rather special position in the classification of corpuscular radiation. Its particles are electrically neutral and possess a considerable mass when compared with β -particles. For this reason they are able to penetrate an atom and reach the nucleus where they can cause nuclear transformations. This is technically how radionuclides are made. The irradiated substance itself becomes radioactive. Neutron emission does not occur when a radionuclide decays spontaneously but takes place during nuclear fission or in the wake of other, externally induced reactions.

Recently, various kinds of particles, such as π -mesons, accelerated ions and small nuclides have been used in cancer therapy.

The process of energy dissipation that occurs during the absorption of electromagnetic radiation by matter follows the same pattern as that during the absorption of corpuscular radiation in that it is not continuous but takes place in steps i.e. in quantum leaps. The size of energy quanta absorbed in this way depends on the type of interaction. In general, three types of absorption

processes may be distinguished according to the relation between the required ionization energy and the available quantum energy:

1. With low quantum energies, the *photo effect* is produced. In this, a γ -quantum (also called a *photon*) is absorbed and this causes a displacement of an orbiting electron from the shell of an atom. The excess energy, above that which is required for the ionization process, serves to accelerate this so called *photoelectron*.
2. The *Compton effect* occurs at a quantum energy of about 10^5 eV. In this case a *Compton electron* is ejected from the atom and is accompanied by the scattering of a secondary γ -radiation. However, this latter radiation has a quantum energy lower than that which was originally absorbed.
3. If the quantum energy of the γ -radiation is above 1.02 MeV then *electron-pair formation* can occur. The quantum energy disappears producing a negative and a positive electron. Such formation of an ion pair can only take place close to the nucleus of an atom i.e. it can only occur in an absorbing material and not spontaneously in a vacuum.

The radiation energy that is absorbed per unit mass is measured as energy dose in Gray units (abbreviated to Gy):

$$1 \text{ Gy} = 1 \text{ J kg}^{-1}.$$

In earlier papers the units Roentgen (R) and rad (rd) can be found. These units can be converted as follows:

$$1 \text{ rd} = 10^{-2} \text{ Gy}.$$

In water and tissues:

$$1 \text{ R} = 0.93\text{--}0.98 \text{ rd}.$$

In addition, as measure of the dose rate, the unit Gy s^{-1} is used.

For considerations of radiation protection a further parameter is used which additionally takes into account a factor which marks the relative biological effectiveness of the radiation. So, for example, irradiation by neutrons or positrons are 10–20 times as effective as γ -rays or X-rays. For α -rays this factor amounts 20. This parameter is the dose equivalent with the unit “Sievert” (Sv). This means that the following relations occur: for X-rays: $1 \text{ Gy} = 1 \text{ Sv}$, for α -rays, however: $1 \text{ G} = 20 \text{ Sv}$.

4.5.2

Primary Processes of Radiation Chemistry

The action of ionizing radiation of biological systems initiates a complicated sequence of reactions as depicted in Fig. 4.21. The biological reaction starts with

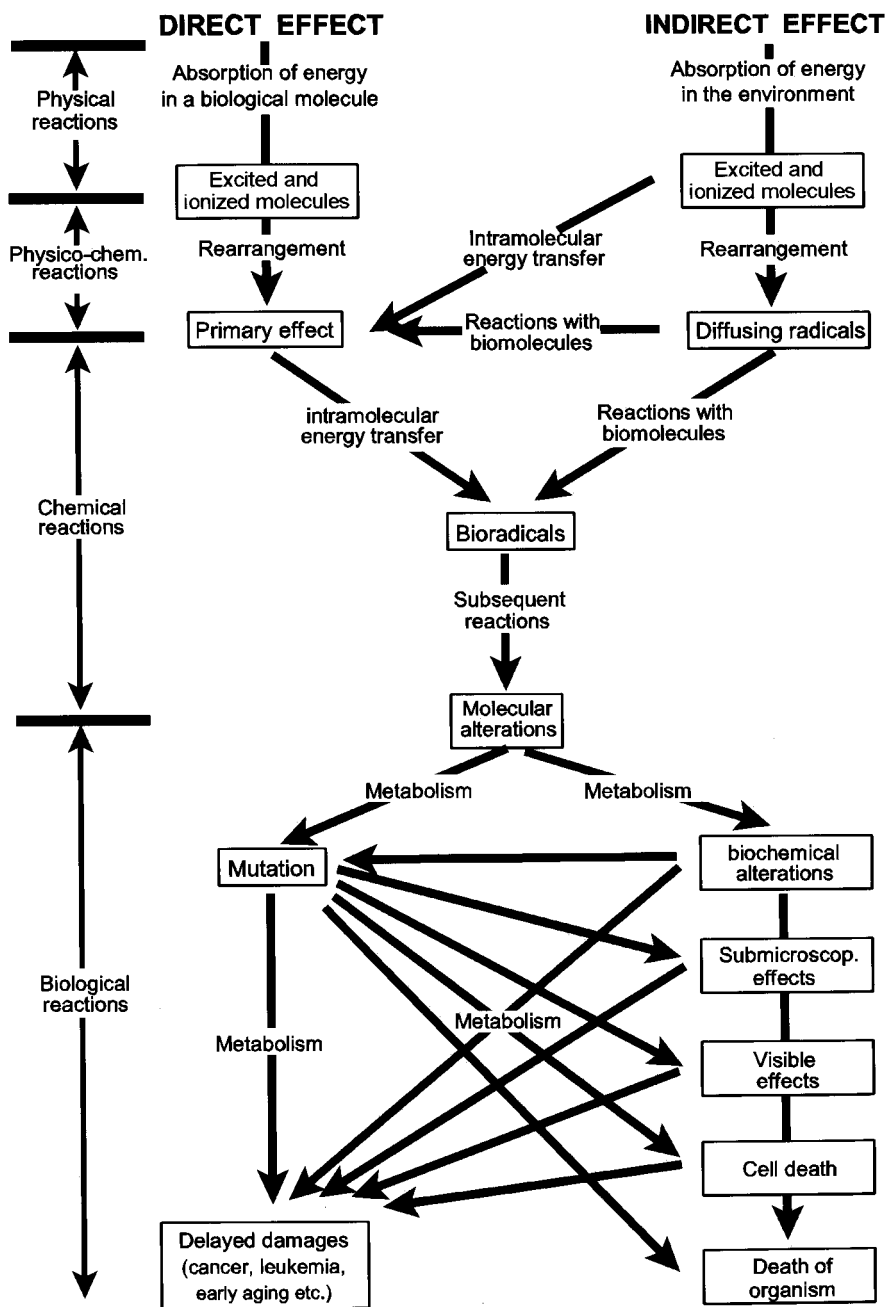
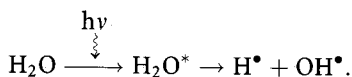


Fig. 4.21. Schematic representation of various steps of the interaction of ionizing radiation on biological systems. (After Dertinger and Jung 1969, redrawn)

an alteration of a biological molecule such as an enzyme or a DNA. Such a change may be induced in two ways:

- By a direct interaction of the energy quantum with the appropriate molecule (*direct effect*);
- Through an interaction of a biologically important molecule with any other product of radiolysis (*indirect effect*).

Water is by far the most common molecule in biological systems and as such plays a special role in indirect effects of radio-biological reactions. In Fig. 4.22 some steps of radiolysis of water are illustrated schematically. In a first step within a time span of 10^{-18} – 10^{-16} s an electron is pulled out of the molecular binding by an energy input of 12.56 eV. This primary process of absorption of the energy quantum $h\nu$ can be expressed by the following reaction:



Both of these primary reaction products undergo further reactions. Greatly simplified, the H_2O^+ ion breaks up as follows:



The radical $\text{OH}^\bullet$ has oxidizing properties and can be regarded as one of the most important of the radiolysis products that give rise to secondary reactions. It can act through electron transfer, by addition of OH or by H-transfer.

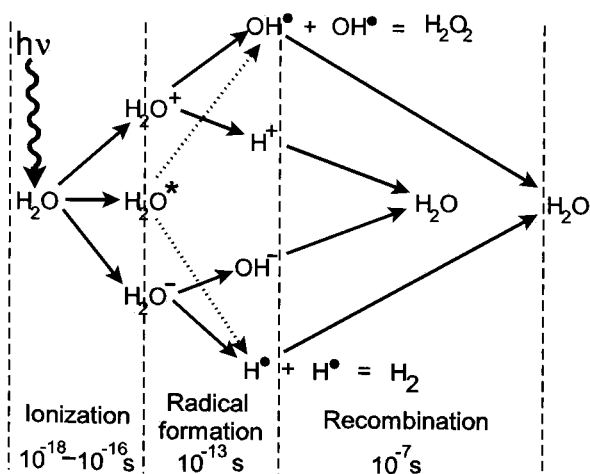
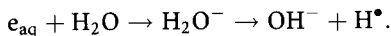


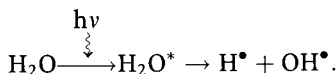
Fig. 4.22. Schematic illustration of the most important reactions of water radiolysis (* – excited states, • – radicals)

The released electron (e_{aq}) becomes hydrated, similar to ions in aqueous solutions (Sect. 2.4.2). For this, in Fig. 4.22 it is depicted as H_2O^- . Because of this shell of bound water, its life span, depending on the pH in the solution, can achieve 600 μs . It can interact with various other molecules. Furthermore, reactions with other water molecules in the hydration shell are possible in the following way:



In this way a new product of radiolysis appears, the H-radical which indicates strongly reducing properties.

It can also be shown that an excited water molecule can be formed, a process that requires an energy uptake of 7 eV only. This excited molecule can directly break up into its free radicals:



Summarizing this, three direct products of water radiolysis are obtained: e_{aq} , $H^\bullet$, and $OH^\bullet$. The probability of these products appearing can be expressed by a so-called *G value*. It is defined as the number of molecules which become altered per 100 eV absorbed energy. At a pH of the solution in the range between 3–10, the following G values are obtained:

$$G(e_{aq}) = 2.65; G(H^\bullet) = 0.55; G(OH^\bullet) = 2.7.$$

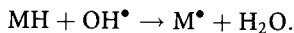
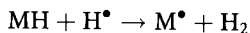
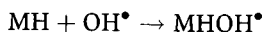
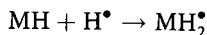
It has been known for a long time that as the dose rate of the radiation and the ionization density are increased, hydrogen peroxide and molecular hydrogen are formed in growing amounts. The following recombination reactions are assumed to take place provided that locally, for a sufficient time a sufficient number of free radicals are present.



H_2O_2 is well known to be toxic but the amount generated radiolytically is generally not very great. On the other hand, these reactions eliminate aggressive radicals, thus the indirect effect of radiation will be lessened.

Recombination of these radicals with oxygen, solved in the aqueous phase, is very important for further reactions (*oxygen effect*). This oxygen forms various products with the hydrated electrons as well as with hydrogen radicals, leading eventually to additional production of H_2O_2 . Oxygen-rich tissue therefore suffers more from irradiation than that with low oxygen content.

The most important reactions between $H^\bullet$ and $OH^\bullet$ radicals and organic molecules (MH) can be summarized as follows:



The hydrated electron reacts primarily with those parts of an organic molecule that have high affinity for electrons such as SH^- , CO^- , NO_2^- or NO -groups. Either it simply becomes attached to the molecule, thus imparting a negative charge to it, or a dissociative electron capture takes place which, similar to the instances described above, leads again to the formation of a free radical.

In addition to the indirect effects of radiation initiated by the radiolysis products of water, biologically important molecules can also become ionized directly. Amino acids generally become deaminated. Aromatic amino acids such as tyrosine, phenylalanine and tryptophan can be destroyed by cleavage of the benzene ring. Amino acids bearing an $-SH$ group (cysteine) and their derivatives (for example glutathione) are particularly sensitive. These groups are oxidized to become disulfide groups. When aqueous solutions of amino acids or proteins are irradiated then both the direct and the indirect mechanisms are involved in the effect that ensues.

The particular sensitivity of SH -bearing amino acids to ionizing radiation is retained even after they have been incorporated into large protein molecules. In such molecules energy transfer is of considerable importance because energy can pass from one monomer to the next until a sensitive group is reached.

Radiolysis of DNA is possible via sequence of many reaction steps, whereas direct as well as indirect effects via water radicals can occur. Single strand breaks which do not necessarily lead to the break of the whole molecule appear, as well as direct breaks of the double helix. Furthermore, the loss of single bases is possible. Irradiation can also lead to formation of whole denaturated zones as well as to intermolecular cross-links with neighboring macromolecules. The double strand helix can be considered as a construction of higher stability against irradiation in relation to the single stranded DNA helix. If the hydrogen bridges between the two molecular filaments are broken by placing the DNA in a solution of urea, then fragmentation of the DNA by irradiation is easier.

Using the example of radiolysis of DNA the particular property of UV-radiation must be pointed out. Even in the region of frequencies in which the quantum energy is not sufficient to lead to radiolysis of water, alterations in the base structure of DNA can be observed. Irradiation with UV-light with wavelength of about 260 nm with a quantum energy of 4.8 eV leads to formation of base dimers and some other products of photolysis. This is the region on the borderline of UVB and UVC. This process mostly concerns dimers of pyrimidine, thymine and cytosine, and purine to a lesser extent. In these

reactions two neighboring bases of the same strand are interconnected, via opening of the 5-6-double bonds and formation of a cyclobutyl ring structure. Dimer formation of two different bases is possible too. This process is reversible.

Further reading: Farhataziz and Rodgers 1987; Kiefer 1990

4.5.3

Radiobiological Reactions

In this section we leave the narrow region of biophysics and come to the biological reactions according to Fig. 4.21. It can be easily understood how the processes of radiolysis as described in the previous section may interact with biological processes in various ways. As shown in Fig. 4.21, this kind of interaction is marked by a multitude of different paths.

A significant starting point for radiobiological reactions, of course, are alterations in the genome. This kind of reaction is considered as a *stochastic effect* of irradiation. Although DNA molecules play a dominant role in primary reactions of irradiation, this does not mean that every damage of an DNA molecule must lead to the manifestation of a mutation. Thanks to a multitude of repair mechanisms, only a very low percentage of damaged DNA molecules will finally lead to biologically important alterations. Of course the repair process itself can be influenced by irradiation. This, by the way, is a mechanism which is hypothesized for the possible carcinogenic influence of non-ionizing irradiation. Furthermore, these repair processes themselves can lead to incorrectly repaired molecules.

To determine the genetic defects caused by ionizing radiation, the dose of irradiation must be integrated over the whole life span of an individual or even over the whole population. The reason for this is not a biophysical, but a genetic mechanism of accumulation. Recessive mutations especially, i.e. such which become effective only when two of them occasionally meet by arbitrary combinations, become dangerous if they become accumulated in the population. The same holds for radiation-induced cancer as result of somatic mutations.

Mutations caused by irradiation are of particular importance for those cells that are proliferating even in the adult body at a fast rate. These are found, for example, in the blood forming organs and in the gonads. It is therefore understandable that following irradiation, drastic changes in the blood picture can be seen. In this context the reduction of the number of lymphocytes is particularly dangerous.

The different phases in the life cycle of cells show different sensitivities towards radiation. The phase when DNA is being synthesized, the S-phase of the cell cycle, is particularly sensitive. Even the process of mitosis can be disturbed by several effects. As a result of irradiation of cells during metaphase, chromosome aberrations may occur. Disturbances that arise during the anaphase and telophase of mitosis are mainly attributable to the effects of radiation on the spindle apparatus. The result is an incomplete separation of the chromosomes. The chromosomes that are left behind as well as some parts of chromosomes form para-nuclei that are not viable.

When tissues are irradiated while they are proliferating intensively, then changes in the incidence of mitosis can be observed over a period of time. Immediately after irradiation by a sub-lethal dose, the number of dividing cells is reduced. After a certain recovery time it returns to its normal value. This is obviously due to the fact that the further development of cells that are at a particular stage in their life cycle, is inhibited. Later, these cells undergo mitosis together with the undamaged cells, sometimes leading to a small overshoot of the number of proliferating cells.

Considering the facts that have been outlined up to now it is understandable that juvenile organisms, particularly those in the embryonic phase, are especially sensitive to irradiation. Damage to the genetic and mitotic apparatus can lead to severe malformations or even death of the individual. There are some phases during the development of an organism that are particularly critical in relation to radiation damage. One such phase is during the early stages of cleavage of the ovum up to the morula formation. Radiation damage in these states, however, does not usually lead to malformations because the injured cells either die or are successful in repairing the damage. Irradiation during the stage when the organs are being laid down is substantially more critical. Exposure during these stages frequently causes malformations. In this context, the development of the central nervous system is particularly sensitive, in contrast to the completely developed brain which is one of the most resistant organs in relation to ionizing radiation.

In addition to the influence on the genetic or epigenetic systems of the cell, other effects of radiation may occur, caused by other mechanisms. A particular role is played by radiochemical alterations of enzymes. Their sensibilities to radiation show great variations. Among the most sensitive enzymes are ATPases and catalases. This sensitivity of proteins in general depends on their composition. We already mentioned that proteins with a high percentage of sulfur-containing amino acids are particularly sensitive to radiation. These proteins which are to be found in substantial amounts in the epidermis as well as its products (hairs, nails, horn) are characterized by a high content of cysteine. Radiolysis of these proteins leads to loss of hair, to skin damage and some other reactions.

One process that can be attributed to an alteration of protein structures as a consequence of irradiation is an increase in the permeability of biological membranes. This reaction, however, occurs only at relatively high doses of radiation, but may have a profound influence on physiological processes. These alterations, for example, are responsible for radiation effects in the central nervous system.

These circumstances help us to understand the course of acute radiation sickness. Figure 4.23 shows the average life span of some animals. All of the curves first show a decline at doses below 10 Gy, then there is a plateau followed by a second decrease at about 80–100 Gy. The shape of these curves reflects two different types of radiation sickness. At low doses the radiation sickness is mainly caused by pathologic alterations in blood-forming organs, which is reflected in dramatic changes in the blood picture. A considerable deficiency of lymphocytes occurs which reduces the resistance of the body to bacterial

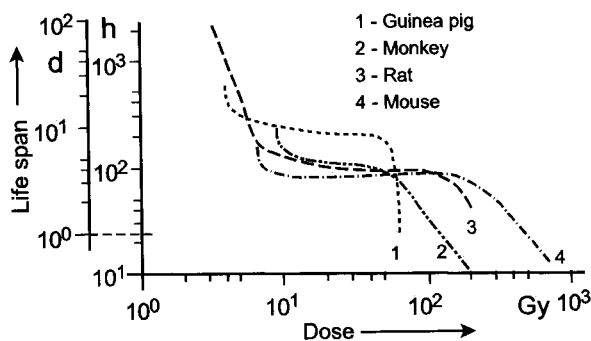


Fig. 4.23. Average life span of various mammals as a function of the dose of a single whole-body irradiation with X-rays. (After Grodzenskij 1966, redrawn)

infections. This can eventually lead to a situation where the intestinal bacteria become pathogenic. This type of radiation sickness lasts from several days up to weeks and can lead to the death of the organism.

The pathological course of this type of radiation sickness is maximal at a dose of about 10 Gy. A further increase in radiation intensity does not accelerate this process. If the animals are irradiated with doses over 80 Gy, the central nervous system becomes damaged directly. In this case the animals die in a much shorter period of time.

Because of the complexity of radio-biological reactions it is difficult to find bench mark figures as a measure of the resistance of different organisms to irradiation. Similar to the characterizations of poisonous drugs, the parameter LD_{50}^{30} (mean lethal doses) is used. This means the dose of radiation that causes 50% of the irradiated animals to die within 30 days.

From the values shown in Table 4.4 the following conclusions can be drawn: the sensitivity of radiation varies over several orders of magnitude for different species. A comparison of the sensitivity at different ages and stages in the development of a given species clearly shows the embryonic organism to be more sensitive than the adult one.

Further reading: Kiefer 1990; Nias 1990

4.5.4

Some Aspects of Radiation Protection

Radiation protection means on the one hand the protection against irradiation from the environment, and on the other hand possible irradiation emitted by incorporated radionuclides must be considered. Besides naturally occurring radionuclides in the body, mostly ^{40}K , and to some extent ^{14}C and some others beginning with the nuclear age, many technically produced radioisotopes are incorporated. The protection of man against incorporation of radionuclides from the environment is the task of *radioecology*. This discipline concerns the distribution of radionuclides in the ecosystem, its accumulation in various products of the food chain and finally their degree of resorption, accumulation

Table 4.4. Examples for the mean lethal dose (LD_{50}^{30}) of various organisms after single irradiation with X-rays. (After Grajevski and Šapiro 1957)

Organism	LD_{50}^{30} (Gy)	Organism	LD_{50}^{30} (Gy)
Algae		Low vertebrata	
Chlorella spec.	180	Goldfish	6.7
Chroococcus spec.	80	Frog	7.0
Ancistrodesmus spec.	110	Turtle	15
Synechococcus spec.	1000	Serpend	820
Protozoa		Birds	
Amoeba spec.	1000	Chicken	10
Colpidium colpoda	330	Mammals	
Paramecium caud.	350	Mouse	4–6.5
Crustacea		Rat	6–10
Calliopius laevisuc	10	Guinea pig	1.7–4.1
Artemia salina (egs)	800	Rabbit	7.5–8.3
Insects		Hamster	7.25
<i>Drosophila</i> (adults)	950	Dog	2.75
<i>Drosophila</i> (larvs)	1.3	Donkey	6.5
<i>Drosophila</i> (eggs)	1.5	Monkey	5.0
<i>Calliphora</i> (eggs)	0.4	Man	4.0–5.0

and excretion by man. In fact it is much more complicated to calculate the dose or irradiation caused by accumulated radionuclides, than to calculate intensity of irradiation coming from external sources. Radionuclides emitting α -, or weak β -rays are most dangerous. Empirically obtained coefficients are listed indicating concentration limits of these nuclides in air, water and various foods, as well as in humans themselves.

The protection of man against ionizing radiation is based on recommendations of the International Commission of Radiological Protection (ICRP), a non-governmental organization founded in 1928. The philosophy of these recommendations is the exclusion of any non-stochastic radiation effects and the minimization of inevitable stochastic effects. These stochastic effects really are the critical point of radiation damage. It includes possible genetic aberrations or induction of cancer. In general, one can assume that any increase of irradiation exceeding that of the natural dose of radiation will lead to an increase of the mutation rate. There is no threshold below which no influence of the mutation rate can be expected. It has been evaluated that the doubling of the mutation rate of man occurs at a radiation dose below 1 Sv. The probability of carcinogeneses, evaluated in Japan, is considerably lower than that.

The everyday irradiation of man is caused by natural, as well as by technical sources as products of our civilization. As shown in Table 4.5 as an example, the largest radiation dose of about 2.4 mSv yearly comes from natural sources. Of all technical sources, probably the irradiation caused by medical treatments seems to be dominant. Of course, this is only an example, the numbers may differ depending on cultural, geological as well as individual conditions.

Because of the possibility of stochastic effects becoming integrated over the whole life span, the limits of maximal exposure are formulated over longer periods of time. People with occupational exposure are under special supervision. For occupational exposure, a maximum yearly irradiation dose of 50 mSv is laid down. For the general population 1/30 of this value is considered to be acceptable. As seen in Table 4.5 this is below the level of natural irradiation.

The recommendations of the ICRP states that all exposure should be kept “as low as reasonably achievable”, economic and social factors taken into account. This philosophy is commonly known as ALARA-principle.

Further reading: International Commission of Radiological Protection (1991)

4.5.5

Mathematical Models of Primary Radiobiological Effects

Since the first investigations of biological effects of ionizing radiation as early as the 1920s, efforts were made to formulate theoretical approaches, helping to understand the observed phenomena. This was before anything about molecular genetics or details of primary molecular effects of water radiolysis etc. were known. Even though of limited relevance in modern radiation research, these approaches nevertheless helped to understand various quantum biological processes in general, such as photosynthesis, or scotopic vision.

The starting point of these approaches was the realization that radiation effects are characterized by the following particularities:

Table 4.5. Average effective dose of the population in Germany in 1993 (in mSv). (According to the report of the Federal Office for Radiation Protection)

1. Natural sources	
– Cosmic radiation	approx. 0.3
– External terrestrial radiation	approx. 0.4
– By inhalation of radon-decay products	approx. 1.4
– By ingestion of natural radioactive substances	approx. 0.3
	approx. 2.4
2. Technical sources	
– Nuclear plants	<0.01
– By medical applications of radionuclides and ionizing radiation (rough evaluation)	approx. 1.5
– By application of radionuclides and ionizing radiation in research, techniques and in daily life	<0.01
– Occupational exposition (contribution to average exposure of the population)	<0.01
– By fall-out products of atomic bomb tests	<0.02
	approx. 1.55
3. Exposure, caused by Tschernobyl accident	approx. 0.02

- The effects not only depend on the total amount of the absorbed energy, but additionally on the quality of the applied radiation.
- When comparing biological effects obtained by irradiation with rays of the same quality, a dose dependence was found, but not a proportionality between effects and irradiation dose.

Furthermore, the total amount of energy absorbed in a biological system, even at irradiation with the mean lethal dose (LD_{50}^{35}), is negligible in relation to the thermic energy produced in it under physiological conditions.

These observations have led to the following conclusion: apparently, there are molecules in the living organisms, which must be considered as “neuralgic” points in the system, the destruction of which would produce a sequence of reactions, finally leading to the death of the individual. This sort of primary event which is caused by a reaction with a single energy quantum of radiation was called a “hit” and the corresponding location where the process occurs, a “target”. This leads to the name of the first approach, called *target theory*, firstly applied to a population of single cells.

Let us assume that a population consists of n identical individuals, each of them organized physiologically in such a way that a single “hit” would set off a sequence of reactions, leading to death (*single hit process*). According to the laws of probability, with increasing time of irradiation, an increasing number of individuals would become killed by this *single hit* process. The rate at which the number of living individuals decreases in this way can be expressed by the differential quotient $-dn/dt$. It should be proportional to the number of still living individuals (n), the time of irradiation (t), and an irradiation constant (σ) which depends on quantity and quality of irradiation:

$$\frac{dn}{dt} = -\sigma n. \quad (4.5.1)$$

Using the initial conditions, whereby: $t = 0$; $n = n_0$ it results in:

$$n = n_0 e^{-\sigma t}. \quad (4.5.2)$$

The reduction of the number of living cells therefore follows a simple exponential function.

In the same way, the number of killed cells (n') can be calculated. Because of the relation: $n' + n = n_0$, we obtain:

$$n' = n_0(1 - e^{-\sigma t}), \quad (4.5.3)$$

or:

$$\frac{n'}{n_0} = 1 - e^{-\sigma t}. \quad (4.5.4)$$

In Fig. 4.24 the curve for $m = 1$ represents this function. At $t = 0$ is $n' = 0$ and therefore $n'/n_0 = 0$. With increasing irradiation time and doses (σt) the value $n'/n_0 = 1$ will be approached.

In a similar way, another model can be calculated. In this case it is assumed that the sequence of processes leading to the death of the individual is only triggered if the organism has received a minimum number (m) of hits. Blau and Altenburger have calculated the following equation describing this *multi-hit process*:

$$n' = n_0 \left\{ 1 - e^{-\sigma t} \left[1 + \sigma t + \frac{(\sigma t)^2}{2!} + \frac{(\sigma t)^3}{3!} + \cdots + \frac{(\sigma t)^{m-1}}{(m-1)!} \right] \right\}. \quad (4.5.5)$$

Some of the curves corresponding to this equation are shown in Fig. 4.24. These functions help to explain experimentally obtained dose dependent effects.

This theory was extended by J. A. Crowther by introducing a *target area* into this concept. This means he transformed this formal target into a concrete area where the "hits" occur. For this he used results of experiments with radiation of different wavelengths. As already stated, radiation leaves an ionization trail as it passes through an object. The density of ionizations along this path depends on the character of the radiation, i.e. of its quantum energy. Suppose a "hit" means an ionization of a quite specific molecule or a molecular configuration, then a certain volume must be sensitive to the radiation, which is called the *target area*.

Figure 4.25 schematically illustrates two experiments where a biological object with a given target area is irradiated by two kinds of ionizing radiation. In case A a radiation is used which induces a lower ionization density than in case B. Let us suppose that the biological reaction in this case is triggered by a single hit process, then in case B the energy will be squandered because the diameter of the target area is larger than the mutual distance between two points of ionization. This theoretical prediction has been verified by experiments, where particular dose-effect curves were obtained when different types of radiation were applied.

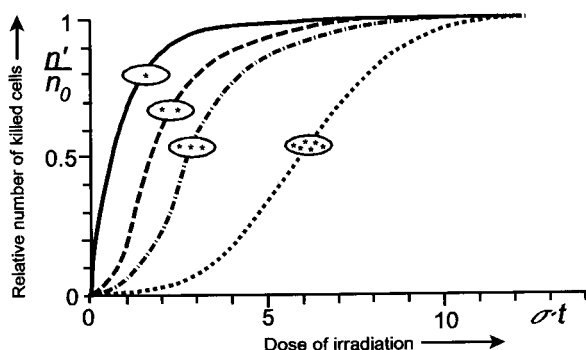
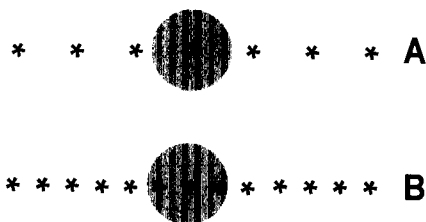


Fig. 4.24. Theoretical curves for one- and multi-hit processes (m corresponds to the number of stars in the "cell"). (After Dessauer 1965)

Fig. 4.25. Schematic illustration of the irradiation of a single target area (*circular area*) with a radiation of lower (A), and higher (B) ionization density. The points mark the locations of ionizations



In a similar way the target theory has been modified step by step in adaptation to experimental results. Meanwhile, a considerable number of additional factors have been taken into account influencing the course of radiobiological reactions. For example, it is possible to calculate a multi-hit process, the targets of which have different sensibility. Furthermore, it is also possible to take into consideration some processes of repair. For example, one can assume that in the course of a multi-hit process the efficiency of subsequent “hits” depends on the time period in between. Processes of energy transfer, e.g. indirect effects, can be included. Of course any extension of the model leads to the introduction of additional parameters. This finally leads to ambiguous interpretation of experimental results. This seems to be the limit of the extension of these sort of calculations.

Nevertheless, the target theory has achieved a certain importance in the analysis of various kinds of radiobiological effects. So, for example, mortality curves for bacteria have been characterized as multi-hit processes. Induction of mutations as a result of irradiation has also been analyzed by this formalism. Some mutations, even those of chromosomes, could be characterized as single-hit processes.

These considerations have also been extended to the influence of non-ionizing radiation. So, for example, scotopic vision at dawn could be explained by curves of three to seven hit processes. The process of photosynthesis has also been investigated in the same way.

Further reading: Dessauer 1964; Hug and Kellerer 1969; Kiefer 1990; Zimmer 1961

The Kinetics of Biological Systems

This section provides an introduction to systems theory as applied to living organisms, which is an important part of theoretical biophysics. Originally, thermodynamics as a universal theory of the energetic basis of all movement in nature on the one hand, and kinetics, as a theory to predict the time course of processes on the other hand, were considered two separate directions in science. Thermodynamics gives an answer to the questions: what is the cause, the driving force, and the direction of a movement, and finally what is the equilibrium situation which will be eventually arrived at? Kinetics, on the other hand, just studies the course of a given reaction, its time constants and rates, its possible stationary states and oscillations.

After introducing time as a new variable into the framework of thermodynamics, starting with the thermodynamics of irreversible processes, i.e. nonequilibrium thermodynamics, these two directions became more and more linked. A new name was even proposed to characterize this connection: it was called *synergetics* (see Hagen 1983).

The idea of *isomorphism* still forms the basis of system analysis. This means that quite different processes in nature, technology or society can formally be described by similar mathematical approaches and therefore show similarities in their properties. This is also the basic idea of cybernetics.

5.1

Some Foundations of Systems Theory

The system analysis concerns consequences of movements, and interplay of various processes determining the behavior of the system as a whole. Therefore, we will follow the definition of the term "system" which was given in Section 2.3.3 and the general discussion on system properties in Section 3.1.1. The definitions of stationary states and stability, which have been discussed from the thermodynamic point of view in Section 3.1.4 are important here as well.

5.1.1

Problems and Approaches of System Analysis

The main concern of systems theory is to investigate the behavior of a system and the interplay between its elements. Formally, one considers all kinds of

parameters influencing the system as *input*. This leads to a response of the system, which is named the *output* of the system. The transformation of the input into an output signal is performed by the *input-output law* or: *law of system behavior* (see Fig. 5.1). This law in simple systems can be expressed by a mathematical equation.

In general, there are three kinds of questions which are directed to system analysis:

1. The input and the input-output law of the system are known. It is necessary to find the output (*direct problem*). This is the problem that the practicing doctor has to solve before deciding on a therapeutic method.
2. The input-output law and the output of the system are known. The input has to be determined (*indirect problem*). This is the problem of finding the cause for an alteration in a known system. To give another medical example: it is the problem of diagnosing a disease.
3. The input and the output of the system are known. What is input-output law of the system (*black-box problem*)? This third question is typical of the problems facing the scientist and is, at the same time, the most difficult of the three. The "black box" must be converted into a "white box", the law of which should be clarified.

The cognitive process usually follows an iterative course, continuously setting up hypotheses and testing them by special experiments on the system. The more complex the system is, the more tedious and ambiguous this problem will be. Studies on such systems often require that they be broken down into smaller subsystems that will still allow individual measurements to be made. Then the laws governing the behavior of these subsystems must be analyzed. Subsequently, an attempt must be made to reintegrate the whole system. A functional diagram can be drawn indicating the interaction of the individual components (Fig. 5.2C), each of which is similar to that shown in Fig. 5.1. Even if one knows the rules governing the function of each of the subsystems, it is not possible to determine the behavior of the system as a whole without more information because usually the arrangement of the elements in the system is not known.

The approaches used to analyze the system behavior are based on differential equations containing time derivatives of the corresponding parameters. To describe complex systems, usually a system of many differential equations is required. It is usually complicated to find sensible solutions for these equations. It is therefore reasonable to attempt to simplify the system, taking into account the time hierarchy of the individual processes. This approach has already been

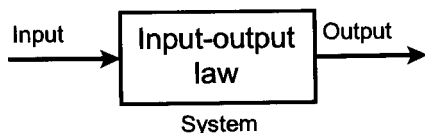
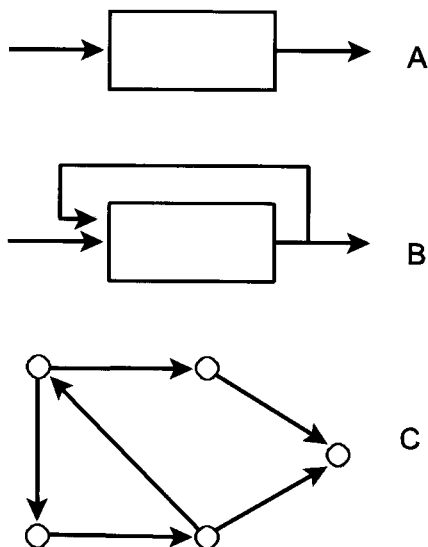


Fig. 5.1. Schematic illustration of behavior of a simple system

Fig. 5.2. Schematic representation of various types of systems: **A** system without feedback, **B** system with feedback, **C** directed graph; the points in **C** mean subsystems of type **A** or **B** with corresponding inputs and outputs



considered in Section 3.1.4 in connection with Fig. 3.8. We will discuss this problem in more detail in Section 5.2.2 in the context of biochemical reactions.

Schemes of reactions such as those shown in Fig. 5.2C are called *directed graphs*. *Theory of graphs*, or *network theory* which was established by Leonhard Euler in the eighteenth century enables general laws to be derived to explain their behavior and their possibilities of transformations.

Further reading: Doucet and Sloep 1992; Rashevsky 1960; Haken 1983

5.1.2

General Features of System Behavior

Systems theory is concerned with the behavior of the parameters of a system as a function of time. The theoretical approaches therefore describe the time dependence of these parameters in the form of ordinary differential quotients. In some instances, such as for example the propagation of organisms as a function of time in an ecological model, or where there are problems of growth, or changes in the distribution in space taking place at the same time, approaches are necessary which contain partial differential equations.

In general, the system behavior will be characterized by the order and the degree of the corresponding differential equations. The *order* of a differential equation is the order of its highest derivative. The *degree* of an equation is determined by the highest power to which one of its variables or their derivatives is raised, provided that the equation has been so arranged that only positive powers of the dependent variables or their derivatives occur. The product of two variables has to be considered as a kind of second power.

The thermodynamic characteristics of non-linear processes have already been discussed in Section 3.1. We will complete these considerations by introducing kinetics. To illustrate this, let us consider again a simple mechanical system, namely the extension of a spring. We will now calculate the time characteristics of these systems which we discussed in Section 3.6.3 only qualitatively.

Suppose there is a spring hanging from a rigid support. At the lower end of the spring is a pointer which marks the length (l) of the spring. A certain force (F) holds the spring under tension and is in equilibrium with the pulling force of the spring all the time. If the force is suddenly changed, the extension of the spring is correspondingly altered. The relation between these two variables is given by the following equation:

$$F = k_1 l, \quad (5.1.1)$$

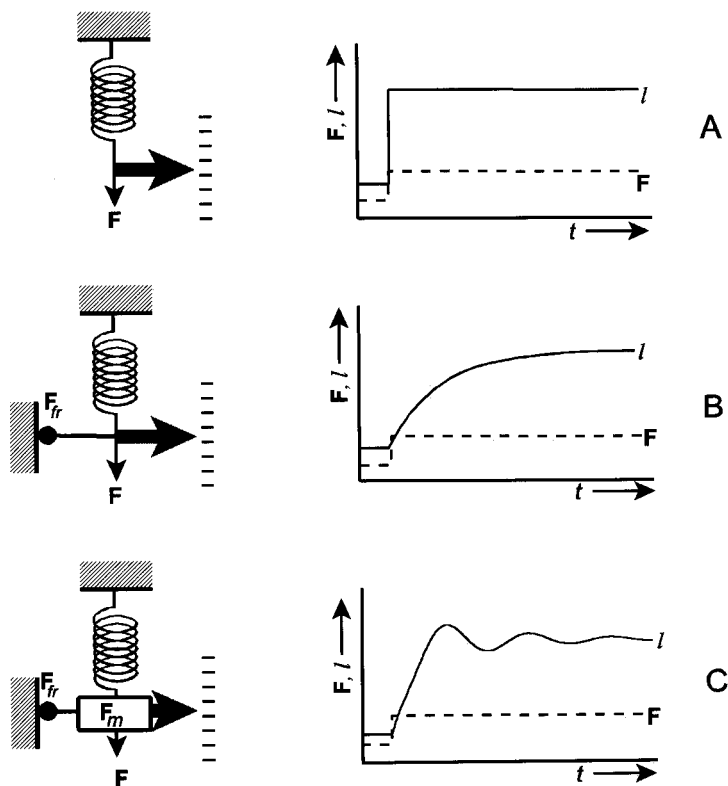


Fig. 5.3. Three mechanical constructions illustrating the kinetic behavior of systems of zero (A), first (B), and second (C) order as response to a rectangular function of the extending force F

where k_1 is a constant containing the Young's modulus. Equation (5.1.1) is a simple algebraic equation. If it were to be included in the category of differential equations, it could formally be classified as of zero order, because it does not contain a derivative of the variable l . The curve $l(t)$ shows a sudden jump corresponding to the alteration of the force (F).

The example in Fig. 5.3B has a brake incorporated into the system. The frictional force of which (F_{fr}) is taken to be proportional to the rate of displacement (dl/dt):

$$F_{fr} = k_2 \frac{dl}{dt}, \quad (5.1.2)$$

where k_2 , again, is a proportionality factor. In this case the action of the force (F) is compensated by the frictional force (F_{fr}) as well as by the force of extension:

$$F = k_1 l + k_2 \frac{dl}{dt}. \quad (5.1.3)$$

This is already a first order differential equation. The behavior of this system shows a correspondingly complex time course. A sudden alteration of the force (F) leads to an extension of the spring according to a simple exponential function.

The third example (Fig. 5.3C) additionally has a mass (m) built into it, which imparts a certain amount of inertia to the system. The inertial force (F_m) is, according to Newton's law, proportional to the acceleration. Acceleration, however, is the second time derivative of a distance:

$$F_m = k_3 \frac{d^2 l}{dt^2}. \quad (5.1.4)$$

Considering this additional force, we get a differential equation of the second order:

$$F = k_1 l + k_2 \frac{dl}{dt} + k_3 \frac{d^2 l}{dt^2}. \quad (5.1.5)$$

It is easy to imagine that a second order system like this will show oscillations as it settles down, like pairs of scales. After practical acquaintance with such scales, we know that these oscillations can be made to slow down at different rates by use of various damping devices.

Non-linear systems have further possibilities for kinetic behavior. Figure 5.4 shows in principle how a given parameter c in a system can change when it is displaced by an amount Δc from its stationary value. When the system is stable it then returns to its original state. This can occur without oscillations increasing the time constant up to an aperiodic limiting condition, or by oscillations that die away asymptotically. Under certain conditions non-damped

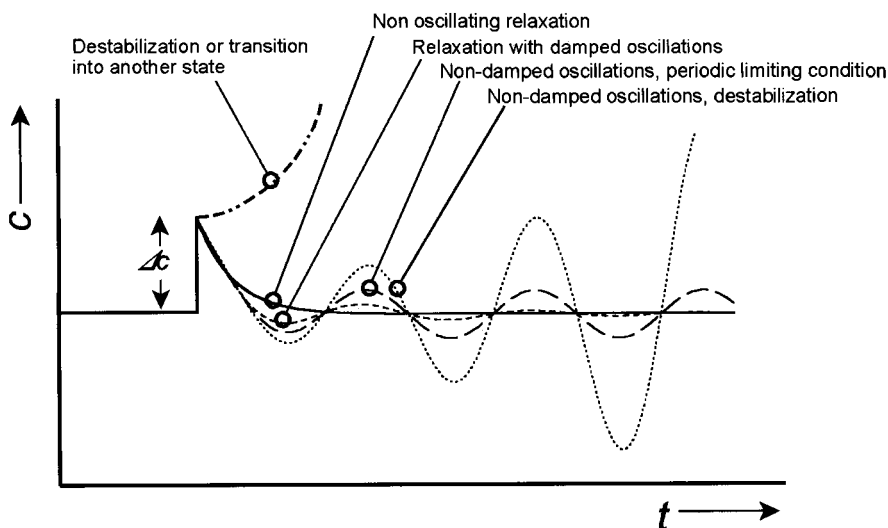


Fig. 5.4. Possible kinds of system behavior, illustrated as time course of the parameter $c(t)$ of a system after a particular displacement Δc

oscillations can occur (periodic limiting condition). Disturbances of the system, however, may also lead to instabilities, possibly leading to another stationary state.

Each state of a system with n independent parameters can be characterized as a point in an n -dimensional phase space. We will illustrate this for example in the biochemical reaction model of Fig. 5.15, or in the mixed culture of two species of paramecium (Fig. 5.18). Let this system be determined by the two variables c_1 and c_2 . It is possible to characterize the state of this system by plotting both curves $c_1(t)$, and $c_2(t)$ separately (Fig. 5.5, left part). At any moment (t) a value of the parameter c_1 corresponds to a particular value of c_2 . The state of the system therefore can also be represented by a graph, where c_2 is plotted against c_1 . At the time t_B , for example, the state of the system corresponds to the point B on the right graph. Following the functions $c_1(t)$, and $c_2(t)$ on the left plot from the moments t_A , t_B , t_C , t_D , t_E ... it is possible to construct a trajectory following the points A, B, C, D, E ... in the graph at the right side of Fig. 5.5. When there are only two variables, the trajectory is just a line in the plane. The arrow indicates where the system proceeds. In systems with more parameters, this is a function in an n -dimensional space.

Figure 5.6 shows such trajectories in a two dimensional parameter space. These curves were obtained in the same way as the trajectory in the right part of Fig. 5.5. This illustration is a kinetic completion of the energetic schemes in Figs. 3.5 and 3.6. At first we can distinguish between stable systems (curves A, D, F) and unstable ones (curves B, C, E), whereby the curve F represents a system with stable permanent oscillations. The stability of a system in this plot is

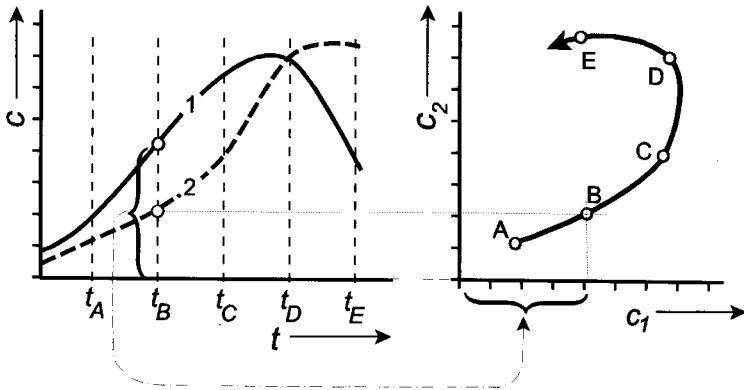
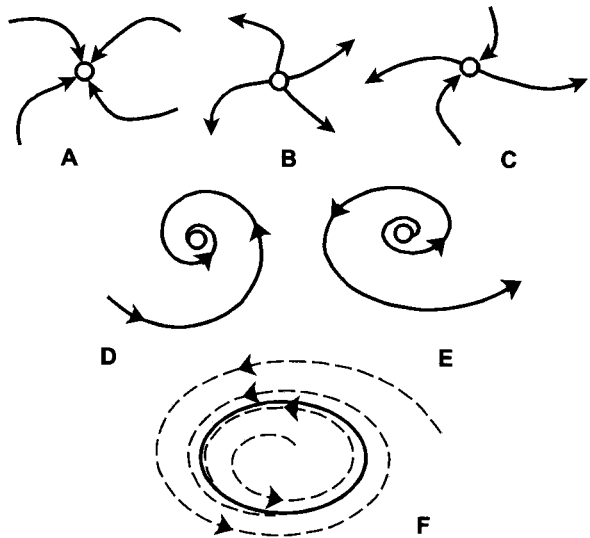


Fig. 5.5. Plot of the functions $c_1(t)$ (curve 1), and $c_2(t)$ (curve 2) on the left, and the corresponding trajectory of these values $c_2(c_1)$ on the right side

Fig. 5.6. Trajectories in a two-dimensional space to characterize various states of the system: A stable node, B unstable node, C saddle point, D stable focus, E unstable focus, F stable limit cycle



characterized by a trajectory leading to a fixed point or a limit circle, whereas in case of instability, the arrows lead away. Metastable states (see Fig. 3.5), which are not plotted in Fig. 5.6, would be characterized by an arrow leading from one point to another.

Furthermore, in this figure different ways are indicated for systems used in order to attend the stable states. The aperiodic approximation to this state, as shown in Fig. 5.4, corresponds to the case A in Fig. 5.6. The situation with damped oscillations in Fig. 5.4 is represented in case D of Fig. 5.6, and the

undamped oscillation corresponds to case F in Fig. 5.6. The distortion of the circle into an ellipse (case F) simply depends on the chosen coordinates. In the case of oscillations superimposed by several frequencies, this ellipse will be deformed further.

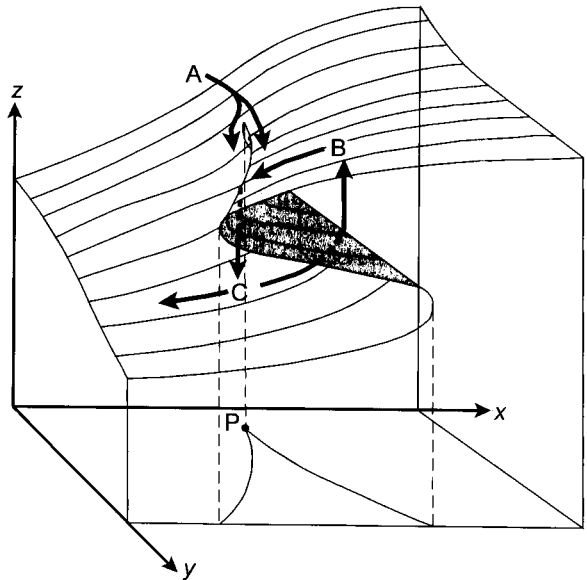
In this way, the set of differential equations describing the behavior of the whole system determines a vector space, with the n variables as coordinates. This means that at each point in this n -dimensional vector space a signpost is located, indicating the direction of further system development. There are some "watersheds", or borderlines where the directions become bifurcated, like the point in case C (Fig. 5.6). This would correspond to the energy maximum between two metastable states (Fig. 3.5).

System trajectories can be plotted in 2-dimensional coordinates only if the systems are simple. Generally, multidimensional spaces are required to describe alterations in the state of more complex systems in mathematical terms. In 1969 the French mathematician, Rene Thom developed a comprehensive topological theory of dynamic systems which primarily enables the occurrence of multi-stationary states to be systematized. He called a sudden change from one state to another a catastrophe, which has led to the name Catastrophe Theory for this entire mathematical approach. Thom was able to demonstrate that for a system which has no more than four controlling parameters, seven different types of control surfaces can be formulated, most of them, however, being 3- or 4-dimensional hypersurfaces.

The basic concepts of this theory will now be elucidated by referring to one of these seven functions, the so-called *cusp catastrophe* which can be demonstrated in a 3-dimensional system of coordinates. Above a control surface (x, y), the function $z(x, y)$ builds up a behavioral surface, indicating a state parameter z for each value in the x, y -control surface. In most cases, there is only one point in the z -coordinates that corresponds to a point in the x, y -plane. The behavioral surface, however, is folded at a particular region. As a result, there are three possible values of z for some points of the x, y -control surface. The system can, therefore, take on three different states for some variables. The value of z in the middle of those three points characterizes an unstable state (shaded area). This can be compared with the thermodynamic picture of a meta-stable state (Fig. 3.5) where there is also an unstable state between the two stable stationary states. The fold in the behavioral surface in Fig. 5.7 starts at the point P which is the point from which the bifurcation lines diverge and can be seen clearly when the fold is projected onto the x, y -control surface.

This picture of a cusp catastrophe can be used to demonstrate the isomorphism, namely the fact that quite different processes can be treated mathematically in a similar way. Take z to be a parameter that characterizes the viscosity or the structure of water. It will depend on pressure (x) and on temperature (y). The value of z at the same time will be an indicator for the corresponding aggregation state of the water. When the temperature falls (trajectory A in Fig. 5.7) the system will reach a bifurcation point at 0 °C at normal pressure. It can either freeze (i.e. move on the lower surface), or remain as a super-cooled liquid (upper surface). The super-cooled liquid, however, will

Fig. 5.7. Functional space of the state function (z) depending on two independent state parameters (x, y) as an example of a cusp catastrophe. A trajectory with bifurcation point (P); B, C trajectories with a phase jump ("catastrophe") near the borderline of stability; shaded area - region of unstable states



spontaneously freeze (trajectory B) if there is any disturbance of the system (alteration of pressure, vibration etc.). This reaction in the trajectory B would be a "catastrophe" in the terminology of this theory.

In Section 5.2.2 we will characterize the behavior of a biochemical reaction chain with such a function (Fig. 5.15B). Furthermore, the electrical potential of a nerve cell was explained in this way. In this case the upper and the lower part of the folded behavioral surface are considered as non-activated and activated nerve membrane, corresponding to its resting potential and action potential. The "catastrophe" in this case was the process of excitation (trajectory B). In this case a transition corresponding to the trajectory C is also possible as the restoring reaction of the nerve.

A quite different example can be taken from behavioral biology: let z be a parameter that characterizes the behavior of a dog, and x and y are parameters corresponding to its disturbance. When these reach a certain size (point P) the dog will be critically irritated, with the result that it either runs away (i.e. the upper surface), or it attacks (lower surface). Either position, once taken, will be maintained and can even be intensified if the degree of irritation is increased. If the flight reaction is suddenly limited (e.g. by an obstruction of some sort), or if the dog is convinced that the attack is senseless, then the dog's behavior may be reversed. The running away in panic changes to an enraged attack or vice versa.

The above example indicates on the one hand the broad field of possible applications of this theory and at the same time the inherent danger of its overuse. Of course, the example from behavioral science is quite graphic, and

similar examples can be stressed taken from human behavior, from some social responses, and even from politics. But of course it is impossible to perform real calculations in these situations. Therefore it remains just an illustration without any new, real insights into the mechanism. It remains a game with words. On the other hand, in cases of simpler systems, in biochemical network theory etc. catastrophe theory really can lead to important results.

In recent decades a kind of system kinetics was investigated which is called *chaotic*. This means an obviously irregular and non predictable behavior, which nevertheless is the result of interaction of deterministic processes. Non-linear systems with several stationary states can jump from one state to another. Sometimes, oscillations are the result, sometimes these oscillations degenerate into chaotic movements. The turbulent flow, which we discussed in Section 3.7, may be taken as an example of a chaotic process. In medicine, the disturbance of the cardiac cycle is analysed by approaches using chaos theory.

Further reading: Doucet and Sloep 1992; Harrison 1993; Heinrich and Schuster 1996; Thom 1969; Zeeman 1976

5.1.3

Cybernetic Approaches to System Analysis

In 1948 Norbert Wiener's book *Cybernetics – or Control and Communication in the Animal and the Machine* was published. This was the signal for a rapid development in bio-cybernetics which, when treated seriously, has made substantial contributions to the understanding of biological regulation and control processes. This theory is largely based on experiences with systems in mechanical technology which are made up of discrete functional units (subsystems) that have a flow of information and energy between them. The analogy with biological systems can be drawn, especially if neuronal processes are involved. In the case of thermal regulation of the body, for example, nervous and hormonal effects are essential, controlling blood flow, muscle contraction etc. This is an area where the application of cybernetic approaches is reasonable. On the other hand, in biochemical and cellular or ecological systems, where the elements of the control system are not clearly separated from each other, direct approaches using the theory of non-linear systems seems to be more appropriate.

The title of Wiener's book indicates two major problems which are included in cybernetics: the control of animals and machines, and the processes of communication. In this section we will discuss only the first of these. Some aspects of information theory have already been dealt with in Section 2.3.2.

The module of cybernetic modeling is the control system (Fig. 5.8). In its simplest form it consists of two subsystems, the *controlled system* itself, and the *controller*. Both of these subsystems process one or more input signals to produce one or more output signals depending on their input-output law (see Fig. 5.1). Cybernetics is mainly concerned with systems that are made up of

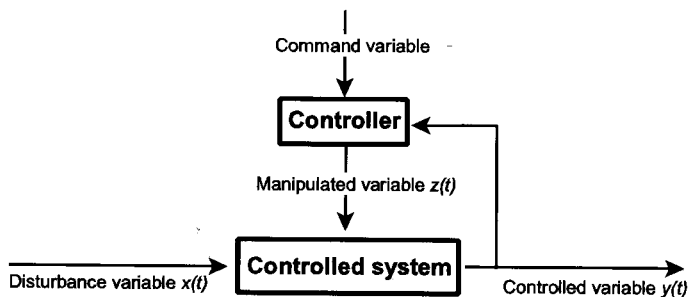


Fig. 5.8. Elements of an automatic control system

linear subsystems in which the system behavior conforms to a linear differential equation of the n th order.

The purpose of a control circuit is to adjust a *controlled variable* $y(t)$ to a given standard which is specified by a *command variable*, in spite of a supplementary effect of a *disturbance variable* $x(t)$. If the command variable remains constant then the network is called a *retaining control circuit*. However, if the command variable is a function of time, which means that the system has an additional control on it, this is a *follow-up control circuit*. The command variable, the controlled variable, and the disturbance variable may differ from each other in their physical properties both in quality and quantity. The regulation of blood pressure (controlled variable), for example, can be controlled through a sequence of nerve impulses acting as the manipulating variable. It could just as well be regulated and controlled by the concentration of a hormone in the blood. Hence, the control signals can be classified as either discrete (sequence of nerve impulses) or analog (hormone concentration).

The difference between the command variable and the controlled variable is called *control deviation* $[\Delta y(t)]$. Depending on the type of the controller, the *manipulated variable* $z(t)$ is calculated in a particular way. According to this, let us consider two types of controllers:

- The proportional controller (*P-controller*):

$$z(t) = k_p \Delta y(t); \quad (5.1.6)$$

- The integral controller (*I-controller*):

$$z(t) = k_I \int \Delta y(t) dt. \quad (5.1.7)$$

In case of the *P-controller*, the modified variable $[z(t)]$ is proportional to the control deviation $[\Delta y(t)]$. This type of controller is not able to completely compensate for an induced disturbance of the system. As illustrated in Fig. 5.9,

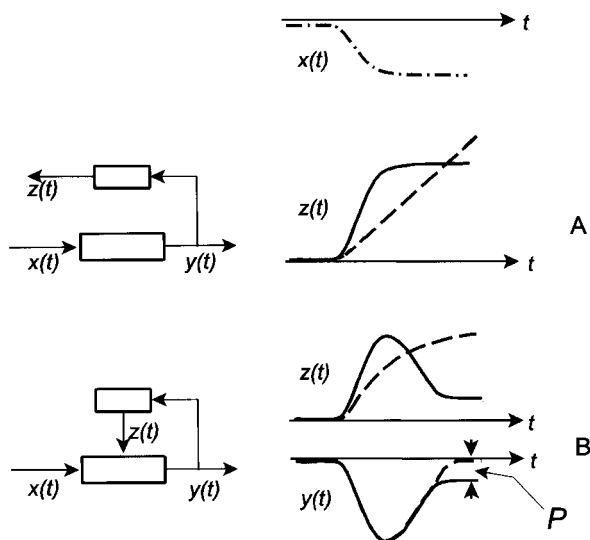


Fig. 5.9. Characteristic responses of a closed (B) and an open (A) control loop with *P*-controller (solid line) and *I*-controller (dashed line) on a disturbance variable $x(t)$. *P*-proportional deviation

there always remains a proportional deviation (*P*) because, if $\Delta y = 0$, then according to Eq. (5.1.6), $z = 0$.

This problem does not arise with the *I*-controller. In this case the modified variable is proportional to the time integral of the control deviation. If the system is brought back to its starting positions then the parameter z in Eq. (5.1.7) becomes constant, which means that an optimal compensation for the disturbance variable is achieved.

In nature, true *P*- or *I*-controllers are hardly ever found. Mostly, the two types exist in combination with each other or are combined with a differential controller (*D*-controller).

For a *D*-controller the following equation holds:

$$z(t) = k_D \frac{d[\Delta y(t)]}{dt}. \quad (5.1.8)$$

This type of controller does not respond to the control deviation but to the rate at which it is changing.

Neither the controller nor the controlled system respond without inertia. In Eqs. (5.1.6)–(5.1.8) terms have to be included that take into account the time dependence of the adjustments. This will result in differential equations of the n th order. The system behaves therefore like those demonstrated as mechanical examples in Fig. 5.3. This means that the curves representing the functions $y(t)$ and $z(t)$ in Fig. 5.9 are, in fact, more complex than shown. Curves like those illustrated in Fig. 5.4 will more likely result. At least there are no multi-stationary states to be expected with linear systems. The amplification depends on the factors k_P , k_I and k_D in Eqs. (5.1.6)–(5.1.8) whereas the attenuation

correlates with the time constants of the rates at which the systems make necessary adjustments.

The sooner and stronger a system reacts to a disturbance, the faster it will achieve the required correction. However, because control circuits are systems of higher order, the compensation can be overshoot. If a system is deflected by a step-wise function then the following types of system behavior can occur as the amplification is increased (see also Fig. 5.4):

- At a low degree of amplification, up to the aperiodic limiting condition, the control variable approaches the target correction value asymptotically without an overshoot.
- As the amplification increases, exceeding the aperiodic limiting condition, the system starts to oscillate. It reaches the target correction value faster but overshoots it and reaches it eventually by damped oscillations. As a result, it may take longer to achieve a stable value than in the first case.
- A further increase in amplification leads to the periodic limiting condition whereby undamped oscillations around the correct parameter occur.
- If the periodic limiting condition is exceeded because of a further increase in the amplification then the system becomes unstable. The oscillations get bigger and bigger and cause the system to become increasingly more deflected.

Such instabilities can lead to the system destroying itself unless the appropriate safety measures are provided. The deflection of the system is usually limited by the energy available. However, these observations illustrate the factors involved in optimizing a control circuit. The fastest possible rate of attaining the target correction value, of course, should be considered as an optimal condition. However, as has already been discussed, this cannot be done by simply increasing the amplification but is best achieved by a combination of different control systems.

In order to illustrate this and, at the same time, to demonstrate the application of cybernetic methods to physiology, the example of target-orientated movement of the hand will be discussed. The person to be tested is asked to move a pointer from position A to position B immediately on being given a command. This movement can be recorded as a function of time (Fig. 5.10). After a lag period of about 145 ms, the hand starts to move and reaches the target 500 ms after the signal, without oscillating. This behavior cannot be achieved by a simple *I*-controller. The aperiodic limiting condition, i.e. the fastest possible reaction of this controller without oscillations is shown in the diagram as well. The advantage of the faster movement of the hand in relation to that of the *I*-controller can be explained in the following way: the target-orientated movement of the hand is controlled by two processes. First, the distance the hand has to travel is estimated through the eyes, and a rapid movement with a built-in limit is programmed. This takes up the observed lag period of 145 ms. Only during the last phase of the target-orientated movement, do the eyes correct the hand, a process that can be regarded as being performed

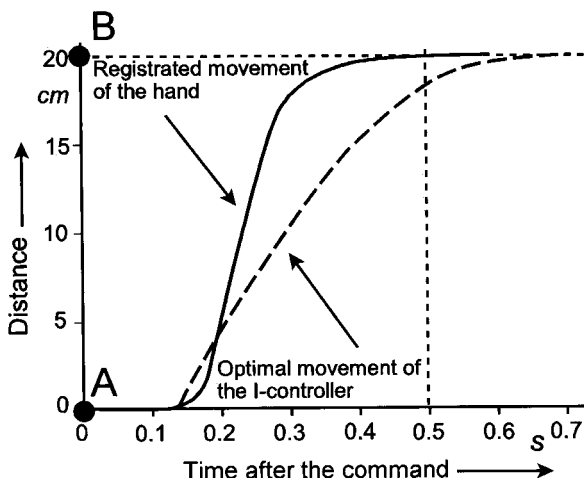


Fig. 5.10. The course of moving the hand (*solid line*) after an order, given at time $t = 0$, to move to a point with a distance of 20 cm from the starting-point. The *dashed line* indicates an optimal reaction of an *I*-controller with a lag period of 145 ms and an amplification up to the aperiodic limit. (Data from Hassenstein 1966)

by an *I*-controller with a time constant of 47 ms. This model is not only mathematically consistent with observations but can also be verified by temporarily preventing the subject from seeing what they are doing during the experiment.

Further reading: Goldman 1960; Morecki et al. 1984

5.2

Systems of Metabolism and Transport

In Section 3.3 flux was introduced as a generalized parameter to explain the transport of matter, of energy and of entropy, and also to calculate the formation of a substance in the course of a chemical reaction. The: “*παντα ρει*” (*panta rhei* = everything is flowing), which was formulated by Aristotle to summarize the dialectics of Heraclitus are related to all levels of biological organization. Propagation of organisms, or their movements in ecological reservoirs can be described mathematically by similar kinetic models, as can biochemical reactions or transport processes across a membrane. In this section we will explain these approaches first for simple examples of fluxes of matter and for biochemical reaction rates. Subsequently, we will extend these applications to more complex processes, such as population dynamics, as well as growth and evolution.

As stated before, systems theory concerns the consequences, but not the energetic reasons for movement. Therefore, in case of kinetic approaches it is not so important whether the considered system is in a true steady state with entropy-producing fluxes (Fig. 3.4B, C), or in a thermodynamic equilibrium with thermic fluctuations (Fig. 3.4A; see also scheme in Fig. 3.6). In Section 3.3.2 (Fig. 3.22) we introduced the net-flux as the difference between two

unidirectional fluxes. The following kinetic approaches will not distinguish whether a unidirectional flux is part of an entropy-producing net flux or whether it just results from thermic fluctuations.

5.2.1

Introduction to Compartmental Analysis

The aim of compartmental analysis is to classify the kinetic terms of exchange processes in various systems. If we consider the flux as a generalized parameter, i.e. as the rate of a transport process, or of a chemical reaction, or as some kind of proliferation rate etc. then we also need a term which formally characterizes the considered system. This term must be applicable to a phase with spatial borders as well as to a component of a chemical reaction and even to populations of cells or organisms.

For this, the concept of the compartment has to be introduced. In principle, a compartment could be a vessel in the hydraulic models of Fig. 3.4. The general definition is as follows: *a compartment is an element of an open system that, with respect to the matter under investigation, has limits which are defined by a specific process and contains no gradients within itself.* The term “matter” is used in a generalized form. It can represent a substance in a reaction system, a cell in a proliferation system, an organism in an ecosystem etc. The “specific process” depends on the type of the considered flux or movement. It could be membrane transport, a biochemical reaction, mitosis of a cell or death of an organism. The absence of gradients inside the compartment means that it is to be considered as homogeneous, or “well mixed”. Otherwise the compartment must be subdivided into smaller compartments, and a multi-compartment system appears. A compartment model is always defined for a special question and is only valid “with respect to the matter under investigation”.

Let us illustrate the polymorphic use of this approach with three examples:

- Investigating the exchange of sodium ions in an organism, the compartment is a space and the process is a flux. The space means the cell when the exchange between the organelles and the cell plasma is much faster than the transport across the cell membrane. In this instance the interior of the cell can be regarded as having no gradients inside. However, the space can also be an organ, such as a muscle, provided that the rate-limiting step in the exchange is the flux through the blood vessel wall and not the transport across the membranes of the individual muscle cells.
- In the compartmental model of radioactive phosphorus (^{32}P) in yeast cells the exchange of P between phosphorus-containing compounds must be taken into account as well as its spatial distribution. In this instance the term “compartment” not only means a spatially limited phase but at the same time it includes the chemical components of biochemical reactions. As a result, the compartments are separated from each other by chemical reaction steps as well as by transport processes.

- Considering a population of cells, for example various blood cells in a suspension, the step defining the borderline of the compartment will probably be the process of cell differentiation, as for example the differentiation of reticulocytes into erythrocytes, or the elimination of erythrocytes by phagocytosis. In this case various states of cell maturation are considered to be the compartments. This example is similar to that used in population kinetics.

It must be mentioned that sometimes in the biochemical literature the term *compartmentation* is used in quite a narrow sense. It means just a spatial separation of components of a chemical reaction leading to additional processes of diffusion.

As in all kinds of systems theory, compartment analysis is realized by an interplay between hypothesis and experiments. The following general questions must be solved:

- What is the number of compartments in the system?
- In which way are these compartments connected to each other?
- What are the rate constants of the individual net fluxes?

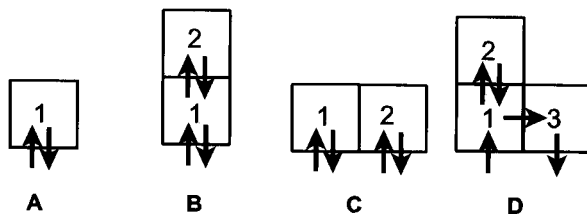
Compartmental analysis requires the use of specific markers. This is necessary in order to measure not only net fluxes, but also unidirectional fluxes (see Fig. 3.22). It is very important to take care that the use of a marker does not alter the behavior of the system. Only the observer, but not the system should "see" the markers. The labelled components of the system must behave identically to the unlabelled.

The techniques of labelling depend on the kind of investigation. Frequently, chemical elements are labelled using their radioactive isotopes. In case of elements with low atomic mass, an *isotope effect* must be considered. This means that the properties of various isotopes may differ from each other. Tritium (^3H), for example, differs in its binding properties in relation to hydrogen (^1H). This difference can be rectified by a correction factor. The isotope effect becomes smaller for elements of higher atomic mass. It is already negligible when using the isotopes of carbon. Of course the isotope effect can be neglected if, for example, tritium is used to label an organic molecule and if reactions of this molecule which are to be studied do not include any modification of this particular binding.

The setting up of a compartmental model starts with the design indicating single compartments and their mutual interactions as well as their connections with the environment. The simplest model is the one-compartment system in exchange with its environment (Fig. 5.11A). Such a model, for example, could represent the exchange of ions in a non-nucleated blood cell, the exchange of phosphorus by a protein etc.

If a suspension of cells is to be analyzed containing two different types of cells, or cells in two different physiological states, then this must be regarded as a two-compartment system where the compartments are arranged in parallel

Fig. 5.11. Various compartmental models: A one-compartment system, B two-compartment system in series, C two-compartment system in parallel, D three-compartment system



without direct connection between them (Fig. 5.11C). In contrast, for cells containing organelles with significant barriers of diffusion or transport, as for example mitochondria, vacuoles etc. then in the simplest case a two-compartment system in series must be applied (Fig. 5.11B). In this case one step is the exchange of matter between the environment and the cytoplasm, another one, the fluxes between cytoplasm and the organelles.

A simple physiological three-compartment system is shown in Fig. 5.11D. This could represent, for example, the transport of a drug in the body: its uptake in the blood (compartment 1), the subsequent accumulation in muscles (compartment 2) and its excretion via the kidneys (compartment 3). We will discuss this model in detail in Section 5.2.3 in the context of problems of pharmacodynamics (see Fig. 5.16).

To test a proposed compartmental model it is necessary to calculate its behavior and to relate the results to measured points of performed experiments. The mathematical representation of such a model starts with the formulation of a system of differential equations using the following basic approach:

$$\frac{dn_m}{dt} = \sum_i J_{im} - \sum_i J_{mi}, \quad \text{where: } i \neq m. \quad (5.2.1)$$

The change of the amount of the substance (n_m) in the compartment m as a function of time (dn_m/dt), therefore, is obtained from the sum of all the unidirectional fluxes (J), where the fluxes with the direction outside-in (J_{im}) carry a positive sign, and the fluxes in the opposite direction (J_{mi}) carry a negative sign. In this equation the symbols J are used instead of J , because these are unidirectional fluxes and not fluxes according to the thermodynamic definitions (see Sect. 3.1.3). Furthermore, these fluxes do not necessarily describe transport per unit of membrane area. This corresponds with the generalized approach discussed above. If necessary, these kinetic fluxes can be converted into fluxes in the thermodynamic sense.

To calculate these fluxes, in the simplest case a linear relationship can be applied:

$$J_{im} = k_{im}n_i. \quad (5.2.2)$$

We used a corresponding approach in Section 3.3.2 [Eq. (3.3.39)]. There we characterized the parameter k_{im} as a rate constant with a unit s^{-1} determining

the flux from the compartment i into the compartment m . Of course, there are other ways to describe fluxes. In Section 5.2.2 we shall use, for example, the Michaelis-Menten equation introducing biochemical reaction rates into this framework of calculations [Eq. (5.2.19)].

Let us demonstrate the formulation of the equations for the three-compartment model as illustrated in Fig. 5.11D:

$$\begin{aligned}\frac{dn_1}{dt} &= k_{01}n_0 + k_{21}n_2 - k_{12}n_1 - k_{13}n_1, \\ \frac{dn_2}{dt} &= k_{12}n_1 - k_{21}n_2, \\ \frac{dn_3}{dt} &= k_{13}n_1 - k_{30}n_3.\end{aligned}\tag{5.2.3}$$

The change of the content of the substance n_i in all the three compartments is expressed by the corresponding differential quotient dn_i/dt . The expressions on the right side of these equations correspond to the sums of Eq. (5.2.1), using the linear approaches of Eq. (5.2.2). The environment is characterized by the subscript "0". The parameter n_0 can be considered as constant.

This approach allows interesting conclusions about the character of the stationary state of this system. In Section 3.1.4 we defined the stationary state as a time independent state. This means for the system of Eqs. (5.2.3) the following conditions:

$$\frac{dn_1}{dt} = 0; \quad \frac{dn_2}{dt} = 0; \quad \frac{dn_3}{dt} = 0.\tag{5.2.4}$$

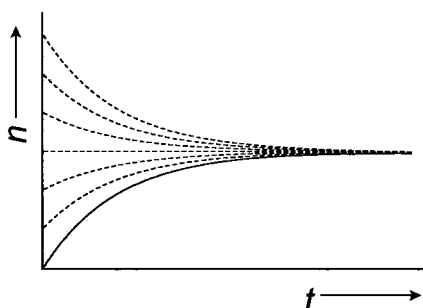
Introducing this in Eqs. (5.2.3) we get a system of simple linear algebraic equations. This makes it possible to calculate the parameters n_1 , n_2 , n_3 for the case of the stationary state without further conditions. As can easily be seen, solutions for the parameters n_1 , n_2 , n_3 can be calculated depending only on the rate constants (k_{nm}). Furthermore, because all equations are linear, only one single solution for these three parameters will be obtained.

This leads to the following general conclusions:

- This system has only one single stationary state.
- The parameters n_i of this stationary state depend only on the rate constants (k_{nm}) and not on the initial conditions of the system (Fig. 5.12).

This property which holds for all linear systems is called *equifinality*. Using the phase diagram illustration which in this case is built up by the three coordinates n_1 , n_2 and n_3 , this means there is only one single point characterizing a stable stationary state. The system approaches this state starting from any point of the phase space. It would correspond to the cases A or D in Fig. 5.6. This, by the way, is the same conclusion as drawn in Fig. 3.6 for the linear steady state and

Fig. 5.12. Illustration of curves of the general solution of the differential equation, Eq. (5.2.5) (---) and its particular solution [Eq. (5.2.9)] (—)



which is expressed thermodynamically by the Prigogine principle of minimal entropy production.

If it is the kinetics, rather than the simple stationary state which is of interest, then it is necessary to solve the differential equations [Eqs. (5.2.3)]. This requires the prediction of initial conditions. Graphically, this means that the point in the n -dimensional phase space where the trajectory starts must be declared. In our case, in a system of three differential equations of the first order, leading in the course of transformation to differential equations of the third order, three initial conditions are required, according also to the 3-dimensional phase space of the coordinates n_1 , n_2 and n_3 .

We will demonstrate this kind of calculation for the simplest example, the behavior of the one-compartment system (Fig. 5.11A). Corresponding to Eq. (5.2.1) one can write:

$$\frac{dn_1}{dt} = J_{01} - J_{10} = J_{01} - k_{10}n_1. \quad (5.2.5)$$

In contrast to the flux J_{10} where the linear approach [Eq. (5.2.2)] is used, flux J_{01} is considered as constant. To integrate this first order differential equation, only a single initial condition is required.

Let us first consider the case of a decorporation experiment, i.e. an experiment where the extrusion of a previously incorporated substance is investigated. This means that at $t = 0$ only the compartment is labelled in contrast to its environment, which is free of label. Considering furthermore that the environment is very large in relation to the volume of the compartment, in this case the influx of labelled substance in the system can be neglected ($J_{01} = 0$). In this case Eq. (5.2.5) becomes simpler:

$$\frac{dn_1}{dt} = -k_{10}n_1. \quad (5.2.6)$$

Supposing that n_1 and k_{10} were always positive, this equation, looked at from a purely geometrical point of view means that the function $n_1(t)$ is negatively inclined ($-dn_1/dt$) and, the larger the value of n_1 , the steeper this becomes.

Equation (5.2.6) can be easily integrated, leading to the following general solution:

$$n_1 = Ke^{-k_{10}t}, \quad (5.2.7)$$

where K is an integration constant which can be determined by the initial condition. Provided that at $t = 0$, $n_1 = n_{10}$ then it results from Eq. (5.2.7):

$$n_1 = n_{10}e^{-k_{10}t}. \quad (5.2.8)$$

This equation, therefore describes the decrease of the amount of labels (n_1) in the compartment as function of time, if at time $t = 0$ the amount n_{10} had been incorporated.

In a similar way it is possible to calculate the kinetics of incorporation, i.e. experiments where the accumulation of a label from the compartment was investigated. In this case the initial conditions are: $t = 0$, $n_1 = 0$. Integrating Eq. (5.2.5) for this case, the following particular solution will be obtained:

$$n_1 = \frac{J_{01}}{k_{10}}(1 - e^{-k_{10}t}). \quad (5.2.9)$$

It can easily be seen that for $t \Rightarrow \infty$ the parameter n_1 will attain the value J_{01}/k_{10} :

$$\frac{J_{01}}{k_{10}} = n_{\infty}. \quad (5.2.10)$$

In Fig. 5.12 a group of general solutions of Eq. (5.2.5) as well as the special course of Eq. (5.2.9) is depicted. At the same time this figure illustrates the term equifinality introduced previously.

The analysis of a multi-compartment system is of course more complicated. Here, we will discuss only some general properties of the corresponding equations. Computer models are mainly used for this kind of analysis. In Section 5.2.3 we will describe a multi-compartment system in context with problems of pharmacodynamics (Fig. 5.16).

The integration of the system of differential equations describing a multi-compartment system allows calculation of a time function of the sum content of all compartments together (n_s), which in fact can be measured in experiments. In the case of a decorporation experiment the following type of exponential function results:

$$n_s = A_1e^{-\alpha_1t} + A_2e^{-\alpha_2t} + A_3e^{-\alpha_3t} + \dots + A_me^{-\alpha_mt}, \quad (5.2.11)$$

where m is the number of compartments. Only when all compartments are arranged in parallel, do the exponents $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_m$ correspond directly to the rate constants of the individual net fluxes. The same holds for the factors A_1 ,

$A_2, A_3, \dots, A_m$. In fact, at $t = 0$ it must be: $A_1 + A_2 + A_3 + \dots + A_m = 0$, but these parameters correspond to the initial contents of the single compartments ($n_{10}, n_{20}, n_{30}, \dots, n_{m0}$) only in the case of a parallel arrangement of all compartments without direct mutual connections. Otherwise, these exponents and factors are more or less complicated functions of all rate constants of the system and furthermore depend on the initial conditions of the system. These particularities can be used to analyze the arrangement of the compartments in the system.

Usually, computer simulations or methods of non-linear regression are used to analyze the results of these kinds of experiments. A better illustration of the character of these curves is given by graphical analysis. Plotting the measured points (n_s) against the time (t), for a one-compartment system, a curve should result according to the functions of Eqs. (5.2.8) or (5.2.9) (Fig. 5.13 left side, above). In the case of a half-logarithmic plot i.e., plotting $\ln n_s$ versus t , the function of Eq. (5.2.8) will show a straight line (Fig. 5.13, right side, above). The crossing point of this line with the ordinate gives the value n_0 . Furthermore, there is the following connection between half-life time ($t_{1/2}$) and rate constant (k):

$$k = \frac{\ln 2}{t_{1/2}} = \frac{0.693}{t_{1/2}}. \quad (5.2.12)$$

If it is impossible to fit the measured parameters in a half-logarithmic plot by using a straight line, then at least a two-compartmental model must be supposed. Frequently, such curves indicate a straight line after an initial curved part (Fig. 5.13 right side, below). This is caused by the fact that after sufficient time only one compartment with the largest half-life ($t_{1/2}$) remains.

This is the point at which the graphic analysis of compartmental systems starts. Supposing that at a certain time only one compartment is determining the behavior of the whole system, then the curve ends with a straight line. One can extrapolate this line back to the point $t = 0$ and evaluate the parameters of one of these exponential terms in the sum [Eq. (5.2.11)]. Now, numerically the difference between these extrapolated values and the parameters in the curved part of the measured function can be calculated. The results will show the behavior of a system containing one compartment less. In the case of a two-compartment system, the rest therefore reflects the behavior of a one-compartment system. Thus, it must indicate a straight line in the half logarithmic plot. If not, the procedure must be repeated. In this way all exponential terms in the sum of Eq. (2.5.11) can be obtained successively. This method however, as well as the method of direct fitting of the measured points to corresponding function, will be successful only if there are not too many compartments, and if the rate constants of these parameters differ significantly from each other.

The analysis of multi-compartment systems for an incorporation experiment can be performed in a similar way, if, instead of the function $n_s(t)$, the parameter $[n_\infty - n_s(t)]$ is used. In this way the curved character of the drawn line from Fig. 5.12 will be transformed into that of Fig. 5.13.

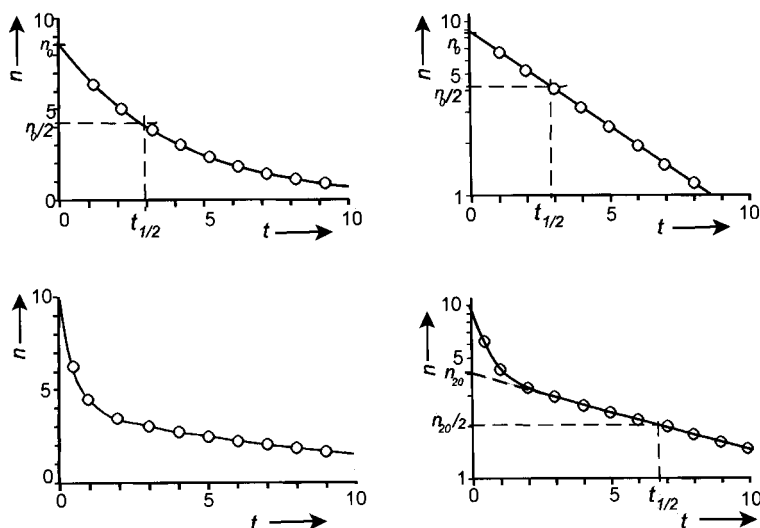


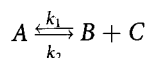
Fig. 5.13. Above: illustration of points following the function $n = n_0 e^{-\alpha t}$ (for: $n_0 = 8.6$ and $\alpha = 0.25$); below: function $n = n_{10} e^{-\alpha t} + n_{20} e^{-\beta t}$ (for: $n_{10} = 5.8$, $n_{20} = 4.2$, $\alpha = 1$, $\beta = 0.1$); left: linear plot; right: half-logarithmic plot (all parameters in relative units)

Further reading: Anderson 1983; Shipley and Clark 1972; Solomon 1960

5.2.2

Models of Biochemical Reactions

The rate of chemical reactions can be described by the time derivative of the advancement number (ξ) which was introduced in Section 3.1.5, as well as directly by differential quotients dn_i/dt or dc_i/dt as shown in the previous section. The individual steps in a chain of biochemical reactions can be *mono-molecular*, or *bi-molecular*, depending on how many molecules react together to bring about the reaction. The reaction



is a mono-molecular reaction when it runs from left to right and a bi-molecular reaction in the opposite direction. Because the rate of reaction is proportional to the probability of a molecular event leading to the reaction, the unidirectional flux (J_A) for the mono-molecular reaction can be formulated as follows:

$$J_A = k_1 c_A. \quad (5.2.13)$$

The unidirectional fluxes J_B and J_C as a result of bi-molecular reactions are, however,

$$J_B = J_C = k_2 c_B c_C. \quad (5.2.14)$$

Only Eq. (5.2.13) therefore corresponds to a linear approach, similar to the equations which we used in compartmental analysis in Section 5.2.1 [Eqs. (5.2.2) and (5.2.3)]. Furthermore, the rate constants k_1 and k_2 have different units. Eqs. (5.2.13) show that k_1 have the usual unit for a reaction rate: s^{-1} . From Eq. (5.2.14) one can derive the unit of k_2 , which is $\text{m}^3 \text{s}^{-1} \text{mol}^{-1}$ (or $\text{l s}^{-1} \text{mol}^{-1}$).

Equation (5.2.14) leads to a non-linear, first order differential equation because it contains the product of two variables ($c_B \cdot c_C$). Correspondingly a non-linear behavior of these systems is to be expected. The chemists call reactions that follow the kinetics described by equations of the type of Eq. (5.2.14) second order reactions. This discrepancy between the nomenclature of systems analysis and chemistry sometimes may lead to misunderstandings.

Bi-molecular reactions do not always lead to non-linear kinetic approaches. If one of the two reactants is present in vast excess, then the rate of the reaction is limited only by the concentration of the other reactant. In this case a bi-molecular reaction behaves like a first order reaction. Such conditions arise, for example, in all biochemical reactions where water is one of the reactants.

Biochemical reactions are usually catalyzed by enzymes. In its simplest form an enzymatic reaction may be described as follows:



A substrate S forms a complex ES with the enzyme E . This complex then breaks down to give the product P and release the enzyme E .

Usually, the rate limiting step of this sequence of reactions is the breakdown of the enzyme-substrate complex. Therefore, the rate of this reaction can be described in the following way:

$$J = k c_{ES}, \quad (5.2.15)$$

where k is the rate constant of this breakdown reaction. Assuming further that the first step of this reaction is always in thermodynamic equilibrium, then by the mass action law:

$$K_M = \frac{c_S c_E}{c_{ES}}. \quad (5.2.16)$$

The total concentration of the enzyme (c_{E0}) must remain constant:

$$c_{E0} = c_E + c_{ES}. \quad (5.2.17)$$

Combining Eqs. (5.2.16) and (5.2.17) one obtains:

$$c_{ES} = \frac{c_S c_{E0}}{K_M + c_S}. \quad (5.2.18)$$

Introducing this into Eq. (5.2.15) one gets the Michaelis-Menten equation which describes the reaction rate (J) of simple enzymatic processes as a function of the substrate concentration (c_S):

$$J = \frac{kc_S c_{E0}}{K_M + c_S}. \quad (5.2.19)$$

K_M , as the dissociation constant of the enzyme-substrate complex, is known as the *Michaelis constant*. It has the units of concentration and corresponds to the concentration of the substrate when the reaction rate is at half of its maximum value (see Fig. 5.14). It usually lies somewhere between 10^{-2} and 10^{-5} M.

As shown in Fig. 5.14 there are two particular regions in this function where some simplifications are allowed. At high substrate concentrations ($c_S \gg K_M$) a maximum reaction rate $J_{\max}$ is attained. For this case Eq. (5.2.19) leads to:

$$J_{\max} = kc_{E0}. \quad (5.2.20)$$

In this case, the flux is independent from substrate concentration (c_S). $J_{\max}$ can be introduced as a constant parameter into the general system of equations [Eq. (5.2.1)].

In contrast, at very low substrate concentrations ($c_S \ll K_M$), Eq. (5.2.19) becomes:

$$J \approx \frac{kc_S c_{E0}}{K_M} = \frac{J_{\max}}{K_M} c_S \quad \text{for: } c_S \ll K_M. \quad (5.2.21)$$

In this case a linear flux equation similar to Eqs. (5.2.2) or (5.2.15) can be used. In many particular situations, therefore, simple approaches are possible even in the case of enzymatic reactions.

In all other cases the Michaelis-Menten equation must be directly introduced into the system of flux equations. Obviously, this leads to non-linear equations. This seems to be the typical case for enzymatic reactions and for flux systems where such reactions are included. Furthermore, in many cases additional

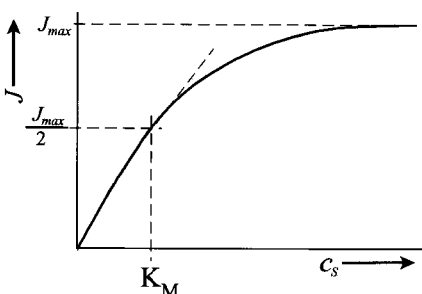


Fig. 5.14. The reaction rate (J) as a function of the substrate concentration (c_S) of an enzymatic reaction with constant enzyme concentration

interactions of enzyme activation or inhibition occur. In the case of allosteric reactions the enzyme contains two binding places, the occupation of which leads to conformational changes in the molecule and alterations of its function. These reactions are governed by equations which are more complicated than the Michaelis-Menten equation.

In Section 5.1.2 we discussed the general behavior of non-linear systems with particular reference to metastable states. We will illustrate this now by using an example of a simple chain of biochemical reactions (Fig. 5.15).

A product (P) is formed by an enzymatic reaction from a substrate (S). The substrate (S) itself is formed by a first order reaction from a substance A . It can, however, break down with the same kind of kinetics to Z . The time dependence of the concentration of the substrate (c_S) can be formulated using Eq. (5.2.1) in the following way:

$$\frac{dc_S}{dt} = J_{AS} - J_{SZ} - J_{SP}. \quad (5.2.22)$$

Let the flux J_{SP} , which denotes the rate of the enzymatic destruction, be dependent on c_S in a way that corresponds to the Michaelis-Menten equation (substrate inhibition), because of a special kind of regulation. The function $J_{SP}(c_S)$ is illustrated in Fig. 5.15A. The fluxes J_{AS} and J_{SP} shall be determined by

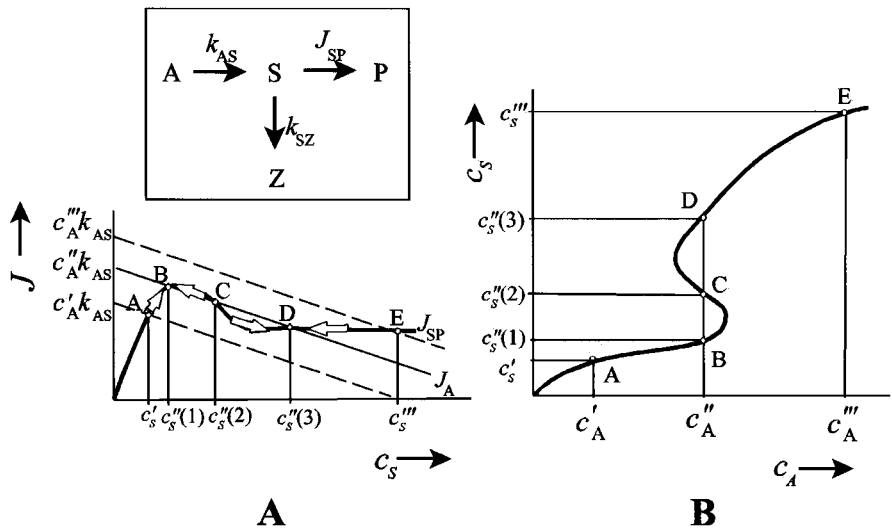


Fig. 5.15. A kinetic model of a simple biochemical reaction chain. **A** Reaction rate (J) as function of the substrate concentration (c_S) for the reaction $S \Rightarrow P$ (J_{SP}), and for the difference of fluxes: $J_A = J_{AS} - J_{SZ}$ at various concentrations of the substance A (c'_A , c''_A , c'''_A). The intersections of the curves (A, B, C, D, E) indicate stationary states. The arrows are directed towards stable states. **B** Trajectory of the stationary states of the left picture

the linear approaches according to Eq. (5.2.2). In this case Eq. (5.2.22) can be transformed in the following way:

$$\frac{dc_S}{dt} = k_{AS}c_A - k_{SZ}c_S - J_{SP}. \quad (5.2.23)$$

Formally introducing a flux:

$$J_A = J_{AS} - J_{SZ} = k_{AS}c_A - k_{SZ}c_S, \quad (5.2.24)$$

then Eq. (5.2.23) transforms into:

$$\frac{dc_S}{dt} = J_A - J_{SP}. \quad (5.2.25)$$

For the case of a stationary state the following must apply:

$$\frac{dc_S}{dt} = 0 \quad \text{and therefore} \quad J_A = J_{SP}. \quad (5.2.26)$$

The function $J_A(c_S)$, as defined by Eq. (5.2.24), is a straight line with a slope of $-k_{SZ}$, intersecting the ordinate at a point $k_{AS}c_A$ (Fig. 5.15A). The points where the straight line cuts the curve of the function $J_{SP}(c_S)$ denote stationary states according to Eq. (5.2.26).

Let us first consider the case $c_A = c_A''$. This straight line intersects with the function $J_{SP}(c_S)$ at points B, C and D. This means that there are three stationary states for one single value of c_A . However, these points are not equivalent to each other. Only states B and D are stable, C being unstable. This can easily be deduced from the curve.

First consider the point B: a small increase in the substrate concentration will lead to the situation where function $J_{SP} > J_A$. Reference to the balance equation [Eq. (5.2.25)] will show that this will bring about a reduction in c_S . The system therefore is moving towards its initial state. If c_S'' is reduced slightly, then $J_{SP} < J_A$ and this results in an increase in c_S . Small deviations from the stationary value B will thus be corrected. This behavior is indicated by the arrows on the function drawn in Fig. 5.15A.

Near the point C, the system behaves differently. Small reductions in $c_S''(2)$ cause J_{SP} to become larger than J_A . The substrate concentration will continue to fall, until point B is reached. An increase in the value of $c_S''(2)$ by a small value will cause the system to slide in the direction of state D which is stable like the state B.

The stationary states in this system can be summarized by a trajectory. In Fig. 5.15B the function of the substrate concentration (c_S) versus the concentration of the substance A (c_A) is shown. First, the three stationary values for c_A'' shown in Fig. 5.15A are plotted. A change in c_A causes a parallel shift of the curve J_A in Fig. 5.15A (for example the dotted lines). The three stationary values occur only within a relatively narrow range of c_A , or better for the condition: $J_A = c_A k_A$. If

all of the stationary states of c_S for different values of c_A are plotted in the coordinates of Fig. 5.15B then an S-shaped trajectory is formed. This corresponds to a section through the behavioral surface of the cusp catastrophe which is shown in Fig. 5.7. This particular example clearly shows that the points on the shaded area of the surface corresponding to the point C in Fig. 5.15, are unstable states.

For the curve Fig. 5.15B the flux J_{SZ} is taken to be constant. Any variation in the rate at which the substrate (S) is used, i.e. a modification of the coefficient k_{SZ} will change the slope of the straight lines representing the function J_A in Fig. 5.15A [according to Eq. (5.2.24)]. It can easily be seen that if the slope of this curve exceeds a critical value, a situation will develop where, independent of c_A , there will only be a single possible intercept. When compared with the functional surface in Fig. 5.7, this situation corresponds to a section through the bifurcation point P. In this instance, k_{SZ} (or J_{SZ}) would have to be plotted on the y-axis of the coordinate.

The analysis of metabolic networks of course leads to systems of many differential equations. The integration of them is complicated, not only because most of them are non-linear, but they are also often quite ambiguous. Many possibly occurring stationary states are probably metastable. With small alterations of the parameters of the system, or by changing the initial conditions, the characters of the stationary states may be modified in a not easily predictable way. Also, chaotic situations are possible. These properties not only make calculations difficult, but may also lead to a higher degree of system destabilization.

In Section 3.1.4 we mentioned the time hierarchy of biological systems. In fact, the rate constants of biologically important reactions are distributed over a brought region of magnitudes. In the context of thermodynamic properties of systems we noted that as a consequence of a steady state ($\sigma > 0$) in the whole system, fast reacting equilibria ($\sigma = 0$) of sub-systems can be established. Considering this time hierarchy from the view point of systems analysis, the following conclusion about the behavior of systems can be drawn from mathematical considerations.

Formally, in a system of many differential equations, simplifications occur because differential quotients of fast processes may approach zero. On the one hand, in a non-equilibrium stationary state, some parts of the system, in fact, can be in equilibrium, because their time constants are quite small. Very slow processes, on the other hand, can be taken as non-changing during the considered time scale. Their parameters, therefore, can be taken as constants. These kinds of simplifications do not only help in mathematical modelling, they automatically reduce the number of stationary states and stabilize the system itself. It seems that the established time hierarchy of biological reactions is the result of biological optimization in the course of evolution.

Further reading: Hayashi and Sakamoto 1986; Heinrich et al. 1987; Heinrich and Schuster 1996; Murray 1993; Palmer 1995; Reich and Sel'kov 1981; Segel 1993

5.2.3

Pharmacokinetic Models

Pharmacokinetics, sometimes also called pharmacodynamics is concerned with the time course of the concentration of drugs in the body or at the sites where they are effective. The aim of such studies is to develop dose and application regimes that will result in an optimal therapeutic effect.

In the simplest situation an amount of an intravenously administered drug is rapidly and homogeneously distributed throughout the whole blood volume and is then excreted or metabolized. The decline in the concentration of the drug in the blood as a first approximation corresponds with the release from a one-compartment system as described in Section 5.2.1. This process can therefore be described by the basic equation [Eq. (5.2.5)] and will follow the time function corresponding to Eq. (5.2.7).

The analogous equation in pharmacokinetics is:

$$\frac{dm}{dt} = -C_1 c, \quad (5.2.27)$$

where m is the total amount of the drug administered, c is its concentration in the blood and C_1 is a rate constant, which in contrast to the parameters k which are used in the previous sections, has the unit $\text{m}^3 \text{s}^{-1}$, or more commonly: l h^{-1} . This constant is known as the *clearance constant*. Equation (5.2.27) allows us to define this parameter:

$$C_1 = -\frac{dm}{dt} \cdot \frac{1}{c}. \quad (5.2.28)$$

This definition is also valid when the exchange does not follow the kinetics of a one-compartment system. In this case, of course, the system can no longer be described by a simple exponential function like Eq. (5.2.27) but rather by a sum of several exponential terms [Eq. (5.2.11)]. In this case, the clearance constant according to the definition in Eq. (5.2.28) itself becomes time dependent.

In context with the real compartmental model of the body for a given drug the question arises: what is the volume of distribution of this drug in the body? It can be larger than the volume of the blood; as maximum it can include the whole amount of body water. Therefore, the coefficient of distribution of a drug is calculated as the quotient of the real volume of distribution to the body weight, or body volume.

As a first approximation it can be taken that the concentration of the drug at the receptor side is the same as its concentration in the blood. The amount of drug in the blood which therefore is able to arrive at the receptor side, when expressed as a percentage, is called *bioavailability*. In the case of intravenously administered drugs this has a value of 100%. If the drug is administered orally, the bioavailability is lower and depends on its resorption.

Usually, pharmacokinetics studies are based on linear models. In some instances a concept, known as *capacity-limited elimination* is used. This is the excretion of a drug using saturation kinetics. For the direct binding of the drug to a receptor, non-linear approaches should be used similar to the Michaelis-Menten equation [Eq. (5.2.1)].

In general, however, the linear approaches have proved useful for the overall kinetic analysis of the distribution of drugs in the body. In Section 5.2.1 we depicted a three-compartment model (Fig. 5.11D) which can be regarded as the basic pharmacokinetic model. It can be described by the system of differential equations mentioned there [Eq. (5.2.3)]. In the simplest case it contains the following processes: the incorporation of the drug into the blood stream (compartment 1), its uptake into the tissue (compartment 2) and its excretion by the kidneys (compartment 3). The kinetics of this system depends on the mode of drug administration.

Figure 5.16 shows the concentration curves for different parts of the body following a single oral dose of a drug. It can be seen that the concentration reaches a maximum in the blood and subsequently in the tissues. These curves were obtained by computer simulation of the process. This is a method by which the most promising modes of administration for maintaining the required concentrations of a drug at the desired point of action, for a sufficient time can be determined. It is obvious that repeated oral administration will result in the concentration in the blood showing a series of maxima. The maximum concentration and the subsequent time course will depend on the dose and frequency of administration.

5.3

Model Approaches to Some Complex Biological Processes

In this section biophysical models of processes will be introduced that are among the most complex of their kind. Although rapid progress in this field

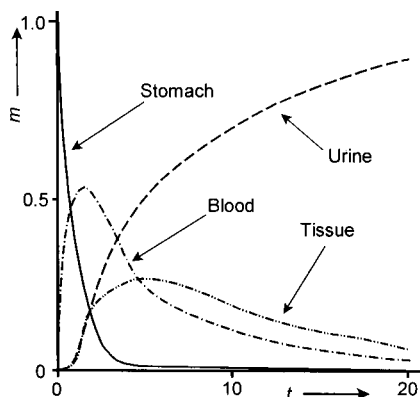


Fig. 5.16. Time course of drug concentration in basic pharmacokinetic model. The normalized parameters according to the system in Fig. 5.11D are: $k_{01} = 1$, $k_{12} = k_{21} = k_{23} = 0.25$. (Data from Knorre 1981)

has been achieved in recent years, not least because of the stormy development of the theory of non-linear systems, such theories have remained at the introductory state. The explanations given in this section should help readers to understand recent approaches and hypotheses.

5.3.1

Models of Propagation and Ecological Interactions

The kinetic models of growth and interactions in biological populations are based on the same kind of mathematical constructions as those of metabolism and transport processes, discussed in the previous section. Let us consider first a population of organisms of a single species in an infinitely large milieu. Let the number of individuals be n , then any changes of this number can be described by the differential quotient dn/dt . Because the external milieu has been taken as infinite, the physiological conditions can be regarded as constant, and thus not influenced by the organisms themselves.

In this case the propagation rate as well as the mortality rate of the organisms can be expressed by time constants which are determined both genetically and physiologically. The rate constant of propagation (k_p) and that of mortality (k_m) can be combined as follows:

$$k = k_p - k_m \quad (k_p, k_m > 0). \quad (5.3.1)$$

In this way the proliferation of a population can be described by the following differential equation:

$$\frac{dn}{dt} = kn. \quad (5.3.2)$$

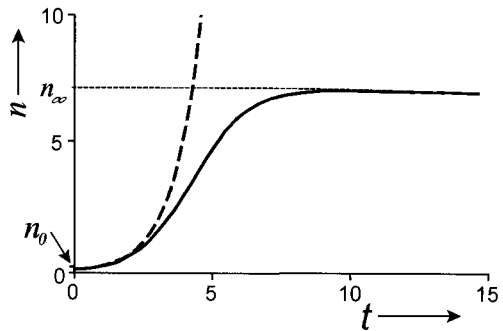
This expression shows in fact that both the rate of proliferation as well as the rate of mortality are proportional to the actual number of individuals in the population. Considering that at $t = 0$ the initial number of individuals is n_0 , then the integration of Eq. (5.3.2) leads to the following equation:

$$n = n_0 e^{kt} \quad (5.3.3)$$

This law of exponential growth was discovered as early as 1798 by the English economist Thomas Robert Malthus. If the propagation coefficient (k_p) preponderates in Eq. (5.3.1), then $k > 0$ and the number of individuals will increase towards infinity. The time course of this function is depicted in Fig. 5.17 (dashed line).

The original assumption that the external milieu is not affected by the organisms themselves is, however, an ideal condition. Only when the number of cells in the suspension is very small, and the time scale is sufficiently short, can this condition apply. In real cell suspensions, this period of growth corresponds to the initial logarithmic period of growth.

Fig. 5.17. The function $n(t)$ according to the Verhulst-Pearl equation [Eq. (5.3.6)] —, and the Malthus equation [Eq. (5.3.3)] - - -. Using arbitrary values in ordinate and abscissa, the parameters are: $n_0 = 0.1$; $k_1 = 1$; $k_2 = 0.14$. For the Malthus curve, $k = k_1$



In most cases a mutual influence of the individuals on one another, for example via competition for food, is to be expected. This circumstance can be taken into account by modifying Eq. (5.3.1). It may be postulated that k_p and k_m are not constants but themselves functions of the population density. So, for example, one can assume that the rate constant of proliferation (k_p) will decrease and the rate constant of mortality (k_m) will increase, when n increases, because of shortage of food. The degree of physiological variability of these parameters, however, will be limited for genetic reasons. The simplest approach is the following linear equation:

$$k = k_1 - k_2 n \quad \text{for: } k_1, k_2 \geq 0. \quad (5.3.4)$$

This takes into account an optimal propagation constant k_1 , determined genetically, in combination with a further constant k_2 leading to a diminution of k_1 with an increase of the population density (n). If $n \Rightarrow 0$ then $k_1 \Rightarrow k$. Introducing Eq. (5.3.4) into Eq. (5.3.2) one obtains:

$$\frac{dn}{dt} = (k_1 - k_2 n)n = k_1 n - k_2 n^2. \quad (5.3.5)$$

This approach therefore leads to a first order non-linear differential equation. The integration of this equation for the same initial conditions as above ($t = 0$; $n = n_0$) gives an equation which in mathematics is known as the *logistic function*. In population kinetics it is called the *Verhulst-Pearl law*:

$$n = \frac{n_0 k_1 e^{k_1 t}}{k_1 + k_2 n_0 (e^{k_1 t} - 1)}. \quad (5.3.6)$$

In contrast to Eq. (5.3.3), in this function n does not approach infinity at $t \Rightarrow \infty$ but rather tends towards a saturation value (n_∞). This saturation value can easily be determined from Eq. (5.3.6) considering that with increasing t the exponential terms increase up to a value where neither the subtraction of 1, nor the addition of k_1 become important:

$$\lim_{t \rightarrow \infty} n = n_{\infty} = \frac{k_1}{k_2}. \quad (5.3.7)$$

Furthermore, it is easy to show that for small values of t , the Eq. (5.3.6) takes the form of Eq. (5.3.3). Both functions for some arbitrarily chosen parameters are illustrated in Fig. 5.17. An example of experimentally obtained data is shown in Fig. 5.18 (above).

If the system contains individuals of several different species then the inter relation between them must be taken into consideration. This can involve competition for the food supply (Fig. 5.18, below) or a hunter-prey relationship (Fig. 5.19). Systems like this were studied in the 1930s by Volterra and Lotka.

The following set of equations describes the simplest case of a hunter-prey relation between two species:

$$\left. \begin{aligned} \frac{dn_p}{dt} &= (k_{pP} - k_{mP}n_H)n_P \\ \frac{dn_H}{dt} &= (k_{pH}n_P - k_{mH})n_H \end{aligned} \right\}. \quad (5.3.8)$$

The subscripts p and m have the same meaning as in Eq. (5.3.1) and the index P in these equations refers to "prey", whereas H means "hunter". For the prey

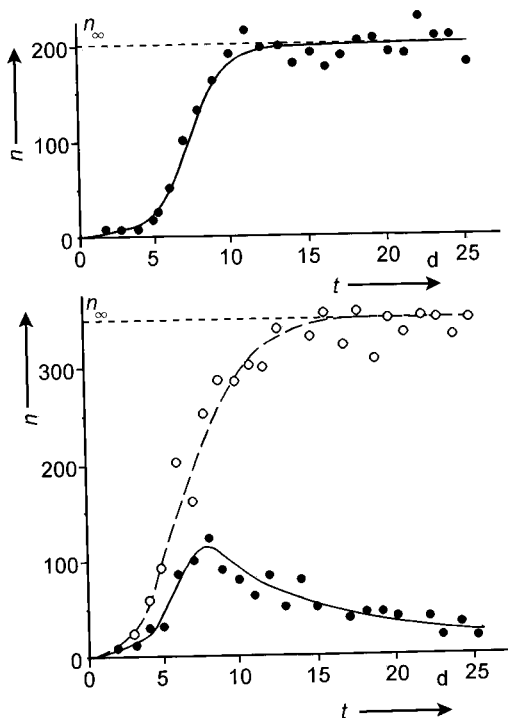


Fig. 5.18. The time dependence of the growth of a pure population of *Paramecium caudatum* (●—● above) and mixed together with a population of *Paramecium aurelia* (○- -○). The ordinate indicates the number of individuals (n) per 0.5 ml medium. The abscissa gives the time in days (d). (Data from Gause 1935)

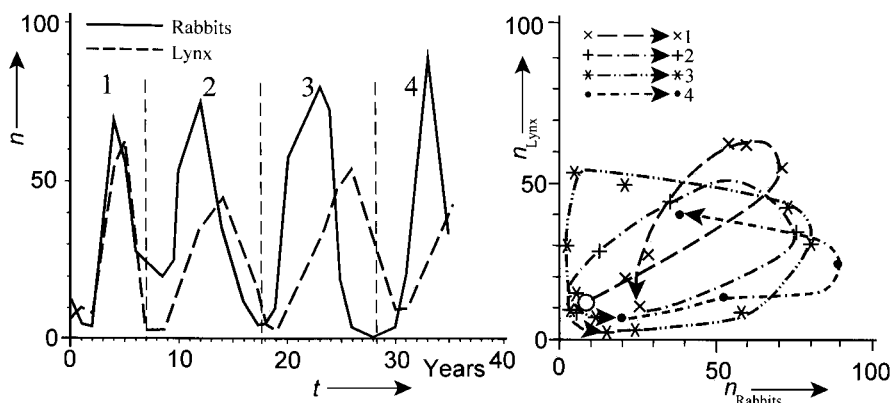


Fig. 5.19. A particular example of population kinetics: the number of pelts, obtained from hunted lynx and snow rabbits in Canada after 1900 (numbers in thousands). On the *left*: time course; on the *right*: the same numbers in a Volterra-plot. The population waves 1–4 of the *left* part of this figure appear as Volterra cycles in the graph on the *right*, starting with the point $\circ$. (Data from Haken 1983)

organisms the mortality ($k_{mP}n_H$) will increase as the number of hunters (n_H) increases. The greater the number of hunters, the shorter the life expectancy of the prey organisms will be. Looked at the other way round, an increasing number of prey organisms means better living conditions for the hunters. For this reason the propagation coefficient of the hunters (k_{PH}) is multiplied by the number of prey organisms (n_P) in the second equation. Removing the brackets shows that these equations resemble those for bimolecular reactions [Eq. (5.2.14)]. In fact, the present “reaction” also depends on the coming together of the two “reaction partners”.

This set of Eqs. (5.3.8) represents an extremely simplified situation. The interaction of individuals of the same species, for example, which was already reflected in Eq. (5.3.5), has been ignored here. This example, however, demonstrates that even the simplest ecological models already lead to complicated systems of non-linear equations. The solution of such systems of equations frequently leads to functions indicating oscillations. This corresponds with experiences where ecological systems oscillate even without additional input of external signals, such as daily or annual oscillations of light and temperature.

These oscillations can either be plotted against time, or represented as trajectories. In the case of Eq. (5.3.8), for example, the functions $n_P(t)$, and $n_H(t)$ can be plotted separately, or combined as the trajectory $n_P(n_H)$. Again, a system trajectory would appear like those shown in Figs. 5.5, 5.6, 5.7 or 5.15B. In the case of periodic oscillations closed curves occur, known as *Volterra cycles*.

Figure 5.19 demonstrates a concrete example of such oscillations. It uses data from the hunting lists of the Hudson-Bay Company in Canada which were kept

from 1845 to 1935. Suppose the numbers of hunted animals were proportional to the number of those really existing, one can roughly use this as a representation of the ecological situation. In Fig. 5.19 the data from 1900–1935 were used indicating the time course with typical oscillations (left) and the same data in a parameter plot showing Volterra cycles (right).

In a similar way to the hunter-prey system, a kind of symbiotic interrelation of two species can be formulated:

$$\left. \begin{aligned} \frac{dn_A}{dt} &= (k'_{pA} - k''_{pA}n_B - k_{mA})n_A \\ \frac{dn_B}{dt} &= (k'_{pB} + k''_{pB}n_A - k_{mB})n_B \end{aligned} \right\} \quad (5.3.9)$$

This set of equations takes into account the fact that when $k'_{pA} > k_{mA}$ and $k'_{pB} > k_{mB}$ then one species of this symbiotic community can survive without the other. The symbiotic cooperation is expressed by the product terms $k''_{pA}n_B$ and $k''_{pB}n_A$.

Mathematical approaches of this kind form the basis for modelling of ecological systems. However, other factors must also be considered such as a large number of trophic levels and complex interrelations, time delays caused by processes of developments, annual rhythms, and also ethological and sociological structures.

Further reading: Ginzburg and Golenberg 1985; Jeffries 1989; Hallam and Levin 1986; Logofet 1993; Renshaw 1990

5.3.2

Models of Growth and Differentiation

The morphology of an organism, its geometric structure and the corresponding morphogenetic mechanisms have provided, time and again, subjects for modelling. In 1917, in his well-known book *On Growth and Form*, D'Arcy Thompson first formulated the mathematical ideas of morphometry. His theory states that different types of body shape, when distorted by geometric projections, can be transformed from one to another. If the shape of an animal or of a part of an animal body is projected onto a system of coordinates, then by means of the mathematically defined distortion, i.e. by a sort of coordinate transformation, one shape can be systematically transformed to another. This method of D'Arcy Thompson has been demonstrated for the shapes of various species of fishes as well as the shape of a crab's, carapace, a mammalian skull and even leaves of plants and many other structures. In this way, for example, phylogenetic changes can be expressed mathematically. Such approaches belong more to the realms of biometry but they do, however, form a starting point for more extensive biophysical modelling.

In the same way, *allometric models* of growth primarily have a purely phenomenological importance. The basic principle of allometry is that both

morphological and functional parameters of similar animals of different sizes can be represented by exponential functions of the body mass. Such functions have been empirically determined (see Fig. 3.64).

The basal metabolic rate (J'_Q) of animals (in $J\ s^{-1}$) can be obtained from the following equation where it is given as a function of the body mass (m , in kg):

$$J'_Q = 3.41 \cdot m^{0.743}. \quad (5.3.10)$$

In this way functions for many parameters of living systems have been formulated based on empirically obtained data. Although such equations rarely explain the reasons for the phenomenon under consideration they do have their practical use. In sport medicine, such relations can be applied to compare the performance parameters of subjects of different sizes; similar information is also useful in anthropology and pharmacology. Furthermore, phenomenological growth curves are important for calculating economic aspects of agriculture.

Biophysical models of morphogenetic processes are based on the following premise. The differentiation of embryonic cells is controlled by growth regulating substances which can be either growth-stimulating, or growth-inhibiting. Such substances, or growth factors, have already been found in many experiments. They are also indicated in classical ligation experiments carried out on embryos and larval stages on animals. It can be assumed that such substances control the morphogenetic movements of cells as well as the activation of the genes that are responsible for differentiation. In this context reference must be made to the discussions on the possible role of electric fields in embryogenesis, leading to electrodiffusion of growth factors (Sect. 3.5.2).

As usual, the models start with the simplest approach. It is supposed that these substances will be formed at particular sites and then distributed by diffusion. In this way complex concentration patterns, with a space-time structure, are set up. If the growth rate, or the mode of differentiation depends on the concentration ratio of these factors, such concentration patterns finally bring about a morphologically visible structure.

There are many ways in which such processes can be modelled. Consider the equation for a simple unidimensional case where the concentration of an activator (c_A) and an inhibitor (c_I) are distributed only in the x -direction. For this case the following set of equations can be formulated:

$$\left. \begin{aligned} \frac{\partial c_A}{\partial t} &= k_A \frac{c_A^2}{c_I} - k'_A c_A + D_A \frac{\partial^2 c_A}{\partial x^2} \\ \frac{\partial c_I}{\partial t} &= k_I c_I^2 - k'_I c_I + D_I \frac{\partial^2 c_I}{\partial x^2} \end{aligned} \right\}. \quad (5.3.11)$$

This is a system of partial, non-linear differential equations, because the variables c_A and c_I are differentiated both with respect to time and position. These equations correspond in principle to the scheme of the flux balance given in Eq. (5.2.1).

In particular, the following assumptions are made in this model. The rate of formation of the activating factor A is determined by a process that is

autocatalytically promoted by c_A . This corresponds to the term: $k_A(c_A^2/c_I)$. The inactivation of the activator is expressed by the simple linear term $-k'_A c_A$. The diffusion along the x -axis $D_A(\partial^2 c_A/\partial x^2)$ is determined by the second Ficks law [Eq. (3.3.8)]. The differential equation for the concentration of the inhibitor (I) differs from that for the activator (A) only in the fact that there is not an inhibition but an autocatalytic activation ($k_I c_I^2$).

The behavior of such a system depends to a great extent on the choice of the constants. The relation of the diffusion coefficients D_A and D_I is of particular importance. If the diffusion constant for the inhibitor (D_I) is much greater than that for the activator (D_A) then the distribution of the factor I can be regarded as being homogeneous.

Now consider the initial state of a likewise homogeneously distributed activator A (Fig. 5.20A). If, in such a system, the concentration of A is raised for a short period (R_1) equally through the whole system, the system can compensate this and will attain the previous state. If, however, the excitation is local (R_2), it may become unstable and take on a new quality. A concentration

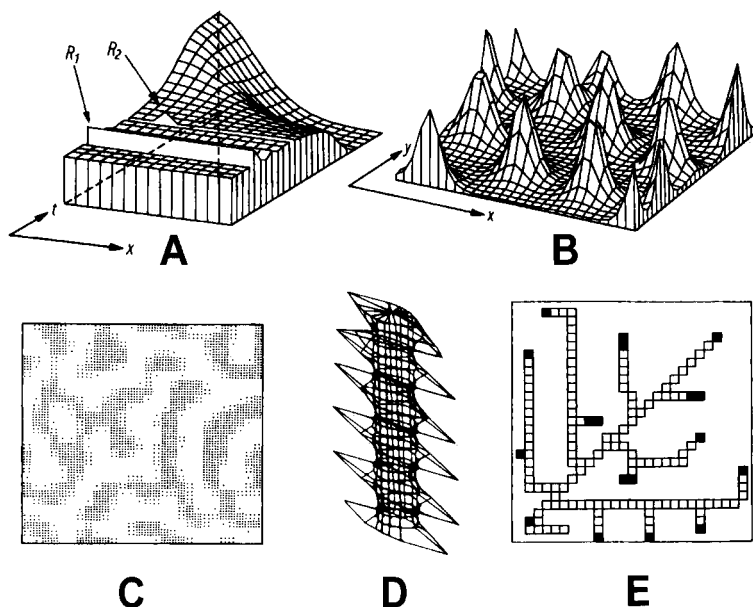


Fig. 5.20. Examples of biophysical models of morphogenesis. A Time course of distribution of an activator in a uni-dimensional model (R_1 and R_2 two kinds of excitation); B Formation of a 2-dimensional pattern of the activator concentration; C Development of a 2-dimensional zebra-pattern, corresponding to a model with stochastic elements; D An axis-symmetrical model with local concentrations of an activator, resembling a botanical system; E A model representing a bifurcation of vessels; After Meinhardt 1982. C Was kindly placed at our disposal by H. Meinhardt

gradient of the activating substance A is set up. Occasionally, local fluctuations are sufficient to induce such changes.

A great number of observed morphogenetic phenomena can be explained on the basis of similar equations. There are systems of equations and particular combinations of parameters that show oscillations with respect to both time and space. If the models are not simply based on a 1-dimensional approach as in the set of Eqs. (5.3.11), but consider diffusion in two or three dimensions then concentration patterns could be modelled on a plane or in a space.

Figure 5.20B shows the genesis of a stationary concentration pattern in a plane of x, y -coordinates. Such a distribution of activators and inhibitors, for example, could result in a distribution pattern of guard cells in the epidermis of a leaf, or distribution of hairs, pigment areas etc. Axial symmetric models may explain the arrangement of leaf nodes on the stem of a plant (Fig. 5.20D). If a term is included in the equation to arrange the mutual catalytic effects of neighbor cells then a model for the development of branching systems of vessels is obtained (Fig. 5.20E). In agreement with the actual situation, the system constants in these cases only determine the principle of branching, whereas the specific configuration of the network is a product of chance.

At the present time, these models still have a predominantly hypothetical character. Nevertheless, there is an increasing number of correlations with the results of experiments in developmental physiology.

Further reading: Britten 1986; Goel 1988; Kaandorp 1994; Meinhardt 1982, 1995

5.3.3

Models of Evolution

The work of Manfred Eigen (1971) stimulated comprehensive investigations on biological evolution on the basis of the theory of non-linear systems. The evolution of life from inorganic matter to its present form can be divided into three, qualitatively different phases:

- *Chemical evolution* – the path from inorganic molecules to the low molecular weight components of biopolymers.
- *Self-organization of biomacromolecules* and the formation of cell-like structures.
- *Evolution of organisms*.

The decisive step in Eigen's theory was his discovery of laws governing pre-biotic evolution. This means that even at the level of simple proteins and nuclear acids, mechanisms of competition and selection occur that reduce the degree of freedom of stochastic combinations. Earlier calculations had indicated that there was only a minute probability that a functional biopolymer could be formed from the chance combination of monomers (Albert Einstein had already

compared this probability with the chance of a complete dictionary resulting from an explosion in a printing office!).

A relatively small biopolymer with an information content of 10^3 bits would result in this way, on average, once in the course of $2^{1000} \approx 10^{300}$ chance combinations. This means that out of 10^{300} resulting molecules, one could be considered as being able to initiate life. Such an incredibly large number might be illustrated as follows: there is enough space in the hydrosphere of the earth to contain about 10^{32} molecules, and about 10^{100} molecules can be accommodated in the entire universe. The earth is "only" 10^{17} s old so, even if the reaction of combination and re-combination were very fast, the possibility of a successful combination is minute (such stochastic considerations were also discussed in Sects. 2.3.2 and 2.3.3).

In fact, it must be assumed that already at the beginning of evolution the principles of the struggle for life controlled the development of particular chains of reactions. The formation of a functional sequence of proteins, therefore, is as unlikely to be a product of chance, as is the functional morphology of an organ according to the principles of Charles Darwin.

These principles of molecular selection can be ascribed to processes of catalysis and auto-catalysis. Auto-catalysis can be regarded in this context as a fore-runner of self-reproduction.

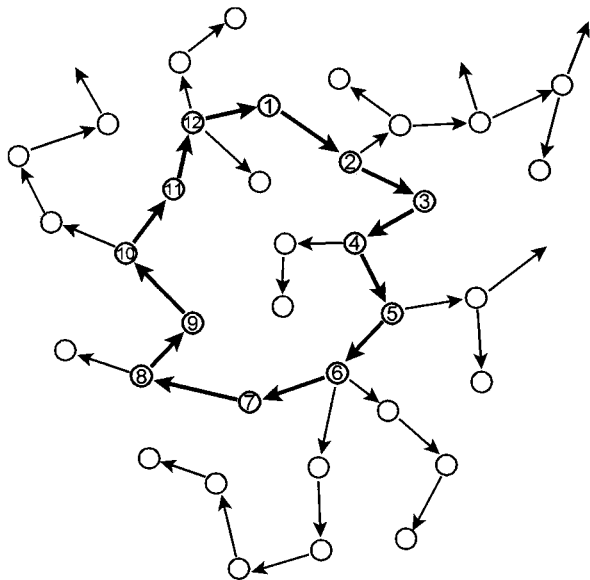
The number of monomers in a given region is limited and this will lead to a form of competition between the different kinds of polymerization reactions. However, such a system would rapidly reach an equilibrium state if it were not exposed to a continuous flow of energy and entropy. It will be continually disturbed by geophysical effects and from the thermodynamic point of view, it is a kind of dissipative structure.

Under these conditions, selection of polymers with high catalytic activity will take place. A network of catalytic relations between the individual molecules is built up (Fig. 5.21). If a closed chain occurs in such a network this will form an auto-catalytic cycle. This is a structure with a good chance for survival. Such cycles sustain themselves and in doing so can change their own structure. Side chains, which are formed through the poly-catalytic activity of several molecules are removed and distensible intermediate linkages are eliminated. In this way the cycle becomes smaller. At the same time, it can be shown that heterogeneous cycles, made up of alternating proteins and nucleic acids possess a selection advantage. Eigen called such structures *hypercycles*.

This concept of evolution therefore possesses a dialectic unity of stochastic and deterministic processes. The stochastic component is derived from the stability of bonds and from the stochastics of molecular contacts in the course of the reaction. At higher levels of development, this stochastic element is realized by the mutation rate or the dependability of self-reproduction.

This theory is backed up by mathematical modelling and computer simulations. It is possible to formulate equations for systems of chemical reactions as well as for processes of reproduction and selection at the highest levels of evolution. With regard to the equations that were given in the previous sections, the following simplified approach shall be demonstrated:

Fig. 5.21. The network of catalytic correlations in a mixture of polymers. The points with numbers connected by *thick arrows* indicate an auto-catalytic cycle



$$\frac{dn_i}{dt} = (k_{pi}q_i - k_{mi} - k'_i)n_i + \sum_{j \neq i} k_{m,ji}n_j. \quad (5.3.12)$$

This equation for the time course of the number of individuals (n_i) of a species i embodies the following postulates: the individuals propagate at a rate that bears a linear relation to the existing number of individuals and is determined by a time factor k_{pi} . This coefficient is, however, reduced by the factor q_i being a little less than one. In this way mutations which disturb the identical reproduction of individuals are taken into account. The coefficient governing the mortality (k_{mi}) is the same as in Eq. (5.3.1). In addition, another coefficient, k'_i is introduced, expressing the possibility that some individuals will leave the area or will diffuse away. The summing term at the end of the equation reflects the possibility of the emergence of an individual of the species i from another of the species j by mutation. In contrast to the ecological equations in Section 5.3.1, n_i is thus capable of increasing even under the initial conditions, when $t = 0$ and $n_i = 0$. This sum term is a prerequisite for the neogenesis of a species.

This is of course a very primitive approach. It presupposes that propagation is based on linear relations, an assumption having limited validity as has already been discussed in Section 5.3.1. The relations between species have not been considered either. Nevertheless, evolution because of competition between species can be demonstrated on the basis of this approach if the following mass balance is used:

$$\sum_{i=1}^m n_i = \text{const} \quad \text{therefore:} \quad \sum_{i=1}^m \frac{dn_i}{dt} = 0. \quad (5.3.13)$$

The advantage for selection of a species will then result from the relations of the chosen system constants, one to another. With more complex evolutionary systems, weighing functions will have to be introduced but these must necessarily be of a speculative nature.

These models for evolution are, of course, still of a highly speculative nature. In spite of this, it is due to them that we have some important fundamental insights into the possible mechanisms of evolution, the highest form of motion in biological systems.

Further reading: Eigen 1971, 1992; Haken 1983; Hoppe et al. 1982; Rowe 1994

5.3.4

Models of Neural Processes

The first models of neural networks used approaches of Boolean algebra, the mathematical theory of binary control elements. As early as the 1940s, McCulloch and Pitts proposed such models. They investigated networks consisting of “formal neurons”. These are elements performing basic logical functions, resembling in some instance nerve cells with synaptic connections. It could be proved that these kinds of networks are able to perform all operations of Boolean algebra.

Such a formal neuron can accept several digital input signals S_i and transform them into an output signal S_o . Figure 5.22 shows this process schematically. The signals may obtain the values: 0 = “non excited” and 1 = “excited”. A weighing factor k_i determines whether the influence is stimulating (+1) or whether it is inhibiting (−1). In the neuron these signals are summarized to get a parameter h . If $h \geq h^*$ this neuron produces an output signal which again has the value 1.

This basic model has been modified in subsequent years. So, for example, the refractory period of nerves has been included. This means that for some milliseconds after an input signal the nerve is not able to accept a further one. This requires a synchronization of all impulses in such a network, and the introduction of a certain sequence of operation. Despite the fact that in the brain neurons actually exist which produce such a sequence, for example cortical cell columns, there is no evidence of an overall synchronization in the brain.

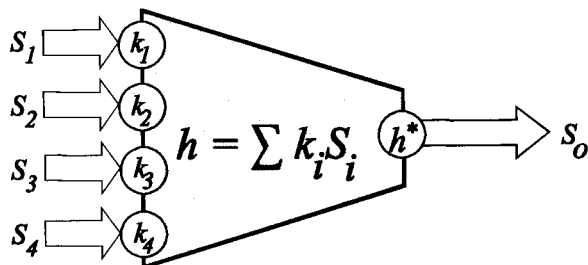


Fig. 5.22. Schematic presentation of the function of a formal neuron according to the theory of McCulloch and Pitts (explanation in text)

Meanwhile, so-called integrate and fire models have been developed to overcome this problem.

Another problem concerns the huge number of possible connections in networks containing a large number of neurons. Even in the case of a ganglion of a nematode with only 300 neurons and about 5000 synapses, the number of possible kinds of interconnections is enormous. The human brain contains about 10^{10} neurons, each of them being connected with others by about 10^4 synapses. The total number of synapses in the brain, therefore, is about 10^{14} . The number of possible permutations of these connections is unimaginably large.

These numbers lead to the following conclusions: a construction of a neuronal network as a copy of the human brain is neither possible nor would it be useful. Furthermore, it is impossible that the pattern of synaptic connections in the brain has been determined genetically. The amount of information which would be required by this exceeds by far the storage capacity of the genome. This means that the problem of neuronal networks is not only a question of its function but additionally of its self organization.

This problem of the immense number of possible realizations, is something which the theory of neuronal networks has in common with other fields of molecular physics. So, for example, approaches of statistical thermodynamics can be used as well as the cooperative behavior of molecules in amorphous materials. In this context, for example, the theory of spin-glasses is of interest.

Spin-glasses are viscous materials containing particles of different magnetic properties. These include ferromagnetic particles, orientating themselves by mutual influences parallel to each other, as well as anti-ferromagnetic particles, orientating anti-parallel (Fig. 5.23). The analogy between spin-glasses and neuronal networks is the direction of the particle and the state of the neuron as well as their mutual interactions. A parallel or anti-parallel orientation of a particle in the spin-glass would correspond to a "silent" or an "excited" cell in a neural network. The ferromagnetic or anti-ferromagnetic interaction of the particles simulates the interaction of the nerve cells.

In this respect, spin-glasses may show a quite complicated pattern of self-organization. Of particular interest are the situations where two opposite kinds of interaction come together at one point. In this case a particle will be influenced at the same time by two forces tending to orientate it in opposite ways. This disturbs the stability of the whole system. This situation is called a *frustrated* state. It can be considered as a sort of metastability and could reflect a kind of memory state of a neuronal network. This is special kind of phase transition which cannot be described like a catastrophe as depicted in Fig. 5.7, but instead shows a more continuous kind of transition. These spin-glass models therefore resemble not only the problem of self-organization but also the function of neuronal networks directly.

The theories of neuronal networks in general are orientated in two directions: on the one hand, neurobiologists are interested in understanding processes of neuronal data processing and memory in the brain, on the other hand there is a branch of modern computer technology with the aim of producing self-organizing systems with so-called artificial intelligence.

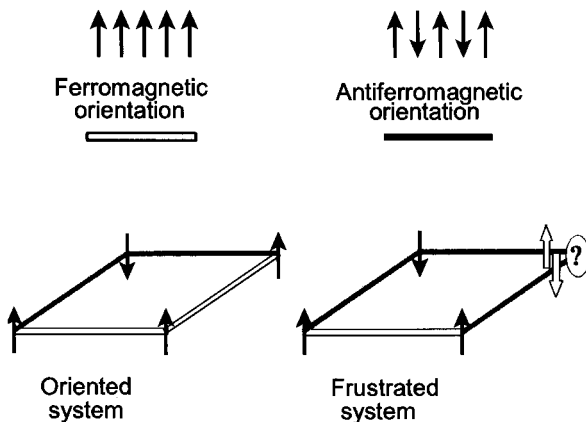


Fig. 5.23. Ferromagnetic and anti-ferromagnetic interactions in spin-glasses and the occurrence of frustrated systems

It is interesting to note that many years ago computer technology started with basic elements resembling a nerve system, like the formal neurons of Fig. 5.22, but so far they are a long way from modelling the living brain. At best, their results are constructions showing some formal similarities with the function of the central nervous system. Despite the importance of the theory of neuronal networks for computer development, it has now become an area which is more or less moving away from neurobiology and biophysics. In some respects research involving big digital networks will support the understanding of neuronal systems, even if there are fundamental differences in the properties of their elements.

The most important difference between brain and computer of course are the time constants of their corresponding elements. In contrast to computers, the elements of which react faster than 10^{-9} s, a nerve cell needs 10^{-3} s for an action potential (Fig. 3.27). Furthermore, the rate of impulse conduction in electronics exceeds that of nerves by about 9 orders of magnitude. The property of the brain to fulfill complicated operations in a short time using these very slowly reacting neurons makes it impossible to explain its function by use of principles of conventional computers.

Furthermore, there are some other properties of the nervous system which underline how they differ from neural networks of the McCulloch type:

- The brain contains neurons and synapses of quite different shapes, properties and functions.
- The neurons in the brain are not connected in a predicted way but connect and disconnect themselves in a process of self-organization controlled by its own function.
- In contrast to computers, the processors, data-stores, data-busses etc. in the brain are not separate elements.
- The loss of neurons which begins at birth in humans is a normal process and does not affect brain function. This process at the beginning of life even runs

parallel with the development of human mental capacity. Even dramatic kinds of brain injury as well as the removal of parts of the brain in some operations mostly do not affect the function of the brain significantly.

In context with these properties, the function of the human brain can be characterized in the terminology of computer science as follows. It is a system

- With parallel information processing,
- With a program hierarchy,
- With self-organizing software as well as hardware (in contrast to computers, hardware and software in the brain cannot be distinguished),
- With non-linear elements, and
- With stochastic properties in some functions.

A number of network theories have already taken these peculiarities into account.

A special problem concerns the large memory capacity of the brain, connected with extremely fast access to stored information. Obviously the memory is not based only on one single mechanism. The discovery of the holographic principle by T. Gabor (1948) and its realization after the development of laser technology raises the idea that the memory in the brain could probably be based on a holographic principle. This has been supported by the fact that a local loss of neurons does not affect the memory, as would be the case in the information storage of technical computers. If information is stored in a holographic way, the loss of a part of the storage elements would only lead to a loss of the resolution of the stored picture, but not to its loss. This is really an intriguing idea, but there is no experimental evidence for this yet, and no proposals for a molecular mechanism.

These considerations show that despite intensive research and some concrete results which are used successfully in computer technology, our models of functions of the central nervous systems are still quite crude.

Further reading: Arbib and Amari 1989; Hertz 1991; Peretto 1992

References

- Able, K. P. (1994) Magnetic orientation and magnetoreception in birds. *Progress in Neurobiology* 42: 449
- Alexander, R. McN. (1983) *Animal Mechanics*. (2nd edn.) Blackwell Scientific Publ., Oxford
- Amman, D. (1986) *Ion-selective Microelectrodes*. Springer, Berlin
- Anderson, D. H. (1983) *Compartmental modeling and tracer kinetics*. Lecture Notes in Biomathematics 50. Springer, Berlin
- Arbib, M. A., and S. Amari (1989) *Dynamic Interactions in Neural Networks: Models and Data*. Research Notes in Neural Computing, Vol. 1, Springer, Berlin
- Atema, J., F. R. Fay, A. H. Popper, and W. N. Tavalja (Eds.) (1987) *Sensory Biology of Aquatic Animals*. Springer, Berlin
- Au, W. W. L. (1994) *Sonar of Dolphins*. Springer, New York
- Azuma, A. (1992) *The Biokinetics of Flying and Swimming*. Springer, Tokyo
- Barber, J. (1992) *The Photosynthesis: Structure, Function and Molecular Biology*. Elsevier, Amsterdam
- Barry, P. H. (1998) Derivation of unstirred-layer transport number equations from the Nernst-Planck flux equations. *Biophys J* 74: 2903
- Bashford, C. L., and C. A. Pasternak (1986) Plasma membrane potential of some animal cells is generated by ion pumping, not by ion gradients. *Trends in Biochemical Sciences* 11: 113
- Batenburg, J. J., and H. P. Haagsman (1998) The lipids of pulmonary surfactant: Dynamics and interactions with proteins. *Prog Lipid Res* 37: 235
- Bauer, J. (Ed.) (1994) *Cell Electrophoresis*. CRC Inc., Boca Raton
- Benz, R., and F. Conti (1986) Effects of hydrostatic pressure on lipid bilayer membranes. *Biophys J* 50: 91, 99
- Bereiter-Hahn, J., O. R. Anderson, and W.-E. Reif (Eds.) (1987) *Cytomechanics. The Mechanical Basis of Cell Form and Structure*. Springer, Berlin
- Bernhardt, I. (1994) Alteration of cellular features after exposure of low ionic strength medium. In: Bauer, J. (Ed.) *Cell Electrophoresis*. CRC Inc., Boca Raton, 163
- Bernhardt, J. H. (1988) The establishment of frequency dependent limits for electric and magnetic fields and evaluation on indirect effects. *Radiat Environ Biophys* 27: 1
- Bernhardt, J. H., R. Matthes, and M. H. Repacholi (Eds.) (1997) *Non-Thermal Effects of Rf Electromagnetic Fields*. International Commission on Non-Ionizing Radiation Protection 3: ISBN 3-9804789-2-0
- Bertalanffy, L. von (1968) *General System Theory*, Braziller, New York
- Biewener, A. A. (1992) *Biomechanics. Structures and Systems – a Practical Approach*. IRL Press, Oxford
- Bingeli, R., and R. Weinstein (1986) Membrane potentials and sodium channels: Hypotheses for growth regulation and cancer formation based on changes in sodium channels and gap junctions. *J Theoret Biol* 123: 377
- Blank, M., and L. Soo (1992) Temperature dependence of electric field effects on Na, K-ATPase. *Bioelectrochem. Bioenergetics* 28: 291

- Bloom, M., E. Evans, and O. G. Mouritsen (1991) Physical properties of the fluid lipid-bilayer component of cell membranes: A perspective. *Quart Rev Biophys* 24: 293
- Blumenfeld, L. A., and A. N. Tikhonov (1994) *Biophysical Thermodynamics of Intracellular Processes*. Springer, Berlin
- Britton, N. F. (1986) *Reaction-diffusion Equations and Their Applications to Biology*. Academic Press, London
- Bucala, R. (1996) Lipid and lipoprotein oxidation: Basic mechanisms and unresolved questions in vivo. *Redox Report* 2: 291
- Buettner, G. R. (1993) The pecking order of free radicals and antioxidants: Lipid peroxidation, α -tocopherol, and ascorbate. *Archiv Biochem Biophys* 300: 535
- Cantor, Ch. R., and P. R. Schimmel (1980) *Biophysical Chemistry III. The Behavior of Biological Macromolecules*. W. H. Freeman and Comp., New York
- Cevc, G. (1990) Membrane electrostatics. *Biochim Biophys Acta* 1031: 311
- Charry, J. M., and R. Kavet (1987) *Air Ions: Physical and Biological Aspects*. CRC Inc., Boca Raton
- Christen, H. R. (1968) *Grundlagen der allgemeinen und anorganischen Chemie*. Diesterweg, Frankfurt
- Collins, K. D., and M. W. Washabaugh (1985) The Hofmeister effect and the behaviour of water at interfaces. *Quat Rev Biophys* 18: 323
- Cook, J. S. (1967) Nonsolvent water in human erythrocytes. *J Gen Physiol* 50: 1311
- Cooper, A. (1976) Thermodynamic fluctuations in protein molecules. *Proc Natl Acad Sci USA* 73: 2740
- Creighton, Th. E. (Ed.) (1992) *Protein Folding*. W. H. Freeman and Company, New York
- Deisenhofer, J., and J. R. Norris (Eds.) (1991) *The Photosynthetic Reaction Center*, Vol. 1, 2, Academic Press, New York
- deNoordhout, A. M. (1998) Cortical magnetic stimulation. *Clin Neurophys* 28: 9
- Dertinger, H., and H. Jung (1969) *Molekulare Strahlenbiologie*. Springer, Berlin
- Dessauer, F. (1964) *Quantenbiologie*. Springer, Berlin
- Dieckmann, S. (1984) Einwirkung mechanischer Schwingungen (Vibrationen) auf den Menschen. In: *Arbeitsmedizin aktuell, ein Loseblattwerk für die Praxis*, Vol. 14, Gustav Fischer Verlag, Stuttgart
- Donath, E., and A. Voigt (1986) Electrophoretic mobility of human erythrocytes. Theory and experimental applicability. *Biophys J* 49: 493
- Doucet, P., and P. B. Sloep (1992) *Mathematical Modeling in the Life Sciences*. Ellis Horwood, New York
- Dunin-Borkowski, R. E., M. R. McCartney, R. B. Frankel, D. A. Bazylinski, M. Posfai, and P. R. Buseck (1998) Magnetic microstructure of magnetotactic bacteria by electron holography. *Science* 282: 1868
- Eigen, M. (1971) Selforganization of matter and the evolution of biological macromolecules. *Naturwiss* 58: 465
- Eigen, M. (1992) *Steps Toward Life*. Oxford University Press, Oxford, England
- Evtodienko, V. Y., Y. N. Antonenko, and L. S. Yaguzhinsky (1998) Increase of local hydrogen ion gradient near bilayer lipid membrane under the conditions of catalysis of proton transfer across the interface. *FEBS Letters* 425: 222
- Farhataziz and Rodgers, M. A. J. (Eds.) (1987) *Radiation Chemistry - Principles and Applications*. Verlag Chemie, Weinheim
- Ferreira, H. G., and M. W. Marshall (1985) *Biophysical Basis of Excitability*. Cambridge University Press, Cambridge
- Fleischer, S., and B. Fleischer (Eds.) (1989) *Transport Theory: Cells and Model Membranes*. Methods in Enzymology, Vol. 171, Academic Press, San Diego
- Franks, F. (1985) *Biophysics and Biochemistry at Low Temperatures*. Cambridge University Press, Cambridge

- Friedman, M. H. (1986) Principles and Models of Biological Transport. Springer, Berlin
- Fritz, M. (1998) Three-dimensional biomechanical model for simulating the response of the human body to vibration stress. *Med Biol Eng Comput* 36: 686
- Freedman, J. C., and J. F. Hoffman (1979) Ionic and osmotic equilibria of human red blood cells treated with nystatin. *J Gen Physiol* 74: 157
- Fuhr, G., and R. Hagedorn (1996) Cell Electrorotation. In: Lynch, P. T., and M. R. Davey (Eds.): *Electrical Manipulation of Cells*. Chapman & Hall, New York: 38
- Fuhr, G., U. Zimmermann, and S. G. Shirley (1996) Cell motion in time-varying fields: principles and potential. In: Zimmermann, U., and G. A. Neil (Eds.): *Electromanipulation of Cells*. CRC Inc., Boca Raton: 259
- Fukada, E. (1983) Piezoelectric properties of biological polymers. *Quart Rev Biophys* 16: 59
- Fung, Y. C. (1984) *Biodynamics*. Springer, New York
- Fung, Y. C. (1993) *Biomechanics* (2nd edn.) Springer, New York
- Gabelnick, H. L., and M. Litt (1973) *Rheology of biological systems*. Charles C. Thomas, Springfield
- Gabriel, C., S. Gabriel, and E. Corthout (1996) The dielectric-properties of biological tissues. I. Literature survey. *Phys Med Biol* 41: 2231
- Galla, H.-J., N. Bourdos, A. von Nahmen, M. Amrein, and M. Sieber (1998) The role of pulmonary surfactant protein C during the breathing cycle. *Thin Solid Films* 327: 632
- Gause, G. F. (1935) Experimentelle Untersuchungen über die Konkurrenz zwischen *Paramecium caudatum* und *Paramecium aurelia*. *Archiv Protistenkunde* 84: 207
- German, B., and J. Wyman (1937) The titration curves of oxygenated and reduced hemoglobin. *J Biol Chem* 117: 533
- George, M. S., Z. Nahas, A. M. Speer, T. A. Kimbrell, E. M. Wassermann, C. C. Teneback, M. Molloy, D. Bohning, S. C. Risch, and R. M. Post (1998) Transcranial magnetic stimulation: A new method for investigating the neuroanatomy of depression. *Advances in Biological Psychiatry* 19: 94
- Georgieva, R., B. Neu, V. M. Shilov, E. Knippel, A. Budde, R. Latza, E. Donath, H. Kiesewetter, and H. Bäuml (1998) Low frequency electrorotation of fixed red blood cells. *Biophys J* 74: 2114
- Giebisch, G., D. C. Tosteson, and H. Using (Eds.) (1979) *Membrane Transport in Biology*. Springer-Verlag, Berlin
- Gimsa, J., and D. Wachner (1999) A polarization model overcoming the geometric restrictions of the Laplace solution for spheroidal cells obtaining new equations for field-induced forces and transmembrane potential. *Biophys J* 77: 1316
- Gimsa, J., and D. Wachner (1998) A unified resistor-capacitor model for impedance, dielectrophoresis, electrorotation, and induced transmembrane potential. *Biophys J* 75: 1107
- Ginsburg, L. R., and E. M. Golenberg (1985) *Lectures in Theoretical Population Biology*. Pentire-Hall
- Glansdorff, P. and I. Prigogine (1971) *Thermodynamic Theory of Structure, Stability and Fluctuations*. Wiley-Interscience Publ., London
- Glaser, R. (1989) *Grundriß der Biomechanik*. (2nd edn.) Akademie-Verlag, Berlin
- Glaser, R. (1996) The electric properties of the membrane and the cell surface. In: Zimmermann, U., and G. A. Neil (Eds.): *Electromanipulation of Cells*. CRC Inc., Boca Raton
- Glaser, R., M. Brumen, and S. Svetina (1980) Stationäre Ionenzustände menschlicher Erythrozyten. *Biol Zbl* 99: 429
- Glaser, R., and J. Donath (1984) Stationary ionic states in human red blood cells. *Bioelectrochem Bioenerget* 13: 71
- Goel, N. S., and R. L. Thompson (1988) *Computer Simulations of Self-organization in Biological Systems*. Croom Helm, London

- Goldman, D. E. (1943) Potential, impedance and rectification in membranes. *J Gen Physiol* 27: 37
- Goldman, St. (1960) *Cybernetic Aspects of Homeostasis*. In: Comar, C. L., and Bronner (Eds.): *Mineral Metabolism*. Academic Press, New York
- Govindjee (Ed.) (1975) *Bioenergetics of Photosynthesis*. Academic Press, New York
- Graevskij, E. J., and N. I. Šapiro (1957) *Sovremennye voprosy radiobiologii*. Nauka, Moskva
- Griffin, D. R. (1958) *Listening in the Dark*. Yale Univ. Press, New Haven
- Grodzenskij, D. E. (1966) *Radiobiologija*. Nauka, Moskva
- Grosse, C., and H. P. Schwan (1992) Cellular membrane potentials induced by alternating fields. *Biophys J* 63: 1632
- Gruher, H. (1995) New insights into directed cell-migration – characteristics and mechanisms. *Nouv Rev Fr Hematol* 37: 255
- Guzelsu, N., and W. R. Walsh (1993) Piezoelectric and electrokinetic effects in bone tissue. *Review Electro- and Magnetobiology* 12: 51
- Haggis, G. H., D. Michie, A. R. Muir, K. B. Roberts and P. M. B. Walker (1965) *Introduction to Molecular Biology*. J Wiley & Sons, New York
- Haken, H. (1983) *Synergetics*, Springer Ser. Synergetics, Vol. 1, 3rd. edn., Springer, Berlin
- Hall, D. O., and K. K. Rao (1994) *Photosynthesis. Studies in Biology* (5th edn.) Cambridge University Press, Cambridge
- Hallam, T. G., and S. A. Levin (Eds.) (1989) *Applied Mathematical Ecology*, Springer-Verlag, Berlin
- Harris, C. M. (1991) *Handbook of Acoustical Measurements and Noise Control*. McGraw-Hill Inc., New York
- Harrison, L. G. (1993) *Kinetic Theory of Living Pattern*. Cambridge University Press, Cambridge
- Harth, O., and P. Vaupeul (1971) Die Verteilung von Na^+ und K^+ unter dem Einfluß von Temperaturgradienten. *Pflügers Arch* 323: 158
- Hassenstein, B. (1966) *Kybernetik und biologische Forschung*. In: Gessner, F. (Ed.): *Handbuch der Biologie*. Bd. 1, 629, Akad. Verlagsgesellschaft Athenaion, Frankfurt/Main 1966
- Hayashi, K., and N. Sakamoto (1986) *Dynamic Analysis of Enzyme Systems*. Springer-Verlag, Berlin
- Heinrich, R., H. G. Holzhütter, and St. Schuster (1987) A theoretical approach to the evolution and structural design of enzymatic networks: Linear enzymatic chains, branched pathways and glycolysis of erythrocytes. *Bull Math Biol* 49: 539
- Heinrich, R., and St. Schuster (1996) *Modeling of Metabolic Systems. Structure, Control and Optimality*. Chapman and Hall, New York
- Heinz, W. F., and J. H. Hoh (1999) Relative surface charge density mapping with the atomic force microscope. *Biophys J* 76: 528–538
- Hertel, U. (1963) *Struktur, Form, Bewegung*. Krauskopf, Mainz
- Hertz, J. (1991) *Introduction to the Theory of Neural Computation*. Addison-Wesley, Redwood City
- Hianik, T., and V. I. Passechnik (1995) *Bilayer Lipid Membranes. Structure and Mechanical Properties*. Kluwer Academic Publishers, Dordrecht
- Higashi, T., A. Yamagishi, T. Takeuchi, N. Kawaguchi, S. Sagawa, S. Onishi, and M. Date (1993) Orientation of erythrocytes in a strong static magnetic field. *Blood* 82: 1328
- Hille, B. (1992) *Ionic Channels of Excitable Membranes* (2nd edn.) Sinauer Ass. Inc., Sunderland
- Hoffmann, E. K., and L. O. Simonsen (1989) Membrane mechanisms in volume and pH regulation in vertebrate cells. *Physiol Rev* 69: 315
- Hoppe, W., W. Lohmann, H. Markl, and H. Ziegler (Eds.) (1983) *Biophysics*. Springer-Verlag, New York

- Horne, R. A. (Ed.) (1972) *Water and Aqueous Solutions. Structure, Thermodynamics, and Transport Processes*. Wiley-Interscience, New York
- Hotary, K. B., and K. H. Robinson (1990) Endogenous electrical currents and the resultant voltage gradients in the chick embryo. *Developmental Biology* 140: 149
- Hudspeth, A. J. (1985) The cellular basis of hearing: The biophysics of hair cells. *Science* 230: 745
- Hudspeth, A. J. (1989) How the ear's works work. *Nature* 341: 397
- Hug, O., and A. M. Kellerer (1966) *Stochastik der Strahlenwirkung*. Springer, Heidelberg
- International Commission on Non-Ionizing Radiation Protection (ICNIRP) (1988) Guidelines for limiting exposure to time-varying electric magnetic, and electromagnetic fields (up to 300 GHz). *Health Physics* 74: 494
- International Commission of Radiological Protection (ICRP) (1991) *Recommendations of the International Commission of Radiological Protection*, Publication No. 60. Pergamon Press, Oxford
- Israelachvili, J. (1994) *Intermolecular and Surface Forces* (2nd edn.) Academic Press, London
- Israelachvili, J. (1974) Van der Waals forces in biological systems. *Quart Rev Biophys* 6: 341
- Jaffe, L. F. (1979) Control of development by ionic currents. In: Cone, R. A., and J. Dowling (Eds.): *Membrane Transduction Mechanisms*. Raven Press, New York
- Jeffries, C. (1989) *Mathematical Modeling in Ecology. A Workbook for Students*. Birkhäuser, Boston
- Jehle, H. (1969) Remarks on the problems of morphogenetic movement in the development of embryos. *Intern J Quantum Chem* 3s: 75
- Kaandorp, J. A. (1994) *Fractal Modelling: Growth And Form In Biology*. Springer, Berlin
- Kalmijn, A. J. (1997) Electric and near-field acoustic detection, a comparative study. *Acta Physiologica Scandinavica* 161: 25
- Kauffman, S. A. (1993) *The Origin of Order*. Oxford University Press, New York
- Katchalsky, A., and P. F. Curran (1965) *Nonequilibrium Thermodynamics in Biophysics*. Harvard Univ. Press, Cambridge
- Keszthelyi, L., and P. Ormos (1989) Protein electric response signals from dielectrically polarized systems. *J Membrane Biol* 109: 192
- Keynes, R. D. (1994) The kinetics of voltage-gated ion channels. *Quart Rev Biophysics* 27: 339
- Kiefer, J. (1990) *Biological Radiation Effects*. Springer, Berlin
- Kirschvink, J. L., D. S. Jones, and B. J. MacFadden (Eds.) (1985) *Magnetite Biomineralization and Magnetoreception in Organisms. A New Biomagnetism*. Plenum Press, New York
- Knorre, W. A. (1981) *Pharmakokinetik*. Akademie-Verlag, Berlin
- König, H. L., A. P. Krueger, S. Lang, and W. Sönning (1981) *Biologic Effects of Environmental Electromagnetism*. Springer, New York
- Kotyk, A., K. Janacek, and J. Koryta (1988) *Biophysical Chemistry of Membrane Function*. J. Wiley and Sons, Chichester
- Lammert, P. E., J. Prost, and R. Bruinsma (1996) Ion drive for vesicles and cells. *J theor Biol* 178: 387
- Lamprecht, I., and A. I. Zotin (Eds.) (1978) *Thermodynamics of Biological Processes*. Walter de Gruyter, Berlin
- Lamprecht, I., and A. I. Zotin (Eds.) (1985) *Thermodynamics and Regulation of Biological Processes*. Walter de Gruyter, Berlin
- Lauffer, M. A. (1975) *Entropy-driven Processes in Biology*. Springer-Verlag, Berlin
- Läuger, P. (1991) *Electrogenic Ion Pumps*. Sinauer Associates Inc. Publishers, Sunderland
- Leff, H. S., and A. F. Rex (1990) *Maxwell's Demon. Entropy, Information, Computing*. Adam Hilger, Bristol
- Leitgeb, N. (1990) *Strahlen, Wellen, Felder. Ursachen und Auswirkungen auf Umwelt und Gesundheit*. Georg Thieme Verlag, Stuttgart

- Leopold, A. C. (Ed.) (1986) *Membranes, Metabolism and Dry Organisms*. Comstock Publ. Assoc. Cornell Univ. Press, Ithaca
- Lerche, D., and H. Bäumler (1984) Moderate heat treatment of only red blood cells shows down the rate of RBC-RBC aggregation in plasma. *Biorheology* 21: 393
- Levitt, D. G., and H. J. Mlekoday (1983) Reflection coefficient and permeability of urea and ethylene glycol in the human red cell membrane. *J Gen Physiol* 81: 239
- Leyton, L. (1975) *Fluid behaviour in biological systems*. Clarendon Press, Oxford
- Logofet, D. O. (1993) *Matrices and Graphs. Stability Problems in Mathematical Ecology*. CRC Inc., Boca Raton
- Low, J., and A. Reed (1992) *Electrotherapy Explained. Principles and Practice*. Butterworth-Heinemann Ltd, London
- Lynch, P. T., and M. R. Davey (Eds.) (1996) *Electrical Manipulation of Cells*. Chapman & Hall, New York
- MacGinitie, L. A. (1995) Streaming and piezoelectric potentials in connective tissues. *Adv Chem Ser* 250: 125
- Machemer-Röhnisch, S., H. Machemer, and R. Bräucker (1996) Electric-field effects on gravikinesis in *Paramecium*. *J Comp Physiol A* 179: 213
- Makarov, V. A., M. Feig, B. K. Andrews, and B. M. Pettitt (1998) Diffusion of solvent around bimolecular solutes: A molecular dynamics simulation study. *Biophys J* 75: 150
- Maniewski, R. (1991) Magnetic studies on mechanical activity of the heart. *Critical Reviews in Biomedical Engineering* 19: 203
- Maret, G., N. Boccara, and J. Kiepenheuer (Eds.) (1986) *Biophysical Effects of Steady Magnetic Fields*. Springer, Berlin
- Markin, V. S., V. F. Pastushenko, and Y. A. Chizmadzhev (1987) *Theory of Excitable Media*. A. Wiley-Interscience Publication, John Wiley & Sons, New York
- Markin, V. S., and A. J. Hudspeth (1995) Modeling the active process of the cochlea phase relations, amplification, and spontaneous oscillation. *Biophys J* 69: 138
- Matthes, R., J. H. Bernhardt, and M. H. Repacholi (Eds.) (1997) *Biological Effects of Static and Elf Electric and Magnetic Fields*. International Commission on Non-Ionizing Radiation Protection, Vol. 4
- Matthew, J. B. (1985) Electrostatic effects in proteins. *Ann Rev Biophys Biophys Chem* 14: 387
- Mazumdar, J. N. (1992) *Biofluid Mechanics*. World Scientific, Singapore
- McLaughlin, St. (1989) The electrostatic properties of membranes. *Ann Rev Biophys Biophys Chem* 18: 113
- Meinhardt, H. (1982) *Models of Biological Pattern Formation*. Academic Press, London
- Meinhardt, H. (1995) *The Algorithmic Beauty of Sea Shell*. Springer, Berlin
- Mohandas, N., and E. Evans (1994) Mechanical properties of the red cell membrane in relation to molecular structure and genetic defects. *Ann Rev Biophys Biomolec Structure* 23: 787
- Møller, H. (1984) *Effects of Infrasound on Man*. Aalborg University Press
- Moore, L. R., M. Zborowski, L. P. Sun, and J. J. Chalmers (1998) Lymphocyte fractionation using immunomagnetic colloid and a dipole magnet flow cell sorter. *J Biochem Biophys Methods* 37: 11
- Morecki, A., J. Ekiel, and K. Fidelus (1984) *Cybernetic Systems of Limb Movement in Man, Animals, and Robots*. Horwood Ed., Halsted Press
- Murray, J. D. (1993) *Mathematical Biology*. (2. Aufl.). Springer, Berlin
- Némethy, G., and A. Scheraga (1962) Structure of water and hydrophobic bounding in proteins. *J Chem Phys* 36: 3382
- Netter, H. (1959) *Theoretische Biochemie*. Springer-Verlag, Berlin
- Nias, A. H. W. (1990) *An Introduction to Radiobiology*. Wiley, Chichester
- Niklas, K. J. (1992) *Plant Biomechanics. An Engineering Approach to Plant Form and Function*. The University of Chicago Press, Chicago

- Nuccitelli, R. (1995) Endogenous electric-fields measured in developing embryos. *Advances in Chemistry Series* 250: 109
- Nyborg, W. L. (Ed.) (1985) *Biological Effects of Ultrasound*. Livingstone New York
- Ogston, A. G. (1966) When is pressure osmotic? *Federation Proceedings* 25: 1112
- Overbeek, J. T. G. (1956) The Donnan equilibrium. *Progress Biophys Biophys Chem* 6: 58
- Olsvik, Ø., T. Popovic, E. Skjerve, K. S. Cudjoe, E. Hornes, J. Ugelstad, and M. Uhlén (1994) Magnetic separation techniques in diagnostic microbiology. *Clinical Microbiology Reviews* 7: 43
- Pain, R. H. (1994) *Mechanism of Protein Folding*. Oxford University Press, Oxford
- Palmer, T. (1995) *Understanding Enzymes*. Prentice Hall, London
- Pauwels, F. (1980) *Biomechanics of the Locomotor Apparatus*. Springer, Berlin
- Pearson, K. (1892) *The Grammar of Science*. Walter Scott, London
- Penzlin, H. (1991) *Lehrbuch der Tierphysiologie*. (5th edn.) Fischer Verlag, Jena
- Péqueux, A. J. R., and R. Gilles (1985) *High Pressure Effects on Selected Biological Systems*. Springer, Berlin
- Peretto, P. (1992) *An Introduction to the Modeling of Neural Networks*. Cambridge University Press, Cambridge
- Pethig, R. (1979) *Dielectric and Electronic Properties of Biological Materials*. John Wiley & Sons, Chichester
- Pethig, R., and D. B. Kell (1987) The passive electrical properties of biological systems: Their significance in physiology, biophysics and biotechnology. *Phys Med Biol* 32: 933
- Petracchi, D., and G. Cercignani (1998) A comment on the sensitivity of fish to low electric fields. *Biophys J* 75: 2117
- Pohl, P., S. M. Saparov, and Y. N. Antonenko (1998) The size of the unstirred layer as a function of the solute diffusion coefficient. *Biophys J* 75: 1403
- Polk, Ch., and E. Postow (1996) *CRC Handbook of Biological Effects of Electromagnetic Fields*. CRC Inc., Boca Raton, 2nd edn
- Pollak, G. D., and J. H. Casseda (1989) *the Neural Basis of Echolocation in Bats*. Springer, New York
- Popper, A. N., and R. R. Fay (Eds.) (1995) *Hearing by Bats*. Springer, New York
- Precht, H., J. Christophersen, and H. Hensel (1955) *Temperatur und Leben*. Springer-Verlag, Berlin
- Prigogine, I. (1967) *Introduction to Thermodynamics of Irreversible Processes*. (3rd edn.) Wiley Interscience, New York
- Pullman, B., and A. Pullman (1963) *Quantum Biochemistry*. J. Wiley & Sons, New York
- Qassem, W., M. O. Othman, and Abdul-Majeed (1994) The effects of vertical and horizontal vibrations on the human body. *Med Eng Phys* 16: 151
- Rajnicek, A. M., K. R. Robinson, and C. D. McCaig (1998) The direction of neurite growth in a weak DC electric field depends on the substratum: Contributions of adhesivity and net surface charge. *Developmental Biology* 203: 412
- Rashevsky, N. (1960) *Mathematical Biophysics, Physico-Mathematical Foundations of Biology*. Dover Publications, New York
- Raudino, A., and D. Mauzerall (1986) Dielectric properties of the polar head group region of zwitterionic lipid bilayers. *Biophys J* 50: 441
- Rayner, J. M. V., and R. J. Wootton (1991) *Biomechanics of Evolution*. Cambridge University Press, Cambridge
- Reich, J. G., and E. E. Selkov (1981) *Energy metabolism of the cell*. Academic Press, London
- Reilly, J. P. (1998) *Applied Bioelectricity. From Electrical Stimulation to Electropathology*. Springer, New York
- Reiter, R. (1992) *Phenomena in Atmospheric and Environmental Electricity*. Elsevier, Amsterdam

- Renger, G. (1994) Biologische Wasserspaltung durch Sonnenlicht im Photosyntheseapparat. *Chemie in unserer Zeit* 28: 118
- Renshaw, E. (1990) *Modelling Biological Populations in Space and Time*. Cambridge University Press, Cambridge
- Repacholi, M. H. (1998) Low-level exposure to radiofrequency electromagnetic fields: Health effects and research needs. *Bioelectromagnetics* 19: 1
- Repacholi, M. H., and B. Greenebaum (1999) Interaction of static and extremely low frequency electric and magnetic fields with living systems: Health effects and research needs. *Bioelectromagnetics* 20: 133
- Riu, P. J., J. Rosell, R. Bragós, and O. Casas (Eds.) (1999) *Electrical Bioimpedance Methods*. Ann NY Acad Sci 873: 1
- Roederer, J. G. (1979) *Introduction to the Physics and Psychophysics of Music*. (2nd edn.) Springer, New York
- Rowbottom, M., and Ch. Susskind (1984) *Electricity and Medicine. History of their Interaction*. San Francisco Press Inc., San Francisco
- Rowe, G. W. (1994) *Theoretical Models in Biology. The Origin of Life, the Immune System, and the Brain*. Clarendon Press, Oxford
- Rumbler, L. (1898) Physikalische Analyse der Lebenserscheinungen der Zelle. *Arch Entw Mech* 7: 103
- Schienze, A., R. Stark, B. Walter, and D. Vaitl (1997) Effects of low-frequency magnetic fields on electrocortical activity in humans: A sferics simulation study. *Intern J Neuroscience* 90: 21
- Schmid-Schönbein, G. W., S. L.-Y. Woo, and B. W. Zweifach (1986) *Frontiers in Biomechanics*. Springer, New York
- Schneck, D. J. (Ed.) (1980) *Biofluid Mechanics*. Plenum Press, New York
- Schrödinger, E. (1944) *What is Life?* Cambridge University Press, London
- Schwan, H. (1957) Electrical properties of tissue and cell suspensions. *Adv Biol and Med Phys* 5: 147
- Segel, I. H. (1993) *Enzyme Kinetics. Behaviour and Analysis of Rapid Equilibrium and Steady-State Enzyme Systems*. John Wiley & Sons Inc., New York
- Seidel, H., and R. Heide (1986) Long-term effects of whole-body vibration: A critical survey of the literature. *Int Arch Occup Environ Health* 58: 1
- Shannon, C. E., and W. Weaver (1962) *The Mathematical Theory of Communication*. University of Illinois Press, Urbana
- Shipley, R. A., and R. E. Clark (1972) *Tracer Methods for in vivo Kinetics*. Academic Press, New York
- Sigworth, F. J. (1994) Voltage gating ion channels. *Quarterly Rev Biophysics* 27: 1
- Silver, B. L. (1985) *The Physical Chemistry of Membranes*. Allen and Unwin, Boston
- Singer, S. J., and G. L. Nicolson (1972) The fluid mosaik model of the structure of cell membranes. *Science* 175: 720
- Skalak, R., and Shu Chien (Eds.) (1987) *Handbook of Bioengineering*. McGraw-Hill Book Company, New York
- Slivinsky, G. G., W. C. Hymer, J. Bauer, and D. R. Morrison (1997) Cellular electrophoretic mobility data. A first approach to a database. *Electrophoresis* 18: 1109
- Solomon, A. K., M. R. Toon, and J. A. Dix (1986) Osmotic properties of human red cells. *J Membrane Biol* 91: 259
- Spanner, D. C. (1954) The active transport of water under temperature gradients. *Symp Soc Exp Biol* 8: 76
- Stanier, M. W., L. E. Mount, and J. Bligh (1984) *Energy balance and temperature regulation. Cambridge Texts in the Physiological Science* 4, Cambridge University Press, Cambridge
- Stein, W. D. (1986) *Transport And Diffusion Across Cell Membrane*. Academic Press, San Diego

- Stein, W. D. (1990) Channels, Carriers, and Pumps. An Introduction to Membrane Transport. Academic Press, San Diego
- Strait, B. J., and T. G. Dewey (1996) The Shannon information entropy of protein sequences. *Biophys J* 71: 148
- Solomon, A. K. (1960) Compartmental methods of kinetic analysis. In: Comar, C. L., and Bronner (Eds.): Mineral Metabolism. Academic Press, New York
- Sukhorukov, V. L., H. Mussauer, and U. Zimmermann (1998) The effect of electrical deformation forces on the electroporabilization of erythrocyte membranes in low- and high-conductivity media. *J Membrane Biol* 163: 235
- Syganov, A., and E. vonKlitzing (1999) (In)validity of the constant field and constant currents assumptions in theories of ion transport. *Biophys J* 76: 768
- Szent-Györgyi, A. (1968) Bioelectronics. A Study in Cellular Regulations, Defense, and Cancer. Academic Press, New York
- Takashima, S. (1963) Dielectric dispersion of DNA. *J Mol Biol* 7: 455
- Takashima, S., A. Casaleggio, F. Giulliana, M. Morando, P. Arrigo, and S. Ridella (1986) Study of bound water of poly-adenine using high frequency dielectric measurements. *Biophys J* 49: 1003
- Talbot, L., and S. A. Berger (1974) Fluid-mechanical aspects of the human circulation. *American Scientist* 62: 671
- Tasaka, M., K. Hanaoka, and Y. Kurosawa (1975) Thermal membrane potential through charged membranes in electrolyte solutions. *Biochem Chem* 3: 331
- Thom, R. (1969) Topological Models in Biology. *Topology* 8: 313
- Thompson, and W. D'Arcy (1966) On Growth and Form. University Press, Cambridge
- Tributsch, H., and L. Pohlmann (1993) Co-operative electron transfer: Superconduction in biomolecular structures? *J Theor Biol* 165: 225
- Vainshtein, M. B., N. E. Suzina, E. B. Kudryashova, E. V. Ariskina, and V. V. Sorokin (1998) On the diversity of magnetotactic bacteria. *Microbiology* 67: 670
- Varma, R., and J. R. Selman (1991) Characterization of Electrodes and Electrochemical Processes. John Wiley & Sons Inc., New York
- Verkman, A. S., and R. J. Alpern (1987) Kinetic transport model for cellular regulation of pH and soluble concentration in the renal proximal tubule. *Biophys J* 561: 533
- Videler, J. (1993) Fish Swimming. Chapman and Hall, London
- Vogel, St. (1994) Life in Moving Fluids. The Physical Biology of Flow. (2nd edn.) Princeton Univ. Press, New Jersey
- Voigt, A., and E. Donath (1989) Cell surface electrostatics and electrokinetics. In: Glaser, R., and D. Gingell (Eds.): Biophysics of the Cell Surface. Springer Series of Biophysics. Springer, Berlin 5: 75
- Volkenstein, M. V. (1994) Physical Approaches to Biological Evolution. Springer, Berlin
- Wassermann, E. M. (1998) Risk and safety of repetitive transcranial magnetic stimulation: Report and suggested guidelines from the international workshop on the safety of repetitive transcranial magnetic stimulation, June 5–7, 1996. *Electroencephalogr Clin Neurophysiol* 108: 1
- Waterman, T. H., and H. J. Morowitz (Eds.) (1965) Theoretical and Mathematical Biology. Blaisdell Publ. Comp., New York
- Webb, P. W., and D. Weihs (Eds.) (1983) Fish – Biomechanics. Praeger, New York
- Westerhoff, H. V., and K. van Dam (1987) Thermodynamics and Control of Biological Free-energy Transduction. Elsevier Science Publ., New York
- Wiggins, P. M. (1990) Role of water in biological processes. *Microbiological Reviews* 54: 432
- Wilson, B. W., R. G. Stevens, and L. E. Anderson (Eds.) (1990) Extremely Low Frequency Electromagnetic Fields: The Question of Cancer. Battelle Press, Columbus
- Wilson, Ch. E. (1989) Noise Control. Measurement, Analysis, and Control of Sound and Vibration. Harper and Row Publ., New York

- Wiltshcko, R., and W. Wiltshcko (1995) *Magnetic Orientation in Animals*. Springer, Berlin
- Yeagle, P. (Ed.) (1992) *The Structure of Biological Membranes*. CRC Inc., Boca Raton
- Yockey, H. P. (1992) *Information Theory and Molecular Biology*. Cambridge University Press, Cambridge
- Zeeman, E. C. (1976) Catastrophe theory. *Scientific American* 234: 65
- Zhao, M., A. Dick, J. V. Forrester, and C. D. McCaig (1999) Electric field-directed cell motility involves up-regulated expression and asymmetric redistribution of the epidermal growth factor receptors and is enhanced by fibronectin and laminin. *Molecular Biology of the Cell* 10: 1259
- Ziemann, F., J. Radler, and E. Sackmann (1994) Local measurements of viscoelastic moduli of entangled actin networks using an oscillating magnetic bead micro-rheometer. *Biophys J* 66: 2210
- Zimmer, K. G. (1961) *Studies of Quantitative Radiation Biology*. Oliver and Boyd, Edinburgh
- Zimmerberg, J., and V. A. Parsegian (1986) Polymer inaccessible volume changes during opening and closing of a voltage-dependent ionic channel. *Nature* 323: 36
- Zimmermann, U. (1978) Physics of turgor and osmoregulation. *Ann Rev Plant Physiol* 29: 121
- Zimmermann, U. (1989) Water relations of plant cells pressure probe techniques. *Methods in Enzymology* 179: 338
- Zimmermann, U., and G. A. Neil (1996) *Electromanipulation of Cells*. CRC Inc., Boca Raton
- Zimmermann, U., and E. Steudle (1978) Physical aspects of water relation of plant cells. *Adv Bot Res* 6: 45

Subject Index

Bold-faced numbers indicate pages, where the subjects are defined or explained explicitly.

- A**
 α -helix
– structure 74, 75
– dipole moment 56
 α -radiation 276
Absorption
– of ionizing radiation 267, 268
– of light 18
Acceptor molecule 20
Acid-base equilibrium 77, 78
Action potential 175, 177–181, 184, 299
Activated state 41–43
Activation energy 39–45
Active transport 131, 167–174, 178
Activity
– chemical 113, 114
– coefficient 114
– measurement 144
Adenosine triphosphate, see: ATP
Admittance 188
Adsorption at surfaces 92
Advancement number of a chemical reaction 131, 312
Affinity, chemical 131
Agglutination 72
Air
– electric conductivity 262
– ions 262, 263
– kinematic viscosity 222
– thermal conductivity 238
ALARA-principle 286
Allometric relations 231, 324, 325
Allosteric interactions 315
Alveoli, surface tension 82
Amino acids
– isoelectric point 78, 79
– sensibility to radiation 281, 282
– sequences in proteins 30, 73–75
– transport 167
Amoeba, movement 83
Ampholytes 77–80
Amplifications in biological systems 235, 264
Ampullae of Lorenzini 264
Angular dependence of hearing sensibility 245
Anisotrope systems 5, 62, 88, 121, 214
Antifield rotation 199, 201
Anti-oxidant defense system 22
Antiport 167
Aperiodic
– limit 295, 296, 303
– structure 34
Arrhenius
– equation 39, 236
– plot 39, 40
Ascorbic acid 22
Atmospherics (=Spherics) 270
Atomic radius, effective 61, 72
ATP
– as energy accumulator 20, 25
– complexation with Ca, Mg 17
– synthetase 25
ATPases 20, 147, 167
Auto-catalysis 328
Avogadro number 78
Axon, propagation of impulses 181
Azimudal quantum number 7, 8
- B**
 β -radiation 276
 β -sheet of protein structure 74, 75
Bacteria, orientation in magnetic field 258–260
Base dimers 281
Basic restrictions 270
Bat, echolocation 251–253
Beat, first and second order 245, 246

- Békésy-theory of hearing 247–249
 Bending of a rod 218
 Benzene ring, orbital 11, 12
 Bernard cells 127, 128
 Bernoulli-equation 229
 Bifurcation 299, 326
 Bilayer couple mechanism 90
 Bilayer of lipids, see: BLM
 Bingham-plastic fluid 206
 Bioavailability of drugs 318
 Biochemical reaction kinetics 312–317
 Bioelectricity 1
 Biomechanics 2, 203, 204
 Bionics 2, 3, 230
 Biophysics
 – subject 1
 – history 1–3
 Biopotential (according to Szent-Györgyi) 21, 22
 Birds
 – area of wings 231, 232
 – echolocation 253
 – flying mechanisms 230–233
 – Reynolds numbers 223
 Bit (binary digit) 30
 BLM (bimolecular lipid membrane)
 – electrical conductivity 86
 – formation 85, 96
 Blood
 – non-Newtonian properties 208, 209, 224
 – passive electrical properties 191
 – pH-measurement 146
 – streaming properties 224–227
 – viscosity 208, 209
 Blood circulation
 – critical Reynolds numbers 227
 – diagnosis by ultrasound 227
 – entrance effect 224, 225
 – Farhaeus-Lindqvist effect 224, 225
 – microcirculation 224
 – Moens-Korteweg equation 225
 – pulsed flow 225
 – regulation 226
 – turbulences 227, 228
 – velocities 227
 Blood vessels
 – branching 226
 – diameter 225–227
 – eddy currents 269
 – Reynolds number 227
 – smooth muscles 226, 227
 – viscoelasticity 25, 226
 Bohr's radius 61, 72
 Boltzmann constant 28, 30
 Boltzmann's distribution law 38, 39, 41, 65, 143
 Boltzmann's equation for entropy 26–28, 36
 Bonding distance 9–13
 Bonds
 – angle of covalent bonds 10, 11
 – by charge-transfer-processes 21
 – by hydrogen bridges 17, 18, 44, 58, 73
 – by Van-der-Waals force 69–73, 94
 – coordinative 15–17
 – covalent 9–13, 16
 – distance of 9–13
 – energy of 12–15
 – hydrophobic 63, 64, 73
 – intermolecular 17, 67–73
 – weak 17
 Bone
 – healing by electric field application 271, 273
 – mechanical stability 215, 216
 – module of elasticity 213
 – passive electrical properties 191
 – piezoelectric properties 23, 184, 185
 – streaming potential 23, 98, 186
 Boundary layers 154, 222
 Brownian motion 48–51, 73, 74, 261
 Buffer properties 78, 79

C
 Cable theory of propagation of nerve impulse 180, 181
 Caisson disease 239
 Calcium, role in processes of cellular control 72, 73, 171
 Calorimetric measurements 126
 Cancer cells, see Tumor cells
 Capacitor equation 57, 175
 Capacity, electric
 – definition 99, 189
 – specific of cell membrane 86, 99, 100, 175, 196
 Capacitor equation 99, 175, 189
 Capillary viscosimeter 207, 208
 Carbon, peculiarity of valence 10
 Carcinogeneses of radiation 282–286
 Cardiac pace maker, interference with electromagnetic fields 261, 262
 Carotenoids 24
 Cartilage
 – piezoelectric properties 185

- viscoelastic properties 209
- Catastrophe theory 298, 299, 331
- Cauchy condition 108, 109
- Cavitation 254, 255
- Cell
 - deformation by electric fields 200
 - deformation by shear stress 205, 209
 - electrophoresis 95-98
 - fusion 198, 202
 - galvanotaxis 185, 265
 - manipulation by electric fields 197-203
 - proliferation, correlation with membrane potential 176
 - separation 203, 203, 261
 - swelling 90, 140, 141
 - viscoelastic properties 209
 - volume regulation 140, 172
- Cell wall
 - ion-exchange properties 98
 - mechanical properties 140
- Cell membrane
 - action potential 175
 - active transport 167-174
 - anisotropy, magnetic 257
 - capacity, specific 86, 99, 100, 175, 196
 - conductivity, (*see* Ohms resistance)
 - dielectric constant 99, 100
 - diffusion potential 143, 163-166, 171-173
 - elasticity modulus 88, 215
 - electric field strength 54, 102, 103, 177
 - electric potential 101-103, 143, 144, 171-177, 193-197, 201, 274
 - electrostatic structure 99-103
 - excitability 177-181
 - fluid-mosaic-model 86
 - fluidity 87
 - functions in general 80
 - induction of electrical potential 193-197
 - ion transport 166-170
 - magnetic anisotropy 257
 - mechanical properties 87-91, 177, 215
 - Ohm's resistance 99, 174, 177, 193-197
 - properties as capacitor 99, 100, 174, 175, 196, 197
 - radiation influence 282, 283
 - structure 83-87, 100
 - surface charge density 100, 101
 - surface charges 86
 - surface dynamics 88, 89
 - surface tension 83
 - temperature gradient 170
 - transport properties 166-179
- viscosity 87, 88, 210
- Center of gravity of macromolecules 46, 47
- Chaos theory 300, 317
- Chaotropic ions 61
- Chaperons 75, 76
- Charge
 - electric 52
 - density 65
 - fluctuations 69
- Charge-transfer
 - complex 21
 - forces 21
- Chelation 15
- Chemical reaction
 - thermodynamic approaches 129-132
 - kinetics 321-327
- Chloride distribution 144, 147, 173
- Chlorophyll 16, 23-25
- Chloroplasts 18, 23-25, 141
- Chromosome
 - aberrations caused by irradiation 282
 - orientation in magnetic field 257
- Clearance constant 318
- Clusters of water structure 59, 60
- Coagulation of particles 72, 73
- Cochlea, viscoelastic properties 247, 248
- Coefficient
 - of activity 113, 114
 - of friction 119
 - of mobility 118
 - of mortality 320
 - of proliferation 320
 - of work 110
- Cofield rotation 199-201
- Coherent oscillations 274
- Collagen, structure 75
- Collisions, thermic 41
- Colloid osmotic pressure 138
- Command variable 301
- Compartment
 - analysis 305-312
 - definition 305
- Compass orientation 259-261
- Complex bonds 15
- Complexation 15-17
- Compton effect 277
- Conductivity
 - electrical 189
 - electrical of human body 263, 269
 - loops in the body 269
 - hydraulic 157
 - thermal 237

- Conformational changes of molecular structure 40
- Conjugated flux-force relation 120
- Conservative System 117
- Constant field theory (Goldman) 160
- Continuum Physics 3, 209
- Control deviation 301
- Controlled variable 301
- Controller, types of 300–303
- Convection, thermal 237
- Coordinate of work 109, 110
- Coordinative bonds 15–17
- Cosmic radiation 186
- Cosmotropic ions 61
- Cotransport 167, 172
- Coulomb's law 13, 14, 53, 67
- Coupling coefficient 120
- Covalent bonds 9–13, 46
- Cryopreservation 238, 239
- Cryoscopy 138
- Crystal radius of ions 60, 61
- Cup catastrophe 289, 299, 317
- Curie-Prigogine principle 121
- Current density
- limit in the body 271, 272
 - near cell surface 184
 - near earth surface 262
- Cyanide 16
- Cybernetics 3, 291, 300–304
- Cytochrome 16, 22, 25
- Cytocrite 206
- Cytolysis 90
- Cytoplasm
- electrical conductivity 193, 198, 201
 - viscosity 51, 209
- Cytoskeleton 86, 88
- D**
- Dark reaction of photosynthesis 24
- Debye
- dispersion 192
 - unit of dipole momentum 56
- Debye-Hückel
- parameter 66, 67, 165
 - theory of electrolytes 64–66, 69
- Decibel (dB) 243
- Deep sea, pressure influence 63, 239, 240
- Degenerated state of electrons 19
- Degree of
- advancement 131
 - freedom 5
- Desoxyribonucleic acid (DNA)
- prebiotic evolution 33, 327, 328
 - aperiodic structure 34
 - content of information 31–33
 - dipole momentum 56
 - double strand breaks 281
 - life span 44–45
 - radiation sensibility 45, 279, 281–283
 - single strand breaks 281
 - structure 48
 - UV-sensibility 281
- Deterministic behavior 5, 6
- Depot lubrication of joints 209
- Diamagnetic properties of biological materials 257
- Diathermy
- caused by electromagnetic fields 267–269
 - caused by ultrasound 255
- Dielectric constant 53
- of cell membrane 99, 100
 - of tissue 191
- Dielectrophoresis 198–203
- Differential
- of work 109, 110
 - total 107, 108
- Differential controller (D-controller) 302
- Differential quotient, partial 107
- Differentiation of biological systems, influence of electric fields 186
- Diffuse double layer 91, 92
- Diffusion
- coefficient 51
 - energy of activation 44
 - lateral in membranes 88, 89, 177, 186, 265
 - layer 154
 - of nonelectrolytes 152–159
 - of ions 159–163
 - potential 143, 163–166, 172–175
- Dilatant fluid 296
- Dipole
- fluctuating 56
 - induced 56, 69
 - interactions 17, 68, 69
 - momentum 12, 55, 79
 - oscillating 20
- Dispersion forces 69, 70
- Dispersion regions of electromagnetic fields 192
- Displacement current 188, 189
- Dissipation function (Rayleigh) 125, 126, 211, 271, 272
- Dissipative structures 33, 34, 106, 125–128, 221

- Disulfide bridges 75
- Divergence (div)-operator 117
- DLVO-theory 69–73
- DNA (*see* Desoxyribonucleic acid)
- Dolphin
 - Reynolds number 223
 - sonar system 253, 254
 - swimming behavior 230
- Donnan
 - equilibrium 146–151, 171
 - osmotic cytolysis 151, 198
 - osmotic pressure 151
 - potential 148–150
 - ratio 148, 149
- Doppler effect
 - in ultrasound diagnosis 254
 - in ultrasound location 253
- Dorn effect 98
- Double bounds 12
- Double layer, electrical 91–99
- Double strand breaks of DNA 281
- E**
- Ear, human
 - anatomical organization 246
 - as frequency filter 246–250
- Echinocytes 90
- Echolocation
 - in bats 251–253
 - in birds 253
 - in dolphins 253–255
 - mechanisms 251–255
- Ecological models 320–324, 329, 330
- Eddy currents in the body 269, 270
- Effective parameters 6, 47, 61, 66, 67, 87, 210
- Einstein equation
 - for diffusion coefficient 51
 - for viscosity of suspensions 205, 206
- Einstein-Smoluchowsky equation of Brownian motion 50, 51
- Eindhoven triangle 187
- Elastin 213, 215
- Elasticity module 88, 211, 218, 226
- Electric break down of cell membranes 198
- Electric charge 52
- Electric field
 - calculation 54
 - field strength 53
 - inhomogeneous 199
 - interaction with biological systems 193–197
 - of natural environment 262
 - trap 202
- Electric organs of fishes 185, 186, 264
- Electric potential 53
- Electrical double layers 91–99, 192
- Electrochemical energy accumulation in cells 171–177
- Electrodes
 - Ag/AgCl/KCl 145, 146
 - ion selective 144–146
 - suspension effect 146
 - tip potential 146
- Electro
 - cardiogram (EKG) 183, 187, 188
 - diffusion 159–163
 - encephalogram (EEG) 183, 186
 - fusion of cells 198
 - injection (=permeabilization) 198
- Electrochemical potential 115
- Electrogenic transport 143, 167, 171–175
- Electrokinetic phenomena 94–99
- Electrolyte flux 159–163
- Electromagnetic field
 - application in therapy 267, 268, 271–275
 - applications in biotechnology 202, 203
 - biological interactions 270–275
 - dosimetry 268, 269
 - intensities in human environment 269, 270
 - limits, *see*: Limits for exposition
 - microdosimetry 272
 - muscle excitations 271
 - of natural origin 270
 - penetration in the body 267–269
 - screening 269
 - spectrum 256, 266
 - window effects 274
- Electromotive force (EMF) 118, 144
- Electron
 - acceptor 20
 - capture 280
 - charge 14
 - clouds 10
 - hydrated 280
 - orbitals 6–13
 - radiation 277
 - spin resonance, *see*: ESR
 - spin 7–9
 - transfer 21, 22
 - transfer potential 22
 - transition, time constants 19, 20
 - volt (eV) 8

- Electronegativity of the elements 12, 17
 Electroneutrality condition 18, 163, 168
 Electroosmosis 95, 98, 265
 Electroporation 198
 Electrophoresis 95–97, 265
 Electrophoretic potential 96
 Electroplax 186
 Electoreception 186, 264, 265
 Electrorotation 198–202
 Electrostatic field
 – around electric fishes 264, 265
 – interaction with body hairs 264
 – of cell membrane 54, 102, 103, 177
 – of natural environment 263, 264
 – receptors 264, 265
 – threshold of reception 264
 Electrostatic unit (eu) 52
 Embryo
 – morphogenic role of electric fields 184, 186
 – sensibility to radiation 283
 Endothermic reaction 131
 Energy
 – acceptor 20
 – assimilation 18
 – chemical 20
 – conservation 36
 – definition 109
 – distribution according to Boltzmann 38, 39, 41, 65
 – electrostatic 53
 – Gibbs, free 107, 111, 130
 – Helmholtz, free 111, 112
 – internal 109
 – kinetic 36
 – of activation 39–45
 – of covalent bonds 12
 – of electrons 8–10
 – of hydrogen bonds 17
 – of ionic bonds 13–15
 – of ionization 8
 – of thermic noise 18
 – of photons 18
 – supplying molecules 20
 – transfer by inductive resonance 20
 – transfer by radiation 20
 – transfer by charged carriers 20, 21
 – units 8
 Enthalpy 111
 Entrance effect 224–225
 Entropy 109
 – account of a system 122
 – connection with information 29–33
 – driven process 62–64, 132
 – elasticity (= rubber elasticity) 48, 213
 – flux 122, 328
 – in statistical thermodynamics 28
 – of activation 43
 – production 122
 Environmental biophysics 1, 235, 236
 Enzyme
 – activation energy 43
 – membrane bound 94
 – reaction kinetics 312–317
 – sensitivity to radiation 283
 – substrate complex 41, 43, 44
 Equifinality 308, 310
 Equilibrium
 – global 125
 – locale 124, 125
 – thermodynamic 36–38, 123–125, 128, 142, 142, 147
 Equilibrium constant 130
 Equipotential lines 53, 54, 194
 Equivalence circuit, electric, for biological systems 188–193
 Equivalence dose of ionizing radiation 277
 Erythrocytes
 – aggregation 208
 – deformability 208, 209
 – ionic conditions 150, 151
 – life span 128, 129
 – magnetic susceptibility 257
 – membrane potential 174
 – osmotic properties 140, 141, 151
 – pressure resistance 140
 – reflection coefficient 139
 – surface charge 86, 97, 101
 – swelling 90
 ESR (electron spin resonance) 87
 Evolution
 – models 327–330
 – prebiotic 327
 Exchange flux 157
 Excitation of electrons 18, 19
 Excitons 24
 Exotherm reaction 131
 Extensive parameters 107
- F**
 Falling sphere viscosimeter 207
 Faraday constant 52, 115
 Farhaeus-Lindqvist effect 224, 225
 Fatty acids 83–85

- Ferromagnetic properties 257–261
 Fick's laws of diffusion 51, 153, 162, 326
 Field lines, see: Equipotential lines
 Filtration coefficient, mechanic 157
 Fish
 – electric organs 185, 186
 – electric sensibility 264
 – Reynolds numbers 223
 – swimming behavior 230
 Flickering clusters of water 59
 Flight mechanisms 230–233
 Flow
 – in a tube 204, 222–225
 – laminar 204, 219–223
 – pulsating 225
 – turbulent 220–223, 226, 227
 Fluctuating dipoles 56
 Fluctuations, see: Thermic noise
 Fluid-mosaic structure of cell membrane 86, 177, 184
 Fluidity 205
 Fluorescence
 – life span 19
 – markers 87, 210, 265
 – role in energy transfer 20
 Flux
 – conservative 117
 – definition 116, 117, 152
 – electrogen 143, 167, 171–175
 – formal approaches 118–120
 – matrix 120
 – of non charged molecules 152–158
 – rheogen 167, 171–175, 184, 186
 – scalar 121
 – unidirectional 161, 306
 Follow-up control circuit 301
 Force, generalized 115–118
 Form pressure drug (mechanical) 228
 Free radicals, see: Radicals
 Frequency
 – analysis of hearing 246–249
 – modulation of orientation sound 252
 "Frozen" dissipative structures 128

G
 γ -radiation 276
 G-value (100eV-gain) 280
 Galvanotaxis 185, 265
 Galvanotropism 265
 Gap junctions, electrical conductivity 271
 Gauss
 – plane of imaginary numbers 190
 – statistics 46
 Genetic effects or radiation 282
 Geophysical properties of environment 258, 259, 264, 265
 Giant axon 177
 Gibbs free energy 42, 76, 107, 111, 130
 Gibbs fundamental equation 109–113, 129
 Gibbs-Duhem equation 111
 Glansdorff-Prigogine principle 127
 Glucose
 – reflection coefficient 139
 – transport 172
 Glycocalyx (=surface coat) 86, 100
 Glycoproteins 86, 100
 Goldman equation
 – of ionic flux 160
 – of liquid junction potential 164, 173, 174
 Gouy-Chapman theory of the electric double layer 96, 97
 Gradient (grad) 54
 Grans (in thylakoids) 24
 Gray (Gy) 277
 Gray's paradox 230
 Grothaus-Draper principle 18, 235
 Growth
 – factors 172, 325, 326
 – influence of electromagnetic fields 184, 185, 265
 – models of 320, 324–327

H
 Hagen-Poiseuille law 224
 Hair cells in inner ear 249, 250
 Half life 311
 Hamaker constant 69
 Health protection, see: Limits for exposition by
 Hearing
 – angular resolution 245
 – frequency dependence 243
 – mechanistic theories 246–250
 Heart
 – generation of electric fields 187, 188
 – muscle fiber, action potential 178, 187
 Heat 109–111
 – conductivity 236–238
 – flux 236–238
 – production of a chemical reaction 113, 131
 Heisenberg's uncertainty principle 7
 Helmholtz
 – double layer 91, 92, 97

- potential 96
- theory of hearing 247
- Hematocrite 206
- Hem-complex 16, 17
- Hemoglobin 16
 - buffer capacity 79, 80, 150
 - concentration in erythrocytes 141, 150
 - isoelectric point 79, 150
 - osmotic coefficient 141
- Hemolysis 151, 209
- Hemorheology 203, 220
- Henderson-Hasselbalch equation 78
- Hierarchy
 - of structural levels 182, 183
 - of time constants 128, 129, 317
- High-energy compounds 20
- High spin state of a molecule 15-17
- „Hit“ process of radiation 286-289
- Hodgkin-Huxley theory of nerve excitation 178-180
- Holography 333
- Homeostatic regulation 171
- Hooke's law 211, 226
- Hot spots in diathermy 272
- Hovering flight 232
- Humming bird, mechanism of flight 231, 232
- Hybrid orbitals 11
- Hydration 60-62
- Hydraulic conductivity 157
- Hydrodynamic radius, see: Stokes' radius
- Hydrogen
 - bonds 17, 18, 44, 58, 73
 - electrode 21
 - orbitals 8, 9
- hydrophobic
 - bounds 63, 64 73
 - regions 83-85
- Hypercycle (Eigen) 328, 329
- I**
- Ice structure of water 59
- Ideal solutions 114, 137, 143, 159
- Immune-magnetic cell separation 261
- Impact pressure 229
- Impedance
 - electric 188, 195, 196, 269
 - mechanic 213, 241, 242, 254, 255
- Impedance transformer (mechanical) 246, 250
- Impulse compression method in sonar systems 253
- Indifferent state 124
- Induction, electric 269
- Influence, electric in the body 263, 264
- Information
 - definition (Shannon) 30
 - of biomacromolecules 31-33
 - semantic 31
 - syntactic 31
 - units 30
- Infrasound 240, 250, 251
- Input-output law of system behavior 292
- Insects
 - flight mechanism 232
 - magnetic orientation 259
 - power requirement 214
 - resistance to radiation 285
 - Reynolds number 223
 - structural proteins 214
 - wing area 231
 - wing loading 231, 232
- Integral controller (I-controller) 301, 304
- Interactions
 - allosteric 315
 - dipole-dipole 17, 68, 69
 - electrostatic 13, 14
 - Van der Waals 69-73, 94
- Intercellular clefts
 - ion transport 167, 168
 - electric field 186
- Interfacial
 - phenomena 80-83
 - tension 81, 82
- Intermolecular interactions 67-73
- Intersystem transitions 19
- Intrinsic
 - pK 80
 - viscosity 205
- Ionic
 - bounds 13, 14
 - cloud 65, 66, 192
 - locale milieu 94, 103
 - pumps 20, 147, 171-175
 - strength 66
 - transport 166-170
- Ionization
 - density 289
 - energy 8
- Ionophores 144, 147
- Iron, complexation 16, 17
- Isoelectric point 79, 80
- Isomerases 75, 76
- Isomorphism 291, 298

Isotonic solution 140, 151

Isotopy effect 306

K

Kidney cell, ion transport 168

Kinetic energy 36

– of electrons 8

Kinetosis 242

Kirchhoff's law 193

L

Laminar flow 204, 220–223

Laminar profile 222, 229

Laplace equation of pressure 82, 226

Laser flash photometry 25

Lateral movement of membrane
compounds 88, 89, 177, 186, 265

LD unit 284, 285, 287

Leak flux 166

Le Chatelier principle 63, 85, 240

Lethal dose of ionizing radiation, see: LD

Life

– origin, theories 327, 330
– peculiarities of emergence 10, 58

Life span

– of fluorescence excitations 19
– of macromolecules 44, 45
– of molecular excitation states 19
– of phosphorescence 19
– of water clusters 59

Ligand field theory 15–17

Light

– absorption 18, 19
– quantum energy 18, 19, 23

Limit

– cycle 297
– of reversible deformation 210

Limiting case

– aperiodic 295, 296, 303
– periodic 286

Limits for electric current in the body 271,
272

Limits for exposition by

– HF-fields 272
– infrasound 251
– ionizing radiation 285, 286
– LF-fields 271, 272
– noise 244
– static magnetic fields 261
– ultrasound 255

Linear force-velocity approach 118

Lipids 85

– behavior at boundaries 84–86

– dielectric constant 100

– lateral diffusion 88

– magnetic anisotropy 261

– phase transitions 40, 91, 102, 261

– shape factor 90

– transversal distribution, flip-flop 87, 88

Liposomes 85

Liquid junction potential, see: Diffusion
potential

Liver, passive electric properties 191

Local concentrations 94, 103, 143, 154, 165,
177

Logistic function 321

London's dispersion forces 69, 70

Long-range interaction forces 68–71

Lorentz force 260, 261

Lorenzini ampullae 264

Loschmidt number (= Avogadro
number) 14

Low spin state 16

Lung

– air flow 225
– diagnostics by ferromagnetic
particles 260
– surfactants 82

Lyso-phospholipids 91

M

Magnetic

– anisotropy 257
– field strength 256, 257
– flow density 256
– mapping 259
– parameters of biological objects 257
– permeability 257
– quantum number 7, 8
– storms 259
– susceptibility 257
– timing 259

Magnetic field

– diagnostic methods 259–260
– influence on biological objects 258–262
– on earth surface 258, 259
– orientation of organisms 259

Magnetites 257, 274

Magneto

– encephalogram 259
– cardiogram 259

Magnetophoresis 261

Magnetophosphenes 261

Magnetosomes 258

- Magnetostatic bacteria 257, 258
- Magnetotaxis 258
- Malthus law of exponential growth 320, 322
- Maximal possible dose, see: Limits or exposure
- Maxwell
 - demon 31, 32, 51, 122
 - element of viscoelasticity 211, 212
 - equations of electromagnetic fields 182, 183, 266
 - equation for velocity distribution 37
- Maxwell-Wagner dispersion 192
- Membrane, (see also: cell membrane)
 - as boundary 80
 - bending 89
 - biological structure 85, 89, 90, 100
 - capacity 99–100
 - electric field strength 54
 - fluidity 87
 - formation 83–86
 - magnetic anisotropy 261
 - mechanical properties 87–91
 - models 85
 - phase transitions 91, 102
 - potential 143, 171–182, 193–197, 274
 - resistance, electrical 99, 177, 189, 193–197
 - self organization 83–86
 - semipermeable 133, 142, 143
 - temperature gradient 170
 - transport properties 166–170
 - viscosity 87–88, 210
- Metastability of systems 124, 125
- Michaelis-Menten equation 314, 315, 319
- Micro electrodes 145, 146, 177
- Microphonic potential of ear 246
- Microphysical behavior 5, 6
- Microtubules, pressure dependence 63
- Microwave therapy 266, 267, 273
- Mitochondria
 - ATP-synthesis 158
 - swelling 141
- Mitosis
 - influence of radiation 282, 283
 - time constants 128, 129
- Mobil telephone, electromagnetic fields 266, 269
- Models
 - of compartment systems 305–312
 - of drug distribution in the body 318
 - of evolution 327–330
 - of excitation 178–180
 - of locomotion 215
 - of morphogenesis 324–327
 - of neuronal networks 330–333
 - of propagation 320–324
- Moens-Korteweg equation 225
- Molecular
 - biophysics, peculiarities 5, 6
 - movement, thermic 45–51
 - orbitals 9–13
 - radius, effective 47, 48
 - volume, partial 63
- Mole fraction 114
- Molten-globe structure of proteins 76
- Monolayer 84
- Morphogenesis
 - models 324–327
 - role of internal electric fields 186, 325
- Multi-compartment system 307–309
- Multi-hit processes of radiation effects 287, 288
- Muscle
 - excitability by AC current 271
 - heat conduction 238
 - magnetic susceptibility 257
 - maximal power 214
 - passive electrical properties 191
- Mutation
 - doubling dose of radiation 285
 - induced by radiation 45, 235, 236, 282, 283
 - role in evolution 329
 - target theoretical approach 288, 289
- Myoglobin 16
- N**
- Nabla-Operator 54
- NADPH (nicotinamid-adenin-dinucleotid phosphate) 22, 24
- Na-K-ATPase 40, 147, 166–168
- Negentropy 30
- Nernst equation 141–143, 173, 174
- Nernst-Planck equation 160, 182, 183
- Nerve
 - external electric field 184, 185
 - Hodgkin-Huxley theory 178–180
 - impulse 178–181, 184
 - regeneration 271, 273
 - refractory period 330
 - sensibility to radiation 283
 - stimulation 177, 178, 271
 - unmyelinated 180
- Net flux 161, 306

- Network theory 393
 Neuraminic acid (=sialic acid) 86, 100
 Neuronal networks 330–333
 Neuston 81
 Neutron radiation 276
 Newtonian fluid 206
 Nitrogen, valence electrons 10
 Nodes of Ranvier 181
 Noise, acoustic
 – legal definition 244
 – Measurement 244, 245
 – protection 44
 Noise, thermic 18, 41, 44, 45, 59, 65, 76, 235, 250
 Non-Newtonian viscosity 206
 Non-radiant energy transfer 20
 Nonequilibrium state 18, 122–125
 Nonlinear systems 106, 221, 321–324
 Nuclear Magnetic Imaging (NMI) 260
 Nuclear Magnetic Resonance (NMR) 260, 261

O
 Ohm's
 – law 118, 189
 – heating of cells in electric fields 198
 Ontogenesis, entropy production 126
 Onsager coefficients 120
 Onsagers law on reciprocal relations 120
 Optimization biological systems 216, 227, 229, 230, 236
 Orbitals of electrons 7–13
 Order
 – of biological systems 34
 – of differential equations 293
 – of chemical reactions 312, 313
 – of systems behavior 293, 294
 Origin of life 327, 330
 Orientation of organisms by
 – electric field 186
 – magnetic field 259
 – ultrasound 251–254
 Oscillations
 – as a kind of dissipative structure 33, 128
 – calculations by nonlinear flux approaches 295, 296, 323
 – damped 296, 297, 303
 – enforced 241
 – mechanical 240, 241
 – of controllers 303
 Osmolarity (osmol) 137
 Osmoregulation 171, 172

 Osmotic pressure 133–141
 – coefficient 137, 138
 – definition 135
 – measurement 138
 Osmotic stability of cells 140, 141
 Ostwald viscosimeter 207
 Oxygen effect of radiation damage 280

P
 Paramagnetic properties 257
 Parameters
 – effective 6, 47, 61, 66, 67, 87
 – extensive 107
 – intensive 107
 – phenomenological 87, 105
 Parameter space 296–299
 Parameter plot of a function 297
 Partial differentiation 107
 Partial mole volume 63, 108, 109, 135, 136
 Passive electrical properties 187–192
 Patch clamp 172
 Pauli principle 7, 9, 10, 15, 19
 PEMF (pulsed electromagnetic fields) 273
 Periodic limiting case 286
 Permeability coefficient 155
 Permittivity 53
 – constant 8, 52
 Pfaffian differential equation 108, 111
 Pfeffer's osmometer 133, 157
 pH
 – electrodes 144–146
 – regulation in cells 172
 Phagocytosis 83, 198
 Pharmacokinetic models 318
 Phase space, n-dimensional 296
 Phase transitions 40, 91, 102, 132
 Phenomenological
 – flux equations 118–120
 – parameters 87, 105
 Phon 243, 244
 Phosphatidylserine 101
 Phospholipids, see: Lipids
 Phosphorescence, life span 19
 Photo
 – effect 277
 – electrons 277
 Photodinesis 26
 Photomorphogenesis 26
 Photonastic movements 26
 Photosynthesis
 – efficiency 23
 – mechanism 23–26

- spectral dependence 18, 19
- Phototaxis 26
- Phototropism 26
- Piezoelectric properties of biological materials 23, 184–186
- pK
 - intrinsic 80
 - value 78
- Planck's constant 8
- Plane power density 269
- Plastoquinone 25
- Poise (P) 205
- Poisson equation 65
- Poisson-Boltzmann equation 65, 66, 91, 92, 101, 182, 183
- Polar moment of area 219
- Polarization of
 - bonds 12–14
 - cells 199–201
 - molecules 56
- Population dynamics 320–324
- Porphyrin 15–17
- Potential
 - chemical 110–115, 113
 - electric, definition 53
 - electrochemical 115
- Power lines, electromagnetic field 266, 269
- Prebiotic evolution 33, 35, 327, 328
- Pressure, hydrostatic in cells 138–140
- Pressure, influence
 - on supramolecular structures 63
 - on biological systems 239, 240
- Prigogine principle of minimal entropy production 125, 127, 225, 309
- Primary quantum number 7
- Principal quantum number 7, 8
- Principles of Thermodynamics
 - first 105, 109
 - second 30, 33, 62, 121, 125, 213
- Probability
 - mathematical 29, 179, 180
 - of presence of electrons 7
 - thermodynamic 28, 29, 36
- Proportional controller (P-controller) 301, 302
- Proportional deviation 301, 302
- Proteins
 - β -leaflet 74, 75
 - behavior near boundaries 83–85
 - charges, pH-dependence 78–81, 150
 - dielectric properties 100
 - dipole momentum 56
 - electric response signals 56
 - fluctuations 76, 77
 - heat capacity 76
 - helix-formation 74, 75
 - in membranes 84–86, 89, 100
 - information contents 31–33
 - isoelectric point 79, 80
 - osmolarity 140, 141
 - prebiotic evolution 33, 35
 - radiolysis 281
 - self assembly 35, 67, 73, 74, 83
 - semiconduction 22, 23
 - stability 76
 - structure 73–77
 - structure information 33
 - volume deviations 76
 - volume compressibility 76
- Proteoglycans 209
- Proton
 - acceptor, donor 20, 61, 77
 - ATPase 24, 172
 - gradient 24
 - transport 24
- Protozoans
 - galvanotaxis 265
 - resistance to radiation 285
- Process
 - entropy driven 62–64
 - irreversible 122
 - non-linear 106, 221
- Pseudoplastic fluid 206
- Pulsed
 - electric fields 197, 198, 273
 - flow in blood circulation 225, 226
- Q**
 - Q₁₀-parameter 40
 - Quantum mechanical
 - approaches 18
 - resonance 274
 - Quantum numbers of electrons 7–9
 - Quasi equilibrium 129, 147
 - Quasi stationary state 129
- R**
 - rad (rd) 277
 - Radiation
 - accumulation of effects 282
 - biology 277–289
 - corpuscular 276
 - disease 282–284
 - effects, direct and indirect 278–282
 - lethal doses (LD) 284, 285

- mathematical model of interaction 286–289
 - protection 284–286
 - repair mechanisms 282
 - sensibility of cell cycle 282, 283
 - of various organisms 285
 - oxygen effect 280
 - stochastic effects 282, 285–289
 - syndrom 284
 - threshold dose 285
 - Radicals, free (see also ROS)
 - in metabolism 22
 - induced by radiation 278–282
 - induced by ultrasound 255
 - Radiolysis 279–281
 - Radioecology 284
 - Radio waves see: Electromagnetic fields
 - Radius
 - of bending 217
 - of gyration of a macromolecule 47
 - Random flight chain (= random coil) 46, 48, 73
 - Rate constant 307, 308, 311
 - Rayleigh's dissipation function 125, 126
 - RC-circuit 191
 - Reaction
 - advancement number 131
 - bimolecular 312
 - coordinates 41
 - enthalpy, free 113
 - entropy 113
 - isotherm (Van 't Hoff) 42
 - kinetics, biochemical 312–317
 - monomolecular 312
 - order of 312
 - rate constant 312
 - theory of absolute reaction rate 41–43
 - Reciprocity relation by Onsager 120
 - Redox
 - potential 21
 - system 22, 24
 - Reference
 - electrodes 144–146
 - intensities of sound 243
 - levels of electromagnetic field interactions 270
 - Reflection coefficient (Staverman) 138, 139, 158
 - Resilience 210, 211
 - Resistance
 - electrical 118
 - phenomenological coefficient 118
 - Resonance frequencies
 - middle ear 246
 - of human body (vibrations) 241, 242
 - of human head (vibrations) 242
 - outer ear 246
 - Resonance transfer of energy 20
 - Retaining control circuit 301
 - Reversibility, different meaning in physics, chemistry and biology 122
 - Reynolds number 221, 223
 - critical 222, 227, 228, 230
 - of animal movement 223
 - of blood circulation 227
 - Reciprocity relation (Onsager) 120
 - Rheogenic ion transport 167, 171–175, 184, 186
 - Rheoplectic viscosity behavior 206
 - Ribonucleic acid see: Desoxyribonucleic acid
 - Ringer's solution, Debye-Hückel-Parameter 66, 67
 - Roentgen (R) 277
 - ROS (reactive oxygen species) 22, 278–282
 - Rotation of molecules 45, 46
 - Rubber elasticity (=entropy elasticity) 48, 213
- S**
- Safety standards, see: Limits for exposure
 - Saltatory conduction 181
 - Salt-out effect 72
 - SAR-value (specific absorption rate) 198, 269, 272
 - Schrödinger equation 7, 8, 182, 183
 - Schulze-Hardy rule 72, 73
 - Schumann resonance 270
 - Scotopic vision 26, 286, 289
 - Second
 - moment of area 218
 - principle of thermodynamics 30, 33, 62, 121, 125, 213
 - quantum number 7
 - Sedimentation potential 95, 98
 - Self assembly 35, 67, 73, 74, 83
 - Self diffusion 60
 - Self organization 35, 327, 311
 - Semiconductor properties of biological molecules 22, 23
 - Semipermeable membrane 133, 142, 143
 - Shannon equation 30
 - Shear
 - rate (=velocity gradient) 204
 - stress 295

- Short-range interactions of particles 71
Sialic acid (=neuraminic acid) 86, 100
Sievert (Sv) 277
Silver-silver chloride electrodes 144, 145
Similarity analysis 222, 225
Single
- hit process of radiation effect 287
- strange breaks of DNA 281
Singlet state of electrons 19
Skin effect 267, 268, 272
Skin friction 228
Skin, heat conduction 238
Smoluchowski equation 96
Sonar system, see: Echolocation
Sound
- audible 243-245
- auditory threshold 243
- for orientation 251-253
- intensity 243, 244
- limit of maximal exposure 244
- pain threshold 243
- pressure 243
- protection 244, 245
Spectrin 86
Spectrum electromagnetic frequencies 256, 266
Spherics see: Atmospherics
Spillover process in photosynthesis 24
Spin
- change by external magnetic field 15
- quantum numbers 7, 8, 15, 16
- labels 87
Spin-glass models of brain 311, 312
SQUID (superconductive quantum interference device) 259
Stability criteria
- of a system 124-125, 296, 303
- of macromolecules 44, 45
Standard enthalpy
- of formation, free molar 113
- of reaction, free molar 113
State
- degenerated 19
- frustrated 331
- metastable 124, 125
- nonequilibrium 18, 122-125
- quasistationary 129
- stationary 122-125
- variables 107
Statistical mechanics 26, 36-40, 44, 46-48
Stationary
- motion 118, 223
- state 123-125
Staverman's reflection coefficient 138, 139, 158
Steady state
- equifinality 308, 310
- definition 123
- kinetics 308
- stability conditions 124, 125
Steel elasticity 213
Stefan-Boltzmann law of heat radiation 237, 238
Stern's double layer 91
Stochastic
- effect of destabilization 221
- effects of radiation 282, 285
- properties of biological systems 5, 6, 26, 78
Stokes'
- law 47, 48, 118, 119, 153
- radius 47, 48, 61
Stop-flow technique 159
Strain 210
Streaming potential 95, 98, 186
Stress, mechanical 210
Stress-strain diagram 210, 211, 214
Structure
- aperiodic 34
- biological 6, 33-36, 182
- dissipative 34
- electric, of organism 182
- in time 33
- information 33, 34
- levels of macromolecules 34
- mathematical definition 34
- of water 57-64
- periodic 34
Structure making ions 60, 61
Sun-spot activity 258
Super helix of proteins 75
Supra conductivity 23
Supraparamagnetic properties 261
Surface
- charge 86, 96, 165
- charge of cell membrane 86, 100, 101
- energy, specific 81, 82
- potential 92-94, 100
- resistance of flow 228
- tension 81, 82
Surfactants, pulmonary 82
Suspension effect of electrodes 146
Susceptibility, magnetic 257
Swimming mechanisms 230

Symbioses, models 324

Synapses 331–333

System

- analysis 291
- anisotropic 5, 62, 88, 121
- closed 107
- conservative 117
- definition 34
- dynamics 106
- isolated 30, 106
- isotropic 5, 121
- mechanical 212, 294, 295
- metastable 224, 317
- nonlinear 34, 106, 293
- of electric units 52
- open 107
- order of 293, 294
- stability 124, 125
- stationary 124, 125
- theory 291, 306

Système International d'Unités (SI) 52

Szent-Györgyi's biopotential 21, 22

T

Target

- area of radiation effect 288, 289
- theory of radiation 286–289

Temperature

- anomalies of water 57, 58
- dependence of biological processes 39–45
- gradient in cell membranes 170
- increase in HF-fields 272, 273
- optimal 236
- regulation 171, 236–238

Tendons, piezoelectric properties 23

TENS (transcutaneous electric nerve stimulation) 271

Terrestrial radiation, see: Geophysical properties of environment

Theory of

- absolute reaction rates 40–42
- biological amplification 235
- graphs 293
- similarity 222, 225
- transmission states 40–42

Thermal

- conductivity 236–238
- convection 237
- flux 236
- radiation 237, 238

Thermic

- energy 41

- effect of electromagnetic fields 269, 272–274

- excitation 19

- movement 26–29, 48, 49, 76, 91

- noise (= thermic fluctuations) 18, 41, 44, 45, 59, 65

Thermo-diffusion 236, 237

Thermodynamics

- irreversible 106, 291
- nonequilibrium 106, 291
- principles, see: Principles of thermodynamics

Thermoreceptor 264

Thixotropy 206

Thylakoid 18, 23–25

Time hierarchy in biological systems 128, 129, 292, 293, 317

Tip potential 146

Titration of ampholytes 78–80

Tons, pure 244

Torsion 219

Toxins 147

Trace elements 15, 16

Trajectory 296, 315, 323

Transcutaneous nerve stimulation (TENS) 271

Translational movement of a molecule 48–51

Transmembrane potential, see: Membrane potential

Transport

- active 131, 167–170
- electrogenic 143, 167, 171–175
- field dependent 172, 179
- frequency, of ions in membrane 168
- of ions 166–170
- of nonelectrolytes 152–159
- rheogenic 167, 171–175

Traveling wave dielectrophoresis 203

Tribology (biotribology) 209

Triplet state of electrons 19, 21

Tumor cells, membrane potential 176

Tumor development 126

Turbulent flow 220–223, 229

Turgor pressure 133, 139

U

Ubbelohde viscosimeter 207

Ultrafiltration coefficient 157

Ultrasound

- diagnostics 227, 254–255

- echolocation 251–254
- influence on biological objects 255
- penetration in tissue 253
- radical formation 255
- risks in medical diagnosis 255

Ultraviolet light

- frequency 275
- interaction with biological systems 281, 282
- quantum energy 18, 275

Unstirred (undisturbed) layer 154, 237

Unidirectional flux 161, 306

Urea

- dipole moment 56
- interaction with hydrogen bridges 281
- reflection coefficient 139

Ussing's equation 163

V

val 52

Valinomycin 174

Van der Waals

- interactions 69–73, 94
- radius 14

Van 't Hoff = s

- equation for osmotic pressure 137
- reaction isotherm 130
- rule for temperature dependence 40

Vapor pressure osmometer 138

Vector space 298

Velocity

- gradient 204
- profile of flow 204

Verhulst-Pearl law 312, 322

Vibration

- electrode 183–185
- energy of molecules 45, 46
- Vibration, mechanical
 - influence on humans 240–242
 - perceiving by humans and animals 242

Vision, sensibility 26

Viscoelasticity 210–215

Viscosimeter 207, 208

Viscosity 205

- kinematic 205
 - methods of measurement 51, 207, 208
 - of suspensions 205, 206
 - reduced 205
 - relative 205
 - specific 205
 - temperature dependence 205
- Voigt element of viscoelasticity 212

Voltage clamp 177

Voltage dependent channels, see: Transport, field dependence

Volterra cycle 322, 323

Volume

- compressibility of proteins 63
 - effective of molecules 79
 - regulation of cells 140, 142
- Vortex street behind a swimming fish 230

W

Wake behind a swimming fish 230

Water

- absorption of ultrasound 253, 254
- cavitation 254, 255
- density 57, 58
- dielectric constant (bulk) 53, 192
- dielectric constant (near surfaces) 62
- dipole moment 56, 58
- electrical dispersion frequency 192, 274
- entropy changes 52–54
- evaporation in the process of heat regulation 238
- ionization energy 21, 256, 275
- kinematic viscosity 222
- magnetic susceptibility 257
- molecular orbital 11
- radical formation at ultrasound interaction 255
- radiolysis 279–281
- structure 57–64, 132
- surface tension 81
- thermal conductivity 238
- temperature anomalies 57, 58
- velocity of sound propagation 253
- viscosity 57–59, 205, 222

Wave function of electrons 7–9

Wave mechanical approaches 7, 8, 17, 33

Weak bonds 17

Weber-Fechner's law 243

Weighting scales of sound 244, 245

Weather, sensitiveness 270

Windows effect of radiation 274

Wolff's law 215

Work differential 13, 109–111

Wound

- healing, electric field effects 126, 271
- potential 184–186, 265

X

Xanthophyll 24

X-rays 256

- Y**
Young's modulus 88, 211, 218, 226, 295
- Z**
Zeta-potential (ζ -potential) 96–98
Zwitterions 79